

TABLE 8-1 Nucleotide and Nucleic Acid Nomenclature

Base	Nucleoside	Nucleotide	Nucleic acid
Purines			
Adenine	Adenosine	Adenylate	RNA
	Deoxyadenosine	Deoxyadenylate	DNA
Guanine	Guanosine	Guanylate	RNA
	Deoxyguanosine	Deoxyguanylate	DNA
Pyrimidines			
Cytosine	Cytidine	Cytidylate	RNA
	Deoxycytidine	Deoxycytidylate	DNA
Thymine	Thymidine or deoxythymidine	Thymidylate or deoxythymidylate	DNA
Uracil	Uridine	Uridylate	RNA

Note: "Nucleoside" and "nucleotide" are generic terms that include both ribo- and deoxyribo- forms. Also, ribonucleosides and ribonucleotides are here designated simply as nucleosides and nucleotides (e.g., riboadenosine as adenosine), and deoxyribonucleosides and deoxyribonucleotides as deoxynucleosides and deoxynucleotides (e.g., deoxyriboadenosine as deoxyadenosine). Both forms of naming are acceptable, but the shortened names are more commonly used. Thymine is an exception; "ribothymidine" is used to describe its unusual occurrence in RNA.

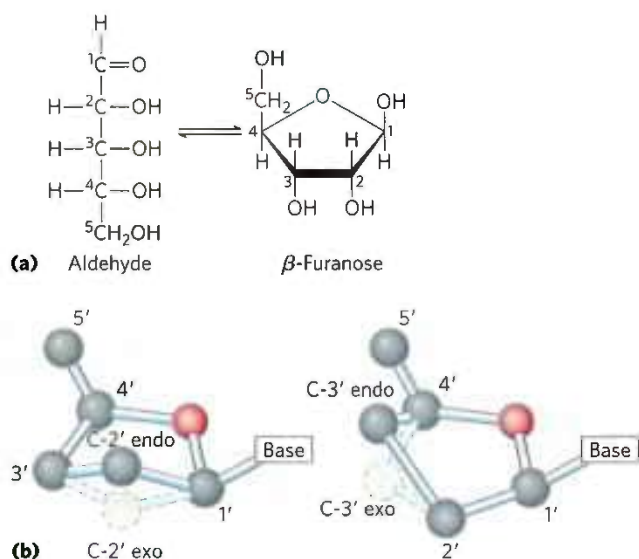


FIGURE 8-3 Conformations of ribose. (a) In solution, the straight-chain (aldehyde) and ring (β -furanose) forms of free ribose are in equilibrium. RNA contains only the ring form, β -D-ribofuranose. Deoxyribose undergoes a similar interconversion in solution, but in DNA it exists solely as β -2'-deoxy-D-ribofuranose. (b) Ribofuranose rings in nucleotides can exist in four different puckered conformations. In all cases, four of the five atoms are nearly in a single plane. The fifth atom (C-2' or C-3') is on either the same (endo) side or the opposite (exo) side of the plane relative to the C-5' atom.

Figure 8-4a gives the structures and names of the four major **deoxyribonucleotides**, the structural units of DNAs, and Figure 8-4b shows the four major **ribonucleotides**, the structural units of RNAs. Deoxyribonucleotides are also referred to as deoxyribonucleoside 5'-monophosphates, deoxynucleotides, and deoxynucleoside triphosphates; ribonucleotides are also called ribonucleoside 5'-monophosphates.

Although nucleotides bearing the major purines and pyrimidines are most common, both DNA and RNA

also contain some minor bases (Fig. 8-5). In DNA the most common of these are methylated forms of the major bases; in some viral DNAs, certain bases may be hydroxymethylated or glucosylated. Altered or unusual bases in DNA molecules often have roles in regulating or protecting the genetic information. Hundreds of different modified bases are also found in RNAs, especially in rRNAs and tRNAs (see Fig. 8-25 and Fig. 26-22). The modifications are usually introduced by enzymes that act after the RNA or DNA is synthesized.

KEY CONVENTION The nomenclature for the minor bases can be confusing. Like the major bases, many minor bases have common names — hypoxanthine, for example, shown as its nucleoside inosine in Figure 8-5. When an atom in the purine ring or the pyrimidine ring is substituted, the usual convention (used here) is simply to indicate the ring position of the substituent by its number — for example, 5-methylcytosine, 7-methylguanine, and 5-hydroxymethylcytosine (shown as the nucleosides in Fig. 8-5). The element to which the substituent is attached (N, C, O) is not identified. The convention changes when the substituted atom is exocyclic (i.e., not within the ring structure), in which case the type of atom is identified, and the ring position to which it is attached is denoted with a superscript. The amino nitrogen attached to C-6 of adenine is N^6 ; similarly, the carbonyl oxygen and amino nitrogen at C-6 and C-2 of guanine are O^6 and N^2 , respectively. Examples of this nomenclature are N^6 -methyladenosine and N^2 -methylguanosine (Fig. 8-5). ■

Cells also contain nucleotides with phosphate groups in positions other than on the 5' carbon (Fig. 8-6). **Ribonucleoside 2',3'-cyclic monophosphates** are isolatable intermediates, and **ribonucleoside 3'-monophosphates** are end products of the hydrolysis of RNA by certain ribonucleases. Other variations are adenosine 3',5'-cyclic

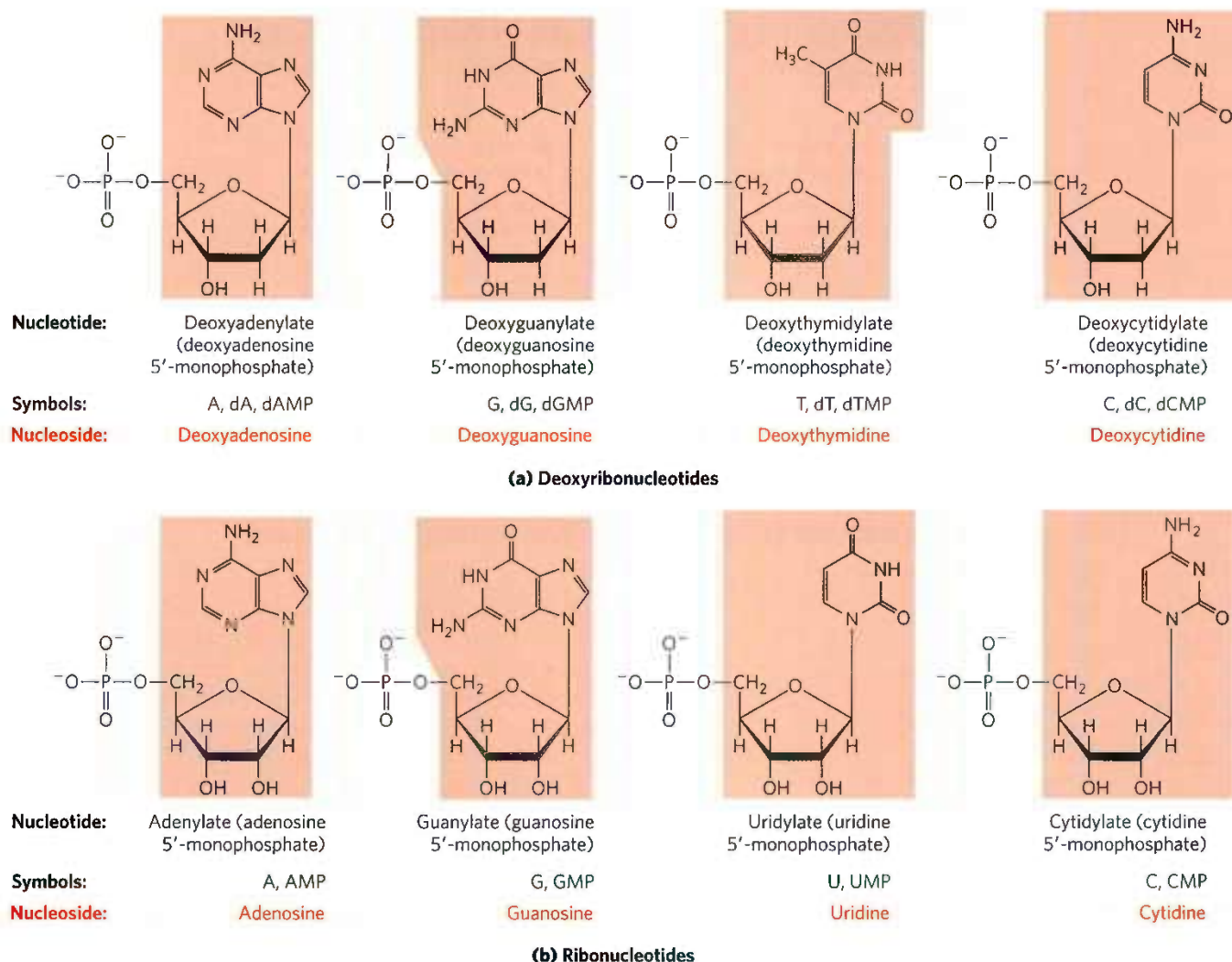


FIGURE 8-4 Deoxyribonucleotides and ribonucleotides of nucleic acids. All nucleotides are shown in their free form at pH 7.0. The nucleotide units of (a) DNA and (b) RNA are shown. For each nucleotide, the more common name is given, followed by the complete name in parentheses

and symbols used to represent them. All abbreviations assume that the phosphate group is at the 5' position. The nucleoside portion of each molecule is shaded in light red. In this and the following illustrations, the ring carbons are not shown.

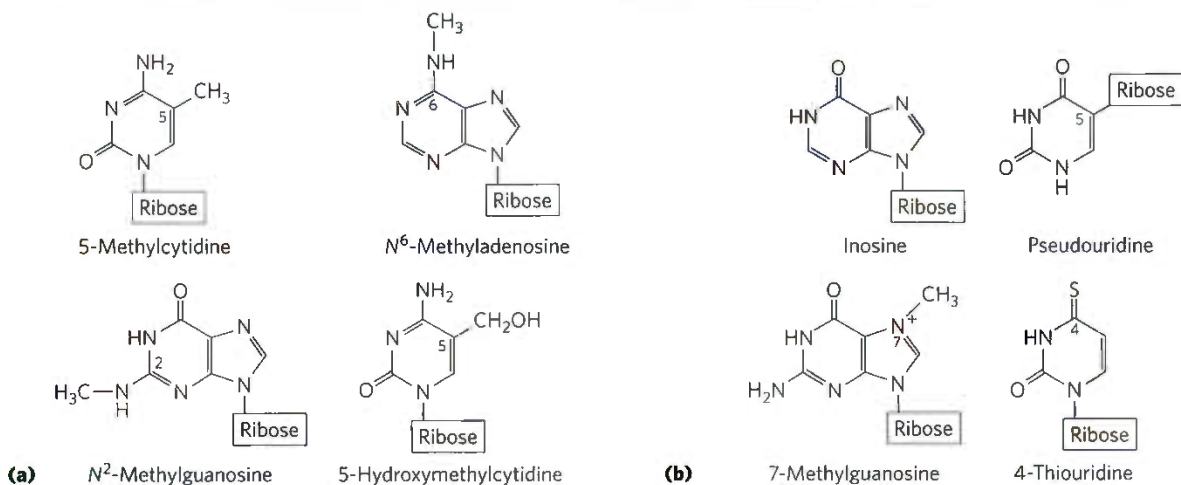


FIGURE 8-5 Some minor purine and pyrimidine bases, shown as the nucleosides. (a) Minor bases of DNA. 5-Methylcytidine occurs in the DNA of animals and higher plants, N⁶-methyladenosine in bacterial DNA, and 5-hydroxymethylcytidine in the DNA of animals and of bacteria infected with certain bacteriophages. (b) Some minor bases

of tRNAs. Inosine contains the base hypoxanthine. Note that pseudouridine, like uridine, contains uracil; they are distinct in the point of attachment to the ribose — in uridine, uracil is attached through N-1, the usual attachment point for pyrimidines; in pseudouridine, uracil is attached through C-5.

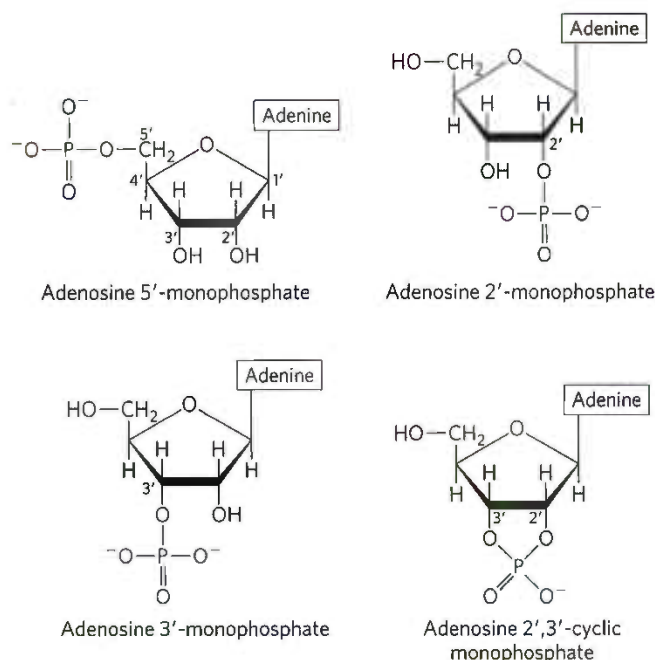


FIGURE 8-6 Some adenosine monophosphates. Adenosine 2'-monophosphate, 3'-monophosphate, and 2',3'-cyclic monophosphate are formed by enzymatic and alkaline hydrolysis of RNA.

monophosphate (cAMP) and guanosine 3',5'-cyclic monophosphate (cGMP), considered at the end of this chapter.

Phosphodiester Bonds Link Successive Nucleotides in Nucleic Acids

The successive nucleotides of both DNA and RNA are covalently linked through phosphate-group “bridges,” in which the 5'-phosphate group of one nucleotide unit is joined to the 3'-hydroxyl group of the next nucleotide, creating a **phosphodiester linkage** (Fig. 8-7). Thus the covalent backbones of nucleic acids consist of alternating phosphate and pentose residues, and the nitrogenous bases may be regarded as side groups joined to the backbone at regular intervals. The backbones of both DNA and RNA are hydrophilic. The hydroxyl groups of the sugar residues form hydrogen bonds with water. The phosphate groups, with a pK_a near 0, are completely ionized and negatively charged at pH 7, and the negative charges are generally neutralized by ionic interactions with positive charges on proteins, metal ions, and polyamines.

KEY CONVENTION All the phosphodiester linkages in DNA and RNA have the same orientation along the chain (Fig. 8-7), giving each linear nucleic acid strand a specific polarity and distinct 5' and 3' ends. By definition, the **5' end** lacks a nucleotide attached at the 5' position, and the **3' end** lacks a nucleotide attached at the 3' position. Other groups (most often one or more phosphates) may be present on one or both ends. The 5'→3' orientation of a strand of nucleic acid refers to the *ends* of the strand and the orientation of individual nucleotides, not the orientation of the individual phosphodiester bonds linking its constituent nucleotides. ■

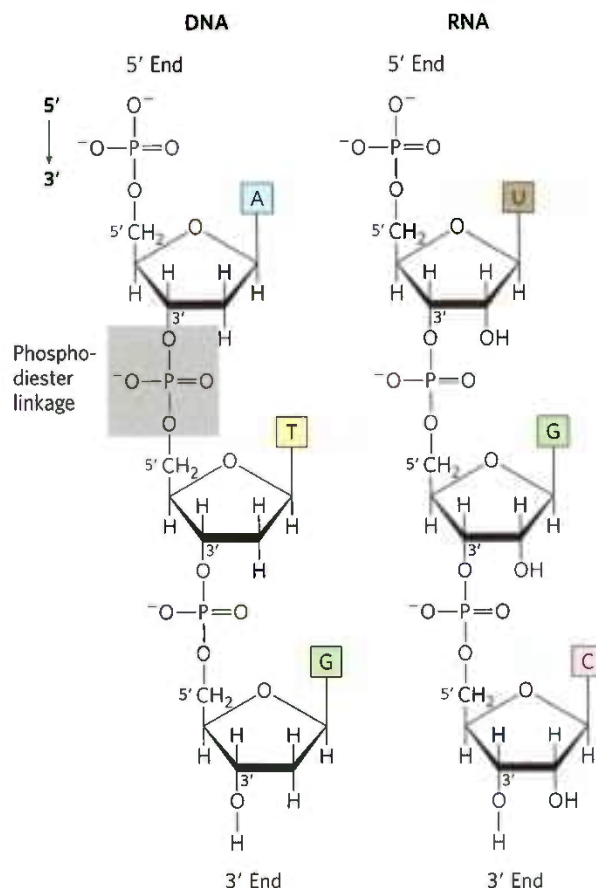
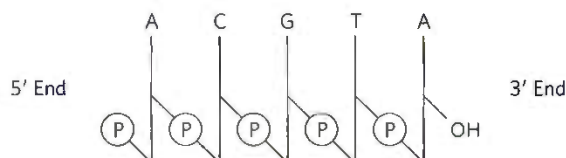


FIGURE 8-7 Phosphodiester linkages in the covalent backbone of DNA and RNA. The phosphodiester bonds (one of which is shaded in the DNA) link successive nucleotide units. The backbone of alternating pentose and phosphate groups in both types of nucleic acid is highly polar. The 5' and 3' ends of the macromolecule may be free or may have an attached phosphoryl group.

The covalent backbone of DNA and RNA is subject to slow, nonenzymatic hydrolysis of the phosphodiester bonds. In the test tube, RNA is hydrolyzed rapidly under alkaline conditions, but DNA is not; the 2'-hydroxyl groups in RNA (absent in DNA) are directly involved in the process. Cyclic 2',3'-monophosphate nucleotides are the first products of the action of alkali on RNA and are rapidly hydrolyzed further to yield a mixture of 2'- and 3'-nucleoside monophosphates (Fig. 8-8).

The nucleotide sequences of nucleic acids can be represented schematically, as illustrated below by a segment of DNA with five nucleotide units. The phosphate groups are symbolized by **P**, and each deoxyribose is symbolized by a vertical line, from C-1' at the top to C-5' at the bottom (but keep in mind that the sugar is always in its closed-ring β -furanose form in nucleic acids). The connecting lines between nucleotides (which pass through **P**) are drawn diagonally from the middle (C-3') of the deoxyribose of one nucleotide to the bottom (C-5') of the next.



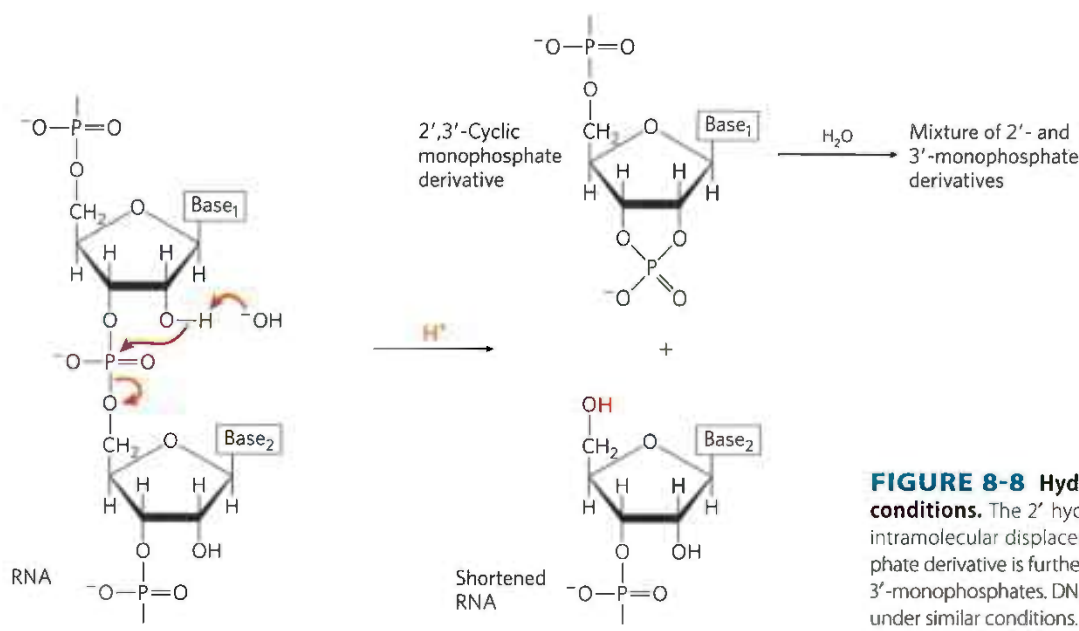


FIGURE 8-8 Hydrolysis of RNA under alkaline conditions. The 2' hydroxyl acts as a nucleophile in an intramolecular displacement. The 2',3'-cyclic monophosphate derivative is further hydrolyzed to a mixture of 2'- and 3'-monophosphates. DNA, which lacks 2' hydroxyls, is stable under similar conditions.

Some simpler representations of this pentadeoxyribonucleotide are pA-C-G-T-A_{OH}, pApCpGpTpA, and pACGTA.

KEY CONVENTION The sequence of a single strand of nucleic acid is always written with the 5' end at the left and the 3' end at the right — that is, in the 5'→3' direction. ■

A short nucleic acid is referred to as an **oligonucleotide**. The definition of “short” is somewhat arbitrary, but polymers containing 50 or fewer nucleotides are generally called oligonucleotides. A longer nucleic acid is called a **polynucleotide**.

The Properties of Nucleotide Bases Affect the Three-Dimensional Structure of Nucleic Acids

Free pyrimidines and purines are weakly basic compounds and thus are called bases. The purines and pyrimidines common in DNA and RNA are aromatic molecules (Fig. 8-2), a property with important consequences for the structure, electron distribution, and light absorption of nucleic acids. Electron delocalization among atoms in the ring gives most of the bonds in the ring partial double-bond character. One result is that pyrimidines are planar molecules and purines are very nearly planar, with a slight pucker. Free pyrimidine and purine bases may exist in two or more tautomeric forms depending on the pH. Uracil, for example, occurs in several readily interconverted forms called **tautomers**—lactam, lactim, and double lactim forms (Fig. 8-9). The structures shown in Figure 8-2 are the tautomers that predominate at pH 7.0. All nucleotide bases absorb UV light, and nucleic acids are characterized by a strong absorption at wavelengths near 260 nm (Fig. 8-10).

The purine and pyrimidine bases are hydrophobic and relatively insoluble in water at the near-neutral pH of the cell. At acidic or alkaline pH, the bases become charged

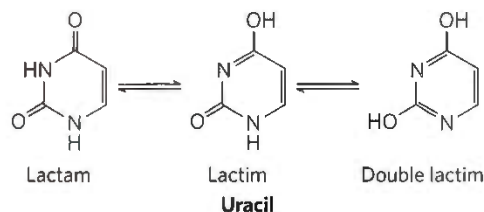


FIGURE 8-9 Tautomeric forms of uracil. The lactam form predominates at pH 7.0; the other forms become more prominent as pH decreases. The other free pyrimidines and the free purines also have tautomeric forms, but they are more rarely encountered.

and their solubility in water increases. Hydrophobic stacking interactions in which two or more bases are positioned with the planes of their rings parallel (like a stack of coins) are one of two important modes of interaction between bases in nucleic acids. The stacking also involves a combination of van der Waals and dipole-dipole interactions between the bases. Base stacking helps to minimize contact of the bases with water, and base-stacking interactions are very important in stabilizing the three-dimensional structure of nucleic acids, as described later.

The functional groups of pyrimidines and purines are ring nitrogens, carbonyl groups, and exocyclic amino groups. Hydrogen bonds involving the amino and carbonyl groups are the most important mode of complementary interaction between two (and occasionally three or four) complementary strands of nucleic acid. The most common hydrogen-bonding patterns are those defined by James D. Watson and Francis Crick in 1953, in which A bonds specifically to T (or U) and G bonds to C (Fig. 8-11). These two types of **base pairs** predominate in double-stranded DNA and RNA, and the tautomers shown in Figure 8-2 are responsible for these patterns. **P1** **P2** It is this specific pairing of bases that permits the duplication of genetic information.

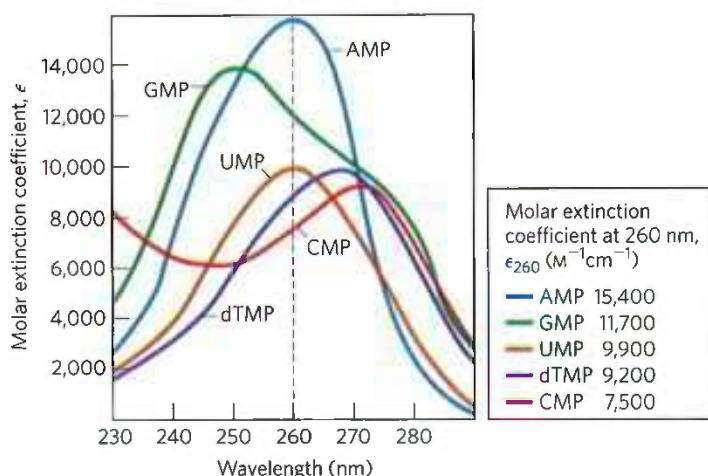
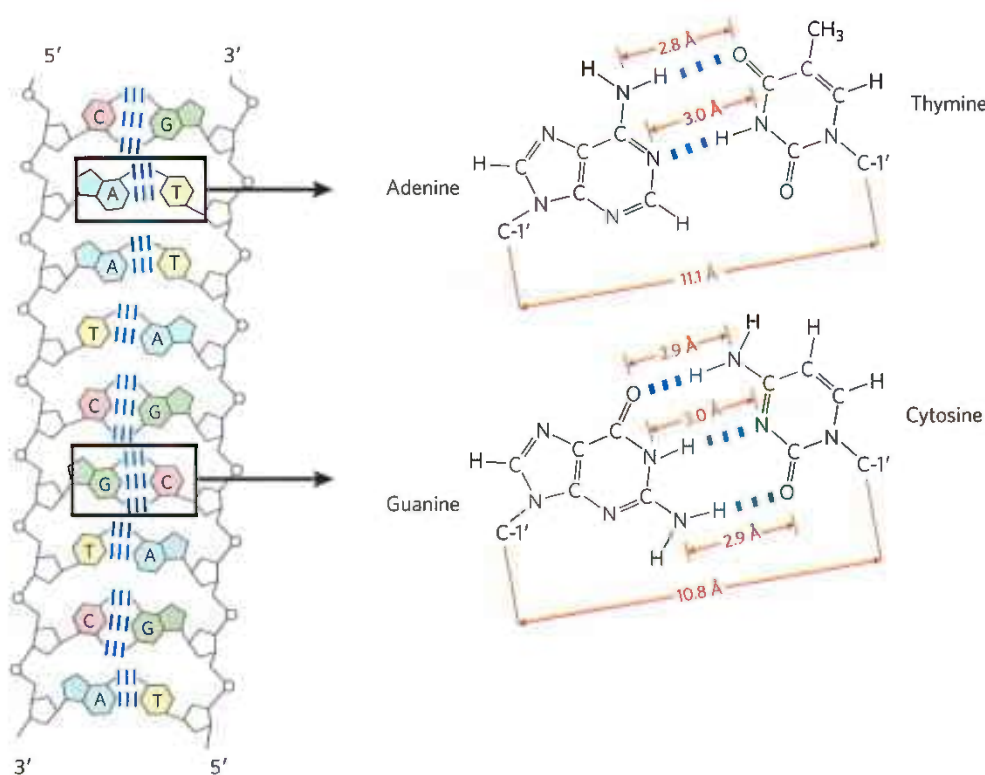


FIGURE 8-10 Absorption spectra of the common nucleotides.

The spectra are shown as the variation in molar extinction coefficient with wavelength. The molar extinction coefficients at 260 nm and pH 7.0 (ϵ_{260}) are listed in the table. The spectra of corresponding ribonucleotides and deoxyribonucleotides, as well as the nucleosides, are essentially identical. For mixtures of nucleotides, a wavelength of 260 nm (dashed vertical line) is used for absorption measurements.



James D. Watson
[Bettmann/Getty Images.]



Francis Crick, 1916–2004
[Bettmann/Getty Images.]

FIGURE 8-11 Hydrogen-bonding patterns in the base pairs defined by Watson and Crick. Here, as elsewhere, hydrogen bonds are represented by three blue lines.

SUMMARY 8.1 Some Basic Definitions and Conventions

- A nucleotide consists of a nitrogenous base (purine or pyrimidine), a pentose sugar, and one or more phosphate groups. If no phosphate is present, the combination of the pentose sugar and the nitrogenous base is a nucleoside.
- Nucleic acids are polymers of nucleotides, joined together by phosphodiester linkages between the 5'-hydroxyl group of one pentose and the 3'-hydroxyl group of the next.
- There are two types of nucleic acid: RNA and DNA. The nucleotides in RNA contain ribose, and the common

pyrimidine bases are uracil and cytosine. In DNA, the nucleotides contain 2'-deoxyribose, and the common pyrimidine bases are thymine and cytosine. The primary purines are adenine and guanine in both RNA and DNA. The nitrogenous bases have a hydrophobic character and interact via base-stacking interactions.

8.2 Nucleic Acid Structure

The discovery of the structure of DNA by Watson and Crick in 1953 gave rise to entirely new disciplines and influenced the course of many established ones. In this section we

focus on DNA structure, some of the events that led to its discovery, and more recent refinements in our understanding of DNA. We also introduce RNA structure.

As in the case of protein structure (Chapter 4), it is sometimes useful to describe nucleic acid structure in terms of hierarchical levels of complexity (primary, secondary, tertiary). The primary structure of a nucleic acid is its covalent structure and nucleotide sequence. Any regular, stable structure taken up by some or all of the nucleotides in a nucleic acid can be referred to as secondary structure. Most structures considered in the remainder of this chapter fall under the heading of secondary structure. The complex folding of large chromosomes within eukaryotic chromatin and bacterial nucleoids, or the elaborate folding of large tRNA or rRNA molecules, is generally considered tertiary structure. DNA tertiary structure is discussed in Chapter 24. RNA tertiary structure is considered briefly in this chapter and more thoroughly in Chapter 26.

DNA Is a Double Helix That Stores Genetic Information

DNA was first isolated and characterized by Friedrich Miescher in 1869. He called the phosphorus-containing substance “nuclein.” Not until the 1940s, with the work of Oswald T. Avery, Colin MacLeod, and Maclyn McCarty, was there any compelling evidence that DNA was the genetic material. Avery and his colleagues found that an extract of a virulent strain of the bacterium *Streptococcus pneumoniae* (causing disease in mice) could be used to transform a nonvirulent strain of the same bacterium into a virulent strain. They were able to demonstrate through various chemical tests that it was DNA from the virulent strain (not protein, polysaccharide, or RNA, for example) that carried the genetic information for virulence. Then in 1952, experiments by Alfred D. Hershey and Martha Chase—in which they studied the infection of bacterial cells by a virus (bacteriophage) with radioactively labeled DNA or protein—removed any remaining doubt that DNA, not protein, carried the genetic information.

Another important clue to the structure of DNA came from the work of Erwin Chargaff and his colleagues in the late 1940s. Examining dozens of species, they found that the four nucleotide bases of DNA occur in different ratios in the DNAs of different organisms. However, the base composition remains constant in different tissues of the same species, and does not vary with age, environment, nutritional state, or generation. Furthermore, regardless of the species, the number of adenosine residues is equal to the number of thymidine residues (that is, $A = T$), and the number of guanosine residues is equal to the number of cytidine residues ($G = C$). From these relationships it follows that the sum of the purine residues equals the sum of the pyrimidine residues; that is, $A + G = T + C$. These quantitative relationships, sometimes called “Chargaff’s rules,” were a key to establishing the three-dimensional structure of DNA.

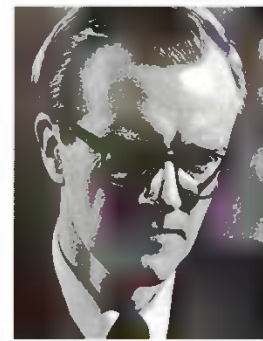


FIGURE 8-12 X-ray diffraction pattern of DNA fibers. The spots forming a cross in the center denote a helical structure. The heavy bands at the left and the right arise from the recurring bases. [Science Source.]

To shed more light on the structure of DNA, in the early 1950s Rosalind Franklin and Maurice Wilkins used the powerful method of x-ray diffraction (see Fig. 4-30) to analyze DNA fibers. Although lacking the molecular definition of diffraction from crystals, the x-ray diffraction pattern generated from the fibers was informative (Fig. 8-12). The pattern revealed that DNA molecules are helical, with two periodicities along their long axis: a primary one of 3.4 Å and a secondary one of 34 Å. The problem then was to formulate a three-dimensional model of the DNA molecule that could account not only for the x-ray diffraction data but also for the specific $A = T$ and $G = C$ base equivalences discovered by Chargaff and for the other chemical properties of DNA.



Rosalind Franklin, 1920–1958
[National Library of Medicine/Science Source]



Maurice Wilkins, 1916–2004
[Bettmann/Getty Images]

James Watson and Francis Crick relied on this accumulated information about DNA to set about deducing its structure. In 1953 they postulated a three-dimensional model of DNA structure that accounted for all the available data. It consists of two helical DNA chains wound around the same axis to form a right-handed double helix. (See Box 4-1 for an explanation of the right- or left-handed sense of a helical structure.) The hydrophilic backbones of alternating deoxyribose and phosphate groups are on the outside of the double helix, facing the surrounding water. The furanose ring of each deoxyribose

is in the C-2' endo conformation. The purine and pyrimidine bases of both strands are stacked inside the double helix, with their hydrophobic and nearly planar ring structures very close together and perpendicular to the long axis. The offset pairing of the two strands creates a **major groove** and a **minor groove** on the surface of the duplex (Fig. 8-13). Each nucleotide base of one strand is paired in the same plane with a base of the other strand. Watson and Crick found that the hydrogen-bonded base pairs illustrated in Figure 8-11, G with C and A with T, are those that fit best within the structure, providing a rationale for Chargaff's rule that in any DNA, $G = C$ and $A = T$. It is important to note that three hydrogen bonds can form between G and C, symbolized $G \equiv C$, but only two can form between A and T, symbolized $A = T$. Pairings of bases other than G with C and A with T tend (to varying degrees) to destabilize the double-helical structure.

When Watson and Crick constructed their model, they had to decide at the outset whether the strands of DNA should be **parallel** or **antiparallel**—whether their 3',5'-phosphodiester bonds should run in the same or opposite directions. An antiparallel orientation produced the most convincing model, and later work with DNA polymerases (Chapter 25) provided experimental evidence that the strands are indeed antiparallel, a finding ultimately confirmed by x-ray analysis.

To account for the periodicities observed in the x-ray diffraction patterns of DNA fibers, Watson and Crick

manipulated molecular models to arrive at a structure in which the vertically stacked bases inside the double helix would be 3.4 Å apart; the secondary repeat distance of about 34 Å was accounted for by the presence of 10 base pairs (bp) in each complete turn of the double helix. The structure in aqueous solution differs slightly from that in fibers, having 10.5 bp per helical turn (Fig. 8-13).

As Figure 8-14 shows, the two antiparallel polynucleotide chains of double-helical DNA are not identical in either base sequence or composition. Instead they are **complementary** to each other. Wherever adenine occurs in one chain, thymine is found in the other; similarly, wherever guanine occurs in one chain, cytosine is found in the other.

The DNA double helix, or duplex, is held together by hydrogen bonding between complementary base pairs (Fig. 8-11) and by base-stacking interactions. The complementarity between the DNA strands is attributable to the hydrogen bonding between base pairs; however, the hydrogen bonds do not contribute significantly to the stability of the structure. The double helix is primarily stabilized by metal cations, which shield the negative charges of backbone phosphates, and by base-stacking interactions between successive base pairs. Base-stacking interactions between successive $G \equiv C$ or $C \equiv G$ pairs are stronger than those between successive $A = T$ and $T = A$ pairs or adjacent pairs including all four bases. Because of this, DNA duplexes with higher $G \equiv C$ content are more stable.

The important features of the double-helical model of DNA structure are now supported by much chemical and biological evidence. Moreover, the model immediately suggested a mechanism for the transmission of

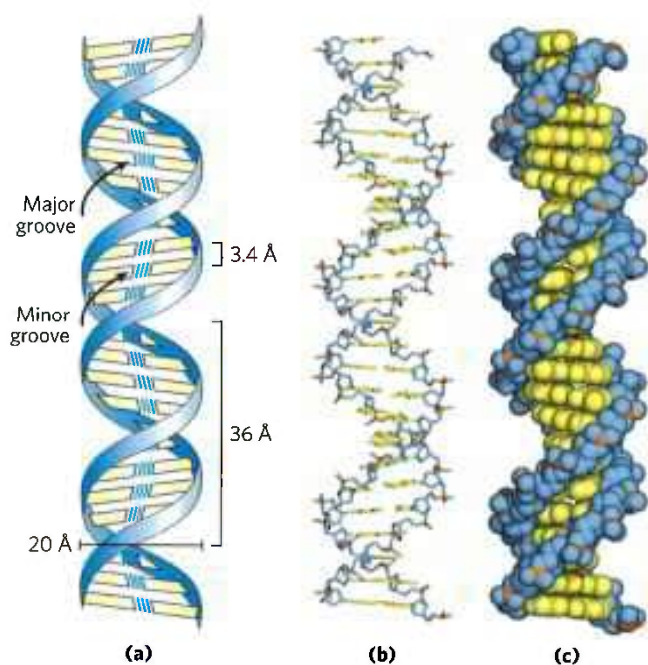


FIGURE 8-13 Watson-Crick model for the structure of DNA. The original model proposed by Watson and Crick had 10 bp, or 34 Å (3.4 nm), per turn of the helix; subsequent measurements revealed 10.5 bp, or 36 Å (3.6 nm), per turn. (a) Schematic representation, showing dimensions of the helix. (b) Stick representation showing the backbone and stacking of the bases. (c) Space-filling model.

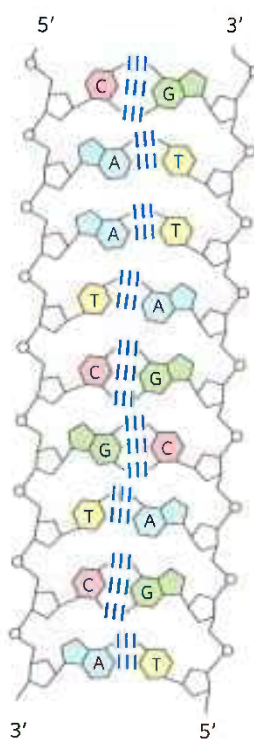


FIGURE 8-14 Complementarity of strands in the DNA double helix.

The complementary antiparallel strands of DNA follow the pairing rules proposed by Watson and Crick. The base-paired antiparallel strands differ in base composition: the left strand has the composition $A_3T_2G_1C_3$; the right strand has $A_2T_3G_3C_1$. They also differ in sequence when each chain is read in the 5'→3' direction. Note the base equivalences: $A = T$ and $G = C$ in the duplex.

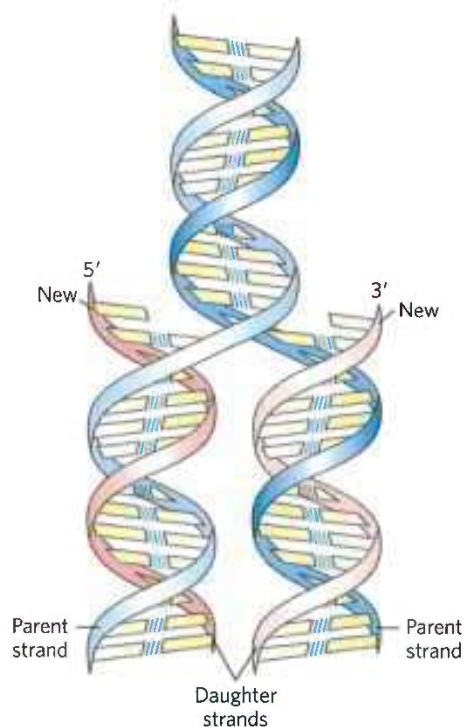


FIGURE 8-15 Replication of DNA as suggested by Watson and Crick. The preexisting or "parent" strands become separated, and each is the template for biosynthesis of a complementary "daughter" strand (in pink).

genetic information. The essential feature of the model was the complementarity of the two DNA strands. As Watson and Crick were able to see, well before confirmatory data became available, this structure could logically be replicated by (1) separating the two strands and (2) synthesizing a complementary strand for each.

P1 **P2** Because nucleotides in each new strand are joined in a sequence specified by the base-pairing rules stated above, each preexisting strand functions as a template to guide the synthesis of one complementary strand (**Fig. 8-15**). These expectations were experimentally confirmed, inaugurating a revolution in our understanding of biological inheritance.

DNA Can Occur in Different Three-Dimensional Forms

DNA is a remarkably flexible molecule. Considerable rotation is possible around several types of bonds in the sugar-phosphate (phosphodeoxyribose) backbone, and thermal fluctuation can produce bending, stretching, and unpairing (melting) of the strands. Many significant deviations from the Watson-Crick DNA structure are found in cellular DNA, some or all of which may be important in DNA metabolism. These structural variations generally do not affect the key properties of DNA defined by Watson and Crick: strand complementarity, antiparallel strands, and the requirement for $A=T$ and $G\equiv C$ base pairs.

Structural variation in DNA reflects three things: the different possible conformations of the deoxyribose, rotation about the contiguous bonds that make up the phosphodeoxyribose backbone (**Fig. 8-16a**), and free

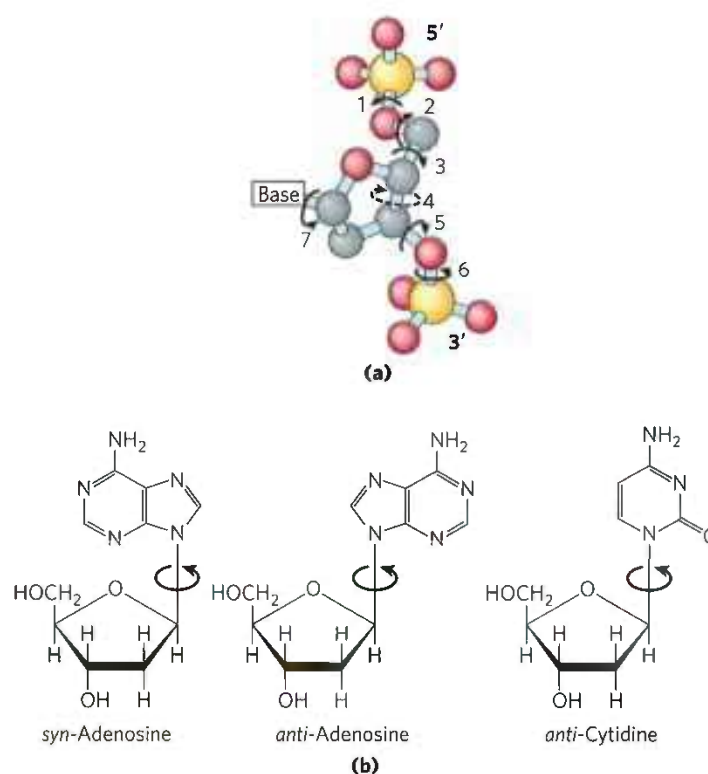
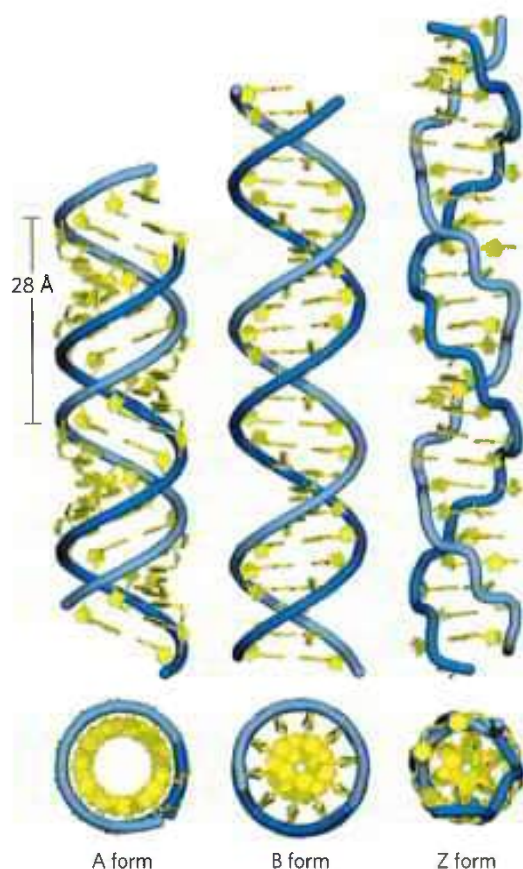


FIGURE 8-16 Structural variation in DNA. (a) The conformation of a nucleotide in DNA is affected by rotation about seven different bonds. Six of the bonds rotate freely. The limited rotation about bond 4 gives rise to ring pucker. This conformation is endo or exo, depending on whether the atom is displaced to the same side of the plane as C-5' or to the opposite side (see Fig. 8-3b). (b) For purine bases in nucleotides, only two conformations with respect to the attached ribose units are sterically permitted: anti or syn. Pyrimidines occur in the anti conformation.

rotation about the C-1'-N-glycosyl bond (**Fig. 8-16b**). Because of steric constraints, purines in purine nucleotides are restricted to two stable conformations with respect to deoxyribose, called syn and anti (**Fig. 8-16b**). Pyrimidines are generally restricted to the anti conformation because of steric interference between the sugar and the carbonyl oxygen at C-2 of the pyrimidine.

The Watson-Crick structure is also referred to as **B-form DNA**, or B-DNA. The B form is the most stable structure for a random-sequence DNA molecule under physiological conditions and is therefore the standard point of reference in any study of the properties of DNA. Two structural variants that have been well characterized in crystal structures are the **A** and **Z forms**. These three DNA conformations are shown in **Figure 8-17**, with a summary of their properties. The A form is favored in many solutions that are relatively devoid of water. The DNA is still arranged in a right-handed double helix, but the helix is wider and the number of base pairs per helical turn is 11, rather than 10.5 as in B-DNA. The plane of the base pairs in A-DNA is tilted about 20° relative to B-DNA base pairs, thus the base pairs in A-DNA are not perfectly perpendicular to the helix axis. These structural changes deepen the major groove while making the minor groove



	A form	B form	Z form
Helical sense	Right-handed	Right-handed	Left-handed
Diameter	~26 Å	~20 Å	~18 Å
Base pairs per helical turn	11	10.5	12
Helix rise per base pair	2.6 Å	3.4 Å	3.7 Å
Base tilt normal to the helix axis	20°	6°	7°
Sugar pucker conformation	C-3' endo	C-2' endo	C-2' endo for pyrimidines; C-3' endo for purines
Glycosyl bond conformation	Anti	Anti	Anti for pyrimidines; syn for purines

FIGURE 8-17 Comparison of A, B, and Z forms of DNA. Each structure shown here has 36 bp. The riboses and bases are shown in yellow. The phosphodiester backbone is represented as a blue rope. Blue is the color used to represent DNA strands in later chapters. The table summarizes some properties of the three forms of DNA.

shallower. The reagents used to promote crystallization of DNA tend to dehydrate it, and thus most short DNA molecules tend to crystallize in the A form.

Z-form DNA is a more radical departure from the B structure; the most obvious distinction is the left-handed helical rotation. There are 12 bp per helical turn, and the structure appears more slender and elongated. The DNA backbone takes on a zigzag appearance. Certain nucleotide sequences fold into left-handed Z helices much more readily than others. Prominent examples are sequences in which pyrimidines alternate with purines, especially alternating C and G (that is, in the helix, alternating C≡G and G≡C pairs) or 5-methyl-C and G residues. To form the left-handed helix in Z-DNA, the purine residues flip to the syn conformation, alternating with pyrimidines in the anti conformation. The major groove is barely apparent in Z-DNA, and the minor groove is narrow and deep.

Whether A-DNA occurs in cells is uncertain, but there is evidence for some short stretches (tracts) of Z-DNA in both bacteria and eukaryotes. These Z-DNA tracts may play a role (as yet undefined) in regulating the expression of some genes or in genetic recombination.

Certain DNA Sequences Adopt Unusual Structures

Other sequence-dependent structural variations found in larger chromosomes may affect the function and metabolism of the DNA segments in their immediate vicinity. For example, bends occur in the DNA helix wherever four

or more adenosine residues appear sequentially in one strand. Six adenosines in a row produce a bend of about 18°. The bending observed with this and other sequences may be important in the binding of some proteins to DNA.

A common type of DNA sequence is a **palindrome**. A palindrome is a word, phrase, or sentence that is spelled identically when read either forward or backward; two examples are ROTATOR and NURSES RUN. In DNA, the term is applied to regions of DNA with **inverted repeats**, such that an inverted, self-complementary sequence in one strand is repeated in the opposite orientation in the paired strand, as in **Figure 8-18**. The self-complementarity within each strand confers the potential to form **hairpin** or **cruciform** (cross-shaped) structures (**Fig. 8-19**). When the inverted repeat occurs within each individual strand of the DNA, the sequence is called a **mirror repeat**. Mirror repeats do not have complementary sequences within the same strand and thus cannot form hairpin or cruciform structures. Sequences of these types are found in almost every large DNA molecule and can encompass a few base pairs or thousands. The extent to which palindromes occur as cruciforms in cells is not known, although some cruciform structures have been demonstrated in vivo in *Escherichia coli*. Self-complementary sequences cause isolated single strands of DNA (or RNA) in solution to fold into complex structures containing multiple hairpins.

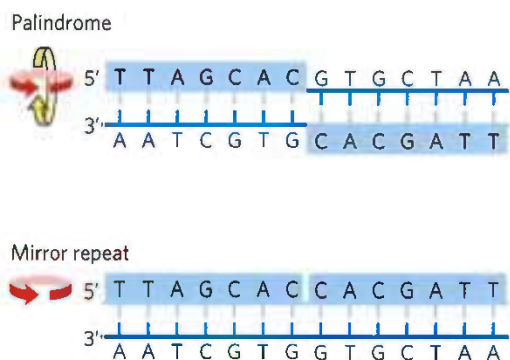


FIGURE 8-18 Palindromes and mirror repeats. Palindromes are sequences of double-stranded nucleic acids with twofold symmetry. To superimpose one repeat (shaded sequence) on the other, it must be rotated 180° about the horizontal axis and then 180° about the vertical axis, as shown by the colored arrows. A mirror repeat, on the other hand, has a symmetric sequence within each strand. Superimposing one repeat on the other requires only a single 180° rotation about the vertical axis.

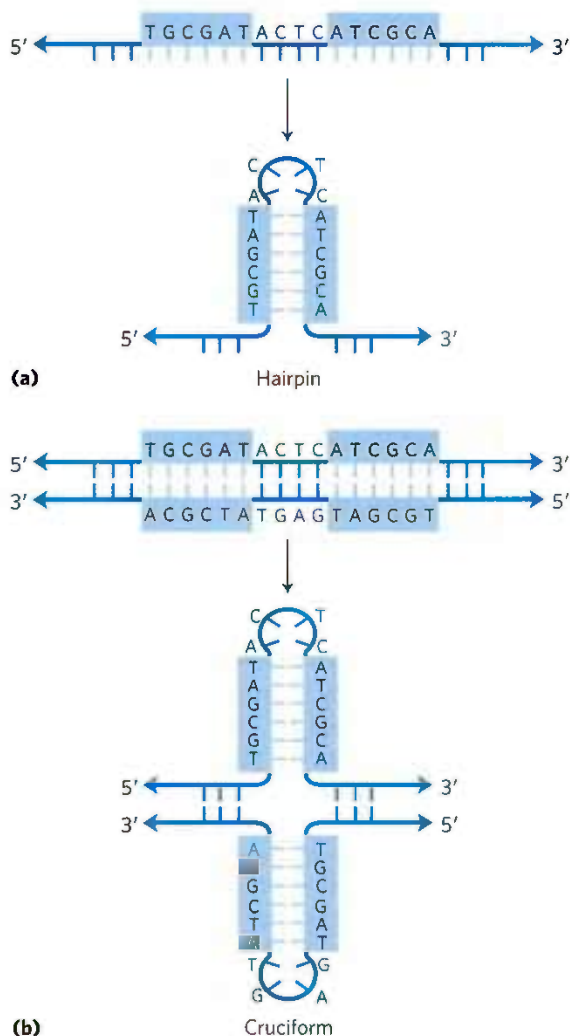


FIGURE 8-19 Hairpins and cruciforms. Palindromic DNA (or RNA) sequences can form alternative structures with intrastrand base pairing. (a) Hairpin structures involve a single DNA or RNA strand. (b) Cruciform structures involve both strands of a duplex DNA. Blue shading highlights asymmetric sequences that can pair with the complementary sequence either in the same strand or in the complementary strand.

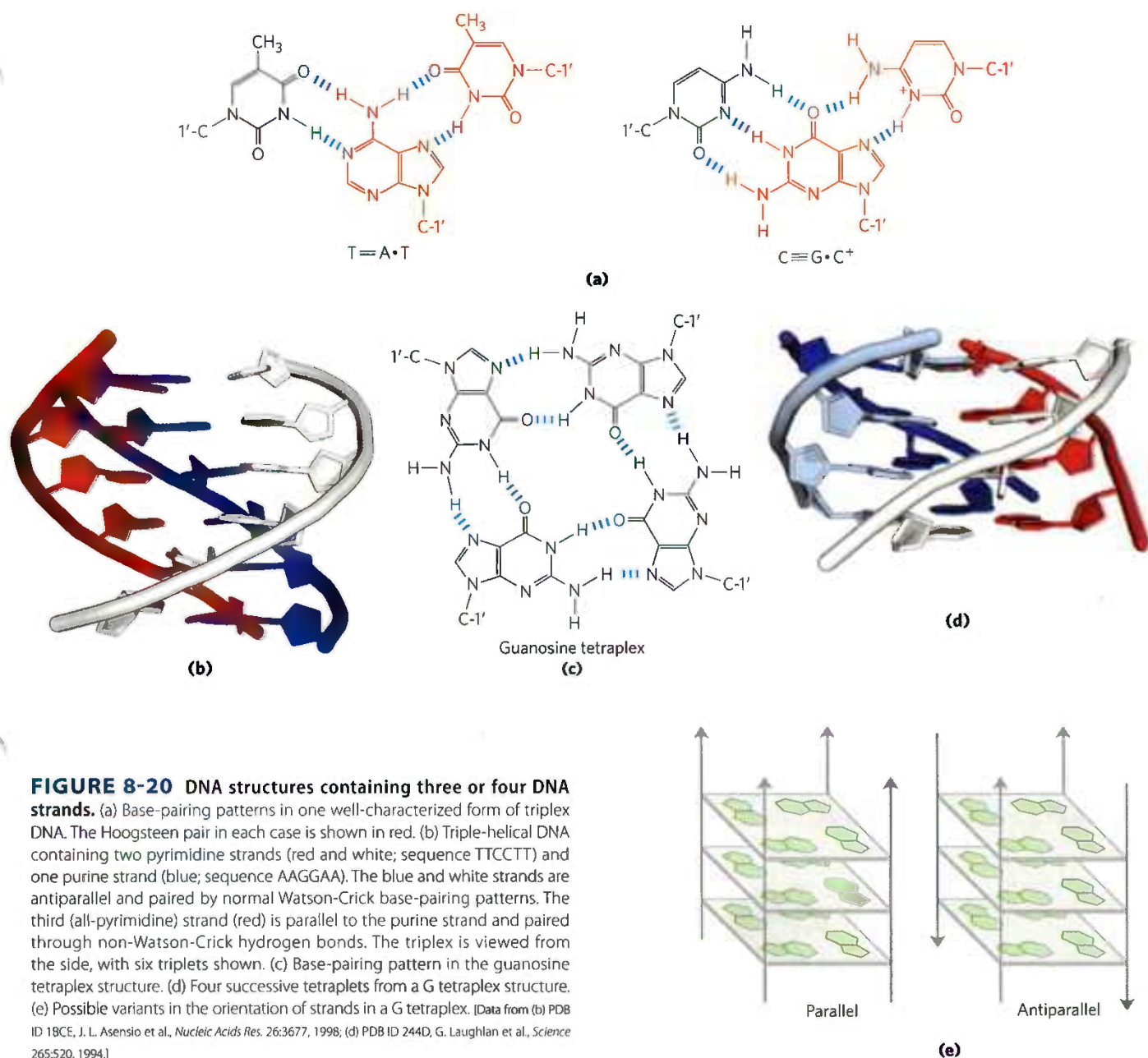
Several unusual DNA structures are formed from three or even four DNA strands. Nucleotides participating in a Watson-Crick base pair (Fig. 8-11) can form additional hydrogen bonds with a third strand, particularly with functional groups arrayed in the major groove. For example, the guanosine residue of a $G \equiv C$ nucleotide pair can pair with a cytidine residue (if protonated) on a third strand (Fig. 8-20a); the adenosine of an $A = T$ pair can pair with a thymidine residue. The N-7, O⁶, and N⁶ of purines, the atoms that participate in the hydrogen bonding with a third DNA strand, are often referred to as **Hoogsteen positions**, and the non-Watson-Crick pairing is called **Hoogsteen pairing**, after Karst Hoogsteen, who in 1963 first recognized the potential for these unusual pairings. Hoogsteen pairing allows the formation of **triplex DNAs**. The triplexes shown in Figure 8-20 (a, b) are most stable at low pH because the $C \equiv G \cdot C^+$ triplet requires a protonated cytosine. In the triplex, the pK_a of this cytosine is >7.5, altered from its normal value of 4.2. The triplexes also form most readily within long sequences containing only pyrimidines or only purines in a given strand. Some triplex DNAs contain two pyrimidine strands and one purine strand; others contain two purine strands and one pyrimidine strand.

Four DNA strands can also pair to form a tetraplex (quadruplex), but this occurs readily only for DNA sequences with a very high proportion of guanosine residues (Fig. 8-20c, d). The guanosine tetraplex, or **G tetraplex**, is quite stable over a broad range of conditions. The orientation of strands in the tetraplex can vary as shown in Figure 8-20e.

In the DNA of living cells, sites recognized by many sequence-specific DNA-binding proteins (Chapter 28) are arranged as palindromes, and polypyrimidine or polypurine sequences that can form triple helices are found within regions involved in the regulation of expression of some eukaryotic genes.

Messenger RNAs Code for Polypeptide Chains

We now turn our attention to the expression of the genetic information that DNA contains. Given that the DNA of eukaryotes is largely confined to the nucleus, whereas protein synthesis occurs on ribosomes in the cytoplasm, some molecule other than DNA must carry the genetic message from the nucleus to the cytoplasm. As early as the 1950s, RNA was considered the logical candidate: RNA is found in both the nucleus and the cytoplasm, and an increase in protein synthesis is accompanied by an increase in the amount of cytoplasmic RNA and an increase in its rate of turnover. These and other observations led several researchers to suggest that RNA carries genetic information from DNA to the protein-synthesizing machinery of the ribosome. In 1961, François Jacob and Jacques Monod presented a unified (and essentially correct) picture of many aspects of this process. They proposed the name “messenger RNA” (mRNA) for that portion of the total



cellular RNA carrying the genetic information from DNA to the ribosomes. **P1** **P2** The mRNAs are formed on a DNA template by the process of **transcription**. Once they reach the ribosomes, the messengers provide the templates that specify amino acid sequences in polypeptide chains. Although mRNAs from different genes can vary greatly in length, the mRNAs from a particular gene generally have a defined size.

In bacteria and archaea, a single mRNA molecule may code for one or several polypeptide chains. If it carries the code for only one polypeptide, the mRNA is **monocistronic**; if it codes for two or more different polypeptides, the mRNA is **polycistronic**. In eukaryotes, most mRNAs are monocistronic. (For the purposes of this discussion, "cistron" refers to a gene. The term itself

has historical roots in the science of genetics, and its formal genetic definition is beyond the scope of this text.) The minimum length of an mRNA is set in part by the length of the polypeptide chain for which it codes. For example, a polypeptide chain of 100 amino acid residues requires an RNA coding sequence of at least 300 nucleotides, because each amino acid is coded by a nucleotide triplet (this and other details of protein synthesis are discussed in Chapter 27). However, mRNAs transcribed from DNA are always somewhat longer than the length needed simply to code for a polypeptide sequence (or sequences). The additional, noncoding RNA includes sequences required to begin and end translation by the ribosome, as well as regulatory sequences. **Figure 8-21** summarizes the general structure of bacterial mRNAs.

nearby self-complementary (palindromic) sequences. Extensive base-paired helical segments are formed in many RNAs (**Fig. 8-24**), and the resulting hairpins are the most common type of secondary structure in RNA. Specific short base sequences (such as UUCG) are often found at the ends of RNA hairpins and are known to form particularly tight and stable loops. Such sequences may act as starting points for the folding of an RNA molecule into its precise three-dimensional structure. Other

contributions are made by hydrogen bonds that are not part of standard Watson-Crick base pairs. For example, the 2'-hydroxyl group of ribose can hydrogen-bond with other groups. Some of these properties are evident in the tertiary structure of the phenylalanine transfer RNA of yeast—the tRNA responsible for inserting Phe residues into polypeptides—and in two RNA enzymes, or ribozymes, whose functions, like those of protein enzymes, depend on their three-dimensional structures (**Fig. 8-25**).

The analysis of RNA structure and the relationship between its structure and its function remains a robust field of inquiry that has many of the same complexities as the analysis of protein structure. The importance of understanding RNA structure grows as we become increasingly aware of the large number of functional roles for RNA molecules.

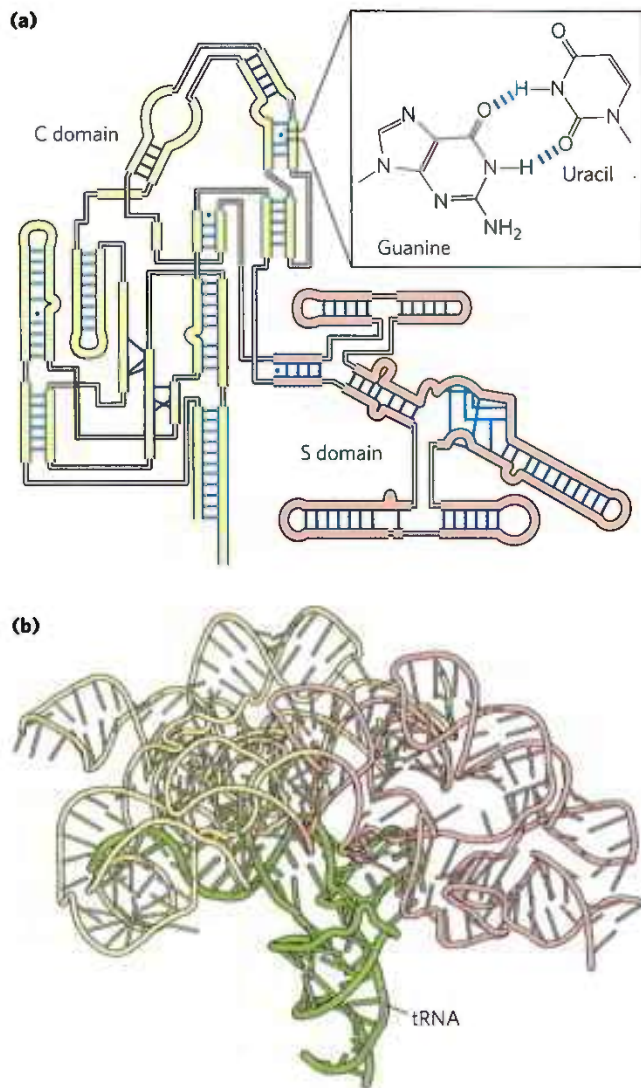


FIGURE 8-24 Base-paired helical structures in an RNA. Shown here are (a) the secondary structure and (b) the three-dimensional structure of the P RNA component of the RNase P of *Thermotoga maritima*. RNase P, which also contains a protein component (not shown), functions in the processing of transfer RNAs. A complexed tRNA is also shown in (b). Separate C (catalytic) and S (specificity) domains are denoted with yellow and light red backbones in both images. The blue dots in (a) indicate non-Watson-Crick G–U base pairs (boxed inset). Note that G–U base pairs are allowed only when presynthesized strands of RNA fold up or anneal with each other. [(a) Information from N. J. Reiter et al., *Nature* 468:784, 2010, Fig. 2a. (b) Data from PDB ID 3Q1R, N. J. Reiter et al., *Nature* 468:784, 2010.]

SUMMARY 8.2 Nucleic Acid Structure

- Many lines of evidence show that DNA bears genetic information. Some of the earliest evidence came from the Avery-MacLeod-McCarty experiment, which showed that DNA isolated from one bacterial strain can enter and transform the cells of another strain, endowing the second strain with some of the inheritable characteristics of the donor. The Hershey-Chase experiment showed that the DNA of a bacterial virus, but not its protein coat, carries the genetic message for replication of the virus in a host cell.
- Putting together the available data, Watson and Crick postulated that native DNA consists of two antiparallel chains in a right-handed double-helical arrangement. Complementary base pairs, A=T and G=C, are formed by hydrogen bonding between chains in the helix. The base pairs are stacked perpendicular to the long axis of the double helix, 3.4 Å apart, with 10.5 bp per turn.
- DNA can exist in several structural forms. Two variations of the Watson-Crick form, or B-DNA, are A- and Z-DNA.
- Some sequence-dependent structural variations cause bends in the DNA molecule. DNA strands with appropriate sequences can form hairpin or cruciform structures or triplex or tetraplex DNA.
- Messenger RNA transfers genetic information from DNA to ribosomes for protein synthesis.
- Transfer RNA and ribosomal RNA are also involved in protein synthesis. RNA can be structurally complex; single RNA strands can fold into hairpins, double-stranded regions, or complex loops. Additional noncoding RNAs have a variety of special functions.

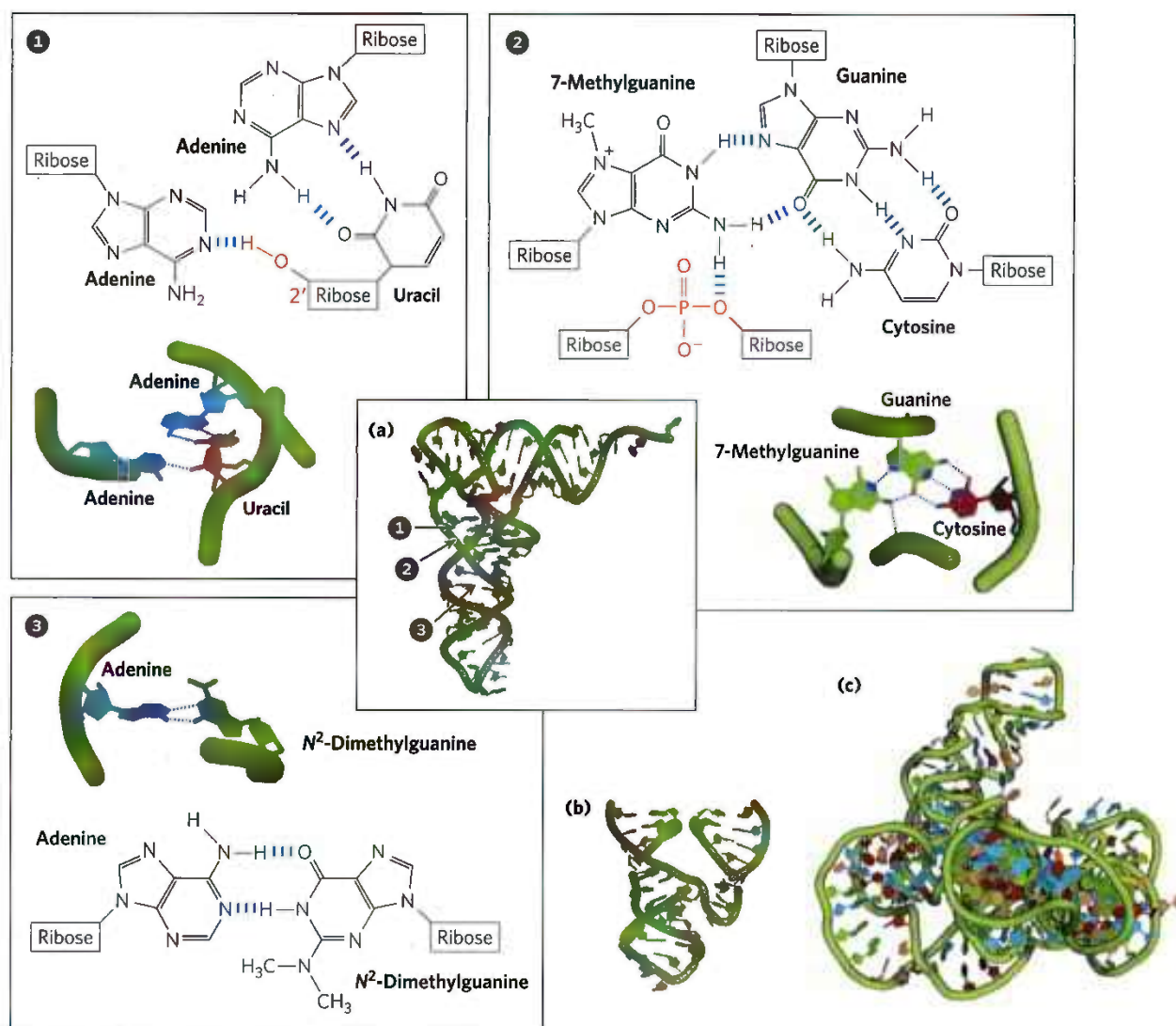


FIGURE 8-25 Three-dimensional structure in RNA. (a) Three-dimensional structure of phenylalanine tRNA of yeast. Some unusual base-pairing patterns found in this tRNA are shown in the numbered insets. Note in **1** a hydrogen bond with a ribose 2'-hydroxyl group and in **2** a hydrogen bond with the oxygen of a ribose phosphodiester (both shown in red). (b) A hammerhead ribozyme (so named because the secondary structure at the active site looks like the head of a hammer), derived from certain plant viruses. Ribozymes, or RNA enzymes, catalyze a variety of

reactions, primarily in RNA metabolism and protein synthesis. The complex three-dimensional structures of these RNAs reflect the complexity inherent in catalysis, as described for protein enzymes in Chapter 6. (c) A segment of mRNA known as an intron, from the ciliated protozoan *Tetrahymena thermophila*. This intron (a ribozyme) catalyzes its own excision from between exons in an mRNA strand (discussed in Chapter 26). [Data from (a) PDB ID 1TRA, E. Westhof and M. Sundaralingam, *Biochemistry* 25:4868, 1986; (b) PDB ID 1MME, W. G. Scott et al., *Cell* 81:991, 1995; (c) PDB ID 1GRZ, B. L. Golden et al., *Science* 282:259, 1998.]

8.3 Nucleic Acid Chemistry

P1 P3 The role of DNA as a repository of genetic information depends in part on its inherent stability. The chemical transformations that do occur are generally very slow in the absence of an enzyme catalyst. The long-term storage of information without alteration is so important to a cell, however, that even very slow reactions that alter DNA structure can be physiologically significant. Processes such as carcinogenesis and aging may be intimately linked to slowly accumulating, irreversible alterations of DNA. Other, nondestructive alterations also occur and are essential to function, such as the strand separation that must precede DNA replication or transcription. In addition to

providing insights into physiological processes, our understanding of nucleic acid chemistry has given us a powerful array of technologies that have applications in molecular biology, medicine, agriculture, and forensic science. We now examine the chemical properties of DNA and a few of these technologies.

Double-Helical DNA and RNA Can Be Denatured

Solutions of carefully isolated, native DNA are highly viscous at pH 7.0 and room temperature (25 °C). When such a solution is subjected to extremes of pH or to temperatures above 80 °C, its viscosity decreases sharply, indicating that the DNA has undergone a physical change. Just

as heat and extremes of pH denature globular proteins, they also cause denaturation, or melting, of double-helical DNA. Disruption of the hydrogen bonds between paired bases and of base-stacking interactions causes unwinding of the double helix to form two single strands, completely separate from each other along the entire length or part of the length (partial denaturation) of the molecule. No covalent bonds in the DNA are broken (Fig. 8-26).

When the temperature or pH is returned to the range in which most organisms live, the unwound segments of the two strands spontaneously rewind, or **anneal**, to yield the intact duplex (Fig. 8-26). However, if the two strands are completely separated, renaturation occurs in two steps. In the first, relatively slow step, the two strands “find” each other by random collisions and form a short segment of complementary double helix. The second step is much faster: the remaining unpaired bases successively come into register as base pairs, and the two strands “zipper” themselves together to form the double helix.

The close interaction between stacked bases in a nucleic acid has the effect of decreasing its absorption of UV light relative to that of a solution with the same concentration of free nucleotides, and the absorption is decreased further when two complementary nucleic acid strands are paired. This is called the **hypochromic effect**. Denaturation of a double-stranded nucleic acid produces the opposite result:

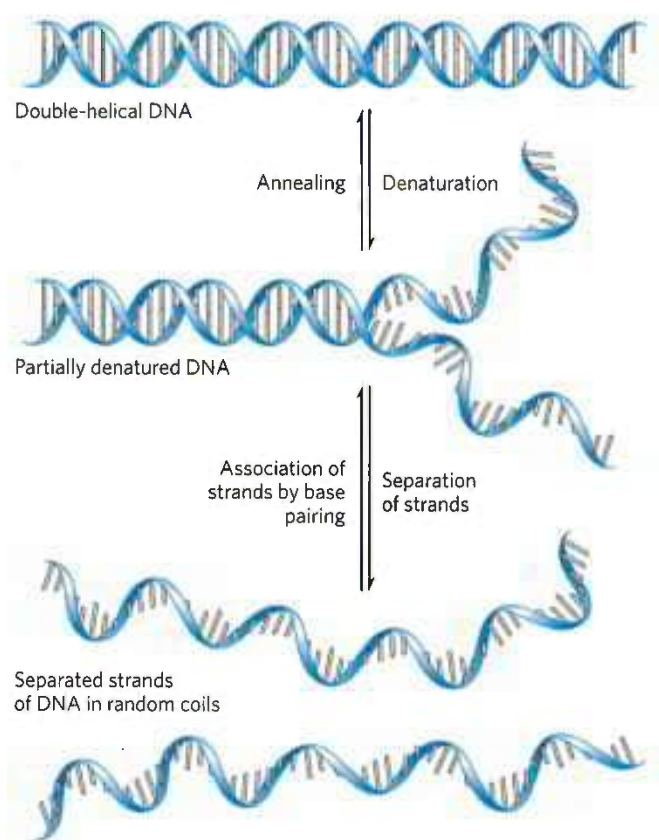


FIGURE 8-26 Reversible denaturation and annealing (renaturation) of DNA.

an increase in absorption called the **hyperchromic effect**. The transition from double-stranded DNA to the denatured, single-stranded form can thus be detected by monitoring UV absorption at 260 nm.

Viral or bacterial DNA molecules in solution denature when they are heated slowly (Fig. 8-27). Each species of DNA has a characteristic denaturation temperature, or melting point (t_m ; formally, the temperature at which half the DNA is present as separated single strands): the higher its content of G≡C base pairs, the higher the melting point of the DNA. This is primarily because, as we saw earlier, G≡C base pairs make greater contributions to base stacking than do A=T base pairs. Thus, the melting point of a DNA molecule, determined under fixed conditions of pH and ionic strength, can yield an estimate of its base composition. If denaturation conditions are carefully controlled, regions that are rich in A=T base pairs will denature while most of the DNA remains double-stranded. Such denatured regions (called bubbles) can be visualized with electron microscopy (Fig. 8-28). In the strand separation of DNA that occurs in vivo during processes

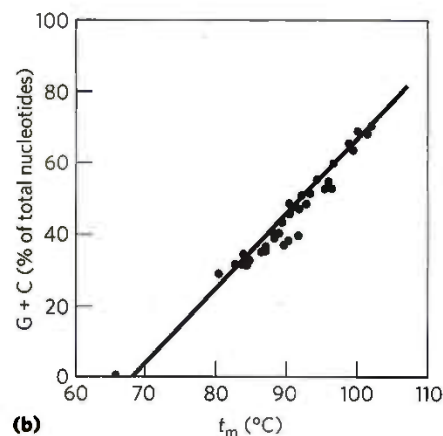
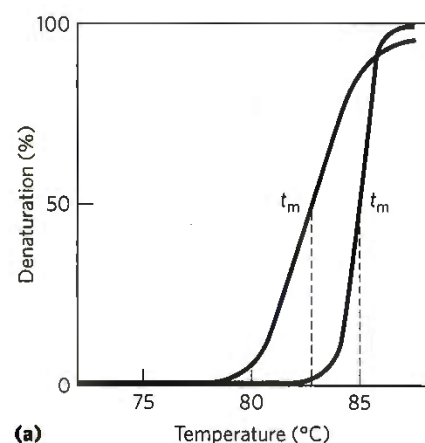


FIGURE 8-27 Heat denaturation of DNA. (a) The denaturation, or melting, curves of two DNA specimens. The temperature at the midpoint of the transition (t_m) is the melting point; it depends on pH and ionic strength and on the size and base composition of the DNA. (b) Relationship between t_m and the G + C content of a DNA. [(b) Data from J. Marmur and P. Doty, *J. Mol. Biol.* 5:109, 1962.]

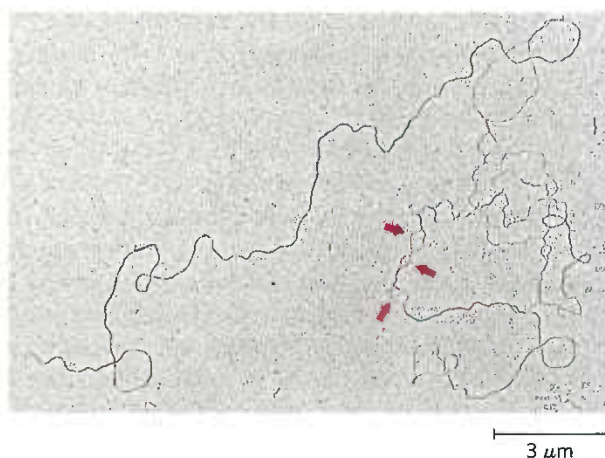


FIGURE 8-28 Partially denatured DNA. This DNA was partially denatured, then fixed to prevent renaturation during sample preparation. Although the shadowing method used to visualize the DNA in this electron micrograph obliterates many details, single-stranded and double-stranded regions are readily distinguishable. The arrows point to some single-stranded bubbles where denaturation has occurred. The regions that denature are highly reproducible and are rich in A=T base pairs. [Ross B. Inman.]

such as DNA replication and transcription, the site where strand separation is initiated is often rich in A=T base pairs, as we shall see.

Duplexes of two RNA strands or of one RNA strand and one DNA strand (RNA-DNA hybrids) can also be denatured. Notably, RNA duplexes are more stable to heat denaturation than DNA duplexes. At neutral pH, denaturation of a double-helical RNA often requires temperatures at least 20 °C higher than those required for denaturation of a DNA molecule with a comparable sequence, assuming that the strands in each molecule are perfectly complementary. The stability of an RNA-DNA hybrid is generally intermediate between that of RNA and DNA duplexes. The physical basis for these differences in thermal stability is not known.

WORKED EXAMPLE 8-1 DNA Base Pairs and DNA Stability

In samples of DNA isolated from two unidentified species of bacteria, X and Y, adenine makes up 32% and 17%, respectively, of the total bases. What relative proportions of adenine, guanine, thymine, and cytosine would you expect to find in the two DNA samples? What assumptions have you made? One of these species was isolated from a hot spring (64 °C). Which species is most likely the thermophilic bacterium, and why?

SOLUTION: For any double-helical DNA, $A = T$ and $G = C$. The DNA from species X has 32% A and therefore must contain 32% T. This accounts for 64% of the bases and leaves 36% as $G \equiv C$ pairs: 18% G and 18% C. The sample from species Y, with 17% A, must contain 17% T, accounting for 34% of the base pairs. The remaining 66% of the bases are thus equally distributed as 33% G and 33% C. This

calculation is based on the assumption that both DNA molecules are double-stranded.

The higher the G+C content of a DNA molecule, the higher the melting temperature. Species Y, having the DNA with the higher G+C content (66%), most likely is the thermophilic bacterium; its DNA has a higher melting temperature and thus is more stable at the temperature of the hot spring.

Nucleotides and Nucleic Acids Undergo Nonenzymatic Transformations



P3 Purines and pyrimidines, along with the nucleotides of which they are a part, undergo spontaneous alterations in their covalent structure. The rate of these reactions is generally *very slow*, but they are physiologically significant because of the cell's very low tolerance for alterations in its genetic information. Alterations in DNA structure that produce permanent changes in the genetic information encoded therein are called **mutations**. In higher organisms, much evidence suggests an intimate link between the accumulation of mutations in an individual and the processes of aging and carcinogenesis.

Several nucleotide bases undergo spontaneous loss of their exocyclic amino groups (deamination) (Fig. 8-29a). For example, under typical cellular conditions, deamination of cytosine (in DNA) to uracil occurs in about one of every 10^7 cytidine residues in 24 hours. This rate of deamination corresponds to about 100 spontaneous events per day, on average, in a mammalian cell. Deamination of adenine and guanine occurs at about 1/100th this rate.

The slow cytosine deamination reaction seems innocuous enough, but it is almost certainly the reason why DNA contains thymine rather than uracil. The product of cytosine deamination (uracil) is readily recognized as foreign in DNA and is removed by a repair system (Chapter 25). If DNA normally contained uracil, recognition of uracils resulting from cytosine deamination would be more difficult, and unrepaired uracils would lead to permanent sequence changes as they were paired with adenines during replication. Cytosine deamination would gradually lead to a decrease in $G \equiv C$ base pairs and an increase in $A = U$ base pairs in the DNA of all cells. Over the millennia, cytosine deamination could eliminate $G \equiv C$ base pairs and the genetic code that depends on them. **P3** Establishing thymine as one of the four bases in DNA may well have been one of the crucial turning points in evolution, making the long-term storage of genetic information possible.

Another important reaction in deoxyribonucleotides is the hydrolysis of the *N*- β -glycosyl bond between the base and the pentose. The base is lost, creating a DNA lesion called an AP (apurinic, apyrimidinic) site or abasic site (Fig. 8-29b). Purines are lost at a higher rate than pyrimidines. As many as one in 10^5 purines

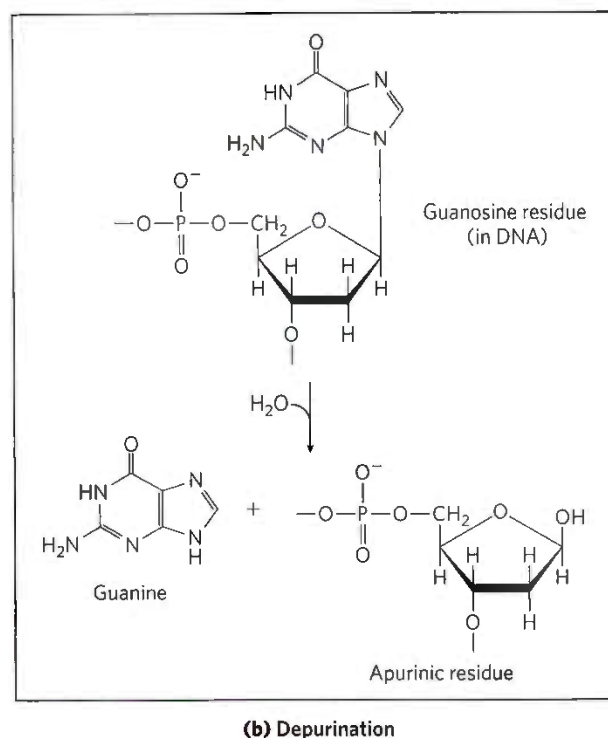
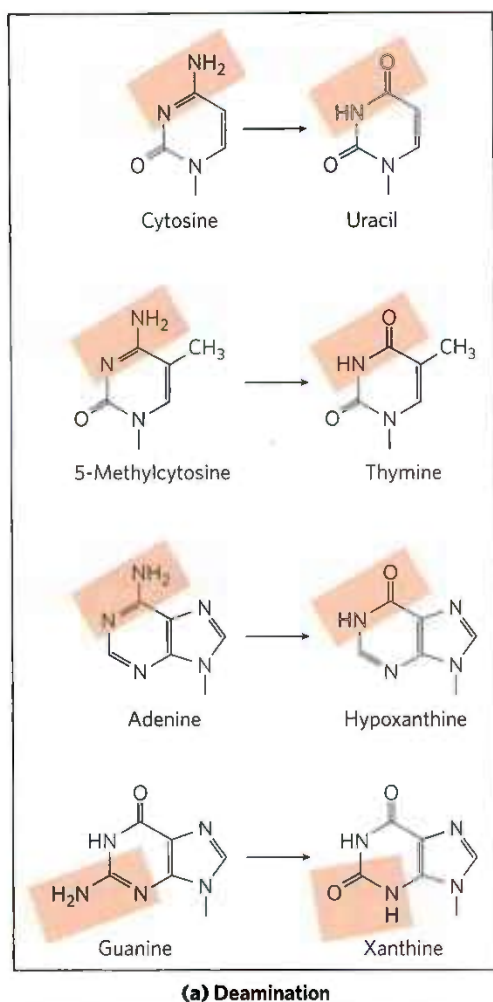


FIGURE 8-29 Some well-characterized nonenzymatic reactions of nucleotides. (a) Deamination reactions. Only the base is shown. (b) Depurination, in which a purine is lost by hydrolysis of the *N*- β -glycosyl bond. Loss of pyrimidines through a similar reaction occurs, but much more slowly. The resulting lesion, in which the deoxyribose is present but the base is not, is called an abasic site or an AP site (apurinic site or, rarely, apyrimidinic site). The deoxyribose remaining after depurination is readily converted from the β -furanose to the aldehyde form (see Fig. 8-3), further destabilizing the DNA at this position. More nonenzymatic reactions are illustrated in Figures 8-30 and 8-31.

(10,000 per mammalian cell) are lost from DNA every 24 hours under typical cellular conditions. Depurination of ribonucleotides and RNA is much slower and less physiologically significant. In the test tube, loss of purines can be accelerated by dilute acid. Incubation of DNA at pH 3 causes selective removal of the purine bases, resulting in a derivative called apurinic acid.

Other reactions are promoted by radiation. UV light induces the condensation of two ethylene groups to form a cyclobutane ring. In the cell, the same reaction between adjacent pyrimidine bases in nucleic acids forms cyclobutane pyrimidine dimers. This happens most frequently between adjacent thymidine residues on the same DNA strand (Fig. 8-30). A second type of pyrimidine dimer, called a 6-4 photoproduct, is also formed during UV irradiation. Ionizing radiation (x-rays and gamma rays) can cause ring opening and fragmentation of bases as well as breaks in the covalent backbone of nucleic acids.

Virtually all forms of life are exposed to energy-rich radiation capable of causing chemical changes in DNA. Near-UV radiation (with wavelengths of 200 to

400 nm), which makes up a significant portion of the solar spectrum, is known to cause pyrimidine dimer formation and other chemical changes in the DNA of bacteria and of human skin cells. We are subjected to a constant field of ionizing radiation in the form of cosmic rays, which can penetrate deep into the earth, as well as radiation emitted from radioactive elements, such as radium, plutonium, uranium, radon, ^{14}C , and ^3H . X-rays used in medical and dental examinations and in radiation therapy of cancer and other diseases are another form of ionizing radiation. It is estimated that UV and ionizing radiations are responsible for about 10% of all DNA damage caused by environmental agents.

DNA also may be damaged by reactive chemicals introduced into the environment as products of industrial activity. Such products may not be injurious per se but may be metabolized by cells into forms that are. There are two prominent classes of such agents (Fig. 8-31): (1) deaminating agents, particularly nitrous acid (HNO_2) or compounds that can be metabolized to nitrous acid or nitrites, and (2) alkylating agents.

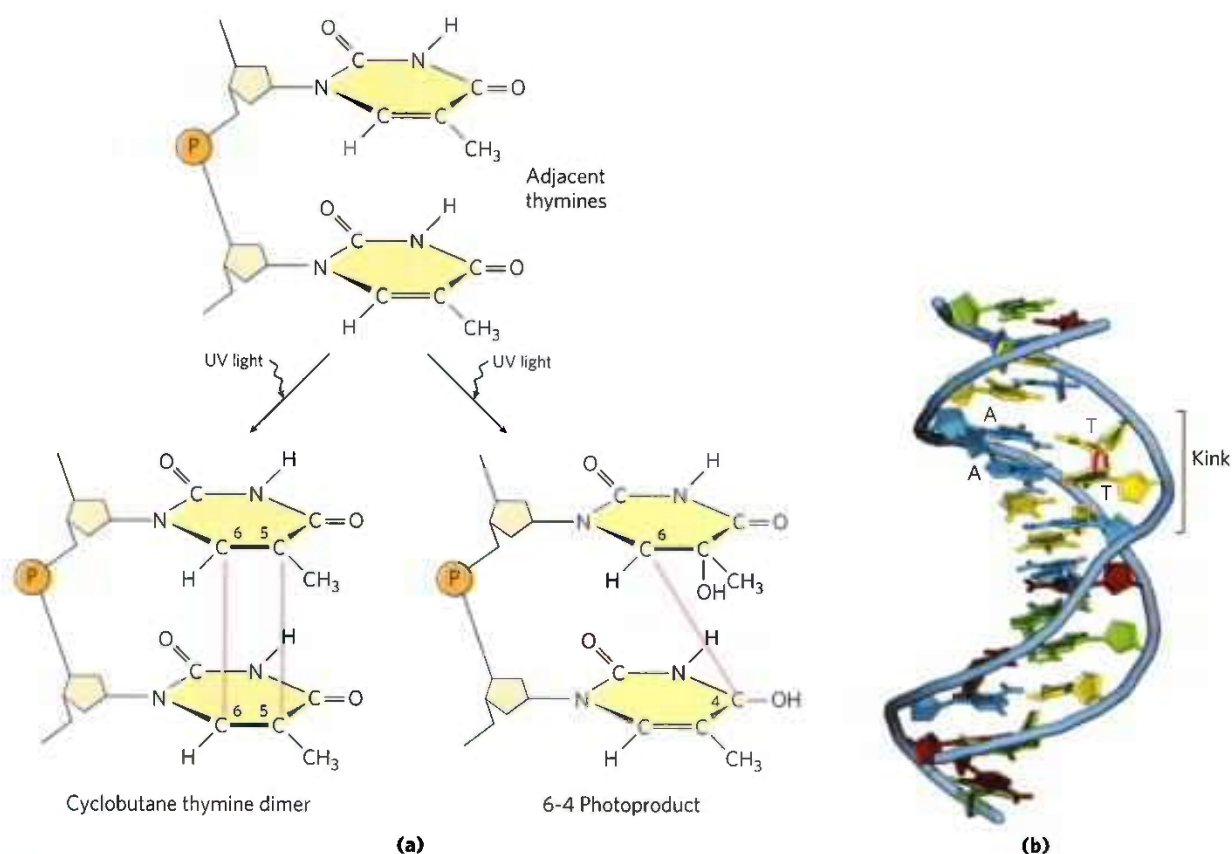
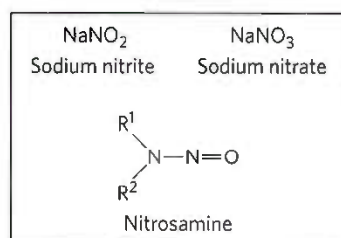


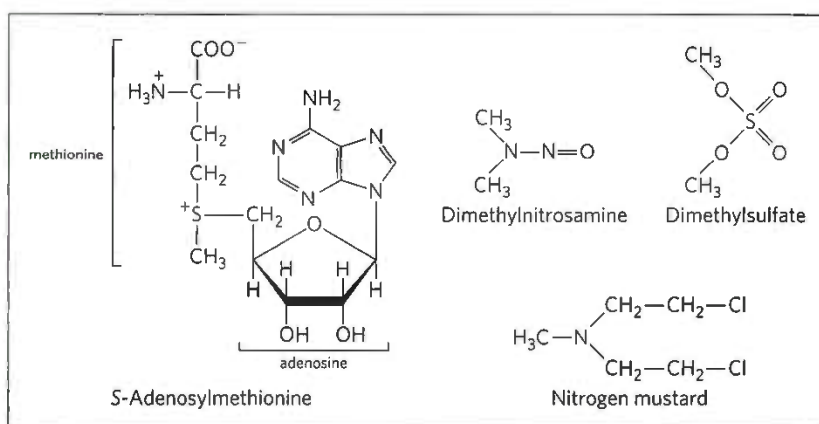
FIGURE 8-30 Formation of pyrimidine dimers induced by UV light. (a) One type of reaction (on the left) results in the formation of a cyclobutyl ring involving C-5 and C-6 of adjacent pyrimidine residues. An alternative reaction (on the right) results in a 6-4 photo-

product, with a linkage between C-6 of one pyrimidine and C-4 of its neighbor. (b) Formation of a cyclobutane pyrimidine dimer introduces a bend or kink into the DNA. [(b) Data from PDB ID 1TTD, K. McAteer et al., *J. Mol. Biol.* 282:1013, 1998.]



(a) Nitrous acid precursors

FIGURE 8-31 Chemical agents that cause DNA damage. (a) Precursors of nitrous acid, which promotes deamination reactions. (b) Alkylating agents. Most generate modified nucleotides nonenzymatically.

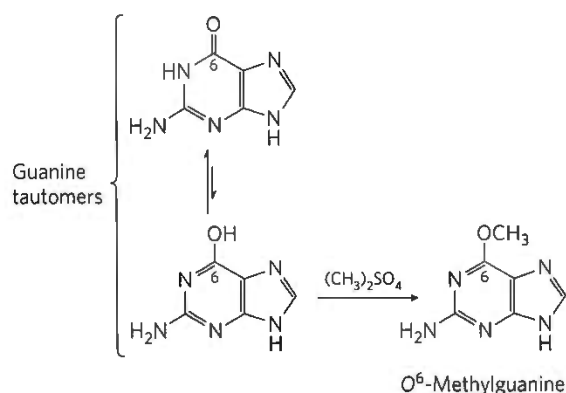


(b) Alkylating agents

Nitrous acid, formed from organic precursors such as nitrosamines and from nitrite and nitrate salts, is a potent accelerator of the deamination of bases. Bisulfite has similar effects. Both agents are used as preservatives in processed foods to prevent the growth of toxic bacteria. They do not seem to increase cancer risks significantly when used in this way, perhaps because they are used in only small amounts and

make only a minor contribution to the overall levels of DNA damage. (The potential health risk from food spoilage if these preservatives were not used is much greater.)

Alkylating agents can alter certain bases of DNA. For example, the highly reactive chemical dimethylsulfate (Fig. 8-31b) can methylate a guanine to yield O^6 -methylguanine, which cannot base-pair with cytosine.



Some alkylation of bases is a normal part of the regulation of gene expression. The enzymatic methylation of certain bases using *S*-adenosyl methionine is one example discussed below.

The most important source of mutagenic alterations in DNA is oxidative damage. Reactive oxygen species such as hydrogen peroxide, hydroxyl radicals, and superoxide radicals arise during irradiation or (more commonly) as a byproduct of aerobic metabolism. These species damage DNA through any of a large, complex group of reactions, ranging from oxidation of deoxyribose and base moieties to strand breaks. Of these species, the hydroxyl radicals are responsible for most oxidative DNA damage. Cells have an elaborate defense system to destroy reactive oxygen species, including enzymes such as catalase and superoxide dismutase that convert reactive oxygen species to harmless products. A fraction of these oxidants inevitably escape cellular defenses, however, and are able to damage DNA. Accurate estimates for the extent of this damage are not yet available, but every day the DNA of each human cell is subjected to thousands of damaging oxidative reactions.

This is merely a sampling of the best-understood reactions that damage DNA. Many carcinogenic compounds in food, water, and air exert their cancer-causing effects by modifying bases in DNA. Nevertheless, the integrity of DNA as a polymer is better maintained than that of either RNA or protein, because DNA is the only macromolecule that has the benefit of extensive biochemical repair systems. These repair processes (described in Chapter 25) greatly lessen the impact of damage to DNA. ■

Some Bases of DNA Are Methylated

Certain nucleotide bases in DNA molecules are enzymatically methylated. Adenine and cytosine are methylated more often than guanine and thymine. Methylation is generally confined to certain sequences or regions of a DNA molecule. In some cases, the function of methylation is well understood; in others, the function remains unclear. All known DNA methylases use *S*-adenosylmethionine as a methyl group donor (Fig. 8-31b). *E. coli* has two prominent DNA methylation systems. One serves in a defense role, allowing the cell to distinguish its DNA

from foreign DNA by marking its own DNA with methyl groups. The cell can then identify as foreign and destroy DNA without the methyl groups (this is known as a restriction-modification system; see p. 303). The other enzyme system methylates adenosine residues within the sequence (5')GATC(3') to *N*⁶-methyladenosine (Fig. 8-5a). Methyl groups are added by the Dam (DNA adenine methylation) methylase shortly after DNA replication, allowing the cell to distinguish newly replicated DNA from older cellular DNA (see Fig. 25-20).

In eukaryotic cells, about 5% of cytidine residues in DNA are methylated to 5-methylcytidine (Fig. 8-5a). Methylation is most common at CpG sequences, producing methyl-CpG symmetrically on both strands of the DNA. The extent of methylation of CpG sequences varies by region in large eukaryotic DNA molecules, affecting DNA metabolism and gene expression.

The Chemical Synthesis of DNA Has Been Automated

An important practical advance in nucleic acid chemistry was the rapid and accurate synthesis of short oligonucleotides of known sequence. The methods were pioneered by H. Gobind Khorana and his colleagues in the 1970s. Refinements by Robert Letsinger and Marvin Caruthers led to the chemistry now in widest use, called the phosphoramidite method (Fig. 8-32). The synthesis is carried out with the growing strand attached to a solid support, using principles similar to those used by Merrifield for peptide synthesis (see Fig. 3-30), and is readily automated. The efficiency of each addition step is very high, allowing the routine synthesis of polymers containing 70 or 80 nucleotides and, in some laboratories, much longer strands. The availability of relatively inexpensive DNA polymers with predesigned sequences revolutionized all areas of biochemistry.

Gene Sequences Can Be Amplified with the Polymerase Chain Reaction

Genome projects, as described in Chapter 9, have given rise to online databases containing the complete genome sequences of thousands of organisms. This archive of sequence information allows researchers to greatly amplify any DNA segment they might be interested in with the **polymerase chain reaction (PCR)**, a process conceived by Kary Mullis in 1983. Even DNA segments with unknown sequences can be amplified if the sequences flanking them are known. The amplified DNA can then be used for a multitude of purposes, as we shall see.

The PCR procedure, shown in Figure 8-33, relies on **DNA polymerases**, enzymes that synthesize DNA strands from deoxyribonucleotides (dNTPs), using a DNA template. DNA polymerases do not synthesize DNA *de novo*, but instead must add nucleotides to the 3' ends of preexisting strands, referred to as **primers** (see Chapter 25). In PCR, two synthetic oligonucleotides are prepared for use as

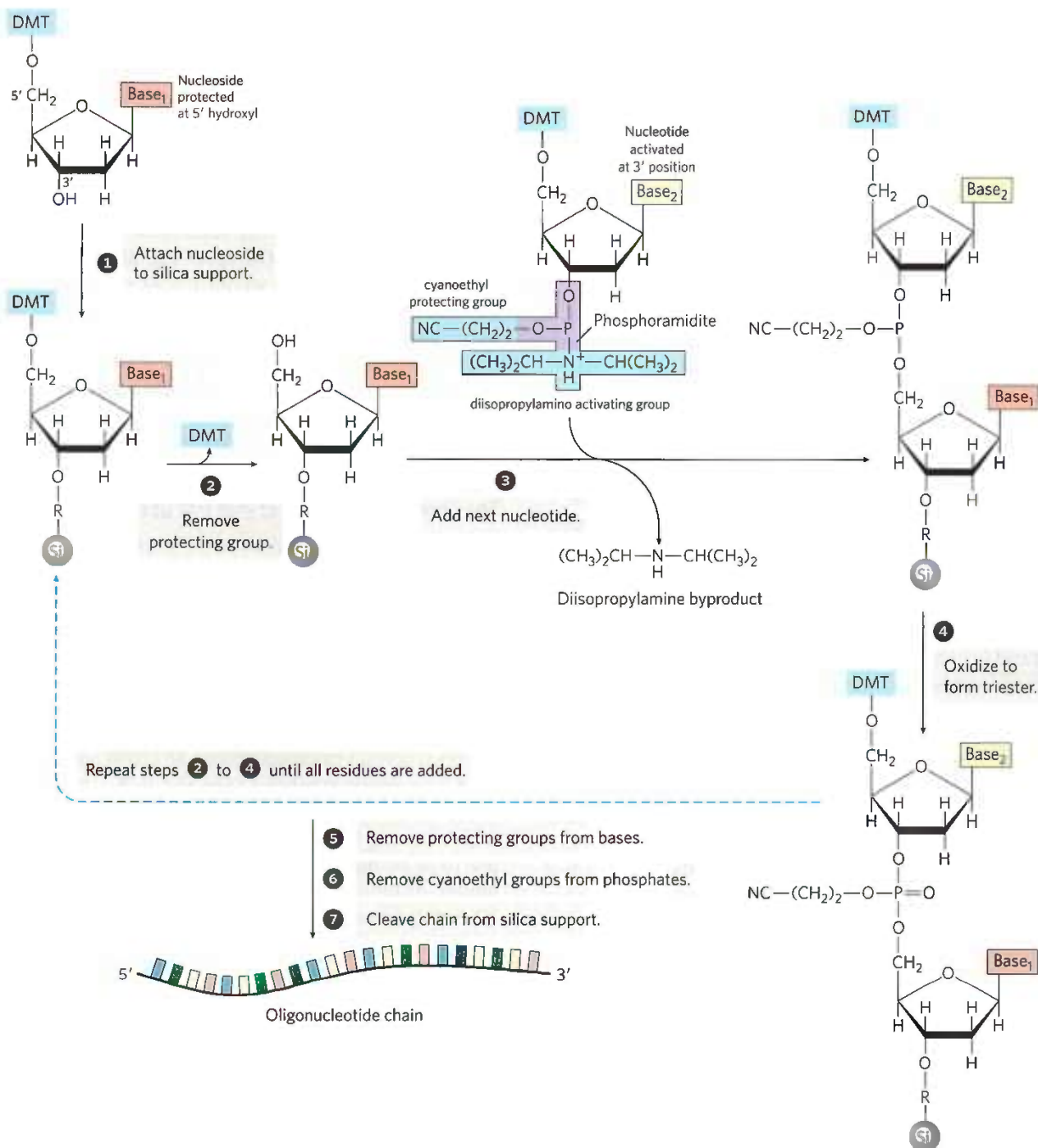


FIGURE 8-32 Chemical synthesis of DNA by the phosphoramidite method. Automated DNA synthesis is conceptually similar to the synthesis of polypeptides on a solid support. The oligonucleotide is built up on the solid support (silica), one nucleotide at a time, in a repeated series of chemical reactions with suitably protected nucleotide precursors. **1** The first nucleoside (which will be the 3' end) is attached to the silica support at the 3' hydroxyl (through a linking group, R) and is protected at the 5' hydroxyl with an acid-labile dimethoxytrityl group (DMT). The reactive groups on all bases are also chemically protected. **2** The protecting DMT group is removed by washing the column with acid (the DMT group is colored, so this reaction can be followed spectrophotometrically). **3** The next nucleotide has a reactive phosphoramidite at its 3' position: a trivalent phosphite (as opposed to the more oxidized pentavalent phosphate normally present in nucleic acids) with one linked oxygen replaced by an

amino group or a substituted amine. In the common variant shown, one of the phosphoramidite oxygens is bonded to the deoxyribose, the other is protected by a cyanoethyl group, and the third position is occupied by a readily displaced diisopropylamino group. Reaction with the immobilized nucleotide forms a 5',3' linkage, and the diisopropylamino group is eliminated. In step **4**, the phosphite linkage is oxidized with iodine to produce a phosphotriester linkage. Reactions 2 through 4 are repeated until all nucleotides are added. At each step, excess nucleotide is removed before addition of the next nucleotide. In steps **5** and **6** the remaining protecting groups on the bases and the phosphates are removed, and in **7** the oligonucleotide is separated from the solid support and purified. The chemical synthesis of RNA is somewhat more complicated because of the need to protect the 2' hydroxyl of ribose without adversely affecting the reactivity of the 3' hydroxyl.

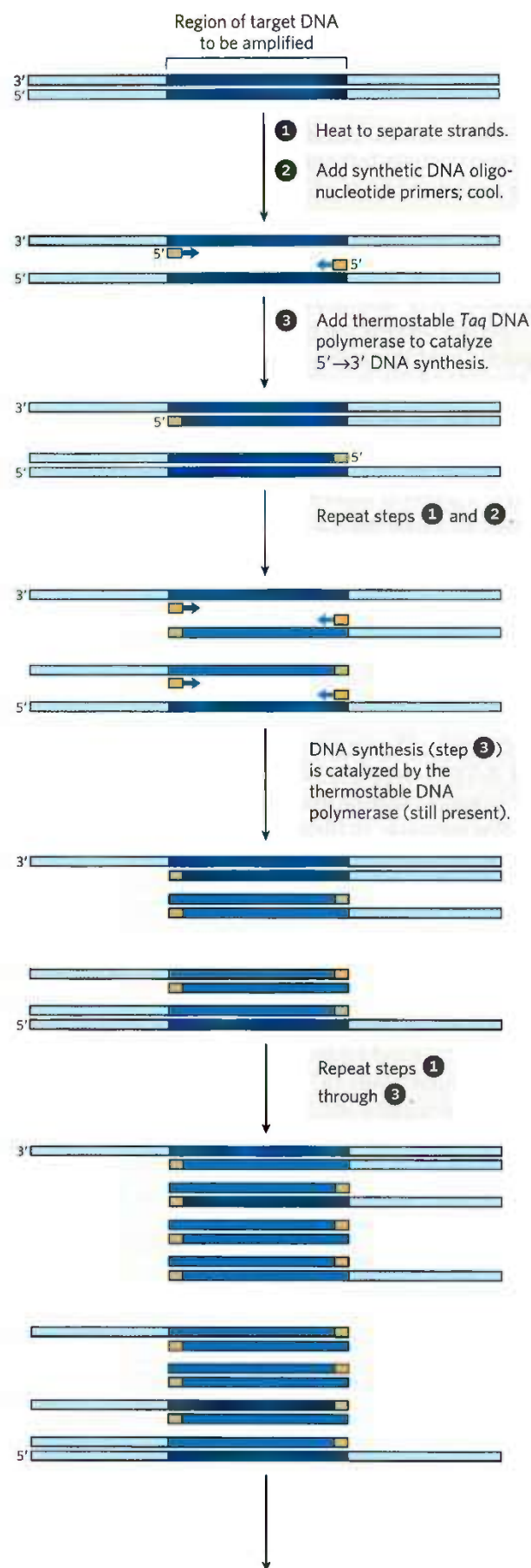


FIGURE 8-33 Amplification of a DNA segment by the polymerase chain reaction (PCR). The PCR procedure has three steps: DNA strands are ① separated by heating, then ② annealed to an excess of short synthetic DNA primers (orange) that flank the region to be amplified (dark blue); ③ new DNA is synthesized by polymerization catalyzed by DNA polymerase. The thermostable *Taq* DNA polymerase is not denatured by the heating steps. The three steps are repeated for 25 or 30 cycles in an automated process carried out in a small benchtop instrument called a thermocycler.

replication primers that can be extended by a DNA polymerase. These oligonucleotide primers are complementary to sequences on opposite strands of the target DNA, positioned so that their 5' ends define the ends of the segment to be amplified, and they become part of the amplified sequence. The 3' ends of the annealed primers are oriented toward each other and positioned to prime DNA synthesis across the targeted DNA segment.

The PCR procedure has an elegant simplicity. Basic PCR requires four components: a DNA sample containing the segment to be amplified, the pair of synthetic oligonucleotide primers, a pool of deoxynucleoside triphosphates, and a DNA polymerase. There are three steps (Fig. 8-33). In step ①, the reaction mixture is heated briefly to denature the DNA, separating the two strands. In step ②, the mixture is cooled so that the primers can anneal to the DNA. The high concentration of primers increases the likelihood that they will anneal to each strand of the denatured DNA before the two DNA strands (present at a much lower concentration) can reanneal to each other. Then, in step ③, the primed segment is replicated selectively by the DNA polymerase, using the pool of dNTPs. The cycle of heating, cooling, and replication is repeated 25 to 30 times over a few hours in an automated process, amplifying the DNA segment between the primers until the sample is large enough to be readily analyzed or cloned (described in Chapter 9).

Each replication cycle doubles the number of target DNA segment copies, so the concentration grows exponentially. The flanking DNA sequences increase in number linearly, but this effect is quickly rendered insignificant. After 20 cycles, the targeted DNA segment has been amplified more than a millionfold (2^{20}); after 30 cycles, more than a billionfold. Step ③ of PCR uses a heat-stable DNA polymerase such as the *Taq* polymerase, isolated from a thermophilic bacterium (*Thermus aquaticus*) that thrives in hot springs where temperatures approach the boiling point of water. The *Taq* polymerase remains active after every heating step (step ①) and does not have to be replenished.

P4 This technology is highly sensitive: PCR can detect and amplify just one DNA molecule in almost any type of sample—including some ancient ones. The double-helical structure of DNA is highly stable, but as we have seen, DNA does degrade slowly over time through various nonenzymatic reactions. PCR has allowed the successful cloning of rare, undegraded DNA segments isolated from samples more than 40,000 years old. Investigators have used the technique to clone DNA

fragments from the mummified remains of humans and extinct animals, such as the woolly mammoth, creating the research fields of molecular archaeology and molecular paleontology. DNA from burial sites has been amplified by PCR and used to trace ancient human migrations (see Fig. 9-31). Epidemiologists use PCR-enhanced DNA samples from human remains to trace the evolution of human pathogenic viruses. Due to its capacity to amplify just a few strands of DNA that might be present in a sample, PCR is a potent tool in forensic medicine (**Box 8-1**). It is also being used to detect viral infections and certain types of cancers before they cause symptoms, as well as in the prenatal diagnosis of genetic diseases.

WORKED EXAMPLE 8-2 *Designing Primers for the Polymerase Chain Reaction*

You set out to amplify the chromosomal sequence between the bases underlined below, using PCR. Only one strand is shown, but keep in mind that it is paired to a complementary strand.

TTGGTAGGCCGAT --- [1,000 bp] --- TAGCTAAGAATCTTTCTCAGAA

Design single-stranded oligonucleotide primers to amplify only those sequences between (and including) the underlined bases. The optimal length for PCR primers is usually 18 to 22 nucleotides. For this example, simply write the first 6 nucleotides of each primer.

SOLUTION:

Left primer GTAGGC

Right primer AAGATT

Remember, (1) the two strands of DNA are antiparallel, (2) DNA synthesis proceeds uniquely in the 5'→3' direction, (3) DNA synthesis must be directed across the region to be amplified, and (4) DNA sequences are always written in the 5'→3' direction. The sequence given is thus oriented 5'→3', left to right, even though no orientation guides are provided. The left primer must be complementary to the strand not shown, which is in the opposite orientation. Thus, the left primer begins at the **G** and is identical to the sequence shown. The right primer must direct DNA synthesis right to left, synthesizing a strand complementary to the strand provided. It will begin with an **A** complementary to the **T**, and then continue with additional nucleotides complementary to the strand shown. Although that sequence is written right to left, it is in the 3'→5' direction and must be flipped to be in the conventional orientation, written 5'→3', left to right. Companies that provide PCR primers expect orders to be written in the conventional 5'→3' direction. Doing otherwise is a common (and expensive) mistake.

The Sequences of Long DNA Strands Can Be Determined

P4 In its capacity as a repository of information, a DNA molecule's most important property is its nucleotide sequence. Until the late 1970s, determining the sequence of a nucleic acid containing as few as 5 or 10 nucleotides

was very laborious. The development of two techniques in 1977 (one by Allan Maxam and Walter Gilbert, the other by Frederick Sanger) made possible the sequencing of larger DNA molecules. Although the two methods are similar in strategy, **Sanger sequencing**, also known as dideoxy chain-termination sequencing, is both technically easier and more accurate (**Fig. 8-34**).

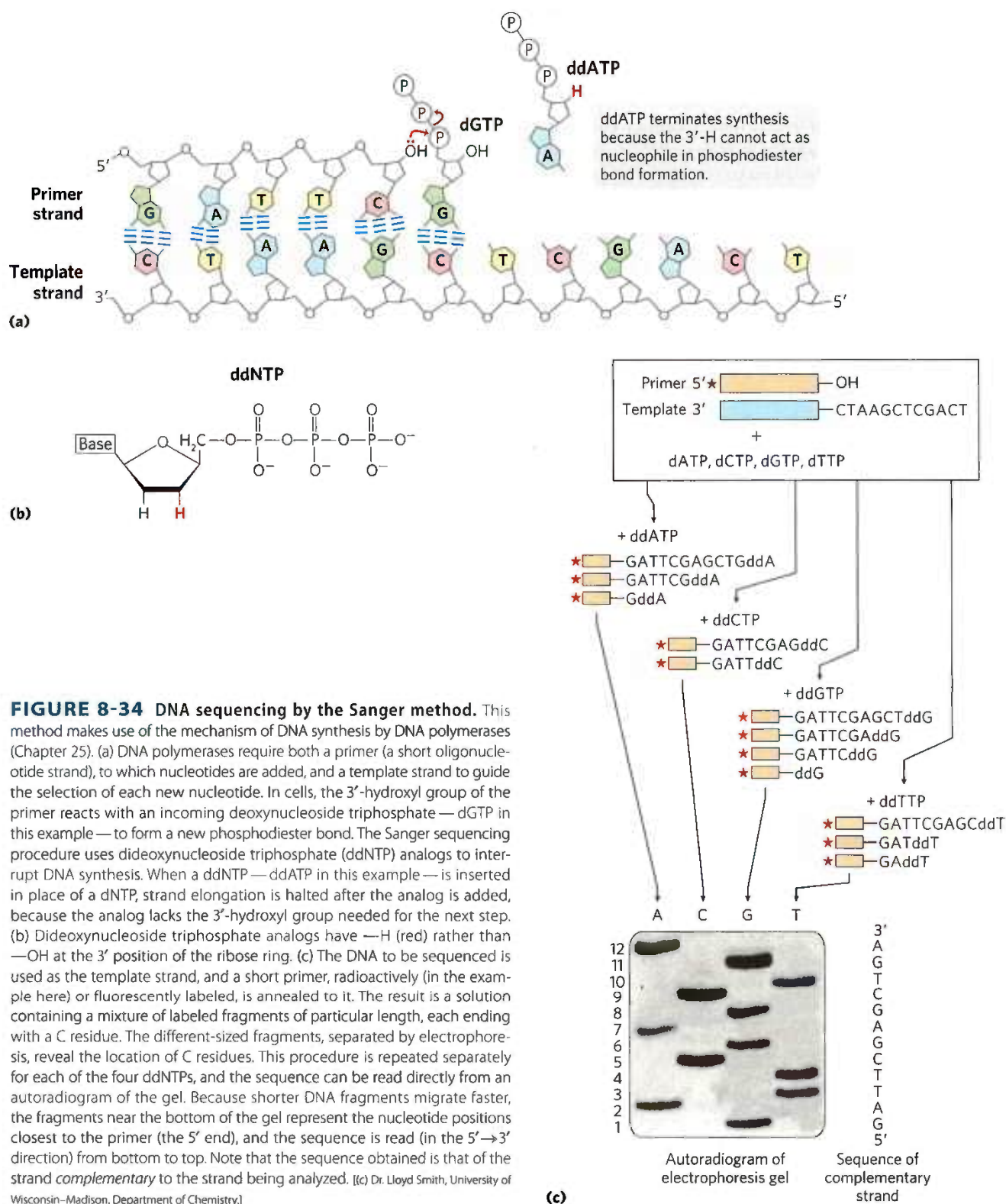
Any protocol for DNA sequencing has two parts. One must first chemically distinguish between G, C, T, and A residues. A second strategy is then needed to determine where the four residues appear in the overall sequence.

Sanger sequencing exploited new (at the time) information about the mechanism of DNA synthesis by DNA polymerase to distinguish between the four nucleotides, making use of nucleotide analogs called dideoxynucleoside triphosphates (ddNTPs) to interrupt synthesis specifically at one or another type of nucleotide. Like PCR, Sanger's method makes use of DNA polymerases and a primer to synthesize a DNA strand complementary to the strand under analysis. Each added deoxynucleotide is complementary, through base pairing, to a base in the template strand. In the reaction catalyzed by DNA polymerase, the 3'-hydroxyl group of the primer reacts with an incoming dNTP to form a new phosphodiester bond (**Fig. 8-34a**). The ddNTPs interrupt DNA synthesis because they bind to the template strand but lack the 3'-hydroxyl group needed to add the next nucleotide (**Fig. 8-34b**). Once a ddNTP is added to a growing strand, that strand cannot be extended further.

For instance, to identify C residues, a small amount of ddCTP is added to a reaction system containing a much larger amount of dCTP (along with the other three dNTPs). A competition then occurs every time the DNA polymerase encounters a G in the template strand. Usually, dC is added, and synthesis of the strand continues. Sometimes, ddC will be added instead, and the strand will be terminated at that position. Thus, a small fraction of the synthesized strands are prematurely terminated at every position where dC would normally be added, opposite each template dG. Given the excess of dCTP over ddCTP, the chance that the analog will be incorporated instead of dC is small. But enough ddCTP is present to ensure that some of the strands will be terminated at each G residue in the template.

The result is a solution containing a mixture of fragments, each ending with a ddC residue. The fragments differ in length, and locating the C residues then relies on precise electrophoretic methods that allow separation of DNA strands differing in size by only one nucleotide residue. (See **Fig. 3-18** for a description of gel electrophoresis.) Note that in most sequencing protocols, the sequence obtained is that of the newly synthesized strand complementary to the template strand being analyzed.

When this procedure was first developed, the process was repeated separately for each of the four ddNTPs. Radioactively labeled primers allowed researchers to



detect the DNA fragments generated during the DNA synthesis reactions. The sequence of the synthesized DNA strand was read directly from an autoradiogram of the resulting gel (Fig. 8-34c). Because shorter DNA fragments migrate faster, the fragments near the bottom

of the gel represented the nucleotide positions closest to the primer (the 5' end), and the sequence was read (in the 5'→3' direction) from bottom to top.

DNA sequencing was first automated by a variation of the Sanger method, in which each of the four ddNTPs

A Potent Weapon in Forensic Medicine

One of the most accurate methods for placing an individual at the scene of a crime is a fingerprint. But with the advent of recombinant DNA technology (see Chapter 9), a much more powerful tool became available: **DNA genotyping** (also called DNA fingerprinting or DNA profiling). As first described by English geneticist Alec Jeffreys in 1985, the method is based on **sequence polymorphisms**, slight sequence differences among individuals — 1 in every 1,000 bp, on average. Each difference from the prototype human genome sequence (the first human genome that was sequenced) occurs in some fraction of the human population; every person has some differences from this prototype.

Forensic work focuses on differences in the lengths of **short tandem repeat (STR)** sequences. An STR locus is a specific location on a chromosome where a short DNA sequence (usually 4 bp long) is repeated many times in tandem. The loci

most often used in STR genotyping are short — 4 to 50 repeats (16 to 200 bp for tetranucleotide repeats) — and have multiple length variants in the human population. More than 20,000 tetranucleotide STR loci have been characterized in the human genome. And more than a million STRs of all types may be present in the human genome, accounting for about 3% of all human DNA.

The length of a particular STR in a given individual can be determined with the aid of the polymerase chain reaction (see Fig. 8-33). The use of PCR also makes the procedure sensitive enough to be applied to the very small samples often collected at crime scenes. The DNA sequences flanking STRs are unique to each STR locus and are identical (except for very rare mutations) in all humans. PCR primers are targeted to this flanking DNA and are designed to amplify the DNA across the

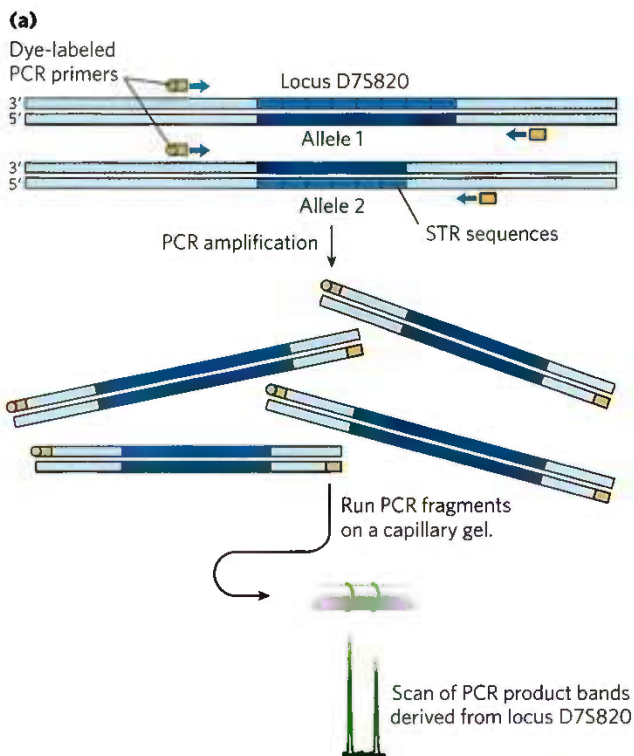
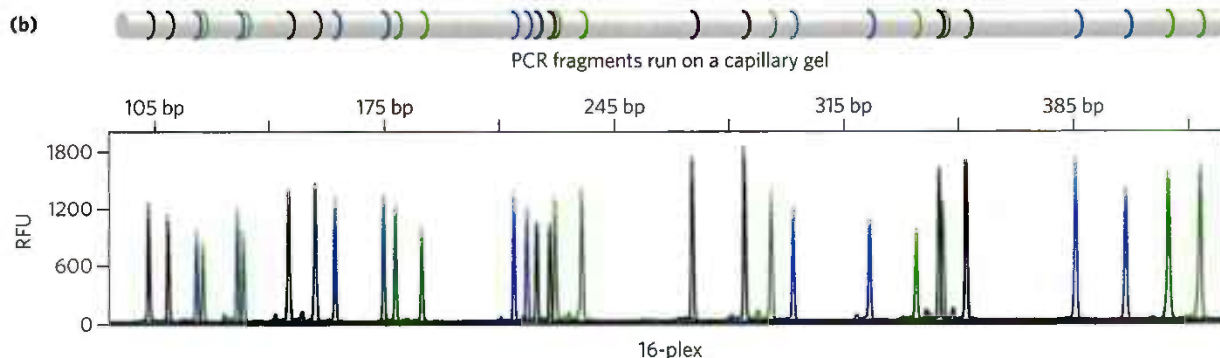


FIGURE 1 (a) STR loci can be analyzed by PCR. Suitable PCR primers (with an attached dye to aid in subsequent detection) are targeted to sequences on each side of the STR, and the region between them is amplified. If the STR sequences have different lengths on the two chromosomes of an individual's chromosome pair, two PCR products of different lengths result. (b) The PCR products from amplification of up to 16 STR loci can be run on a single capillary acrylamide gel (a "16-plex" analysis). Determination of which locus corresponds to which signal depends on the color of the fluorescent dye attached to the primers used in the process and on the size range in which the signal appears (the size range can be controlled by which sequences — those closer to or more distant from the STR — are targeted by the designed PCR primers). Fluorescence is given in relative fluorescence units (RFU), as measured against a standard supplied with the kit. [(b) Information from Carol Bingham, Promega Corporation.]



STR (Fig. 1a). The length of the PCR product then reflects the length of the STR in that sample. Because each human inherits one chromosome of each chromosome pair from each parent, the STR lengths on the two chromosomes are often different, generating two different STR lengths from one individual.

The PCR products are subjected to electrophoresis on a very thin polyacrylamide gel in a capillary tube. The resulting bands are converted into a set of peaks that accurately reveal the size of each PCR fragment and thus the length of the STR in the corresponding allele. Analysis of multiple STR loci can yield a profile that is unique to an individual (Fig. 1b). This is typically done with a commercially available kit that includes PCR primers unique to each locus, linked to colored dyes to help distinguish the different PCR products. PCR amplification enables investigators to obtain STR genotypes from less than 1 ng of partially degraded DNA, an amount that can be obtained from a single hair follicle, a small fraction of a drop of blood, a small semen sample, or samples that might be months or even many years old. When good STR genotypes are obtained, the chance of misidentification is less than 1 in 10^{18} (a quintillion).

The successful forensic use of STR analysis required standardization, first attempted in the United Kingdom in 1995. The U.S. standard, called the Combined DNA Index System (CODIS), established in 1998, was originally based on 13 well-studied STR loci. These continue to be required in

any DNA-typing experiment carried out in the United States (Table 1) and are also used internationally. The amelogenin gene is also used as a marker in the analyses. Present on the human sex chromosomes, this gene has a slightly different length on the X and Y chromosomes. PCR amplification across this gene thus generates different-sized products that can reveal the sex of the DNA donor. By mid-2019, the CODIS database contained more than 18 million STR genotypes and had assisted in nearly 500,000 forensic investigations. As the CODIS database has expanded, the chance for adventitious matches has increased. The CODIS standard was expanded in 2017 to include 20 core loci. The new loci were incorporated with international agreement to ensure compatibility. Loci utilized in commercial kits have since expanded from 16 to 24.

DNA genotyping has been used to both convict and acquit suspects, and to establish paternity with an extraordinary degree of certainty. In the United States, there have been many hundreds of postconviction exonerations based on DNA evidence. The impact of these procedures on court cases will continue to grow as standards are refined and as international STR genotyping databases grow. Even very old mysteries can be solved. In 1996, STR genotyping helped confirm identification of the bones of the last Russian czar and his family, who were assassinated in 1918.

TABLE 1 Properties of the Loci Used for the CODIS Database

Locus	Chromosome	Repeat motif	Repeat length (range) ^a	Number of alleles seen ^b
CSF1PO	5	TAGA	5–16	20
FGA	4	CTTT	12.2–51.2	80
TH01	11	TCAT	3–14	20
TPOX	2	GAAT	4–16	15
VWA	12	[TCTG][TCTA]	10–25	28
D3S1358	3	[TCTG][TCTA]	8–21	24
D5S818	5	AGAT	7–18	15
D7S820	7	GATA	5–16	30
D8S1179	8	[TCTA][TCTG]	7–20	17
D13S317	13	TATC	5–16	17
D16S539	16	GATA	5–16	19
D18S51	18	AGAA	7–39.2	51
D21S11	21	[TCTA][TCTG]	12–41.2	82
Amelogenin ^c	X, Y	Not applicable		

Data from J. M. Butler, *Forensic DNA Typing: Biology, Technology, and Genetics of STR Markers*, 2nd edn, Elsevier, 2005, p. 96.

^aRepeat lengths observed in the human population. Partial or imperfect repeats can be included in some alleles.

^bNumber of different alleles observed as of 2005 in the human population. Careful analysis of a locus in many individuals is a prerequisite to its use in forensic DNA typing.

^cAmelogenin is a gene, of slightly different size on the X and Y chromosomes, that is used to establish gender.

used for a reaction was labeled with a different-colored fluorescent tag (Fig. 8-35). With this technology, all four fluorescent ddNTPs could be introduced into a single reaction. The terminated fragments, each of a different size, could be separated by electrophoresis in a single gel lane. The identity of the residue that terminated each fragment was made evident by its fluorescent color. Researchers could sequence DNA molecules containing thousands of nucleotides in a few hours, and the entire genomes of hundreds of organisms were sequenced in this way. For example, in the Human Genome Project, researchers sequenced all 3.2×10^9 bp of the DNA in a human cell (see Chapter 9) in an effort that spanned nearly a decade and included contributions from dozens of laboratories worldwide. This form of Sanger sequencing is still used for routine analysis of short segments of DNA.

DNA Sequencing Technologies Are Advancing Rapidly

P4 The billions of base pairs in a complete human genome can now be sequenced in a day or two, the millions in a bacterial genome in a few hours. With modest expense, a personal genomic sequence can be routinely included in each individual's medical record. These advances have been made possible by methods sometimes referred to as next-generation, or "next-gen," sequencing. The sequencing strategies have some similarities to the Sanger method. Innovations have allowed a miniaturization of the procedure, a massive increase in scale, and a corresponding decrease in cost. Two widely used approaches, **reversible terminator sequencing** and **single molecule real time (SMRT) sequencing**, both developed commercially, are described.

In both approaches, large genomes are sequenced by first collecting the DNA from many cells of the organism or individual. The DNA is sheared at random locations to generate fragments of a particular average size. The individual fragments — many with overlapping sequences — are immobilized on a solid support, and each is sequenced in place. Fluorescent dyes linked to the nucleotides and powerful optical systems that can detect incorporation of each new nucleotide by DNA polymerase allow the sequencing process to be monitored directly. The fluorescent dyes and the design of the methods solve both problems in DNA sequencing at

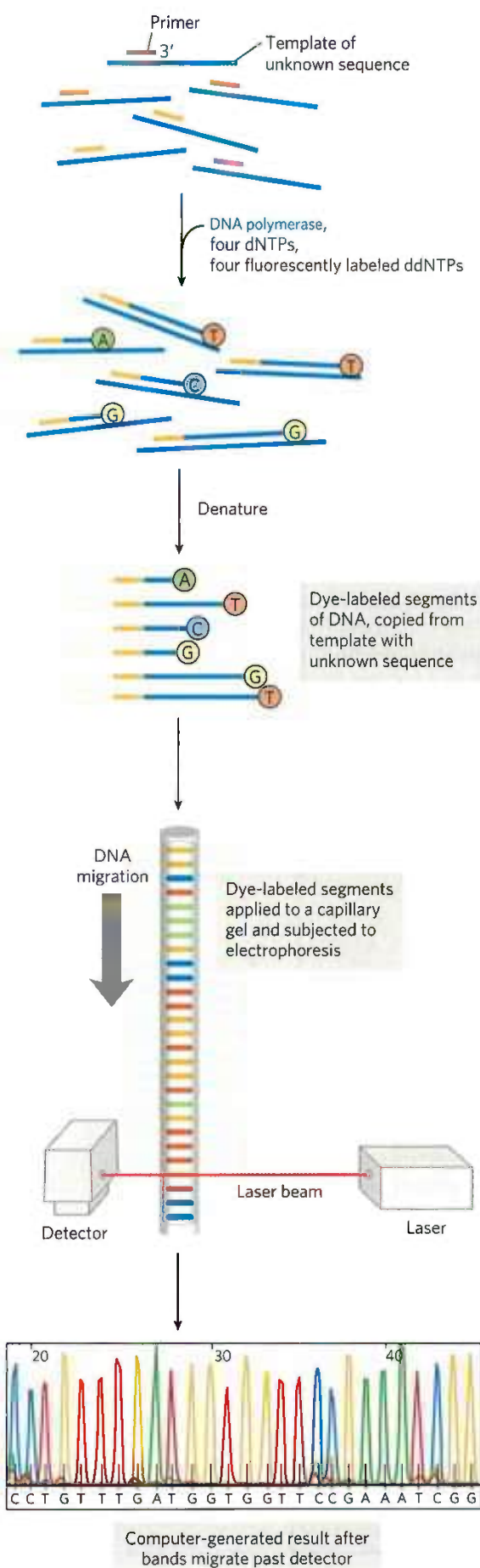


FIGURE 8-35 Automation of DNA sequencing reactions. In the Sanger method, each ddNTP can be linked to a fluorescent (dye) molecule that gives the same color to all the fragments terminating in that nucleotide, with a different color for each nucleotide. All four labeled ddNTPs are added to the reaction mix together. The resulting colored DNA fragments are separated by size in an electrophoretic gel in a capillary tube (a refinement of gel electrophoresis that allows faster separations). All fragments of a given length migrate through the capillary gel together in a single band, and the color associated with each band is detected with a laser beam. The DNA sequence is read by identifying the color sequences in the bands as they pass the detector and feeding this information directly to a computer. The amount of fluorescence in each band is represented as a peak in the computer output.

[Data from Dr. Lloyd Smith, University of Wisconsin–Madison, Department of Chemistry.]

once, identifying each nucleotide and fixing its location in the large sequence. Individual regions of a genome may be sequenced hundreds, even thousands, of times. The entire genomic sequence is reconstructed by computer programs that align the sequences of overlapping fragments.

In the reversible terminator sequencing method developed by Illumina, the genomic DNA to be sequenced is sheared so as to generate fragments a few hundred base pairs long. Synthetic oligonucleotides of known sequence are ligated to each end of each fragment, providing a point of reference on every DNA molecule. The individual fragments are then immobilized on a solid surface, and each is amplified in place by PCR to form a tight cluster of identical fragments. The solid surface is part of a channel on a flow cell that allows liquid solutions to stream over the samples. The result is a solid surface just a few centimeters wide with millions of attached DNA clusters, each cluster containing multiple copies of a single DNA sequence derived from a random genomic DNA fragment. To provide a starting point for DNA polymerase, an oligonucleotide primer is then added that is complementary to the oligonucleotides of known sequence ligated to the various fragment ends. All of these millions of clusters are sequenced at the same time, with the data from each cluster captured and stored by a computer.

The actual sequencing of each cluster employs reversible terminator sequencing (Fig. 8-36). Four different modified deoxynucleotides (A, T, G, and C), each with a particular fluorescent label that identifies the nucleotide by color, are added to the sequencing reaction, along with the DNA polymerase. The labeled nucleotides also incorporate blocking groups attached to their 3' ends that permit only one nucleotide to be added to each strand. The polymerase adds the appropriate nucleotide to the strands in each cluster, giving each cluster a color that corresponds to the added nucleotide. Next, lasers excite all the fluorescent labels, and an image of the entire surface reveals the color (and thus the identity) of the base added to each cluster. The fluorescent label and the blocking groups are then chemically or photolytically removed. The surface goes dark until the solution with labeled nucleotides and DNA polymerase is again introduced to the surface, allowing the next nucleotide to be added to each cluster. The sequencing proceeds stepwise. Read lengths obtained with this technology (that is, the length of individual DNA sequences that can be accurately determined) are typically 100 to 300 nucleotides. Read length is limited by constraints on cluster density on the flow cell surface and by small inefficiencies in the PCR reactions needed to amplify each cluster accurately. Accuracy is high, with error rates as low as 0.1%.

The relatively short read lengths produced by the Illumina technology are problematic in some situations, such as the sequencing of long stretches of DNA where short sequences are repeated over and over. Pacific Biosciences has pioneered the single-molecule

real time (SMRT) sequencing method that allows read lengths averaging up to 30,000 to 40,000 bp (Fig. 8-37). The SMRT technology has a lower throughput, a higher cost, and a higher error rate than the Illumina approach. However, the very long read lengths are essential in some applications, particularly to reconstruct the complete genomes of higher organisms that may contain extensive regions of repeated DNA sequences. They also facilitate the detection of genomic alterations—deletions, duplications, or rearrangements of genomic segments—that arise in some cells, such as those in cancerous tumors (see Box 24-2).

SMRT sequencing utilizes SMRT cells with 150,000 pores, each pore smaller in diameter (~70 nm) than the wavelength of visible light. Attenuated light from an excitation beam penetrates only the lower 20 to 30 nm in each pore, producing a light volume small enough to accommodate just one DNA polymerase molecule (Fig. 8-37a). A single DNA polymerase is immobilized at the bottom of each pore. Genomic DNA is sheared to generate random fragments that are tens of thousands of base pairs in length. Hairpin oligonucleotides are ligated to both ends of each fragment, so that the two DNA strands are joined in one continuous circle (Fig. 8-37b). A primer is annealed to the open single-stranded DNA at one end of the fragment, and this is captured by a DNA polymerase in one of the pores to initiate DNA synthesis.

Fluorescent nucleotides are introduced, with A, T, G, and C each having a specific colored dye attached to the triphosphate. When a nucleotide binds to the DNA polymerase in the pore, it is immobilized long enough to produce a brief fluorescent light pulse that can be read by a detector (Fig. 8-37c). The fluorescent dye is released with the pyrophosphate as each new phosphodiester bond is formed. The error rate is high, about 10% to 15%. However, as the DNA template is a continuous circle, one DNA polymerase can replicate the same fragment over and over. The light pulses continue without interruption as new nucleotides are added in real time, and each pore thus generates a movie of light pulses (sometimes several hours long) that corresponds to the repeated sequencing of the DNA fragment bound in that pore. A computer program deletes the known sequences of the hairpin ends. Error is reduced by compiling a consensus sequence of the fragment by automated alignment of the many repeated sequencing passes and acceptance of the most common nucleotide signal detected at each position.

Translating the sequences of millions of short DNA fragments into a complex and contiguous genomic sequence requires the computerized alignment of overlapping fragments (Fig. 8-38). The number of times that a particular nucleotide in a genome is sequenced, on average, is referred to as the **sequencing depth** or sequencing coverage. In many cases, a sufficiently large number of random fragments are sequenced so that each nucleotide in the genome is sequenced an average of hundreds to thousands of times (100× to 1,000× coverage). Although

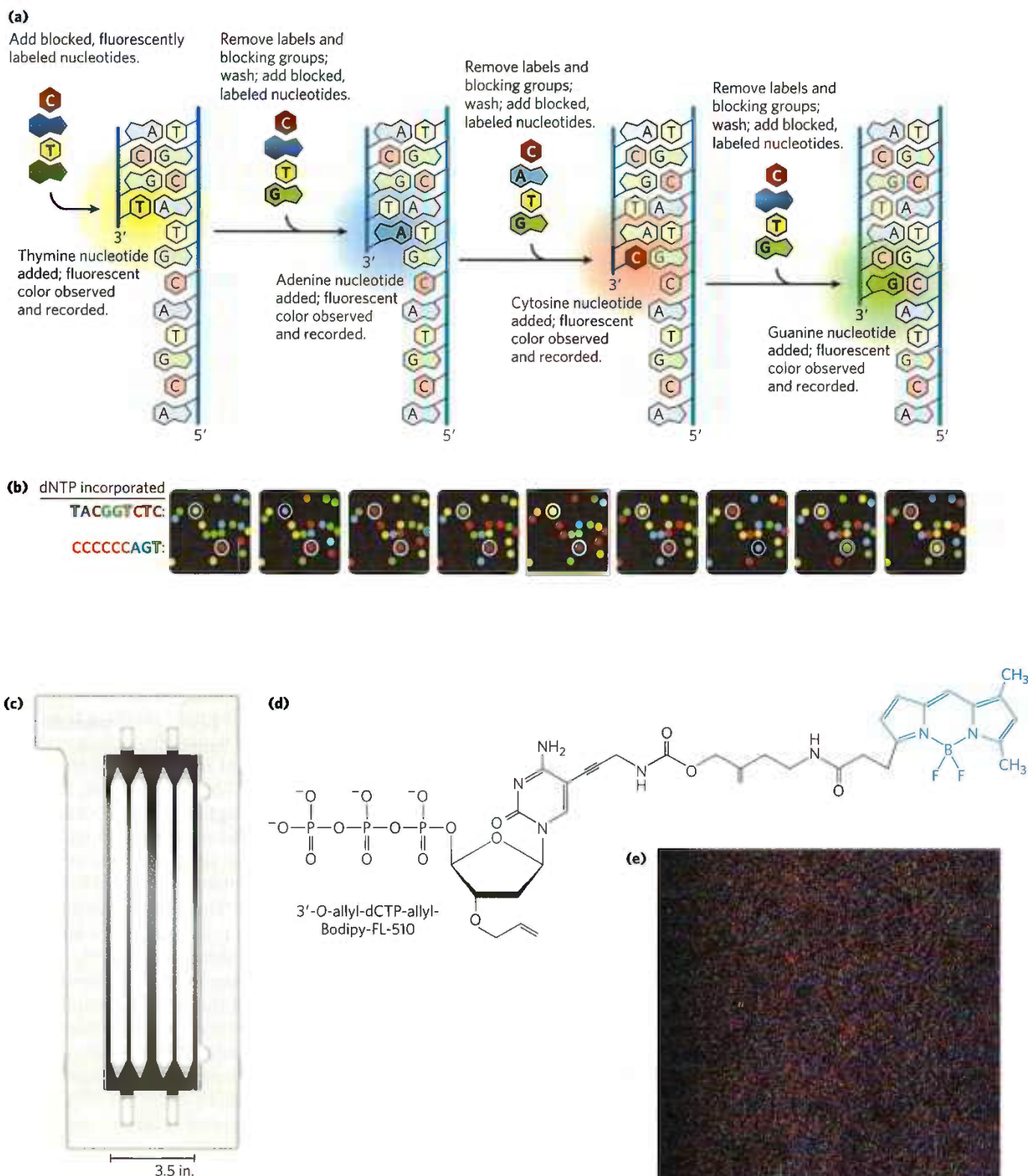


FIGURE 8-36 Next-generation reversible terminator sequencing.

(a) Blocking groups on each fluorescently labeled nucleotide prevent multiple nucleotides from being added in a single cycle. (b) Artist's rendition of nine successive cycles from one very small part of an Illumina sequencing run. Each colored spot represents the location of a cluster of immobilized identical oligonucleotides affixed to the surface of the flow cell. The white-circled spots represent the same two clusters on the surface over successive cycles, with the sequences indicated. Data are recorded and analyzed digitally. (c) Typical flow cell used for a next-generation

sequencer. Millions of DNA fragments can be sequenced simultaneously in each of the four channels. (d) A dCTP molecule modified with a fluorescent dye and a 3' blocking group for use in reversible terminator sequencing. Both the dye and the 3'-end-blocking group can be removed, either chemically or photolytically, leaving a free 3'-OH group for addition of the next nucleotide. The modified nucleotides currently used in reversible terminator sequencing are proprietary. (e) Part of the surface of one channel during a sequencing reaction. [(c, e) Used under license from Illumina, Inc. All Rights Reserved.]

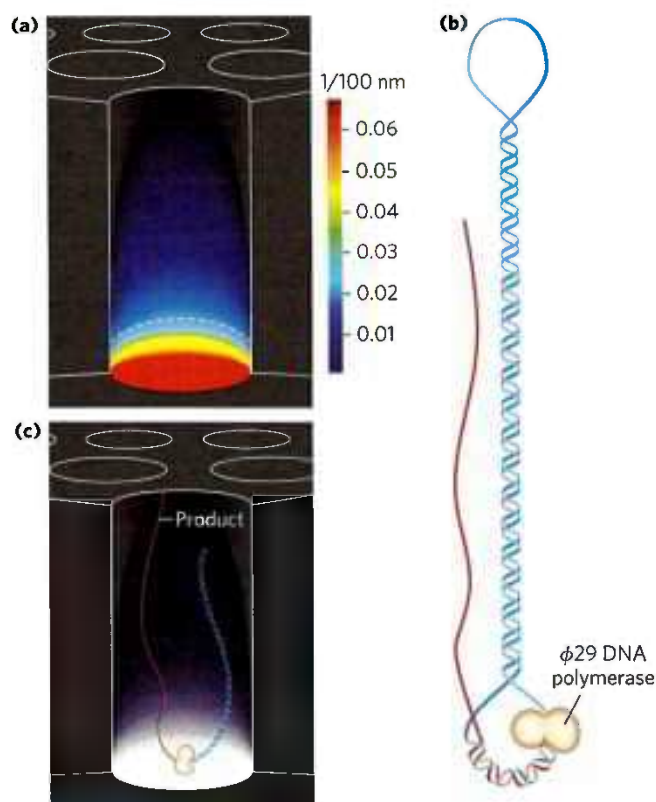


FIGURE 8-37 SMRT sequencing. (a) One pore in an SMRT cell. The pore is smaller in diameter than the wavelength of visible light, so that light projected at the bottom penetrates only a short distance into the pore. (b) DNA fragments sequenced by SMRT technology. Hairpin oligonucleotides are ligated to both ends. A primer for DNA synthesis is annealed to one and sometimes both single-stranded regions in the hairpin ends. (c) DNA synthesis by a DNA polymerase immobilized within a SMRT pore. A fluorescent dye is attached to the triphosphate of the nucleoside triphosphates used. The dye is released from each nucleotide incorporated into the growing DNA strand and its wavelength recorded.

the coverage of particular nucleotides may vary, a high level of coverage ensures that most sequencing errors will be detected and eliminated. The overlaps allow the computer to trace the sequence through a chromosome, from one overlapping fragment to another, permitting the assembly of long, contiguous sequences called **contigs**. In a successful genomic sequencing exercise, many contigs can extend over millions of base pairs.

P4 The rapid evolution of DNA sequencing technologies shows no signs of slowing. As costs plummet and sensitivity increases, new applications come online to enhance medicine, forensics, archaeology, and many other fields. We present some of those applications in Chapter 9.

SUMMARY 8.3 Nucleic Acid Chemistry

- Native DNA undergoes reversible unwinding and separation of strands (melting) upon heating or at extremes of pH. DNAs rich in G≡C pairs have higher melting points than DNAs rich in A=T pairs.

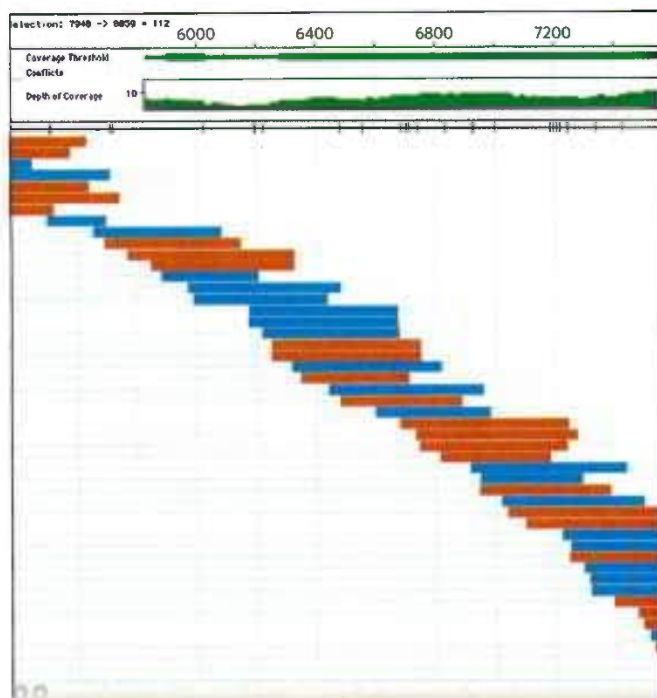


FIGURE 8-38 Sequence assembly. In a genomic sequence, each base pair of the genome is usually represented in multiple sequenced fragments, referred to as reads. This schematic shows how the overlaps between reads are used to assemble a contiguous segment of the genomic sequence, or contig. The numbers at the top represent base-pair positions in the genome, relative to an arbitrarily defined reference point. All sequence fragments come from a particular long contig. The reads are represented by horizontal colored bars. DNA strand segments are sequenced at random, with sequences obtained from one strand (5' to 3', left to right) or the other strand (5' to 3', right to left) represented by blue lines or red lines, respectively. The coverage bar at the top indicates how many times the sequence at a particular position has appeared in a sequenced read, with higher numbers corresponding to increased quality of the output sequence data.

- DNA is a relatively stable polymer. Spontaneous reactions such as deamination of certain bases, hydrolysis of base-sugar *N*-glycosyl bonds, radiation-induced formation of pyrimidine dimers, and oxidative damage occur at very low rates, yet are important because of a cell's very low tolerance for changes in its genetic material.

- DNA is subject to enzymatic modification of nucleotide bases at particular locations. Methylated bases are common.

- Oligonucleotides of known sequence can be synthesized rapidly and accurately.

- The polymerase chain reaction (PCR) provides a convenient and rapid method for amplifying segments of DNA if the sequences of the ends of the targeted DNA segment are known.

- Routine DNA sequencing of genes or short DNA segments is carried out using an automated variation of Sanger dideoxy sequencing.

- DNA sequences, including entire genomes, can be efficiently determined in hours or days using commercial next-gen sequencing technologies.

8.4 Other Functions of Nucleotides

In addition to their roles as the subunits of nucleic acids, nucleotides have cellular functions as energy carriers, components of enzyme cofactors, and chemical messengers.

Nucleotides Carry Chemical Energy in Cells

The phosphate group covalently linked at the 5' hydroxyl of a ribonucleotide may have one or two additional phosphates attached. The resulting molecules are referred to as nucleoside mono-, di-, and triphosphates (Fig. 8-39). Starting from the ribose, the three phosphates are generally labeled α , β , and γ . **P5** Hydrolysis of nucleoside triphosphates provides the chemical energy to drive many cellular reactions. Adenosine 5'-triphosphate, ATP, is by far the most widely used nucleoside triphosphate for this purpose, but UTP, GTP, and CTP are also used in some reactions. Nucleoside triphosphates also serve as the activated precursors of DNA and RNA synthesis, as described in Chapters 25 and 26 (see also Fig. 8-34).

The energy released by hydrolysis of ATP and the other nucleoside triphosphates is accounted for by the structure of the triphosphate group. The bond between the ribose and the α phosphate is an ester linkage. The α , β and β , γ linkages are phosphoanhydrides (Fig. 8-40). Hydrolysis of the ester linkage yields about 14 kJ/mol under standard conditions, whereas hydrolysis of each anhydride bond yields about 30 kJ/mol. ATP hydrolysis often plays an important thermodynamic role in biosynthesis. **P5** When coupled to a reaction with a positive free-energy change, ATP hydrolysis shifts the equilibrium of the overall process to favor product formation (recall the relationship between the equilibrium constant and free-energy change described by Eqn 6-3 on p. 182).

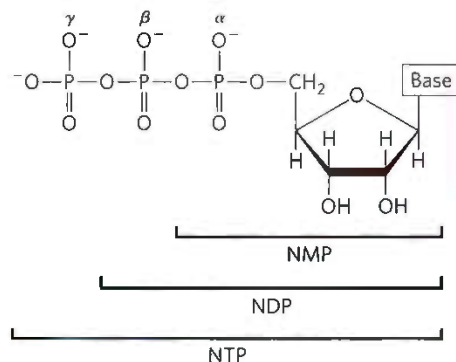


FIGURE 8-39 Nucleoside phosphates. General structure of the nucleoside 5'-mono-, di-, and triphosphates (NMPs, NDPs, and NTPs) and

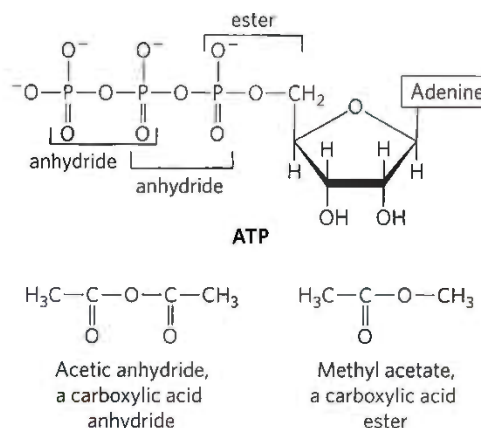


FIGURE 8-40 The phosphate ester and phosphoanhydride bonds of ATP. Hydrolysis of an anhydride bond yields more energy than hydrolysis of the ester. A carboxylic acid anhydride and a carboxylic acid ester are shown for comparison.

Adenine Nucleotides Are Components of Many Enzyme Cofactors

Enzyme cofactors serving a wide range of chemical functions include adenosine as part of their structure (Fig. 8-41). They are unrelated structurally except for the presence of adenosine. In none of these cofactors does the adenosine portion participate directly in the primary function, but removal of adenosine generally results in a drastic reduction of cofactor activities. For example, removal of the adenine nucleotide (3'-phosphoadenosine diphosphate) from acetoacetyl-CoA, the coenzyme A derivative of acetoacetate, reduces its reactivity as a substrate for β -ketoacyl-CoA transferase (an enzyme of lipid metabolism) by a factor of 10^6 . Although this requirement for adenosine has not been investigated in detail, it must involve the binding energy between enzyme and substrate (or cofactor) that is used both in catalysis and in stabilizing

Abbreviations of ribonucleoside 5'-phosphates				Abbreviations of deoxyribonucleoside 5'-phosphates			
Base	Mono-	Di-	Tri-	Base	Mono-	Di-	Tri-
Adenine	AMP	ADP	ATP	Adenine	dAMP	dADP	dATP
Guanine	GMP	GDP	GTP	Guanine	dGMP	dGDP	dGTP
Cytosine	CMP	CDP	CTP	Cytosine	dCMP	dCDP	dCTP
Uracil	UMP	UDP	UTP	Thymine	dTMP	dTDP	dTTP

their standard abbreviations. In the deoxyribonucleoside phosphates (dNMPs, dNDPs, and dNTPs), the pentose is 2'-deoxy-D-ribose.

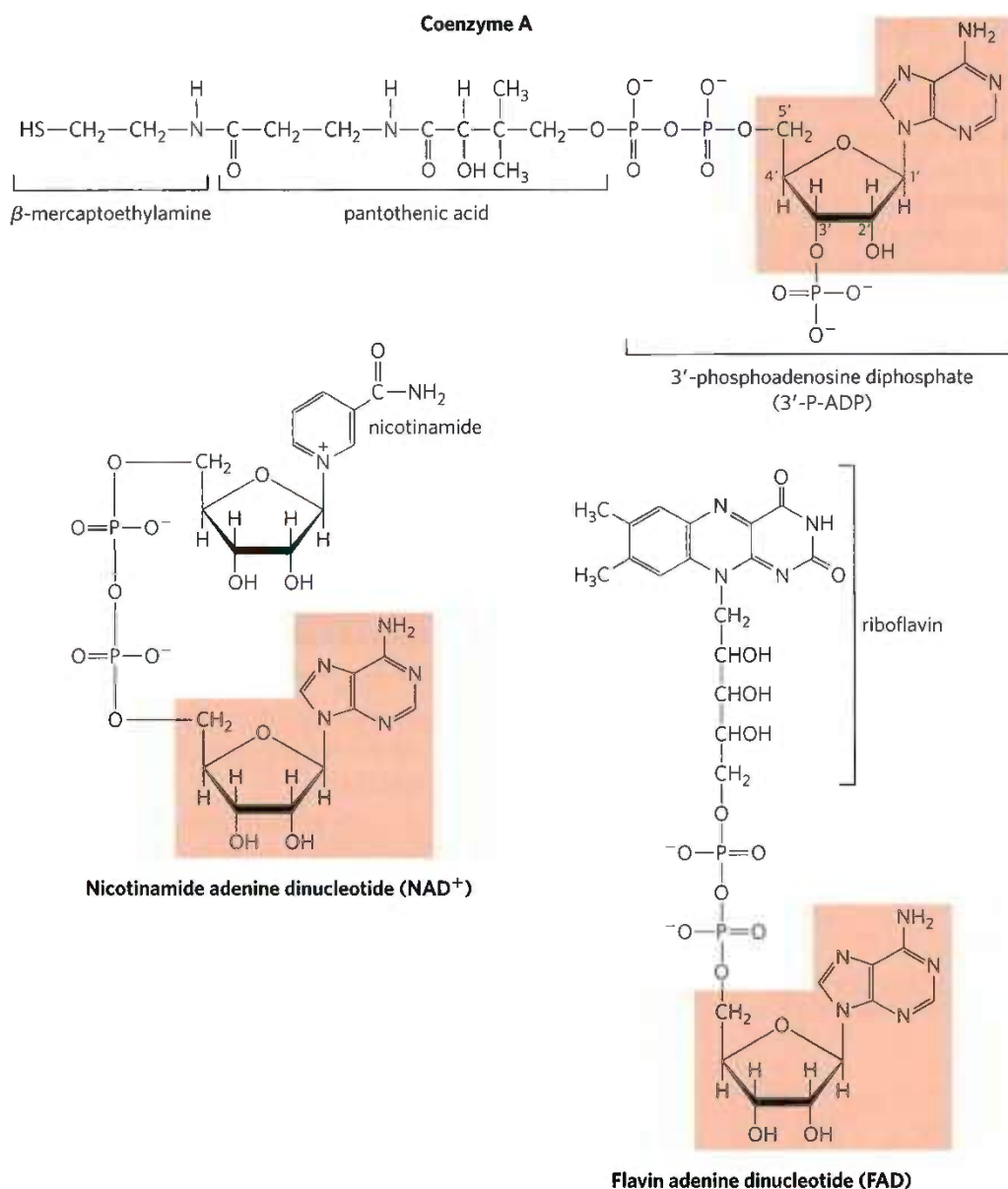


FIGURE 8-41 Some coenzymes containing adenosine. The adenosine portion is shaded in light red. Coenzyme A (CoA) functions in acyl group transfer reactions; the acyl group (such as the acetyl or acetoacetyl group) is attached to the CoA through a thioester linkage to the β -mercaptoethylamine moiety. NAD⁺

functions in hydride transfers, and FAD, the active form of vitamin B₂ (riboflavin), functions in electron transfers. Another coenzyme incorporating adenosine is 5'-deoxyadenosylcobalamin, the active form of vitamin B₁₂ (see Box 17-2), which participates in intramolecular group transfers between adjacent carbons.

the initial enzyme-substrate complex (Chapter 6). In the case of β -ketoacyl-CoA transferase, the nucleotide moiety of coenzyme A seems to be a binding “handle” that helps to pull the substrate (acetoacetyl-CoA) into the active site. Similar roles may be found for the nucleoside portion of other nucleotide cofactors.

Why is adenosine, rather than some other large molecule, used in these structures? The answer here may involve a form of evolutionary economy. Adenosine is certainly not unique in the amount of potential binding energy it can contribute. The importance of adenosine probably

lies not so much in some special chemical characteristic as in the evolutionary advantage of using one compound for multiple roles. Once ATP became the universal source of chemical energy, systems developed to synthesize ATP in greater abundance than the other nucleotides; because it is abundant, it becomes the logical choice for incorporation into a wide variety of structures. The economy extends to protein structure. A single protein domain that binds adenosine can be used in different enzymes. Such a domain, called a **nucleotide-binding fold**, is found in many enzymes that bind ATP and nucleotide cofactors.

Some Nucleotides Are Regulatory Molecules

Cells respond to their environment by taking cues from hormones or other external chemical signals. The interaction of these extracellular chemical signals ("first messengers") with receptors on the cell surface often leads to the production of **second messengers** inside the cell, which in turn leads to adaptive changes in the cell interior (Chapter 12). Often, the second messenger is a nucleotide (Fig. 8-42). One of the most common is **adenosine 3',5'-cyclic monophosphate (cyclic AMP, or cAMP)**, formed from ATP in a reaction catalyzed by **adenylyl cyclase**, an enzyme associated with the inner face of the plasma membrane. Cyclic AMP serves regulatory functions in virtually every cell outside the plant kingdom. Guanosine 3',5'-cyclic monophosphate (cGMP) also has regulatory functions in many cells.

Another regulatory nucleotide, ppGpp (Fig. 8-42), is produced in bacteria in response to a slowdown in protein synthesis during amino acid starvation. This nucleotide inhibits the synthesis of the rRNA and tRNA molecules (see Fig. 28-22) needed for protein synthesis, preventing the unnecessary production of nucleic acids.

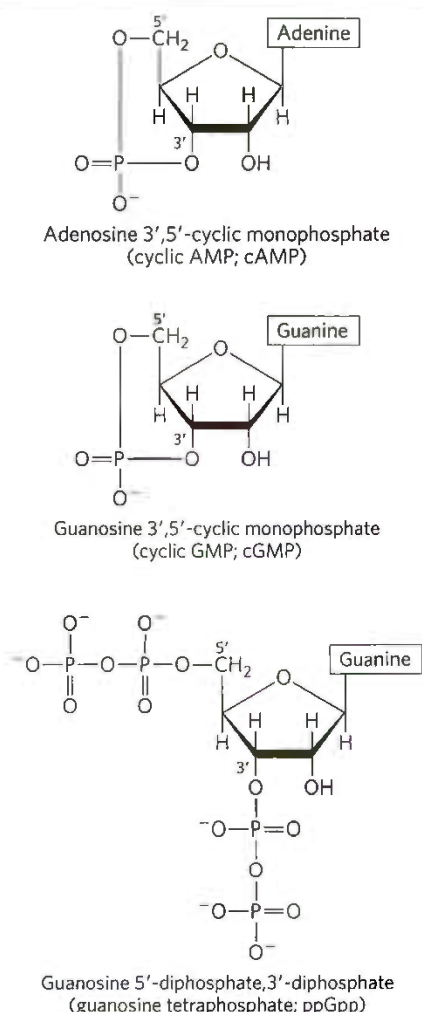


FIGURE 8-42 Three regulatory nucleotides.

Adenine Nucleotides Also Serve as Signals

 ATP and ADP also serve as signaling molecules in many unicellular and multicellular organisms, including humans. In mammals, certain neurons release ATP at synapses. The ATP binds P_{2X} receptors on the postsynaptic cell, triggering changes in membrane potential or the release of an intracellular second messenger that initiates diverse physiological processes, including taste, inflammation, and smooth muscle contraction. One important class of ATP receptors that mediate the sensation of pain is an obvious target for drug development. Extracellular ADP is a signaling molecule that acts through P_{2Y} receptors in sensitive cell types. By preventing ADP from binding the P_{2Y} receptors of platelets, the drug clopidogrel (Plavix) inhibits undesirable blood clotting in patients with cardiac disease. Signaling pathways are discussed in more detail in Chapter 12. ■

SUMMARY 8.4 Other Functions of Nucleotides

- ATP is the central carrier of chemical energy in cells.
- The presence of an adenosine moiety in a variety of enzyme cofactors may be related to binding-energy requirements.
- Cyclic AMP, formed from ATP in a reaction catalyzed by adenylyl cyclase, is a common second messenger produced in response to hormones and other chemical signals.
- ATP and ADP serve as neurotransmitters in a variety of signaling pathways.

KEY TERMS

Terms in bold are defined in the glossary.

deoxyribonucleic acid (DNA) 263	tautomers 268
ribonucleic acid (RNA) 263	base pair 268
gene 263	major groove 271
ribosomal RNA (rRNA) 264	minor groove 271
messenger RNA (mRNA) 264	B-form DNA 272
transfer RNA (tRNA) 264	A-form DNA 272
nucleotide 264	Z-form DNA 272
nucleoside 264	palindrome 273
pyrimidine 264	hairpin 273
purine 264	cruciform 273
deoxyribonucleotides 265	triplex DNA 274
ribonucleotide 265	G tetraplex 274
phosphodiester linkage 267	transcription 275
5' end 267	monocistronic mRNA 275
3' end 267	polycistronic mRNA 275
oligonucleotide 268	mutation 280
polynucleotide 268	polymerase chain reaction (PCR) 283
	DNA polymerases 283
	primer 283
	Sanger sequencing 286

sequence	sequencing depth 291
polymorphisms 288	contig 293
short tandem repeat (STR) 288	nucleotide-binding fold 295
reversible terminator sequencing 290	second messenger 296
single-molecule real-time (SMRT) sequencing 290	adenosine 3',5'-cyclic monophosphate (cyclic AMP, cAMP) 296

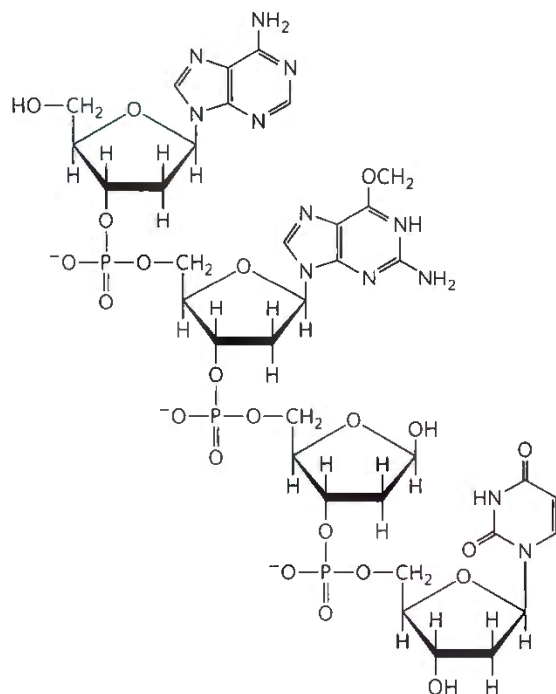
PROBLEMS

1. Nucleotide Structure Which positions in the purine ring of a purine nucleotide in DNA have the potential to form hydrogen bonds but are not involved in Watson-Crick base pairing?

2. Base Sequence of Complementary DNA Strands One strand of a double-helical DNA has the sequence (5')GCG CAATATTTCTCAAAATATTGCGC(3'). Write the base sequence of the complementary strand. What special type of sequence is contained in this DNA segment? Does the double-stranded DNA have the potential to form any alternative structures?

3. DNA of the Human Body If completely unraveled, all of a human's DNA would be able to reach a distance of nearly 3.2×10^5 km, the distance from Earth to the moon. Given that each base pair in a DNA helix extends a distance of 3.4 Å, calculate the number of base pairs found within the entirety of a human's DNA.

4. Nucleic Acids A damaged tetranucleotide structure is shown below. (a) Name each of the nucleotides (or the type of damaged site as appropriate), proceeding from top left to lower right. (b) Indicate which end (upper left or lower right) is the 3' end and which is the 5' end. (c) Is this tetranucleotide DNA or RNA?



5. Distinction between DNA Structure and RNA Structure Secondary structures called hairpins may form at palindromic sequences in single strands of either RNA or DNA. The fully

base-paired portions of hairpins form helices. How do RNA hairpins differ from DNA hairpins?

6. Nucleotide Chemistry The cells of many eukaryotic organisms have highly specialized systems that specifically repair G-T mismatches in DNA. The mismatch is repaired to form a G≡C, not A=T, base pair. This G-T mismatch repair mechanism occurs in addition to a more general system that repairs virtually all mismatches. Suggest why cells might require a specialized system to repair G-T mismatches.

7. Denaturation of Nucleic Acids A duplex DNA oligonucleotide in which one of the strands has the sequence TAATACGACT CACTATAGGG has a melting temperature (t_m) of 59 °C. If an RNA duplex oligonucleotide of identical sequence (substituting U for T) is constructed, will its melting temperature be higher or lower?

8. Spontaneous DNA Damage Hydrolysis of the *N*-glycosyl bond between deoxyribose and a purine in DNA creates an apurinic (AP) site. An AP site is more thermodynamically destabilizing to a DNA molecule than is a mismatched base pair. Examine the structure of an AP site (see Fig. 8-29b) and describe some chemical consequences of base loss.

9. Prediction of Nucleic Acid Structure from Its Sequence A part of a sequenced chromosome has the sequence (on one strand) ATTGCATCCGCGCGTGC GCGCGCGATCCCGT TACTTTCCG. What is the longest part of this sequence that is likely to take up the Z conformation?

10. Nucleic Acid Identity Explain how RNA nucleotides differ from DNA nucleotides.

11. Nucleic Acid Structure Explain why the absorption of UV light by double-stranded DNA increases (the hyperchromic effect) when the DNA is denatured.

12. Solubility of the Components of DNA Draw the structures of deoxyribose, guanine, and phosphate and rate their relative solubilities in water (most soluble to least soluble). How are these solubilities consistent with the three-dimensional structure of double-stranded DNA?

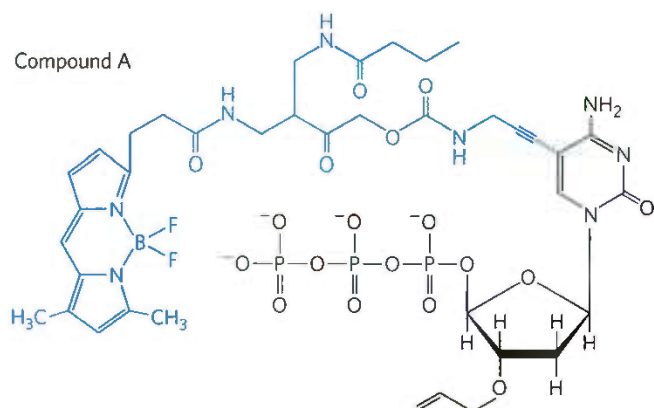
13. Polymerase Chain Reaction An investigator has one strand of a chromosomal DNA whose sequence is shown. She wants to use polymerase chain reaction (PCR) to amplify and isolate the DNA fragment defined by the segment shown in red. Her first step is to design two PCR primers, each 20 nucleotides long, that can be used to amplify this DNA segment. The final PCR product generated from the primers should include no sequences outside the segment in red.

5' --- AATGCCGTCAGCCGATCTGCCTCGAGTCAATCGAT
GCTGGTAACTTGGGGTATAAAGCTTACCCATGGTATCGTAGT
TAGATTGATTGTTAGGTTCTTAGGTTTAGGTTTCTGGTATTG
GTTAGGGTCTTTGATGCTATTAATTGTTGGTTTGGATTG
GTCTTATATGGTTTATGTTTAAAGCCGGGTTTGCTCGG
GATGGTTCGTCTGATGTGCGCGTAGCGTGCGGCG --- 3'

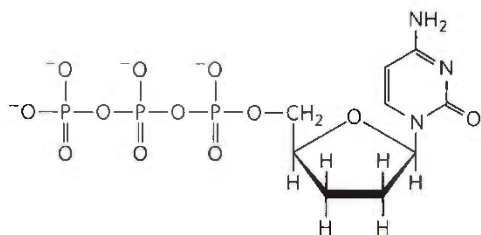
What are the sequences of the investigator's forward primer and reverse primer? Recall that the forward primer binds to the strand of DNA running in the 3' to 5' direction, whereas the reverse primer binds to the opposite strand.

14. DNA Sequencing Reagents Indicate which of the modified cytidine nucleotide triphosphates shown might be used for each procedure: (a) Classical Sanger sequencing

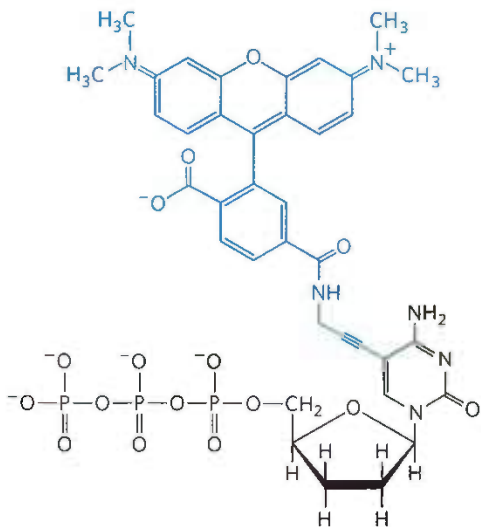
(b) Automated Sanger sequencing (c) Next-generation DNA sequencing (Illumina). Linked fluorescent dyes, where present, are highlighted.



Compound B



Compound C



15. Genomic Sequencing In large-genome sequencing projects, the initial data usually reveal gaps between contigs where no sequence information has been obtained. To close the gaps, DNA primers complementary to the 5'-ending strand at the end of each contig are especially useful. Explain how researchers could use these primers to close the gaps between contigs.

16. Next-Generation Sequencing In reversible terminator sequencing, how would the sequencing process be affected if the 3'-end-blocking group of each nucleotide were replaced with the 3'-H present in the dideoxynucleotides used in Sanger sequencing?

17. Sanger Sequencing Logic In the Sanger (dideoxy) method for DNA sequencing, researchers add a small amount

of a dideoxynucleoside triphosphate, such as ddCTP, to the sequencing reaction along with a larger amount of the corresponding deoxynucleoside, such as dCTP. What result would researchers observe if they omitted CTP from the sequencing reaction?

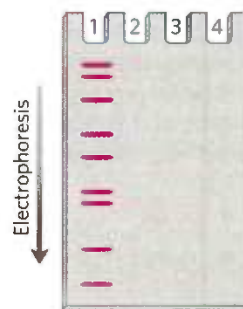
18. DNA Sequencing A researcher used the Sanger method to sequence the DNA fragment shown. The red asterisk indicates a fluorescent label.



She reacted a sample of the DNA with DNA polymerase and each of the four nucleotide mixtures (in an appropriate buffer) listed. Some of the mixtures included dideoxynucleotides (ddNTPs) in relatively small amounts.

1. dATP, dTTP, dCTP, dGTP, ddTTP
2. dATP, dTTP, dCTP, dGTP, ddGTP
3. dATP, dCTP, dGTP, ddTTP
4. dATP, dTTP, dCTP, dGTP

The researcher then separated the resulting DNA by electrophoresis on a polyacrylamide gel and located the fluorescent bands on the gel. The image of the gel shows the band pattern resulting from nucleotide mixture 1. Assuming that all mixtures were run on the same gel, what did the remaining lanes of the gel look like?



19. Snake Venom Phosphodiesterase An exonuclease is an enzyme that sequentially cleaves nucleotides from the end of a polynucleotide strand. Snake venom phosphodiesterase, which hydrolyzes nucleotides from the 3' end of any oligonucleotide with a free 3'-hydroxyl group, cleaves between the 3' hydroxyl of the ribose or deoxyribose and the phosphoryl group of the next nucleotide. It acts on single-stranded DNA or RNA and has no base specificity. This enzyme was used in sequence determination experiments before the development of modern nucleic acid sequencing techniques. What are the products of partial digestion by snake venom phosphodiesterase of an oligonucleotide with the sequence (5')GCGCCAUUGC(3')—OH?

20. Preserving DNA in Bacterial Endospores Bacterial endospores form when the environment is no longer conducive to active cell metabolism. The soil bacterium *Bacillus subtilis*, for example, begins the process of sporulation when one or more nutrients are depleted. The end product is a small, metabolically dormant structure that can survive almost indefinitely with no detectable metabolism. Spores have mechanisms to prevent accumulation of potentially lethal mutations in their DNA over periods of dormancy that can exceed 1,000 years. *B. subtilis* spores are much more resistant than are the organism's growing cells to heat, UV radiation, and oxidizing agents, all of which promote mutations.

(a) One factor that prevents potential DNA damage in spores is their greatly decreased water content. How would this affect some types of mutations?

(b) Endospores have a category of proteins called small acid-soluble proteins (SASPs) that bind to their DNA, preventing formation of cyclobutane-type dimers. What causes cyclobutane dimers, and why do bacterial endospores need mechanisms to prevent their formation?

21. Oligonucleotide Synthesis As shown in the scheme of Figure 8-34, oligonucleotide synthesis involves adding modified bases, one at a time, to a growing chain. The modified bases contain an activated 3' hydroxyl and have a dimethoxytrityl (DMT) group attached to the 5' hydroxyl. What is the function of the DMT group on the incoming base?

BIOCHEMISTRY ONLINE

22. The Structure of DNA Elucidation of the three-dimensional structure of DNA helped researchers understand how this molecule conveys information that can be faithfully replicated from one generation to the next. To see the secondary structure of double-stranded DNA, go to the Protein Data Bank website (www.rcsb.org). Use the PDB identifiers provided in parts (a) and (b) below to retrieve the structure summary for a double-stranded DNA segment. View the 3D structure using JSmol. The viewer select menu is below the right corner of the image box. Once in JSmol, you will need to use both the display menus on the screen and the scripting controls in the JSmol menu. Access the JSmol menu by clicking on the JSmol logo in the lower right corner of the image screen. Refer to the JSmol help links as needed.

(a) Access PDB ID 141D, a highly conserved, repeated DNA sequence from the end of the genome of HIV-1 (the virus that causes AIDS). Set the Style to Ball and Stick. Then use the scripting controls to color by element (Color > Atoms > By Scheme > Element (CPK)). Identify the sugar-phosphate backbone for each strand of the DNA duplex. Locate and identify individual bases. Identify the 5' end of each strand. Locate the major and minor grooves. Is this a right- or left-handed helix?

(b) Access PDB ID 145D, a DNA with the Z conformation. Set the Style to Ball and Stick. Then use the scripting controls to color by element (Main Menu > Color > Atoms > By Scheme > Element (CPK)). Identify the sugar-phosphate backbone for each strand of the DNA duplex. Is this a right- or left-handed helix?

(c) To fully appreciate the secondary structure of DNA, view the molecules in stereo. From the scripting control Main Menu select Style > Stereographic > Cross-eyed viewing or Wall-eyed viewing. (If you have stereographic glasses available, select the appropriate option.) You will see two images of the DNA molecule. Sit with your nose approximately 10 inches from the screen and focus on the tip of your nose (cross-eyed) or on the opposite edges of the screen (wall-eyed). In the background you should see three images of the DNA helix. Shift your focus to the middle image, which should appear three-dimensional. (Note that only one of the authors can make this work.)

DATA ANALYSIS PROBLEM

23. Chargaff's Studies of DNA Structure The main findings of Erwin Chargaff and his coworkers ("Chargaff's rules") are summarized on page 270. In this problem, you will examine the data Chargaff collected in support of his conclusions.

In one paper, Chargaff (1950) described his analytical methods and some early results. Briefly, he treated DNA samples with acid to remove the bases, separated the bases by paper chromatography, and measured the amount of each base with UV spectroscopy. His results are shown in the three tables below. The *molar ratio* is the ratio of the number of moles of each base in the sample to the number of moles of phosphate in the sample—this gives the fraction of the total number of bases represented by each particular base. The *recovery* is the sum of all four bases (the sum of the molar ratios); full recovery of all bases in the DNA would give a recovery of 1.0.

Molar ratios in ox DNA

Base	Thymus			Spleen		Liver
	Prep. 1	Prep. 2	Prep. 3	Prep. 1	Prep. 2	Prep. 1
Adenine	0.26	0.28	0.30	0.25	0.26	0.26
Guanine	0.21	0.24	0.22	0.20	0.21	0.20
Cytosine	0.16	0.18	0.17	0.15	0.17	
Thymine	0.25	0.24	0.25	0.24	0.24	
Recovery	0.88	0.94	0.94	0.84	0.88	

Molar ratios in human DNA

Base	Sperm		Thymus	Liver	
	Prep. 1	Prep. 2	Prep. 1	Normal	Carcinoma
Adenine	0.29	0.27	0.28	0.27	0.27
Guanine	0.18	0.17	0.19	0.19	0.18
Cytosine	0.18	0.18	0.16		0.15
Thymine	0.31	0.30	0.28		0.27
Recovery	0.96	0.92	0.91		0.87

Molar ratios in DNA of microorganisms

Base	Yeast		Avian tubercle bacilli
	Prep. 1	Prep. 2	Prep. 1
Adenine	0.24	0.30	0.12
Guanine	0.14	0.18	0.28
Cytosine	0.13	0.15	0.26
Thymine	0.25	0.29	0.11
Recovery	0.76	0.92	0.77

(a) Based on these data, Chargaff concluded that "no differences in composition have so far been found in DNA from different tissues of the same species." However, a skeptic looking at the data might say, "They certainly look different to me!" If you were Chargaff, how would you use the data to change the skeptic's mind?

(b) The base composition of DNA from normal and cancerous liver cells (hepatocarcinoma) was not distinguishably different. Would you expect Chargaff's technique to be capable of detecting a difference between the DNA of normal and cancerous cells? Explain your reasoning.

As you might expect, Chargaff's data were not completely convincing. He went on to improve his techniques, as described

in his 1951 paper, in which he reported molar ratios of bases in DNA from a variety of organisms.

Source	A:G	T:C	A:T	G:C	Purine:pyrimidine
Ox	1.29	1.43	1.04	1.00	1.1
Human	1.56	1.75	1.00	1.00	1.0
Hen	1.45	1.29	1.06	0.91	0.99
Salmon	1.43	1.43	1.02	1.02	1.02
Wheat	1.22	1.18	1.00	0.97	0.99
Yeast	1.67	1.92	1.03	1.20	1.0
<i>Haemophilus influenzae</i> type c	1.74	1.54	1.07	0.91	1.0
<i>E. coli</i> K-12	1.05	0.95	1.09	0.99	1.0
Avian tubercle bacillus	0.4	0.4	1.09	1.08	1.1
<i>Serratia marcescens</i>	0.7	0.7	0.95	0.86	0.9
<i>Bacillus schatz</i>	0.7	0.6	1.12	0.89	1.0

(c) According to Chargaff, "The base composition of DNA generally varies from one species to another." Provide an argument, based on the data presented so far, that supports this conclusion.

(d) According to Chargaff's rules, "In all cellular DNAs, regardless of the species, . . . $A + G = T + C$." Provide an argument, based on the data presented, that supports this conclusion.

References

Chargaff, E. 1950. Chemical specificity of nucleic acids and mechanism of their enzymatic degradation. *Experientia* 6:201–209.

Chargaff, E. 1951. Structure and function of nucleic acids as cell constituents. *Fed. Proc.* 10:654–659.

DNA-BASED INFORMATION TECHNOLOGIES

- 9.1 Studying Genes and Their Products 302
- 9.2 Exploring Protein Function on the Scale of Cells or Whole Organisms 317
- 9.3 Genomics and the Human Story 326

The complexity of the molecules and systems revealed in this book can sometimes conceal a scientific reality: what we have learned is just a beginning. Novel proteins and lipids and carbohydrates and nucleic acids are discovered every day, and we often have no clue as to their functions. How many have yet to be encountered, and what might they do? Even well-characterized biomolecules continue to challenge researchers with countless unresolved mechanistic and functional questions. A new era, defined by technologies that provide broad access to the entirety of a cell's DNA, the genome, has accelerated progress.

This is a methods chapter, a necessary prelude to much of what comes later in this book. It is organized around just a few straightforward principles:

P1 An organism's DNA—its genome—is the ultimate source of biological information.

Genomic information is a resource of unparalleled importance for investigators studying any aspect of biology. Genomes vary in size but all are large enough to direct all aspects of an organism's structure and function. To approach them often requires tools to break them into small parts that are experimentally digestible.

P2 Genomic information is accessible. Advances in DNA sequencing (Chapter 8) are being matched by new approaches to understanding how chromosomal information is expressed and regulated on a genomic and cellular scale. Important clues to protein function are embedded in the sequences of the genes that encode them.

P3 Genomic information is malleable. We can not only elucidate cellular genomic information; we can also change it. That capacity provides a path to altering any aspect of cellular metabolism, structure, or function.

The word “genome,” coined by German botanist Hans Winkler in 1920, was derived from the Greek words *genesis* and *soma* to describe a body of genes. A **genome** today is defined as the complete haploid genetic complement of an organism. In essence, a genome is one copy of the hereditary information required to specify the organism. For sexually reproducing organisms, the genome includes one set of autosomes and one of each type of sex chromosome. When cells have organelles that also contain DNA, the genetic content of the organelles is not considered part of the nuclear genome. Mitochondria, found in most eukaryotic cells, and chloroplasts, in the light-harvesting cells of photosynthetic organisms, each have their own distinct genome. For viruses, which can have genetic material composed of DNA or RNA, the genome is a complete copy of the nucleic acid required to specify the virus.

As objects of study, DNA molecules present a special problem: their size. Chromosomes are far and away the largest biomolecules in any cell. How do researchers find the information they seek when it is just a small part of a chromosome that can include millions or even billions of contiguous base pairs? Decades of advances by thousands of scientists working in genetics, biochemistry, cell biology, and physical chemistry came together in the laboratories of Paul Berg, Herbert Boyer, and Stanley Cohen to yield the first techniques for locating, isolating, preparing, and studying small segments of DNA derived from much larger chromosomes. The science of **genomics** is dedicated to the study of DNA on a cellular scale. In turn, genomics contributes to **systems biology**, the study of biochemistry on the scale of whole cells and organisms.

The methods described in this chapter were built on advances in our understanding of DNA and RNA

metabolism that are not presented in this text until Part III. Fundamental concepts of DNA replication, RNA transcription, protein synthesis, and gene regulation are intrinsic to an appreciation for how these methods work. Yet all facets of modern biochemistry rely on these same methods to such an extent that a current treatment of any aspect of the discipline becomes very difficult without a proper introduction to them. By presenting these technologies early in the book, we acknowledge that they are inextricably interwoven with both the advances that gave rise to them and the newer discoveries they now make possible. The background we necessarily provide makes the discussion here not just an introduction to technology but also a preview of many of the fundamentals of DNA and RNA biochemistry encountered in later chapters.

We begin by outlining the principles of DNA cloning, then illustrate the range of applications and the potential of many newer technologies that support and accelerate the advance of biochemistry.

9.1 Studying Genes and Their Products

A researcher has isolated a new enzyme that she knows is the key to a human disease. She hopes to isolate large amounts of the protein to crystallize it for structural analysis and to study it. She wants to alter amino acid residues at its active site so that she can understand the reaction it catalyzes. She plans an elaborate research program to elucidate how this enzyme interacts with, and is regulated by, other proteins in the cell. All of this, and much more, becomes possible if she can isolate the gene encoding her enzyme. Unfortunately, that gene consists of just a few thousand base pairs within a human chromosome with a size measured in hundreds of millions of base pairs. How does she isolate the small segment that she needs and then study it? The answer lies in DNA cloning and methods developed to manipulate cloned genes.

Genes Can Be Isolated by DNA Cloning

A *clone* is an identical copy. This term originally applied to cells of a single type, isolated and allowed to reproduce to create a population of identical cells. When applied to DNA, a clone represents many identical copies of a particular gene segment. **P1** In brief, our researcher must separate the gene from the larger chromosome, attach it to a much smaller piece of carrier DNA, and allow microorganisms to make many copies of it. This is the process of **DNA cloning**. The result is selective amplification of a particular gene or DNA segment so that its genetic information may be studied and utilized. Classically, the cloning of DNA from any organism entails five general procedures:

1. *Obtaining the DNA segment to be cloned.* Enzymes called restriction endonucleases act as precise molecular scissors, recognizing specific sequences

in DNA and cleaving genomic DNA into smaller fragments suitable for cloning. Alternatively, genomic DNA can be sheared randomly into fragments of a desired size. Since the sequence of targeted genomic regions is often known (available in databases), DNA segments to be cloned are most often amplified by the polymerase chain reaction (PCR) or are simply synthesized (both methods are described in Chapter 8).

2. *Selecting a small molecule of DNA capable of autonomous replication.* These small DNAs are called **cloning vectors** (a vector is a carrier or delivery agent). Most cloning vectors used in the laboratory are modified versions of naturally occurring small DNA molecules found in bacteria or eukaryotes. Viral DNAs may also play this role.
3. *Joining two DNA fragments covalently.* The enzyme DNA ligase links the cloning vector to the DNA fragment to be cloned. Composite DNA molecules of this type, comprising covalently linked segments from two or more sources, are called **recombinant DNAs**.
4. *Moving recombinant DNA from the test tube to a host organism.* The host organism provides the enzymatic machinery for DNA replication.
5. *Selecting or identifying host cells that contain recombinant DNA.* The cloning vector generally has features that allow the host cells to survive in an environment in which cells lacking the vector would die. Cells containing the vector are thus “selectable” in that environment.

The methods used to accomplish these and related tasks are collectively referred to as **recombinant DNA technology** or, more informally, **genetic engineering**.

Much of our initial discussion focuses on DNA cloning in the bacterium *Escherichia coli*, the first organism used for recombinant DNA work and still the most common host cell. *E. coli* has many advantages: its DNA metabolism (like many other of its biochemical processes) is well understood; many naturally occurring cloning vectors associated with *E. coli*, such as plasmids and bacteriophages (bacterial viruses; also called phages), are readily available; and techniques are available for moving DNA expeditiously from one bacterial cell to another. The principles discussed here are broadly applicable to DNA cloning in other organisms, a topic discussed more fully later in the section.

Restriction Endonucleases and DNA Ligases Yield Recombinant DNA

A set of enzymes (**Table 9-1**) made available through decades of research on nucleic acid metabolism is indispensable for generating and propagating a recombinant DNA molecule (**Fig. 9-1**). First, **P1** **restriction endonucleases** (also called restriction enzymes)

TABLE 9-1 Some Enzymes Used in Recombinant DNA Technology

Enzyme(s)	Function
Type II restriction endonucleases	Cleave DNA molecules at specific base sequences
DNA ligase	Joins two DNA molecules or fragments
DNA polymerase I (<i>E. coli</i>)	Fills gaps in duplexes by stepwise addition of nucleotides to 3' ends
Reverse transcriptase	Makes a DNA copy of an RNA molecule
Polynucleotide kinase	Adds a phosphate to the 5'-OH end of a polynucleotide to label it or to permit ligation
Terminal transferase	Adds homopolymer tails to the 3'-OH ends of a linear duplex
Exonuclease III	Removes nucleotide residues from the 3' ends of a DNA strand
Bacteriophage λ exonuclease	Removes nucleotides from the 5' ends of a duplex to expose single-stranded 3' ends
Alkaline phosphatase	Removes terminal phosphates from the 5' end or 3' end (or both)

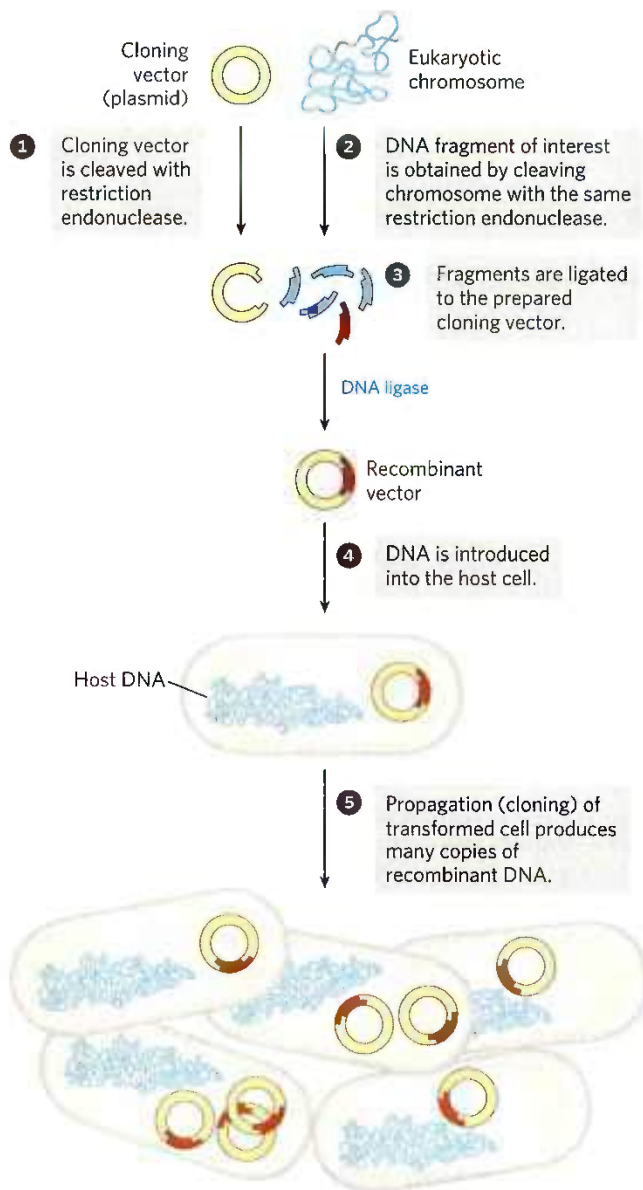


FIGURE 9-1 Schematic illustration of DNA cloning. A cloning vector and eukaryotic chromosomes are separately cleaved with the same restriction endonuclease. (A single chromosome is shown here for simplicity.) The fragments to be cloned are then ligated to the cloning vector. The resulting recombinant DNA (only one recombinant vector is shown here) is introduced into a host cell, where it can be propagated (cloned). Note that this drawing is not to scale: the size of the *E. coli* chromosome relative to that of a typical cloning vector (such as a plasmid) is much greater than depicted here.

recognize and cleave DNA at specific sequences (recognition sequences or restriction sites) to generate a set of smaller fragments. Second, the DNA fragment to be cloned is joined to a suitable cloning vector by using **DNA ligases** to link the DNA molecules together. The recombinant vector is then introduced into a host cell, which amplifies the fragment in the course of many generations of cell division.

Restriction endonucleases are found in a wide range of bacterial species. As Werner Arber discovered in the early 1960s, the biological function of restriction endonucleases is to recognize and cleave foreign DNA (the DNA of an infecting virus, for example); such DNA is said to be *restricted*. In the host cell's DNA, the sequence that would be recognized by one of its own restriction endonucleases is protected from digestion by methylation of the DNA, catalyzed by a specific DNA methylase. The restriction endonuclease and the corresponding methylase are sometimes referred to as a **restriction-modification system**.

There are three types of restriction endonucleases, designated I, II, and III. Types I and III are generally large, multisubunit complexes containing both the endonuclease and methylase activities. **Type II restriction endonucleases**, first isolated by Hamilton Smith in 1970, are simpler, require no ATP, and catalyze the hydrolytic cleavage of particular phosphodiester bonds in the DNA within the recognition sequence itself. The extraordinary utility of this group of restriction endonucleases was demonstrated by Daniel Nathans, who first used them to develop novel methods for mapping and analyzing genes and genomes.

Thousands of type II restriction endonucleases have been discovered in different bacterial species, and more than 100 different DNA sequences are recognized by one or more of these enzymes. The recognition sequences are usually 4 to 6 bp long and are palindromic (see Fig. 8-18). **Table 9-2** lists sequences recognized by a few type II restriction endonucleases.

Some restriction endonucleases make staggered cuts on the two DNA strands, leaving two to four

TABLE 9-2 Recognition Sequences for Some Type II Restriction Endonucleases

BamHI	$\begin{array}{c} \downarrow \quad \quad * \\ (5') \text{ G G A T C C } (3') \\ \quad \quad \quad \uparrow \\ \text{C C T A G G} \\ \quad \quad \quad * \end{array}$	HindIII	$\begin{array}{c} \downarrow \\ (5') \text{ A A G C T T } (3') \\ \quad \quad \quad \uparrow \\ \text{T T C G A A} \end{array}$
ClaI	$\begin{array}{c} \downarrow \quad \quad * \\ (5') \text{ A T C G A T } (3') \\ \quad \quad \quad \uparrow \\ \text{T A G C T A} \\ \quad \quad \quad * \end{array}$	NotI	$\begin{array}{c} \downarrow \\ (5') \text{ G C G G C C G C } (3') \\ \quad \quad \quad \uparrow \\ \text{C G C C G G C G} \end{array}$
EcoRI	$\begin{array}{c} \downarrow \quad \quad * \\ (5') \text{ G A A T T C } (3') \\ \quad \quad \quad \uparrow \\ \text{C T T A A G} \\ \quad \quad \quad * \end{array}$	PstI	$\begin{array}{c} \downarrow \quad \quad * \\ (5') \text{ C T G C A G } (3') \\ \quad \quad \quad \uparrow \\ \text{G A C G T C} \\ \quad \quad \quad * \end{array}$
EcoRV	$\begin{array}{c} \downarrow \\ (5') \text{ G A T A T C } (3') \\ \quad \quad \quad \uparrow \\ \text{C T A T A G} \end{array}$	PvuII	$\begin{array}{c} \downarrow \\ (5') \text{ C A G C T G } (3') \\ \quad \quad \quad \uparrow \\ \text{G T C G A C} \end{array}$
HaeIII	$\begin{array}{c} \downarrow * \\ (5') \text{ G G C C } (3') \\ \quad \quad \quad \uparrow \\ \text{G G C C} \\ \quad \quad \quad * \end{array}$	Tth111I	$\begin{array}{c} \downarrow \\ (5') \text{ G A C N N N G T C } (3') \\ \quad \quad \quad \uparrow \\ \text{C T G N N N C A G} \end{array}$

Note: Arrows indicate the phosphodiester bonds cleaved by each restriction endonuclease. Asterisks indicate bases that are methylated by the corresponding methylase (where known). N denotes any base. Note that the name of each enzyme consists of a three-letter abbreviation of the bacterial species from which it is derived, sometimes followed by a strain designation and roman numerals to distinguish different restriction endonucleases isolated from the same bacterial species. Thus BamHI is the first (I) restriction endonuclease characterized from *Bacillus amyloliquefaciens*, strain H.

nucleotides of one strand unpaired at each resulting end. These unpaired strands are referred to as **sticky ends** (Fig. 9-2a) because they can base-pair with each other or with complementary sticky ends of other DNA fragments. Other restriction endonucleases cleave both strands of DNA straight across, at opposing phosphodiester bonds, leaving no unpaired bases on the ends, often called **blunt ends** (Fig. 9-2b).

P2 The gene or DNA segment to be cloned is most often generated by the polymerase chain reaction. Careful design of the primers used for PCR (see Fig. 8-33) can alter the amplified segment by the inclusion, at each end, of additional DNA not present in the chromosome that is being targeted. For example, including restriction endonuclease cleavage sites can facilitate the subsequent cloning of the amplified DNA (Fig. 9-2c).

P3 After the target DNA fragment is prepared and digested with the appropriate restriction enzyme, DNA ligase can be used to join it to a vector digested by the *same* restriction endonuclease; a fragment generated by EcoRI, for example, generally will not link to a fragment generated by BamHI. As described in more detail in Chapter 25 (see Fig. 25-15), DNA ligase catalyzes the formation of new phosphodiester bonds in a reaction that uses ATP or a similar cofactor. The base pairing of complementary sticky ends greatly facilitates the ligation reaction (Fig. 9-2a). Blunt ends can also be ligated, albeit less efficiently. Researchers can create new DNA sequences for a wide range of purposes by inserting synthetic DNA fragments, called **linkers**,

to bridge the ends that are being ligated. An inserted DNA fragment with multiple recognition sequences for restriction endonucleases (often useful later as points for inserting additional DNA by cleavage and ligation) is called the **multiple cloning site (MCS)** (Fig. 9-2d).

The effectiveness of sticky ends in selectively joining two DNA fragments was apparent in the earliest recombinant DNA experiments. Before restriction endonucleases were widely available, some investigators found they could generate sticky ends by the combined action of the bacteriophage λ exonuclease and terminal transferase (Table 9-1). The fragments to be joined were given complementary homopolymeric tails. Peter Lobban and Dale Kaiser used this method in 1971 in the first experiments to join naturally occurring DNA fragments. Similar methods were used soon after in Paul Berg's laboratory to join DNA segments from simian virus 40 (SV40) to DNA derived from bacteriophage λ , thereby creating the first recombinant DNA molecule with DNA segments from different species.

Cloning Vectors Allow Amplification of Inserted DNA Segments

The factors that govern the delivery of recombinant DNA in clonable form to a host cell, and its subsequent amplification in the host, are well illustrated in three popular cloning vectors: plasmids and bacterial

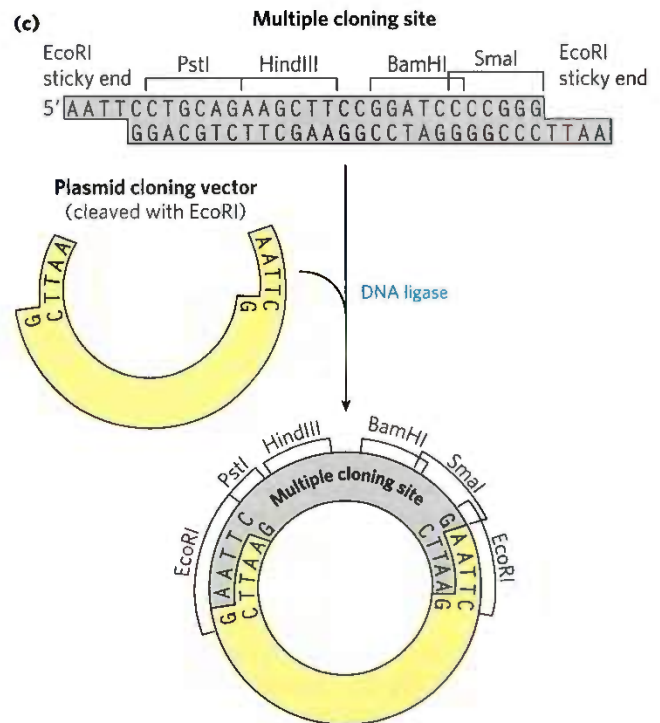
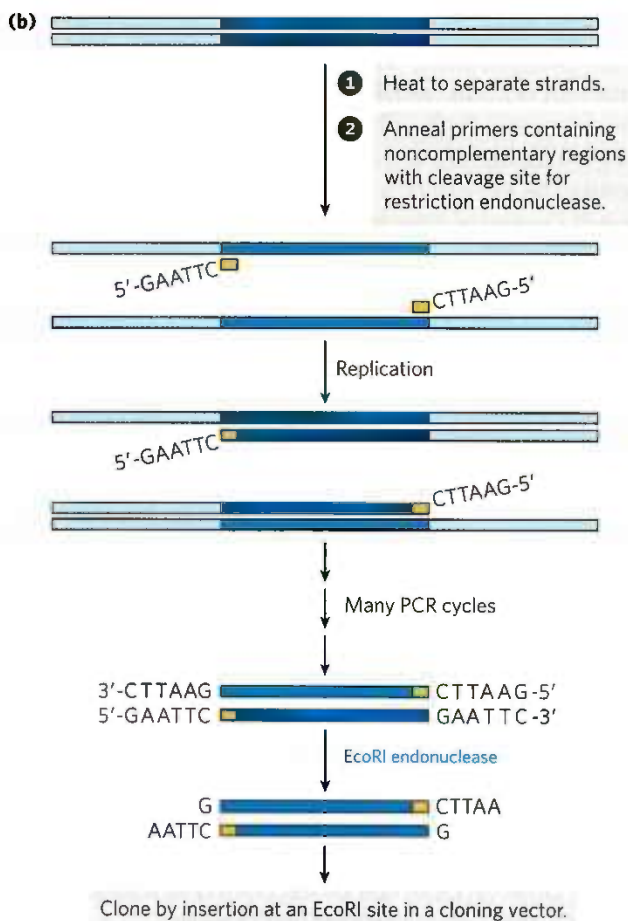
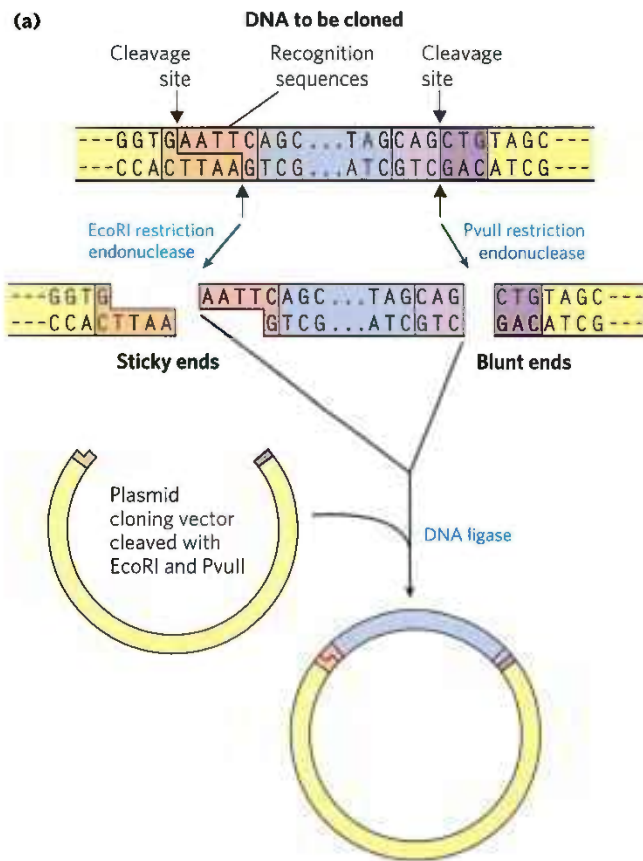


FIGURE 9-2 Use of restriction endonucleases in cloning.

(a) Restriction endonucleases recognize and cleave only specific sequences, leaving either sticky ends (with protruding single strands) or blunt ends. Fragments can be ligated to other DNAs, such as the cleaved cloning vector (a plasmid) shown here. This reaction is facilitated by the annealing of complementary sticky ends. Ligation is less efficient for DNA fragments with blunt ends than for those with complementary sticky ends, and DNA fragments with different (noncomplementary) sticky ends generally are not ligated. (b) DNA that has been amplified by the polymerase chain reaction (see Fig. 8-33) can be cloned. The primers can include noncomplementary ends that have a site for cleavage by a restriction endonuclease. Although these parts of the primers do not anneal to the target DNA, the PCR process incorporates them into the DNA that is amplified. Cleavage of the amplified fragments at these sites creates sticky ends, used in ligation of the amplified DNA to a cloning vector. (c) A synthetic DNA fragment with recognition sequences for several restriction endonucleases can be inserted into a plasmid that has been cleaved by a restriction endonuclease. The insert is called a linker; an insert with multiple restriction sites is generally called a multiple cloning site (MCS).

artificial chromosomes, used in experiments with *E. coli*, and a vector used to clone large DNA segments in yeast.

Plasmids A **plasmid** is a circular DNA molecule that replicates separately from the host chromosome. Naturally occurring bacterial plasmids range in size from 5,000 to 400,000 bp. Many of the plasmids found in bacterial populations are little more than molecular parasites, similar to viruses but with a more limited capacity to transfer from one cell to another. To survive in the host cell, plasmids incorporate several specialized sequences that enable them to make use of the cell's resources for their own replication and gene expression.

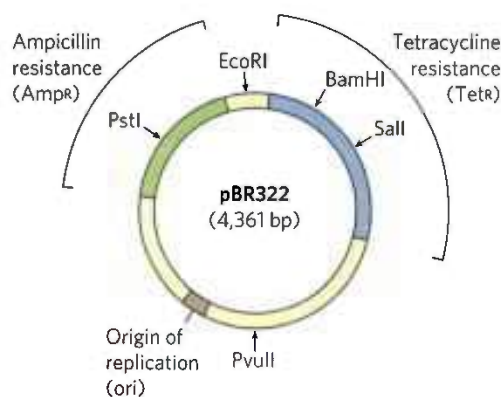


FIGURE 9-3 The constructed *E. coli* plasmid pBR322. Notice the location of some important restriction sites, for PstI, EcoRI, BamHI, Sall, and PvuII; genes for ampicillin and tetracycline resistance (Amp^R and Tet^R); and the replication origin (ori). Constructed in 1977, this was one of the early plasmids designed expressly for cloning in *E. coli*.

Naturally occurring plasmids usually have a symbiotic role in the cell. They may provide genes that confer resistance to antibiotics or that perform new functions for the cell. For example, the Ti plasmid of *Agrobacterium tumefaciens* allows the host bacterium to colonize the cells of a plant and make use of the plant's resources. The same properties that enable plasmids to grow and survive in a bacterial or eukaryotic host are useful to molecular biologists who want to engineer a vector for cloning a specific DNA segment. Constructed in 1977, one of the first recombinant vectors—*E. coli* plasmid pBR322—illustrates some key features that define a useful cloning vector (Fig. 9-3):

1. The plasmid pBR322 has an **origin of replication**, or **ori**, a sequence where replication is initiated by cellular enzymes (see Chapter 25). This sequence is required to propagate the plasmid. An associated regulatory system is present that limits replication to maintain pBR322 at a level of 10 to 20 copies per cell.
2. The plasmid contains genes that confer resistance to the antibiotics ampicillin (Amp^R) and tetracycline (Tet^R), allowing the selection of cells that contain the intact plasmid or a recombinant version of the plasmid (discussed below).
3. Several unique recognition sequences in pBR322 are targets for restriction endonucleases (PstI, EcoRI, BamHI, Sall, and PvuII), providing sites where the plasmid can be cut to insert foreign DNA.
4. The small size of the plasmid (4,361 bp) facilitates its entry into cells and the biochemical manipulation of the DNA. This small size was the result of trimming away many DNA segments from a larger, parent plasmid—sequences that the biochemist does not need.

The replication origins inserted in common plasmid vectors were originally derived from naturally occurring

plasmids. As in pBR322, each of these origins is regulated to maintain a particular plasmid copy number. Depending on the origin used, the plasmid copy number can vary from one to hundreds or thousands per cell, providing many options for investigators. Two different plasmids cannot function in the same cell if they use the same origin of replication, because the regulation of one will interfere with the replication of the other. Such plasmids are said to be incompatible. When a researcher wants to introduce two or more different plasmids into a bacterial cell, each plasmid must have a different replication origin.

In the laboratory, small plasmids can be introduced into bacterial cells by a process called **transformation**. The cells (often *E. coli*, but other bacterial species are also used) and plasmid DNA are incubated together at 0 °C in a calcium chloride solution, then are subjected to heat shock by rapidly shifting the temperature to between 37 °C and 43 °C. For reasons not well understood, some of the cells treated in this way take up the plasmid DNA. Some species of bacteria, such as *Acinetobacter baylyi*, are naturally competent for DNA uptake and do not require the calcium chloride–heat shock treatment. In an alternative method, called **electroporation**, cells incubated with the plasmid DNA are subjected to a high-voltage pulse, which transiently renders the bacterial membrane permeable to large molecules.

Regardless of the approach, relatively few cells take up the plasmid DNA, so a method is needed to identify those that do. The usual strategy is to utilize one of two types of genes in the plasmid, referred to as selectable and screenable markers. A **selectable marker** either permits the growth of a cell (positive selection) or kills the cell (negative selection) under a defined set of conditions. The plasmid pBR322 provides markers for both positive and negative selection (Fig. 9-4). A **screenable marker** is a gene encoding a protein that causes the cell to produce a colored or fluorescent molecule. Cells are not harmed when the gene is present, and the cells that carry the plasmid are easily identified by the colored or fluorescent colonies they produce.

Transformation of typical bacterial cells with purified DNA (never a very efficient process) becomes less successful as plasmid size increases, and it is difficult to clone DNA segments longer than about 15,000 bp when plasmids are used as the vector.

To illustrate the use of a plasmid as a cloning vector, consider the bacterial gene encoding a recombinase called the RecA protein (see Chapter 25). In most bacteria, the gene encoding RecA is one of the thousands of genes on a chromosome millions of base pairs long. The *recA* gene is just over 1,000 bp long. A plasmid would be a good choice for cloning a gene of this size. As described later, the cloned gene can be altered in a variety of ways, and the gene variants can be expressed at high levels to enable purification of the encoded protein.

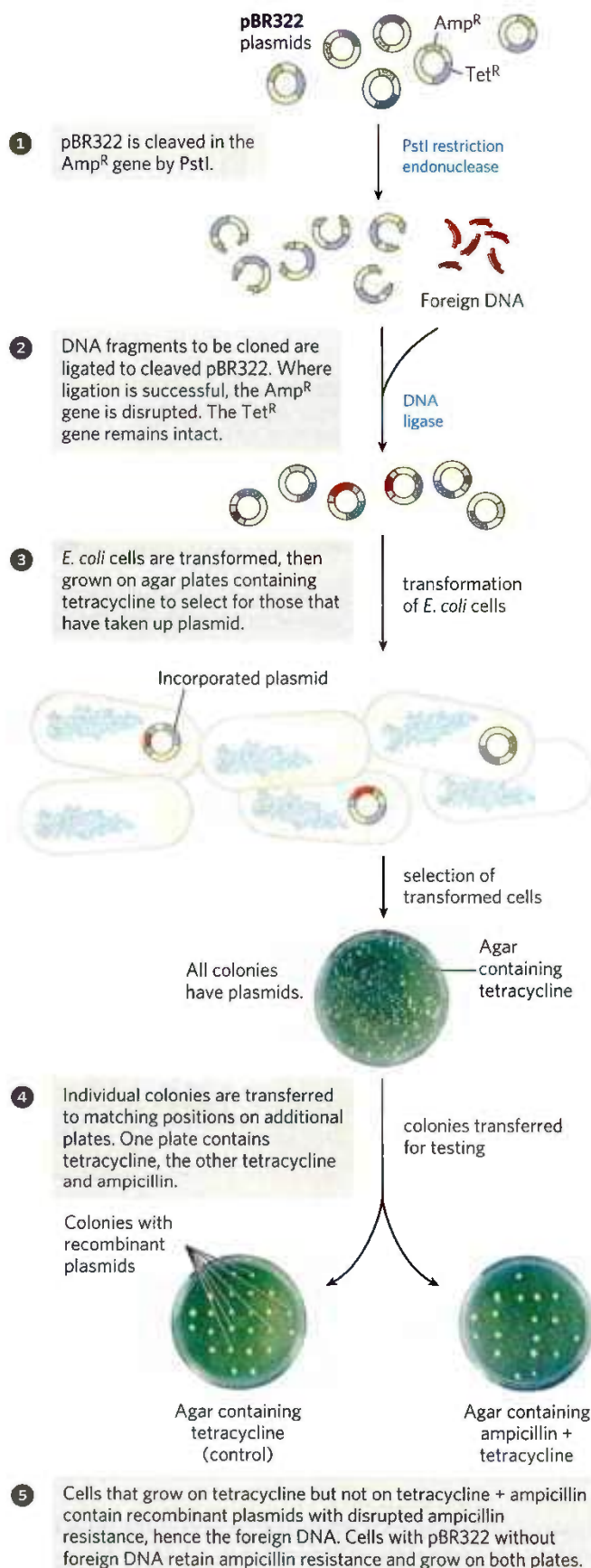


FIGURE 9-4 Use of pBR322 to clone foreign DNA in *E. coli* and identify cells containing the DNA. [Courtesy Elizabeth A. Wood, University of Wisconsin–Madison, Department of Biochemistry.]

Bacterial Artificial Chromosomes Researchers sometimes want to clone much longer DNA segments than can typically be incorporated into standard plasmid cloning vectors such as pBR322. To meet this need, plasmid vectors have been developed with special features that allow the cloning of very long segments (typically 100,000 to 300,000 bp) of DNA. Once such large segments of cloned DNA have been added, these vectors are large enough to be thought of as chromosomes and are known as **bacterial artificial chromosomes**, or **BACs** (Fig. 9-5).

A BAC vector (without any cloned DNA inserted) is a relatively simple plasmid, generally not much larger than other plasmid vectors. To accommodate very long segments of cloned DNA, BAC vectors have stable origins of replication that maintain the plasmid at one or two copies per cell. The low copy number is useful in cloning large segments of DNA, because it limits the opportunities for unwanted recombination reactions that can unpredictably alter large cloned DNAs over time. BACs also include *par* genes, derived from a type of plasmid called an F plasmid. The *par* genes encode proteins that direct the reliable distribution of the recombinant chromosomes to daughter cells at cell division, thereby increasing the likelihood of each daughter cell carrying one copy, even when few copies are present.

The BAC vector includes both selectable and screenable markers. The BAC vector shown in Figure 9-5 contains a gene that confers resistance to the antibiotic chloramphenicol (Cam^R). Vector-containing cells can be selected by growing them on agar plates containing this antibiotic—a positive selection, as the cells with the vector survive. A *lacZ* gene, required for production of the enzyme β -galactosidase, is a screenable marker that can reveal which cells contain plasmids—now chromosomes—that incorporate the cloned DNA segments. The β -galactosidase catalyzes conversion of the colorless molecule 5-bromo-4-chloro-3-indolyl- β -D-galactopyranoside (more simply, X-gal) to a blue product. If the gene is intact and expressed, the colony containing it is blue. If gene expression is disrupted by the introduction of a cloned DNA segment, the colony is white.

Yeast Artificial Chromosomes As with *E. coli*, yeast genetics is a well-developed discipline. Research on large genomes and the associated need for high-capacity cloning vectors led to the development of **yeast artificial chromosomes**, or **YACs** (Fig. 9-6). As with BACs, YAC vectors can be used to clone very long segments of DNA. In addition, **P3** the DNA cloned in a YAC can be altered to study the function of specialized sequences in chromosome metabolism, mechanisms of gene regulation and expression, and many other aspects of eukaryotic molecular biology.

The genome of *Saccharomyces cerevisiae* contains only 14×10^6 bp (less than four times the size of the *E. coli* chromosome), and its entire sequence is known.

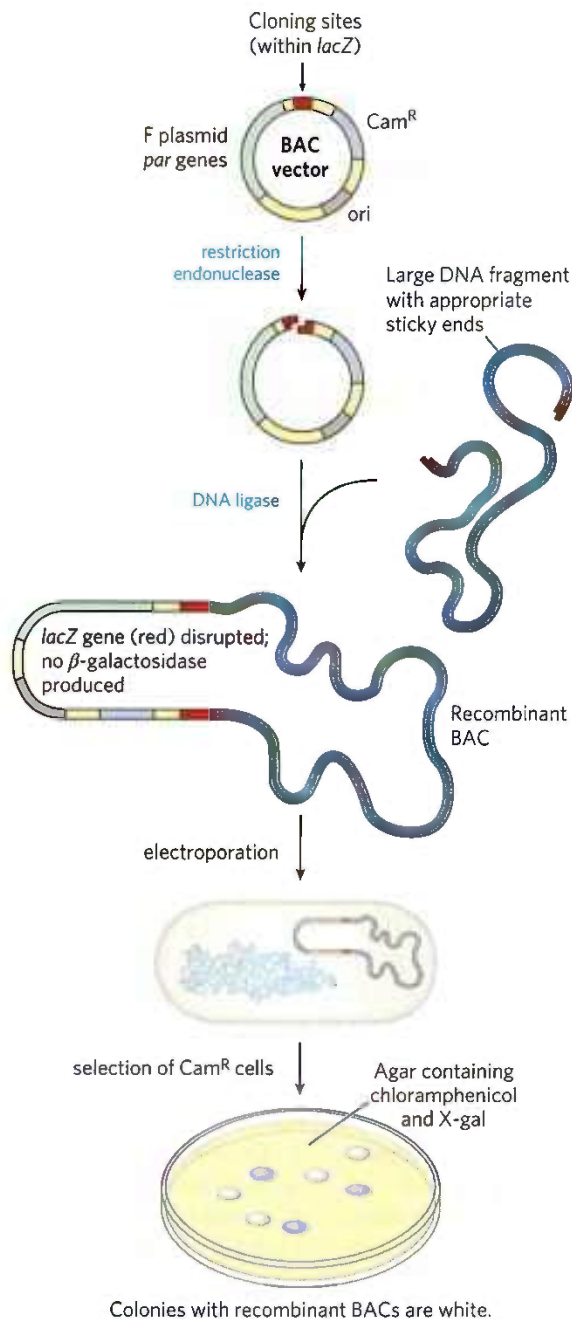


FIGURE 9-5 Bacterial artificial chromosomes (BACs) as cloning vectors. The vector is a relatively simple plasmid, with a replication origin (ori) that directs replication. The *par* genes assist in the even distribution of plasmids to daughter cells at cell division. This increases the likelihood of each daughter cell carrying one copy of the plasmid, even when few copies are present. The low number of copies is useful in cloning large segments of DNA, because this limits the opportunities for unwanted recombination reactions that can unpredictably alter large cloned DNAs over time. The BAC includes selectable markers. A *lacZ* gene (required for the production of the enzyme β-galactosidase) is situated in the cloning region such that it is inactivated by cloned DNA inserts. Introduction of recombinant BACs into cells by electroporation is promoted by the use of cells with an altered (more porous) cell wall. Recombinant DNAs are screened for resistance to the antibiotic chloramphenicol (Cam^R). Plates also contain X-gal, a substrate for β-galactosidase that yields a blue product. Colonies with active β-galactosidase, and hence no DNA insert in the BAC vector, turn blue; colonies without β-galactosidase activity, and thus with the desired DNA inserts, are white.

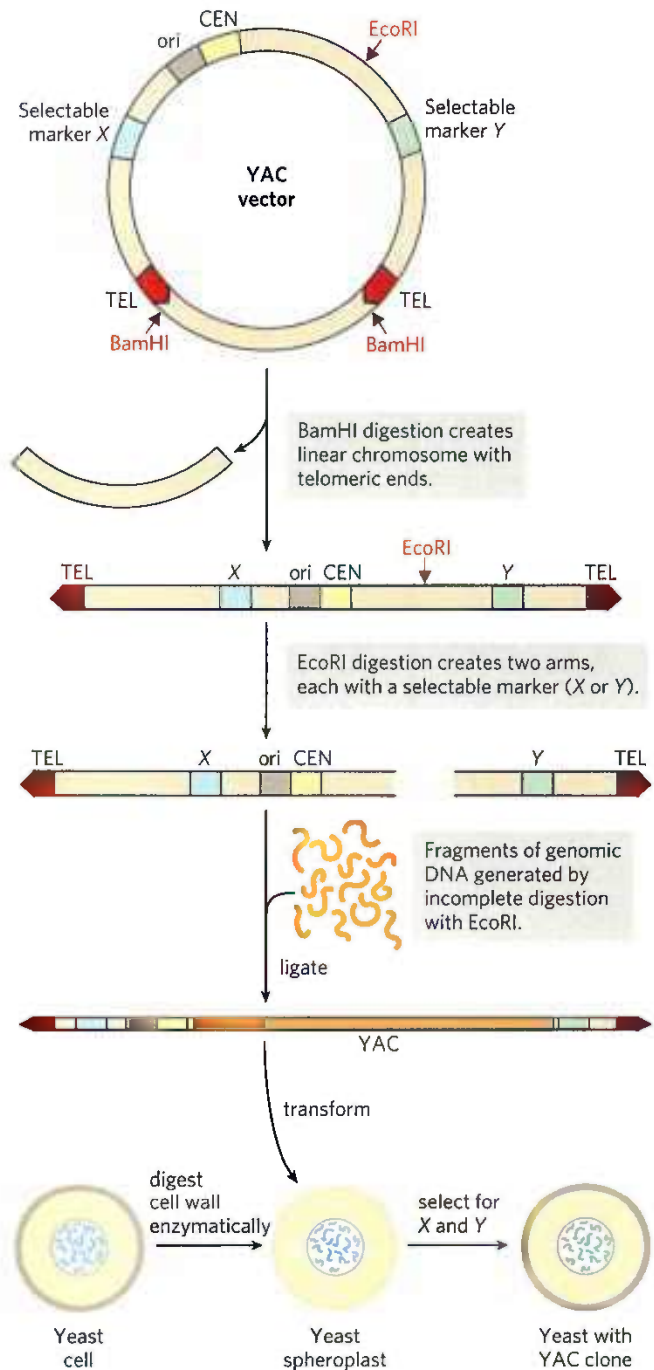


FIGURE 9-6 Construction of a yeast artificial chromosome (YAC). A YAC vector includes an origin of replication (ori), a centromere (CEN), two telomeres (TEL), and selectable markers (X and Y). Digestion with BamHI and EcoRI generates two separate DNA arms, each with a telomeric end and one selectable marker. A large segment of DNA (e.g., up to 2 × 10⁶ bp from the human genome) is ligated to the two arms to create a yeast artificial chromosome. The YAC transforms yeast cells (prepared by removal of the cell wall to form spheroplasts), and the cells are selected for X and Y; the surviving cells propagate the DNA insert.

Yeast is also very easy to maintain and grow on a large scale in the laboratory. Plasmid vectors have been constructed for insertions into yeast cells, employing the same principles that govern the use of *E. coli* vectors. Convenient methods for moving DNA into and out of

yeast cells permit the study of many aspects of eukaryotic cell biochemistry. Some recombinant plasmids incorporate multiple replication origins and other elements that allow them to be used in more than one species (e.g., in yeast and in *E. coli*). Plasmids that can be propagated in cells of two or more species are called **shuttle vectors**.

YAC vectors contain all the elements needed to maintain a eukaryotic chromosome in the yeast nucleus: a yeast origin of replication, two selectable markers, and specialized sequences (derived from the telomeres and centromere) needed for stability and proper segregation of the chromosomes at cell division (see Chapter 24). In preparation for its use in cloning, the vector is propagated as a circular bacterial plasmid and then isolated and purified. Cleavage with a restriction endonuclease (BamHI in Fig. 9-6) removes a length of DNA between two telomere sequences (TEL), leaving the telomeres at the ends of the linearized DNA. Cleavage at another internal site (by EcoRI in Fig. 9-6) divides the vector into two DNA segments, referred to as vector arms, each with a different selectable marker.

Genomic DNA to be cloned is prepared by partial digestion with restriction endonucleases to obtain a suitable fragment size. Genomic fragments are then separated by **pulsed field gel electrophoresis**, a variation of gel electrophoresis (see Fig. 3-18) that segregates very large DNA segments. DNA fragments of appropriate size (up to about 2×10^6 bp) are mixed with the prepared vector arms and ligated. The ligation mixture is then used to transform yeast cells (pretreated to partially degrade their cell walls) with these very large DNA molecules—which now have the structure and size to be considered yeast chromosomes. Culture on a medium that requires the presence of both selectable marker genes ensures the growth of only those yeast cells that contain an artificial chromosome with a large insert sandwiched between the two vector arms (Fig. 9-6). The stability of YAC clones increases with the length of the cloned DNA segment (up to a point). Those with inserts of more than 150,000 bp are nearly as stable as normal cellular chromosomes, whereas those with inserts of fewer than 100,000 bp are gradually lost during mitosis (so, generally, there are no yeast cell clones carrying only the two vector ends ligated together or vectors with only short inserts). YACs that lack a telomere at either end are rapidly degraded.

Cloned Genes Can Be Expressed to Amplify Protein Production

P1 Frequently, the product of a cloned gene, rather than the gene itself, is of primary interest—particularly when the protein has commercial, therapeutic, or research value. Proteins are encoded by genes in DNA; alter the DNA in a gene, and one can alter the protein

product of that gene. Biochemists use purified proteins for many purposes, including to elucidate protein function, study reaction mechanisms, generate antibodies to the proteins, reconstitute complex cellular activities in the test tube with purified components, and examine protein binding partners. With an increased understanding of the fundamentals of DNA, RNA, and protein metabolism and their regulation in a host organism such as *E. coli* or yeast, investigators can manipulate cells to express cloned genes in order to study their protein products. The general goal is to alter the sequences around a cloned gene to trick the host organism into producing the protein product of the gene, often at very high levels. This overexpression of a protein can make its subsequent purification much easier.

We'll use the expression of a eukaryotic protein in a bacterium as an example. Eukaryotic genes have surrounding sequences needed for their transcription and regulation in the cells they are derived from, but these sequences do not function in bacteria. Thus, eukaryotic genes lack the DNA sequence elements required for their controlled expression in bacterial cells: promoters (sequences that instruct RNA polymerase where to bind to initiate mRNA synthesis), ribosome-binding sites (sequences that allow translation of the mRNA to protein), and additional regulatory sequences. Appropriate bacterial regulatory sequences for transcription and translation must be inserted in the vector DNA at the correct positions relative to the eukaryotic gene.

Cloning vectors with the transcription and translation signals needed for the regulated expression of a cloned gene are called **expression vectors**. The rate of expression of the cloned gene is controlled by replacing the gene's normal promoter and regulatory sequences with more efficient and convenient versions supplied by the vector. Generally, a well-characterized promoter and its regulatory elements are positioned near several unique restriction sites for cloning, so that genes inserted at the restriction sites will be expressed from the regulated promoter elements (Fig. 9-7). Some of these vectors incorporate other features, such as a bacterial ribosome-binding site to enhance translation of the mRNA derived from the gene (Chapter 27) or a transcription termination sequence (Chapter 26). In some cases, cloned genes are so efficiently expressed that their protein product represents 10% or more of the cellular protein. At these concentrations, some foreign proteins can kill the host cell (usually *E. coli*), so expression of the cloned gene must be limited to the few hours before the planned harvesting of the cells.

Many Different Systems Are Used to Express Recombinant Proteins

Every living organism has the capacity to express genes in its genomic DNA; thus, in principle, any organism can serve as a host to express proteins from a different

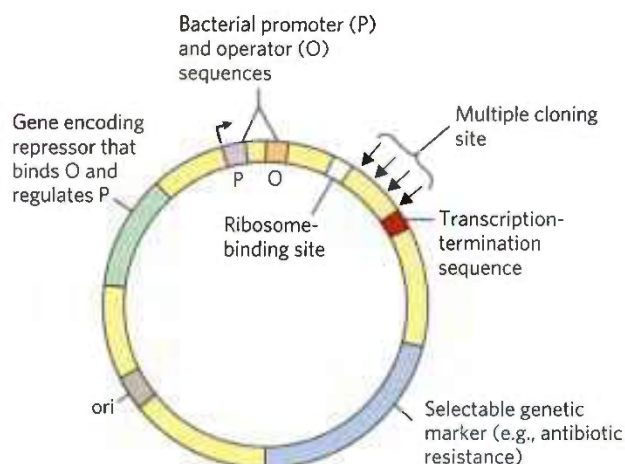


FIGURE 9-7 DNA sequences in a typical *E. coli* expression vector.

The gene to be expressed is inserted into one of the restriction sites in the MCS, near the promoter (P), with the end of the gene encoding the amino terminus of the protein positioned closest to the promoter. The promoter allows efficient transcription of the inserted gene, and the transcription-termination sequence sometimes improves the amount and stability of the mRNA produced. The operator (O) permits regulation by a repressor that binds to it. The ribosome-binding site provides sequence signals for the efficient translation of the mRNA derived from the gene. The selectable marker allows the selection of cells containing the recombinant DNA.

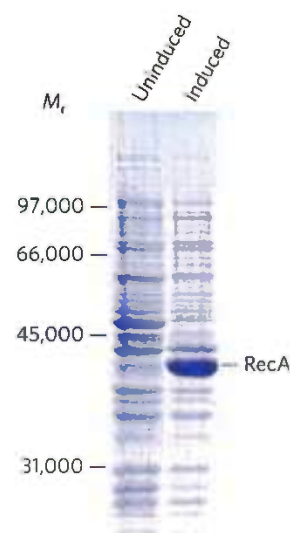


FIGURE 9-8 Regulated expression of RecA protein in a bacterial cell.

The gene encoding the RecA protein, fused to a bacteriophage T7 promoter, is cloned into an expression vector. Under normal growth conditions (uninduced), no RecA protein appears. When the T7 RNA polymerase is induced in the cell, the *recA* gene is expressed, and large amounts of RecA protein are produced. The positions of standard molecular weight markers that were run on the same gel are indicated. [Courtesy Rachel Britt, University of Wisconsin–Madison, Department of Biochemistry.]

(heterologous) species. Almost every sort of organism has, indeed, been used for this purpose, and each host type has a particular set of advantages and disadvantages.

Bacteria Bacteria, especially *E. coli*, remain the most common hosts for protein expression. The regulatory sequences that govern gene expression in *E. coli* and many other bacteria are well understood and can be harnessed to express cloned proteins at high levels. Bacteria are easy to store and grow in the laboratory, on inexpensive growth media. Efficient methods also exist to get DNA into bacteria and extract DNA from them. Bacteria can be grown in huge amounts in commercial fermenters, providing a rich source of the cloned protein.

Problems do exist, however. When expressed in bacteria, some heterologous proteins do not fold correctly, and many do not undergo the posttranslational modifications or proteolytic cleavage that may be necessary for their activity. Certain features of a gene sequence also can make a particular gene difficult to express in bacteria. For example, intrinsically disordered regions are more common in eukaryotic proteins. When expressed in bacteria, many eukaryotic proteins aggregate into insoluble cellular precipitates called **inclusion bodies**. For these and many other reasons, some eukaryotic proteins are inactive when purified from bacteria or cannot be expressed at all. To help address some of these problems, researchers are regularly developing new bacterial host strains that include enhancements such as the engineered presence of eukaryotic protein chaperones or enzymes that modify eukaryotic proteins.

There are many specialized systems for expressing proteins in bacteria. The well-characterized promoter and regulatory sequences associated with the lactose operon (see Chapter 28) are often fused to the gene of interest to direct transcription. The cloned gene will be transcribed when lactose is added to the growth medium. However, regulation in the lactose system is “leaky”: it is not turned off completely when lactose is absent—a potential problem if the product of the cloned gene is toxic to the host cells. Transcription from the Lac promoter is also not efficient enough for some applications.

An alternative system uses the promoter and RNA polymerase of a bacterial virus called bacteriophage T7. If the cloned gene is fused to a T7 promoter, it is transcribed, not by the *E. coli* RNA polymerase, but by the T7 RNA polymerase. The gene encoding this polymerase is separately cloned into the same cell in a construct that affords tight regulation (allowing controlled production of the T7 RNA polymerase). The polymerase is also very efficient and directs high levels of expression of most genes fused to the T7 promoter. This system has been used to express the RecA protein in bacterial cells (Fig. 9-8).

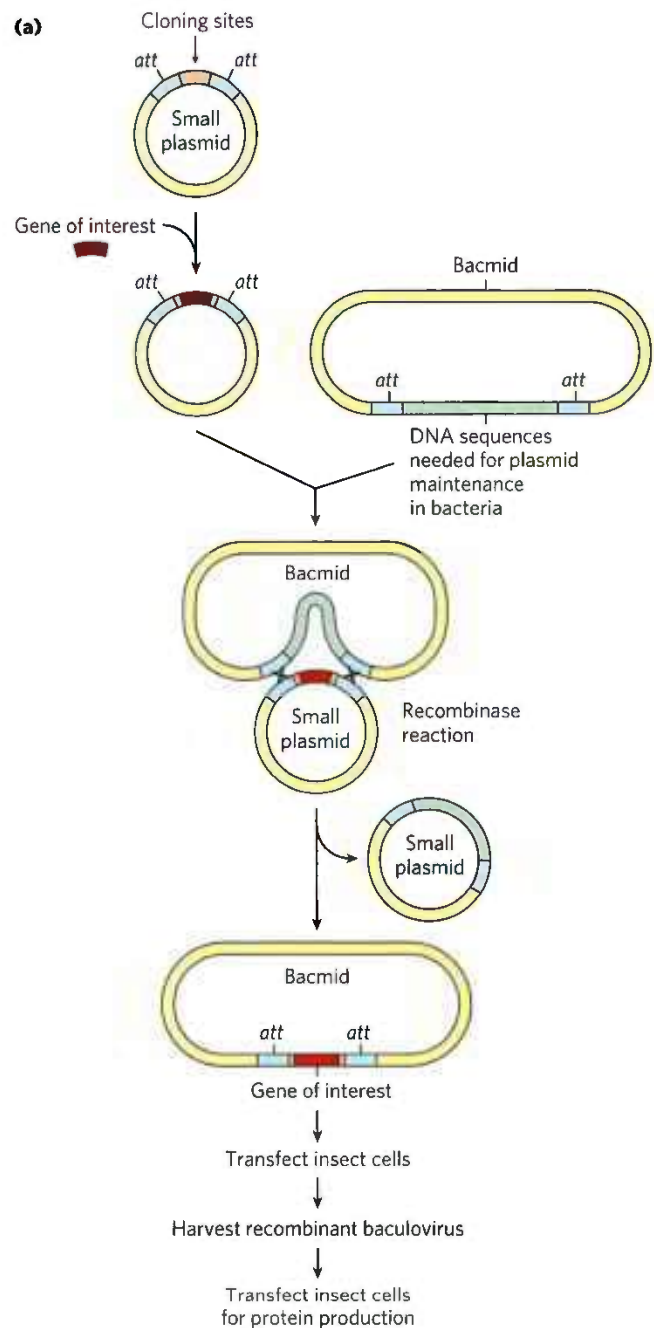
Yeast *Saccharomyces cerevisiae* is probably the best understood eukaryotic organism. The principles underlying the expression of a protein in yeast are the same as those for bacteria. Cloned genes must be linked to promoters that can direct high-level expression in yeast cells. For example, the yeast *GAL1* and *GAL10* genes (encoding enzymes involved in galactose metabolism)

are under cellular regulation such that they are expressed when yeast cells are grown in media with galactose but shut down when the cells are grown in glucose. Thus, if a heterologous gene is expressed using these same regulatory sequences, the expression of that gene can be controlled simply by choosing an appropriate medium for cell growth.

Some of the same problems that accompany protein expression in bacteria also occur with yeast. Heterologous proteins may not fold properly, yeast may lack the enzymes needed to modify the proteins to their active forms, or certain features of the gene sequence may hinder expression of a protein. However, because *S. cerevisiae* is a eukaryote, the expression of eukaryotic genes (especially yeast genes) is sometimes more efficient in this host than in bacteria. As yeast possess many of the same protein chaperones and modification systems of higher eukaryotes, protein products may also be folded and modified more accurately than are proteins expressed in bacteria.

Insects and Insect Viruses **Baculoviruses** are insect viruses with double-stranded DNA genomes. When baculoviruses infect their insect larval hosts, they act as parasites, killing the larvae and turning them into factories for virus production. Late in the infection process, the viruses produce large amounts of two proteins (p10 and polyhedrin), neither of which is needed for production of viruses in cultured insect cells. The genes for both of these proteins can be replaced with the gene for a heterologous protein. When the resulting recombinant virus is used to infect insect cells or larvae, the heterologous protein is often produced at very high levels — up to 25% of the total protein present at the end of the infection cycle.

Autographa californica multicapsid nucleopolyhedrovirus (AcMNPV; *A. californica* is a moth species that it infects) is the baculovirus most often used for protein expression. It has a large genome (134,000 bp), too large for direct cloning. Virus purification is also cumbersome. These problems have been solved by the creation of **bacmids**, large circular DNAs that include the entire baculovirus genome along with sequences that allow replication of the bacmid in *E. coli* (Fig. 9-9). The gene of interest is cloned into a smaller plasmid and combined with the larger plasmid by site-specific recombination



(b)



FIGURE 9-9 Cloning with baculoviruses. (a) Shown here is the construction of a typical vector used for protein expression in baculoviruses. The gene of interest is cloned into a small plasmid (top left) between two sites (*att*) recognized by a site-specific recombinase, then is introduced into the baculovirus vector by site-specific recombination. This generates a circular DNA product that is used to infect the cells of an insect larva. The gene of interest is expressed during the infection cycle, downstream of a promoter that normally expresses a baculovirus coat protein at very high levels. (b) The photographs show larvae of the cabbage looper moth. The larva on the left is uninfected; the larva on the right was infected with a recombinant baculovirus vector expressing a protein that produces a red color. [(b) USDA-ARS.]

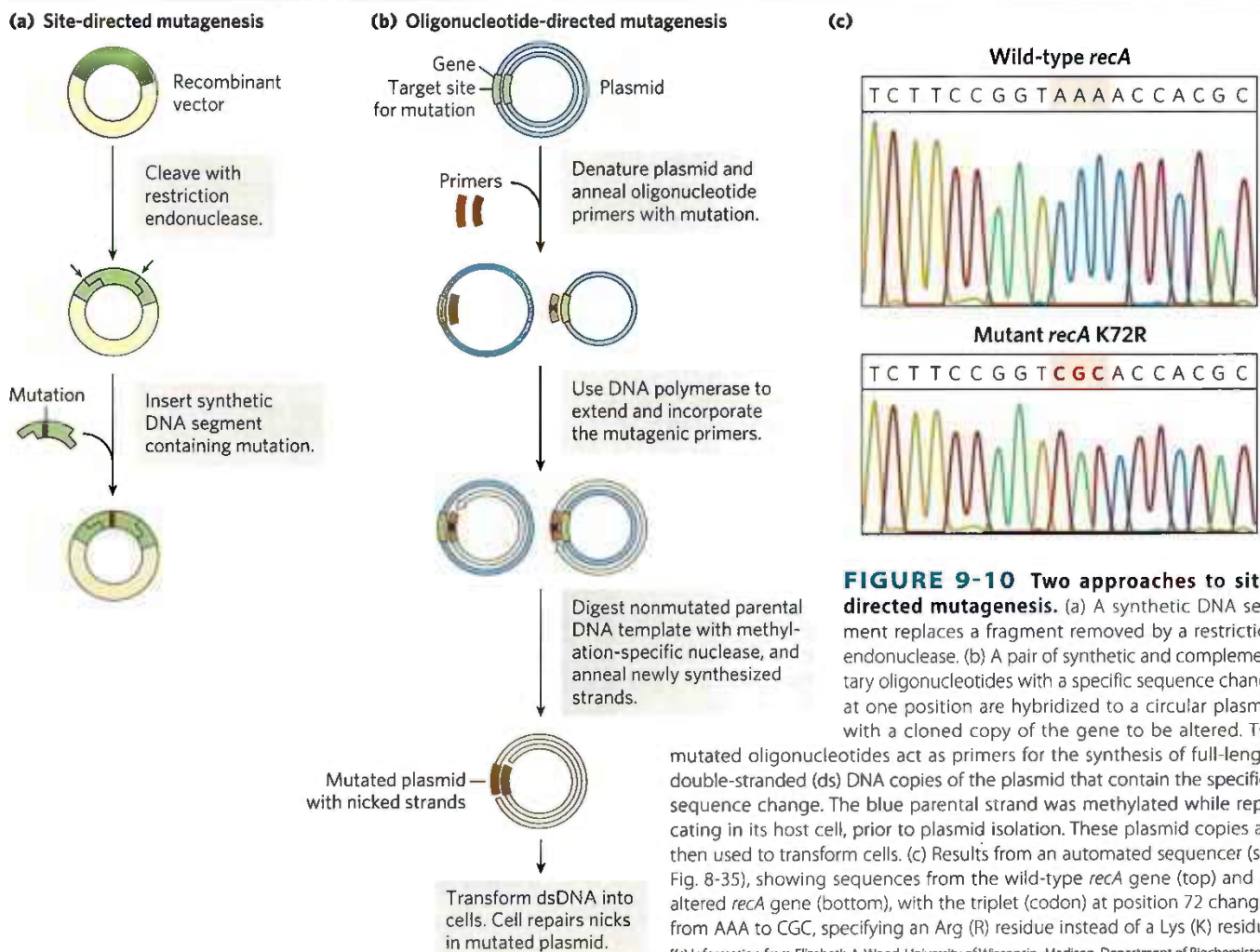
in vivo (see Fig. 25-37). The recombinant bacmid is then isolated and transfected into insect cells (the term **transfection** is used when the DNA used for transformation includes viral sequences and leads to viral replication), followed by recovery of the protein once the infection cycle is finished. A wide range of bacmid systems are available commercially. Baculovirus systems are not successful with all proteins. However, with these systems, insect cells sometimes successfully replicate the protein-modification patterns of higher eukaryotes and produce active, correctly modified eukaryotic proteins.

Mammalian Cells in Culture The most convenient way to introduce cloned genes into a mammalian cell is with viruses. This method takes advantage of the natural capacity of a virus to insert its DNA or RNA into a cell, and sometimes into the cellular chromosome. A variety of engineered mammalian viruses are available as vectors, including human adenoviruses and retroviruses. The gene of interest is cloned so that its expression is controlled by a virus promoter. The virus uses its natural infection mechanisms to introduce the recombinant genome into cells, where the cloned protein is expressed. One advantage of these systems is that proteins can be expressed either transiently (if the viral DNA is maintained separately from the host cell genome and eventually

degraded) or permanently (if the viral DNA is integrated into the host cell genome). With the correct choice of host cell, the proper posttranslational modification of the protein to its active form can be ensured. However, the growth of mammalian cells in tissue culture is very expensive, and this technology is generally used to test the function of a protein in vivo rather than to produce a protein in large amounts.

Alteration of Cloned Genes Produces Altered Proteins

P1 Cloning techniques can be used not only to overproduce proteins but also to produce proteins that are altered, subtly or dramatically, from their native forms. Specific amino acids may be replaced individually by **site-directed mutagenesis**. This approach has greatly enhanced research on proteins by allowing investigators to make specific changes in the primary structure and examine the effects of these changes on the protein's folding, three-dimensional structure, and activity. The amino acid sequence of the protein is changed by altering the DNA sequence of the cloned gene. If appropriate restriction sites flank the sequence to be altered, researchers can simply remove a DNA segment and replace it with a synthetic one, identical to the original except for the desired change (Fig. 9-10a).



When suitably located restriction sites are not present, **oligonucleotide-directed mutagenesis** can create a specific DNA sequence change (Fig. 9-10b). The cloned gene is denatured, separating the strands. Two short, complementary synthetic DNA strands, each with the desired base change, are annealed to opposite strands of the cloned gene within a suitable circular DNA vector. The mismatch of a single base pair in 30 to 40 bp does not prevent annealing. The two annealed oligonucleotides serve to prime DNA synthesis in both directions around the plasmid vector, creating two complementary strands that contain the mutation. After several cycles of selective amplification by the polymerase chain reaction (PCR; see Fig. 8-33), the mutation-containing DNA predominates in the population and can be used to transform bacteria. Most of the transformed bacteria will have plasmids carrying the mutation.

For an example, we go back to the bacterial *recA* gene. The product of this gene, the RecA protein, has several activities (see Section 25.3) including the hydrolysis of ATP. The Lys residue at position 72 in RecA (a 352 residue polypeptide) is involved in ATP hydrolysis. Changing Lys⁷² to an Arg creates a variant of RecA protein that will bind, but not hydrolyze, ATP (Fig. 9-10c). The engineering and purification of this variant RecA protein has facilitated research into the roles of ATP hydrolysis in the functioning of this protein.

Changes can be introduced into a gene that involve far more than one base pair. Large parts of a gene can be deleted by cutting out a segment with restriction endonucleases and ligating the remaining portions to form a smaller gene. For example, if a protein has two domains, the gene segment encoding one of the domains can be removed so that the gene now encodes a protein with only one of the original two domains. Parts of two different genes can be ligated to create new combinations; the product of such a fused gene is called a **fusion protein**. Researchers have ingenious methods to bring about virtually any genetic alteration in vitro. After reintroducing the altered DNA into the cell, they can investigate the consequences of the alteration.

Terminal Tags Provide Handles for Affinity Purification

Affinity chromatography is one of the most efficient methods for purifying proteins (see Fig. 3-17c). Unfortunately, many proteins do not bind a ligand that can be conveniently immobilized on a column matrix. However, the gene for almost any protein can be altered to express a fusion protein that can be purified by affinity chromatography. The gene encoding the target protein is fused to a gene encoding a peptide or protein that binds a simple, stable ligand with high affinity and specificity. The peptide or protein used for this purpose is referred to as a **tag**. Tag sequences can be added to genes such that the resulting proteins have tags at their amino terminus or carboxyl terminus. **Table 9-3** lists some of the peptides or proteins commonly used as tags.

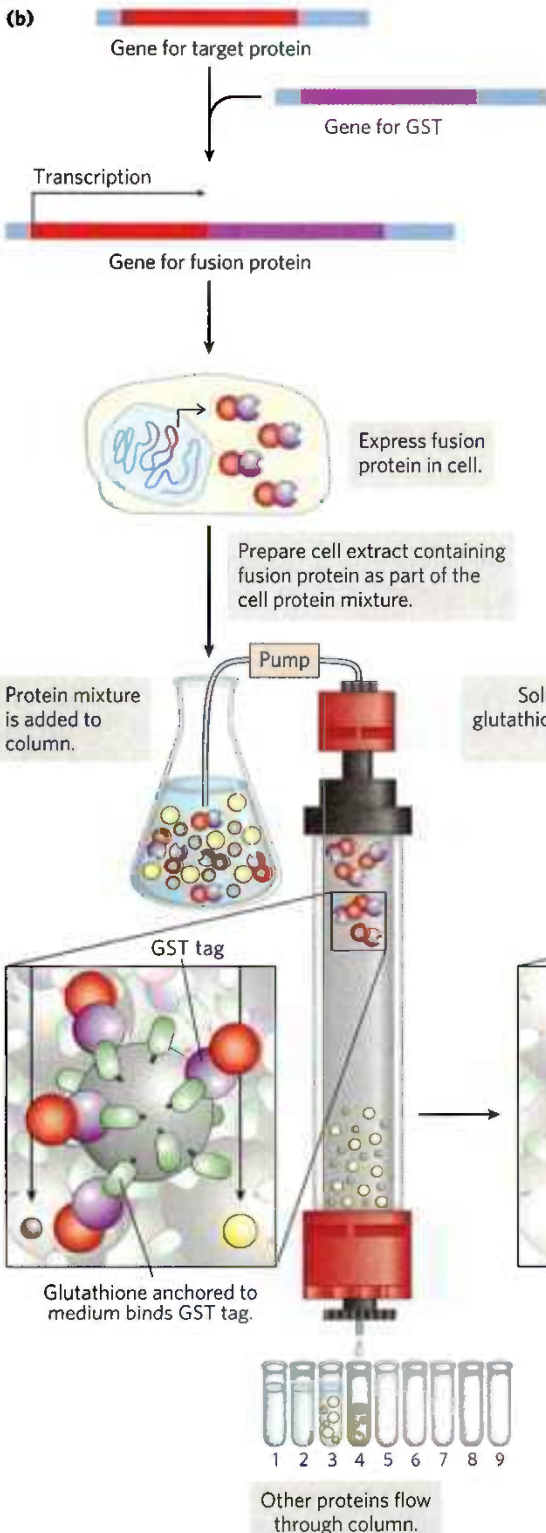
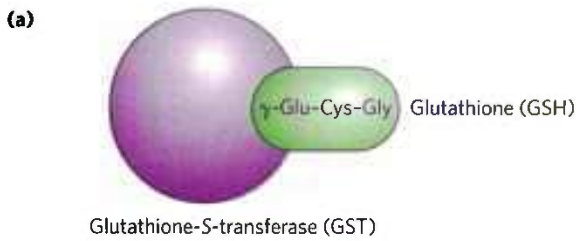
TABLE 9-3 Commonly Used Protein Tags

Tag protein/peptide	Molecular mass (kDa)	Immobilized ligand
Protein A	59	Fc portion of IgG
(His) ₆	0.8	Ni ²⁺
Glutathione-S-transferase (GST)	26	Glutathione
Maltose-binding protein	41	Maltose
β -Galactosidase	116	<i>p</i> -Aminophenyl- β -D-thiogalactoside (TPEG)
Chitin-binding domain	5.7	Chitin

The general procedure can be illustrated by focusing on a system that uses the glutathione-S-transferase (GST) tag (**Fig. 9-11**). GST is a small enzyme (M_r 26,000) that binds tightly and specifically to glutathione. When the GST gene sequence is fused to a target gene, the fusion protein acquires the capacity to bind glutathione. The fusion protein is expressed in a host organism such as a bacterium, and a crude extract is prepared. A column is filled with a porous matrix consisting of the ligand (glutathione) immobilized on microscopic beads of a stable polymer such as cross-linked agarose. As the crude extract percolates through this matrix, the fusion protein becomes immobilized by binding the glutathione. The other proteins in the extract are washed through the column and discarded. The interaction between GST and glutathione is tight but noncovalent, allowing the fusion protein to be gently eluted from the column with a solution containing either a higher concentration of salts or free glutathione to compete with the immobilized ligand for GST binding. The fusion protein is often obtained with good yield and high purity. In some commercially available systems, the tag can be entirely or largely removed from the purified fusion protein by a protease that cleaves a sequence near the junction between the target protein and its tag.

A shorter tag with widespread application consists of a simple sequence of six or more His residues. These histidine tags, or His tags, bind tightly and specifically to nickel ions. A chromatography matrix with immobilized Ni²⁺ can be used to quickly separate a His-tagged protein from other proteins in an extract. Some of the larger tags, such as maltose-binding protein, provide added stability and solubility, allowing the purification of cloned proteins that are otherwise inactive due to improper folding or insolubility.

Affinity chromatography using terminal tags is powerful and convenient. The tags have been successfully used in thousands of published studies; in many cases, the protein would be impossible to purify and study without the tag. However, even very small tags can affect the properties of the proteins they are attached to, thereby influencing the study results. For example, the tag



may adversely affect protein folding. Even if the tag is removed by a protease, one or a few extra amino acid residues can remain behind on the target protein, which may or may not affect the protein's activity. The types of experiments to be carried out, and the results obtained from them, should always be evaluated with the aid of well-designed controls to assess any effect of a tag on protein function.

The Polymerase Chain Reaction Offers Many Options for Cloning Experiments

Many adaptations of PCR have increased its utility in cloning. **P1** For example, sequences in RNA can be amplified if the first PCR cycle uses reverse transcriptase, an enzyme that works like DNA polymerase (see Fig. 8-33) but uses RNA as a template (Fig. 9-12a). After the DNA strand is made from the RNA template, the remaining cycles can be carried out with DNA polymerases, using standard PCR protocols. This **reverse transcriptase PCR (RT-PCR)** can be used, for example, to detect sequences derived from living cells (which are transcribing their DNA into RNA) as opposed to dead tissues.

PCR protocols can also be used to estimate the relative copy numbers of particular sequences in a sample, an approach called **quantitative PCR (qPCR)** or **real-time PCR**. If a DNA sequence is present in higher

FIGURE 9-11 Use of tagged proteins in protein purification. (a) Glutathione-S-transferase (GST) is a small enzyme that binds glutathione. (b) The GST tag is fused to the carboxyl terminus of the protein by genetic engineering. The tagged protein is expressed in the cell and is present in the crude extract when the cells are lysed. The extract is subjected to affinity chromatography through a matrix with immobilized glutathione.

than usual amounts in a sample—for example, if certain genes are amplified in tumor cells—qPCR can reveal the increased representation of that sequence. In brief, the PCR is carried out in the presence of a probe that emits a fluorescent signal when the PCR product is present

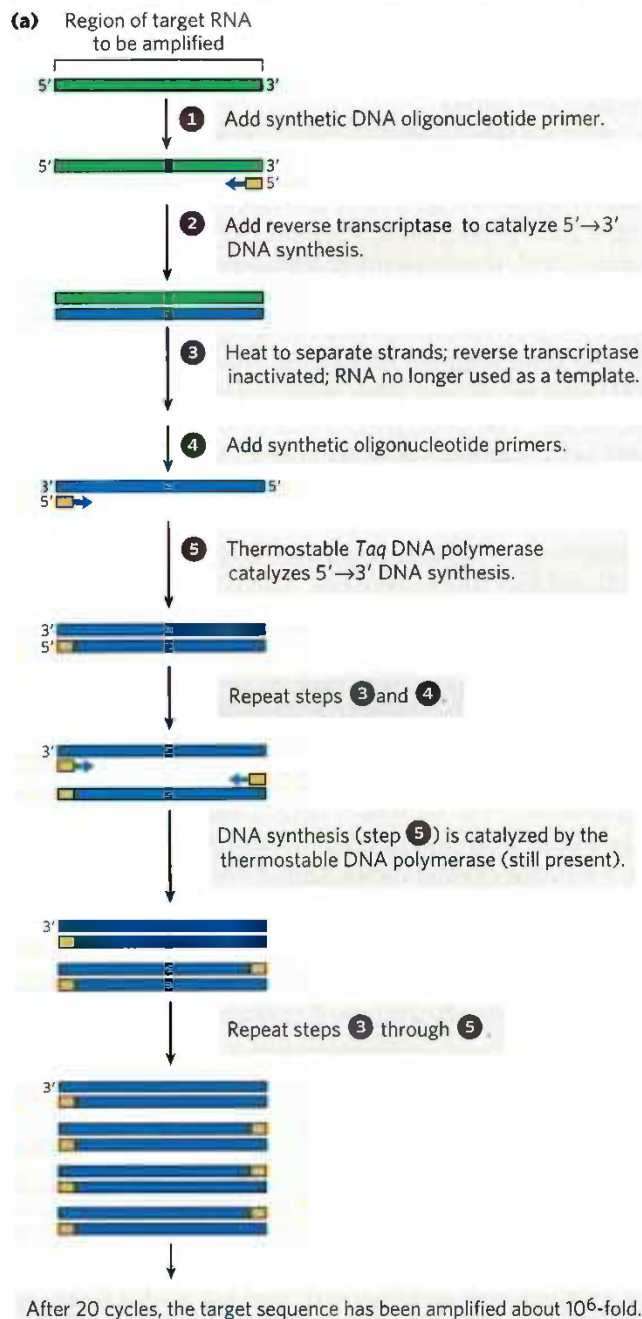
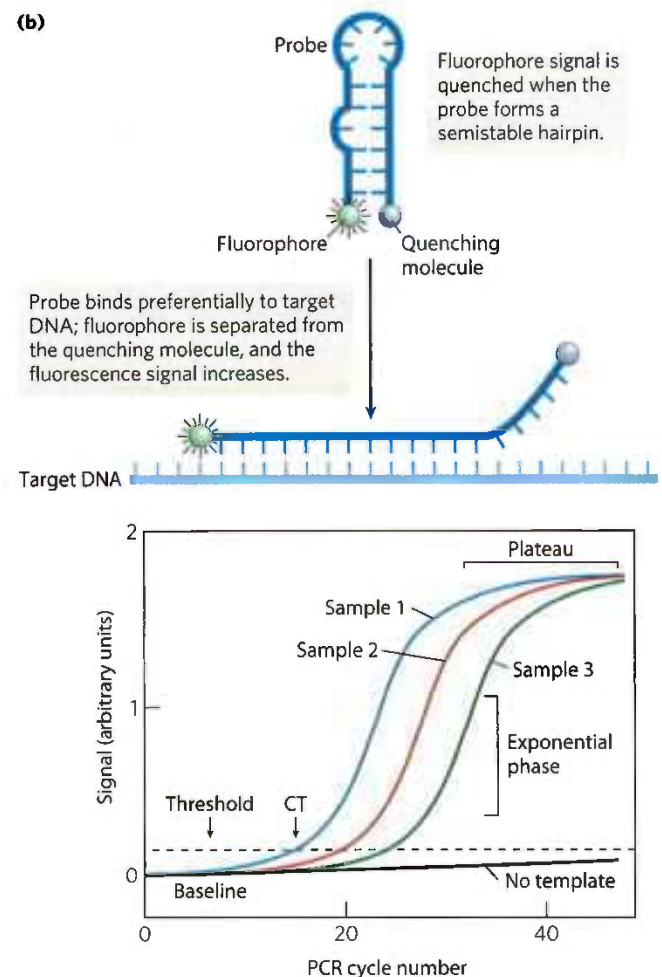


FIGURE 9-12 Some applications of PCR. (a) In reverse transcriptase PCR, or RT-PCR, RNA molecules are amplified by using reverse transcriptase in the first two cycles. (b) In quantitative PCR, or qPCR, careful monitoring of the progress of a PCR amplification allows one to determine when a DNA segment has been amplified to a specified threshold level. The amount of PCR product present is determined by measuring the level of a fluorescent probe attached to a reporter oligonucleotide complementary to the DNA segment that is being amplified. Probe fluorescence is not detectable initially, due to a fluorescence quencher attached to the same oligonucleotide. When the reporter oligonucleotide pairs with its complement in a copy of the amplified

(Fig. 9-12b). If the sequence of interest is present at higher levels than other sequences in the sample, the PCR signal will reach a predetermined threshold faster. Reverse transcriptase PCR and qPCR can be combined to determine the relative concentrations of a particular mRNA molecule in a cell, and thereby monitor gene expression under different environmental conditions.

DNA Libraries Are Specialized Catalogs of Genetic Information

In some instances, it is useful to clone many genes or genomic segments rather than a particular one. A **DNA library** is a collection of DNA clones, usually gathered for purposes of gene discovery or the determination of gene or protein function. The library can take a variety



DNA segment, the fluorophore is separated from the quenching molecule and fluorescence results. As the PCR reaction proceeds, the amount of the targeted DNA segment increases exponentially, and the fluorescent signal also increases exponentially as the oligonucleotide probes anneal to the amplified segments. After many PCR cycles, the signal reaches a plateau as one or more reaction components become exhausted. When a segment is present in greater amounts in one sample than another, its amplification reaches a defined threshold level earlier. The "No template" line follows the slow increase in background signal observed in a control that does not include added sample DNA. CT is the cycle number at which the threshold is first surpassed.

of forms, depending on the source of the DNA and the ultimate purpose of the library.

An example is a library that includes only the genes that are transcribed into RNA—*expressed*—in a given organism or even just in certain cells or tissues. Such a library lacks any genomic DNA that is not transcribed. The researcher first extracts mRNA from an organism, or from specific cells of an organism, and then prepares the **complementary DNAs (cDNAs)**. Like RT-PCR, this multistep reaction (**Figure 9-13a**) relies on reverse transcriptase, which synthesizes DNA from a template

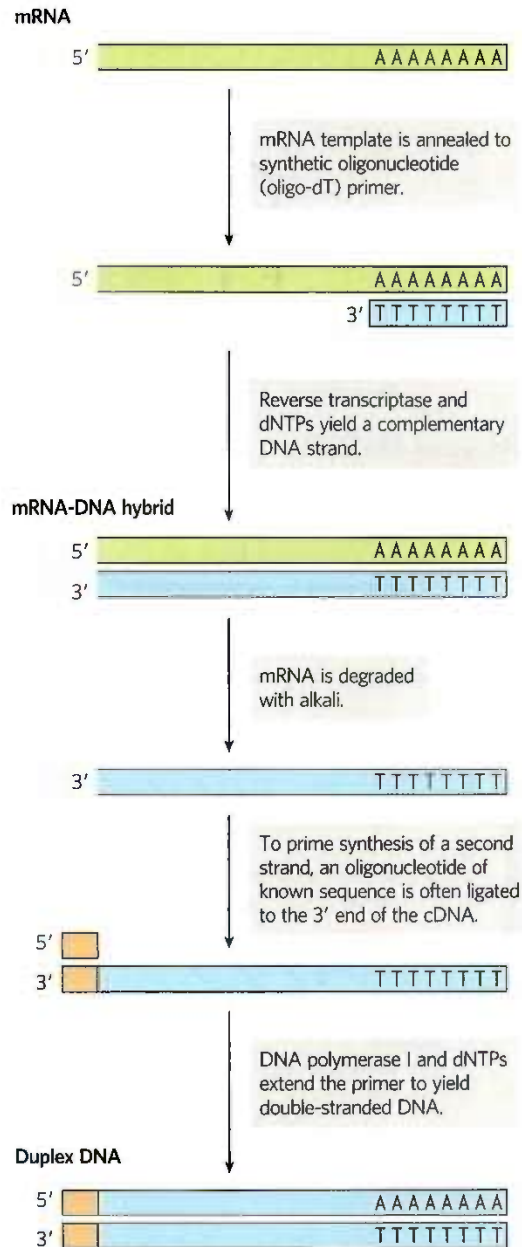


FIGURE 9-13 Building a cDNA library from mRNA. A cell's total mRNA content includes transcripts from thousands of genes, and the cDNAs generated from this mRNA are correspondingly heterogeneous. Reverse transcriptase can synthesize DNA on an RNA or a DNA template. To prime the synthesis of a second DNA strand, oligonucleotides of known sequence are ligated to the 3' end of the first strand, and the double-stranded cDNA so produced is cloned into a plasmid.

RNA. The resulting double-stranded DNA fragments are inserted into a suitable vector and cloned, creating a population of clones called a **cDNA library**. If the library host is a bacterium like *E. coli*, each cell in the population will carry one particular cloned sequence. The library will encompass many millions of cells with millions of different cloned segments. The presence of a gene for a particular protein in such a library implies that this gene is expressed in the cells and under the conditions used to generate the library.

Another type of library, called a **combinatorial gene library** or simply a **gene library**, focuses on sequence variants within one gene. For example, beginning with the cloned gene of enzyme X, a segment of the gene could be replaced with nearly identical fragments synthesized with a slight imprecision so that each clone had one or two random base pair changes relative to the original. For example, the gene segment of interest could be amplified by PCR using an altered DNA polymerase that was slightly inaccurate. The library of clones would then consist of many cells, many of which harbored a different variant of the gene for enzyme X. Investigators could use the library to select for variants of enzyme X with enhanced catalytic properties or could simply determine which changes were functional and which were not. The possibilities are limited only by the imagination of the researcher.

SUMMARY 9.1 Studying Genes and Their Products

- DNA cloning and genetic engineering involve the cleavage of DNA and assembly of DNA segments in new combinations—recombinant DNA. Cloning entails generating a DNA fragment of interest, inserting the fragment into a suitable cloning vector, transferring the vector with the DNA insert into a host cell for replication, and identifying and selecting cells that contain the DNA fragment.
- Key enzymes in gene cloning include restriction endonucleases (especially the type II enzymes) and DNA ligase.
- Cloning vectors include plasmids and, for the longest DNA inserts, bacterial artificial chromosomes (BACs) and yeast artificial chromosomes (YACs).
- Cloned genes can be expressed in a host cell by incorporating them into expression vectors that have the sequence signals needed for transcription and translation.
- Proteins can be expressed in different types of cells using expression systems with various useful features and advantages.
- Genetic engineering techniques can alter cloned genes as required by the investigator.
- Proteins or peptides can be attached to a protein of interest by altering its cloned gene, creating a fusion protein. The additional peptide segments can be used to detect the protein or to purify it, using convenient affinity chromatography methods.
- The polymerase chain reaction (PCR) permits the amplification of chosen segments of DNA or RNA for

cloning and can be adapted to determine gene copy number or to monitor gene expression quantitatively.

■ DNA libraries consist of many clones, encompassing many genomic segments or many variants of a particular gene.

9.2 Exploring Protein Function on the Scale of Cells or Whole Organisms

Protein function can be described on three levels. **Phenotypic function** describes the effects of a protein on the entire organism. For example, loss of the protein may lead to slower growth of the organism, an altered development pattern, or even death. **Cellular function** is a description of the network of interactions a protein engages in at the cellular level. Identifying interactions with other proteins in the cell can help define the kinds of metabolic processes in which the protein participates. Finally, **molecular function** refers to the precise biochemical activity of a protein, including details such as the reactions that an enzyme catalyzes or the ligands that a receptor binds. In response to the challenge of understanding these functions of the thousands of proteins in a typical cell, scientists have developed a variety of techniques in the broader discipline of genomics. We can apply these techniques to determine when a particular protein is expressed, what other proteins it might be related to, where it is located in the cell, what other cellular components it interacts with, and what happens to the cell when the protein is missing.

A variety of interrelated methods broadly probe a cell's RNA or protein content. The entire complement of transcribed RNAs present at a given moment in a cell is defined as the cellular **transcriptome**. As introduced in Chapters 1 and 3, the entire complement of proteins present at a given moment in a cell is defined as that cell's **proteome**. Studies of transcriptomes and studies of proteomes are referred to as **transcriptomics** and **proteomics**, respectively. Changes in these cellular macromolecules that occur when a particular gene or its expression is altered can provide important additional clues about protein function, as we will see. The methods we cover here are summarized in Table 9-4. The list is by no means comprehensive, but it serves to illustrate important approaches.

Sequence or Structural Relationships Can Suggest Protein Function

P2 One important reason to sequence many genomes is to provide databases that can be used to assign gene functions by genome comparisons, an enterprise referred to as **comparative genomics**.

A genome sequence is simply a very long string of A, G, T, and C residues, all meaningless until interpreted. **Genome annotation** yields information about the location and function of genes and other critical sequences. Genome annotation converts the sequence into information that any researcher can use, and it is typically

TABLE 9-4 Methods for Discovering New Proteins and Exploring Their Functions

Clue	Method to Apply
What is a protein's function?	
What other proteins of known function have similar sequences?	Comparative genomics
What known sequence motifs does the protein possess?	Comparative genomics
Under what conditions is the gene encoding the protein expressed?	RNA-Seq
How much of the protein is present in the cell under different conditions?	Mass spectrometry
Where is the protein located in the cell?	Microscopy with fusion proteins and immunofluorescence
What does the protein interact with?	Immunoprecipitation; tandem affinity purification; yeast two-hybrid analysis
What happens to the cell when the protein is missing or altered?	CRISPR/Cas9 or other mutagenic methods
What genes (some unknown) are involved in a process?	Large-scale screening

focused on genomic DNA encompassing genes that encode RNA and protein, the most common targets of scientific investigation. Every newly sequenced genome includes many genes—often 40% or more of the total—about which little or nothing is known.

Using online tools that apply computational power to comparative genomics, scientists can define gene locations and assign tentative gene functions (where possible) based on similarity to genes previously studied in other genomes. The classic BLAST (Basic Local Alignment Search Tool) algorithm allows a rapid search of all genome databases for sequences related to one that a researcher is exploring, and it is especially valuable for investigating the function of a particular gene. BLAST is one of many resources available at the NCBI (National Center for Biotechnology Information) site (www.ncbi.nlm.nih.gov), sponsored by the National Institutes of Health, and the Ensembl site (www.ensembl.org), cosponsored by the EMBL-EBI (European Molecular Biology Laboratory–European Bioinformatics Institute).

Comparative genomics is made possible by evolutionary biology. Sometimes a newly discovered gene is related by sequence homologies to a previously studied gene in another or the same species, and its function can be entirely or partly defined by that relationship. Genes that occur in different species but have a clear sequence and functional relationship to each other are called **orthologs**. Genes similarly related to each other within a single species are called **paralogs**. We introduced these terms in Chapter 3 in the context of proteins. As with proteins, information about the function of a gene in one species can be used to at least tentatively assign function to the orthologous gene found in a second species. The correlation

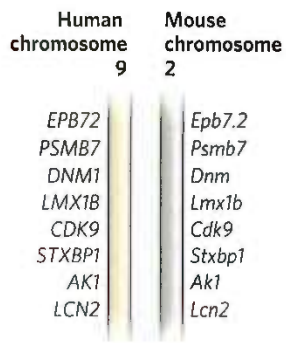


FIGURE 9-14 Synteny in the human and mouse genomes. Large segments of the two genomes have closely related genes aligned in the same order on the chromosomes. In these short segments of human chromosome 9 and mouse chromosome 2, the genes show a very high degree of homology, as well as the same gene order. The different lettering schemes for the gene names simply reflect the different naming conventions for the two species.

[Information from T. G. Wolfsberg et al., *Nature* 409:824, 2001, Fig. 1.]

is easiest to make when comparing genomes from relatively closely related species, such as mouse and human, although many clearly orthologous genes have been identified in species as distant as bacteria and humans. Sometimes even the order of genes on a chromosome is conserved over large segments of the genomes of closely related species (Fig. 9-14). Conserved gene order, called **synteny**, provides additional evidence for an orthologous relationship between genes at identical locations within the related segments.

Alternatively, certain amino acid sequences associated with particular structural motifs (Chapter 4) may be identified within a protein. **P2** The presence of a structural motif may help to define molecular function by suggesting that a protein, say, catalyzes ATP hydrolysis, binds to DNA, or forms a complex with zinc ions. These relationships are determined with the aid of sophisticated computer programs, limited only by the current information on gene and protein structure and by our capacity to associate sequences with particular structural motifs. Sequences at an enzyme active site that have been highly conserved during evolution are typically associated with catalytic function, and their identification is often a key step in defining an enzyme's reaction mechanism. The reaction mechanism, in turn, provides information needed to develop new enzyme inhibitors that can be used as pharmaceutical agents.

When and Where a Protein Is Present in a Cell Can Suggest Protein Function

If a protein is involved in a reaction or process, it must be present at the location and at the moment that reaction or process occurs. This aspect of protein function can now be explored at multiple levels and with ever-increasing precision.

RNA-Seq and Transcriptomics The RNAs that are transcribed from a genome under a given set of conditions can be determined using DNA sequencing methods described in

Chapter 8. The approach is called **RNA-Seq** (Fig. 9-15). RNA is first isolated from a tissue or a population of cells. The RNA is fragmented and converted to double-stranded DNA using reverse transcriptase (see Fig. 9-12a). This DNA is then subjected to deep DNA sequencing, which reveals both the RNAs that are present and the relative abundance of each (if more copies of one RNA are present, they will give rise to more DNA sequencing reads). This method is sensitive enough to apply to single cells, an approach called **single cell RNA-Seq**, or **scRNA-Seq**. It allows investigators to catalog the RNAs being transcribed in different parts of a tissue.



The transcriptional state of a human cell or tissue can be diagnostic of conditions ranging from diabetes to cancer. If a particular gene under study is expressed in a certain tissue or under particular metabolic conditions, the result provides a new functional clue. Detailed knowledge about what genes are expressed in a given tumor may eventually help guide treatment options. RNA-Seq can reveal patterns of gene regulation

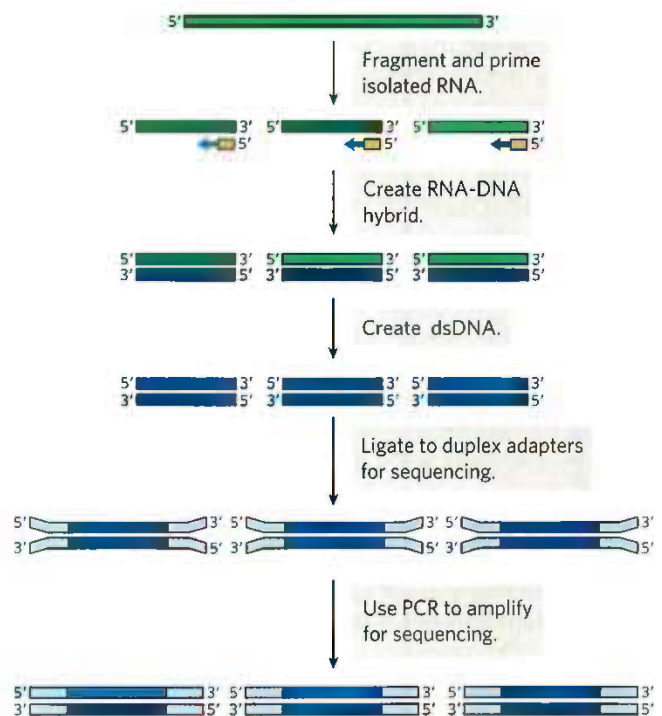


FIGURE 9-15 RNA-Seq. To define a cellular transcriptome, the first step is to isolate cellular RNA. Because many RNAs, particularly mRNAs and rRNAs, are quite long, the RNA is then fragmented to an average size commensurate with the DNA sequencing platform to be used. The RNA is converted to DNA using reverse transcriptase. Hexameric DNA oligonucleotides of random sequence are used to prime the reverse transcriptase if all RNA is to be included in the transcriptome. RNA-DNA hybrids are more stable than DNA-DNA hybrids, so hexameric duplexes are sufficient for this task. If the transcriptome is focused on expression of protein-coding genes, beads coated with poly(dT) can be used to hybridize to the poly(A) tails of eukaryotic mRNAs, allowing their precipitation and enrichment relative to other RNAs. After reverse transcription, the DNA fragments are ligated to duplex adapters that provide a universal priming location for the DNA sequencing as well as sequences that allow annealing to anchors on the sequencing flow cell (see Fig. 8-36). This is followed by deep DNA sequencing and data analysis.

and expression. It has special importance in studying cancerous tumors, in which rapid evolution triggered by genome instability creates a range of cell types. The mRNAs that are present in tumor cells provide a clue to the proteins that may be present, although not all mRNAs are immediately translated into protein. RNA-Seq also reveals the presence of many types of noncoding RNAs (described in Chapter 26) that are now being defined. ■

Cellular Proteomes and Mass Spectrometry A more direct way to establish protein presence or absence is to assess the cellular proteome. Mass spectrometry (Chapter 3) can accurately catalog and quantify the thousands of proteins present in a typical cell. This approach is a complement to RNA-Seq, as it provides a comprehensive list of the genes that are both transcribed and translated into protein. Mass spectrometry also provides information about how those proteins are modified, in turn allowing an assessment of their regulatory state.

Fusion Proteins and Immunofluorescence **P2** Often, an important clue to the function of a gene product comes from determining its location within the cell. For example, a protein found exclusively in the nucleus could be involved in processes that are unique to that organelle, such as transcription, replication, or chromatin condensation. Researchers often engineer fusion proteins for the purpose of locating a protein in the cell or organism. Some of the most useful fusions are the attachment of marker proteins that signal the location by direct visualization or by immunofluorescence.

A particularly useful marker is the **green fluorescent protein (GFP)** (Fig. 9-16), discovered by Osamu Shimomura. As subsequently shown by Martin Chalfie, a target gene (encoding the protein of interest) fused to the GFP gene generates a fusion protein that is highly fluorescent—it literally lights up when exposed to blue light—and can be visualized directly in a living cell. GFP is a protein derived from the jellyfish *Aequorea victoria* (Fig. 9-16a).

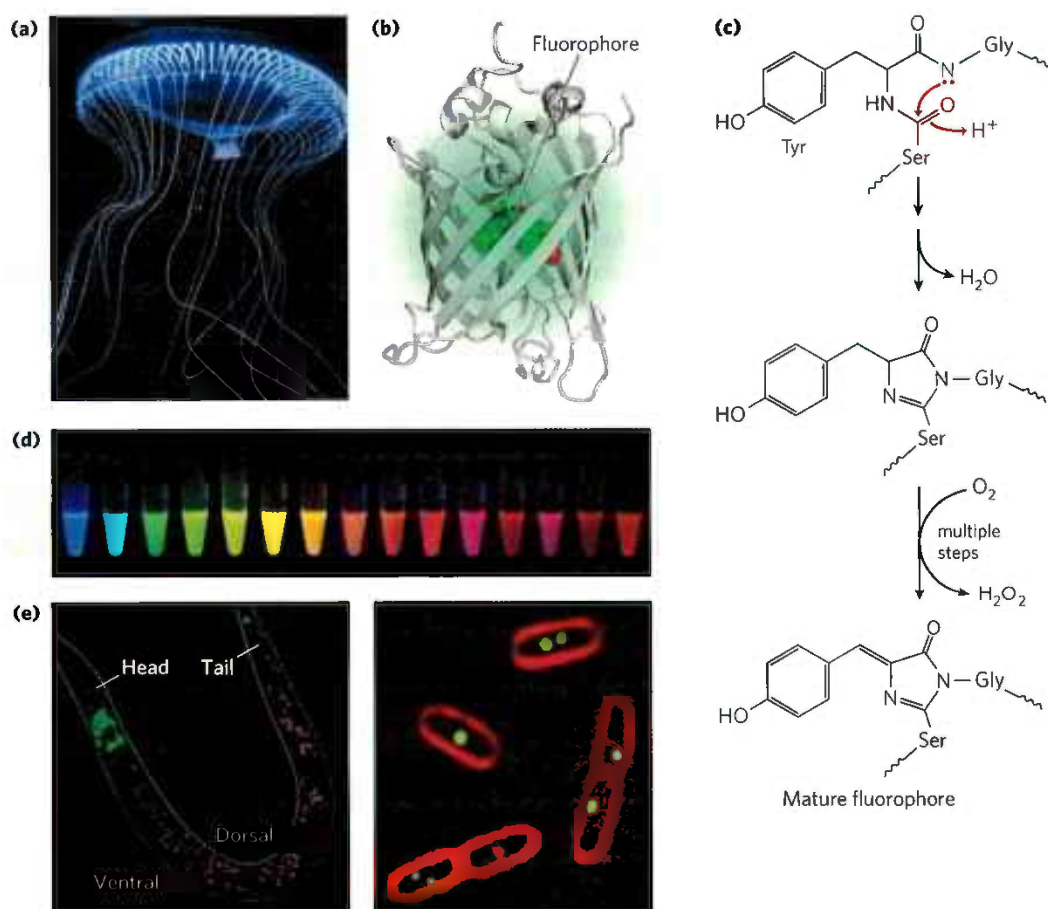


FIGURE 9-16 Green fluorescent protein (GFP). (a) GFP is derived from the jellyfish *Aequorea victoria*. (b) The protein has a β -barrel structure; the fluorophore is in the center of the barrel. (c) The fluorophore in GFP is derived from a sequence of three amino acids: $\text{Ser}^{65}\text{-Tyr}^{66}\text{-Gly}^{67}$. The fluorophore achieves its mature form through an internal rearrangement, coupled to a multistep oxidation reaction. An abbreviated mechanism is shown here. (d) Variants of GFP are now available in almost any color of the visible spectrum. (e) A GLR1-GFP fusion protein fluoresces bright green in *C. elegans*, a nematode worm (left). GLR1 is a glutamate receptor of nervous tissue. (In this photograph, autofluorescing fat droplets

are false-colored in magenta.) The membranes of *E. coli* cells (right) are stained with a red fluorescent dye. The cells are expressing a protein that binds to a resident plasmid, fused to GFP. The green spots indicate the locations of plasmids. [(a) Chris Parks/ImageQuest Marine. (b) Data from PDB ID 1GFL, F. Yang et al., *Nature Biotechnol.* 14:1246, 1996. (c) Information from Roger Tsien, University of California, San Francisco, Department of Pharmacology, and Paul Steinbach. (d) Courtesy of Roger Tsien and Paul Steinbach, University of California, San Diego, Department of Pharmacology. (e) (left) Courtesy Penelope J. Brockie and Andres V. Maricq, Department of Biology, University of Utah; (right) Courtesy Joseph A. Pogliano, from J. Pogliano et al. (2001), Multicopy plasmids are clustered and localized in *Escherichia coli*, *Proc. Natl. Acad. Sci. USA* 98:4486–4491.]

The protein has a β -barrel structure with a fluorophore (the fluorescent component of the protein) in the center (Fig. 9-16b). The fluorophore is derived from a rearrangement and oxidation of three amino acid residues (Fig. 9-16c). Because this reaction is autocatalytic and requires no proteins or cofactors other than molecular oxygen, GFP is readily cloned in an active form in almost any cell. Just a few molecules of this protein can be observed microscopically, allowing the study of its location and movements in a cell.

Careful protein engineering by Roger Tsien, coupled with the isolation of related fluorescent proteins from other marine coelenterates, has made variants of these proteins available in an array of colors (Fig. 9-16d) and other characteristics (brightness, stability). If fusion to GFP does not impair the function or properties of a protein one wishes to study, the fusion protein can be used to reveal the protein's location in the cell under a range of conditions and to detect interactions with other labeled proteins. With this technology, for example, the protein GLR1 (a glutamate receptor of nervous tissue) has been visualized as a GLR1-GFP fusion protein in the nematode *Caenorhabditis elegans* (Fig. 9-16e).

In some cases, the GFP fusion protein may be inactive or may not be expressed at sufficient levels to allow visualization. **Immunofluorescence** is an alternative approach for visualizing the endogenous (unaltered) protein. This approach requires fixation (and thus death) of the cell. The protein of interest is sometimes expressed as a fusion protein with an **epitope tag**, a short protein sequence that is bound tightly by a well-characterized, commercially available antibody. Fluorescent molecules (fluorochromes) are attached to this antibody. More commonly, the target protein is unaltered and is bound by an antibody that is specific for the protein. Next, a second antibody is added that binds specifically to the first one, and it is the second antibody that has the attached fluorochrome(s) (Fig. 9-17). A variation of this indirect approach to visualization is to attach biotin molecules to the first antibody, then add streptavidin (a bacterial protein closely related to avidin, a protein that binds biotin; see Table 5-1) complexed with fluorochromes. The interaction between biotin and streptavidin is one of the strongest and most specific known, and the potential to add multiple fluorochromes to each target protein gives this method great sensitivity. In all of these cases, the end product is a microscopic view of a cell in which a spot of light (a focus) reveals the location of the protein.

Knowing What a Protein Interacts with Can Suggest Its Function

Another key to defining the function of a particular protein is to determine its biochemical playmates. **P2** In the case of protein-protein interactions, the association of a protein of unknown function with one whose function is known can compellingly imply a functional relationship. The techniques used in this effort are quite varied.

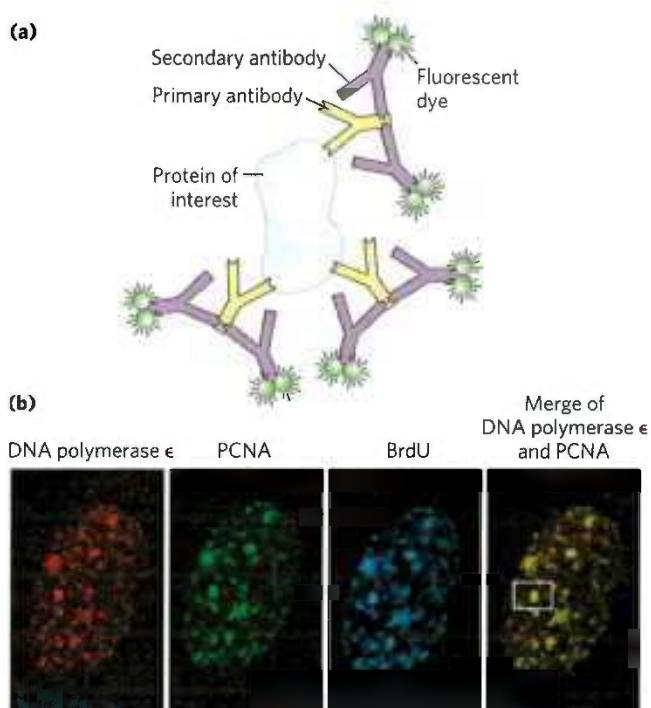


FIGURE 9-17 Indirect immunofluorescence. (a) The protein of interest is bound to a primary antibody, and a secondary antibody is added; this second antibody, with one or more attached fluorescent groups, binds to the first. Multiple secondary antibodies can bind the primary antibody, amplifying the signal. If the protein of interest is in the interior of the cell, the cell is fixed and permeabilized, and the two antibodies are added in succession. (b) The end result is an image in which bright spots indicate the location of the protein or proteins of interest in the cell. The images here show a nucleus from a human fibroblast, successively stained with antibodies and fluorescent labels for DNA polymerase ϵ ; for PCNA, an important polymerase accessory protein; and for bromo-deoxyuridine (BrdU), a nucleotide analog. The BrdU, added as a brief pulse, identifies regions undergoing active DNA replication. The patterns of staining show that DNA polymerase ϵ and PCNA co-localize to regions of active DNA synthesis (rightmost image); one such region is visible in the white box. [(b) Fuss, J. and Linn, S., 2002, "Human DNA Polymerase ϵ Colocalizes with Proliferating Cell Nuclear Antigen and DNA Replication Late, but Not Early, in S Phase," *J. Biol. Chem.* 277:8658–8666. Courtesy Jill Fuss, University of California, Berkeley.]

Purification of Protein Complexes By fusing the gene encoding a protein under study with the gene for an epitope tag, investigators can precipitate the protein product of the fusion gene by complexing it with the antibody that binds the epitope. This process is called **immunoprecipitation** (Fig. 9-18). If the tagged protein is expressed in cells, other proteins that bind to it precipitate with it. Identifying the associated proteins reveals some of the intracellular protein-protein interactions of the tagged protein. There are many variations of this process. For example, a crude extract of cells that express a tagged protein is added to a column containing immobilized antibody (see Fig. 3-17c for a description of affinity chromatography). The tagged protein binds to the antibody, and proteins that interact with the tagged protein are sometimes also retained on the column. The connection between the protein and the tag is cleaved with a specific protease. The protein complexes are eluted from the column, and the proteins in them are

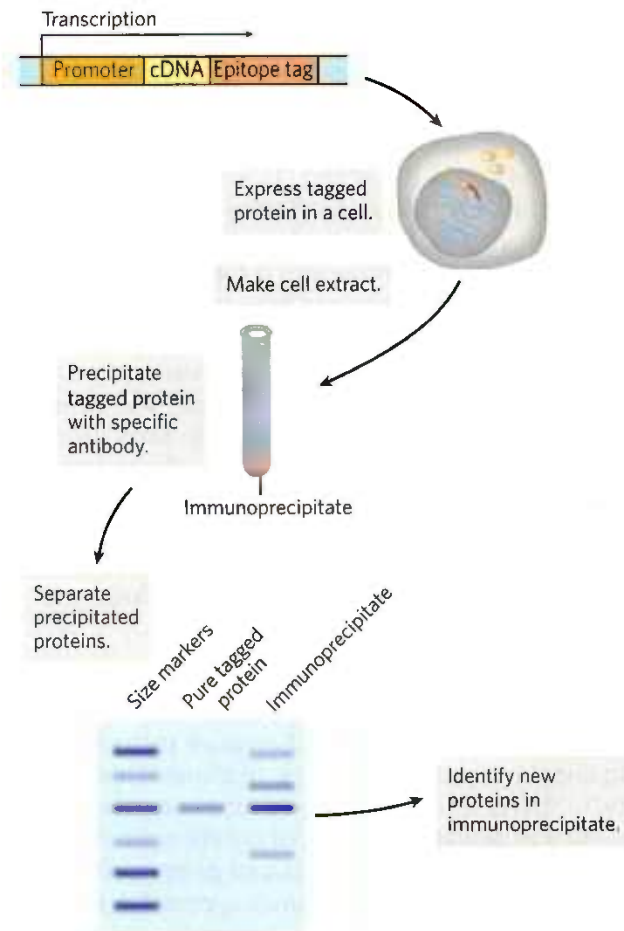


FIGURE 9-18 The use of epitope tags to study protein-protein interactions. The gene of interest is cloned next to a gene for an epitope tag, and the resulting fusion protein is precipitated by antibodies to the epitope. Any other proteins that interact with the tagged protein also precipitate, thereby helping to elucidate protein-protein interactions.

identified by mass spectrometry. Researchers can use these methods to define complex networks of interactions within a cell. In principle, the chromatographic approach to analyzing protein-protein interactions can be used with any type of protein tag (His tag, GST, etc.) that can be immobilized on a suitable chromatographic medium.

The selectivity of this approach can be enhanced with **tandem affinity purification (TAP) tags**. Two consecutive tags are fused to a target protein, and the fusion protein is expressed in a cell (Fig. 9-19). The first tag is protein A, a protein found at the surface of the bacterium *Staphylococcus aureus* that binds tightly to mammalian immunoglobulin G (IgG). The second tag is often a calmodulin-binding peptide. A crude extract containing the TAP-tagged fusion protein is passed through a column matrix with attached IgG antibodies that bind protein A. Most of the unbound cellular proteins are washed through the column, but proteins that normally interact with the target protein in the cell are retained. The first tag is then cleaved from the fusion protein with a highly specific protease, TEV protease, and the shortened fusion target protein and any proteins associated noncovalently

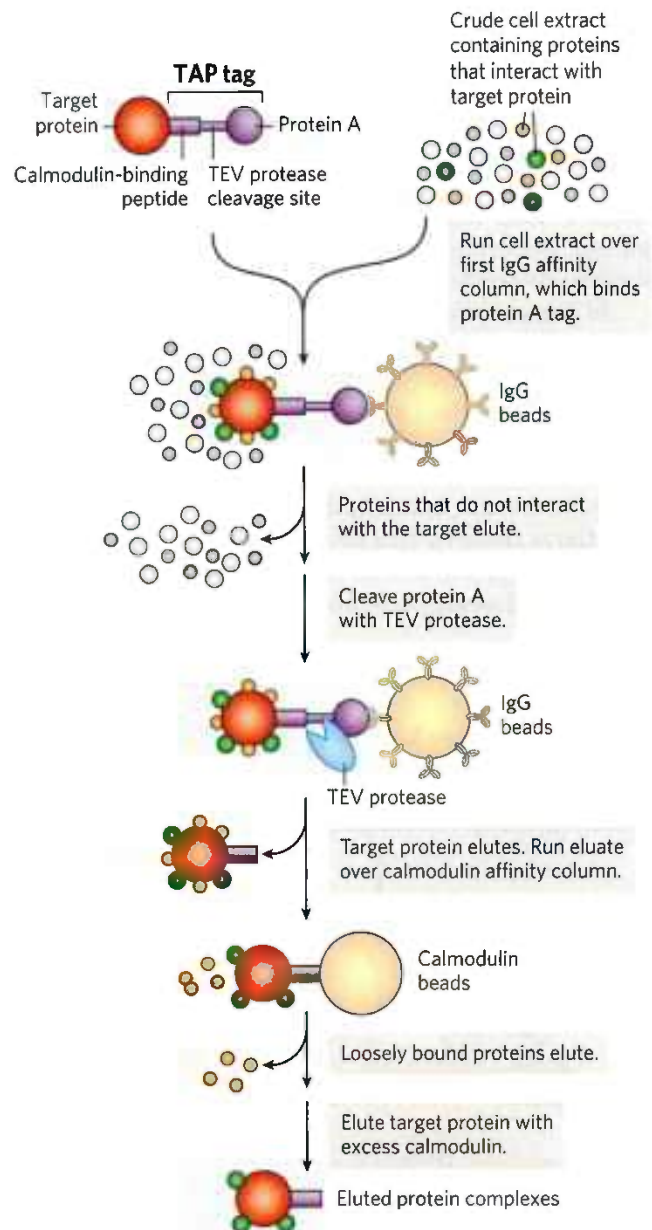


FIGURE 9-19 Tandem affinity purification (TAP) tags. A TAP-tagged protein and associated proteins are isolated by two consecutive affinity purifications, as described in the text.

with the target protein are eluted from the column. The eluate is then passed through a second column containing a matrix with attached calmodulin that binds the second tag. Loosely bound proteins are again washed from the column. After the second tag is cleaved, the target protein is eluted from the column with its associated proteins. The two consecutive purification steps eliminate most weakly bound contaminants. False positives are minimized, and protein interactions that persist through both steps are likely to be functionally significant.

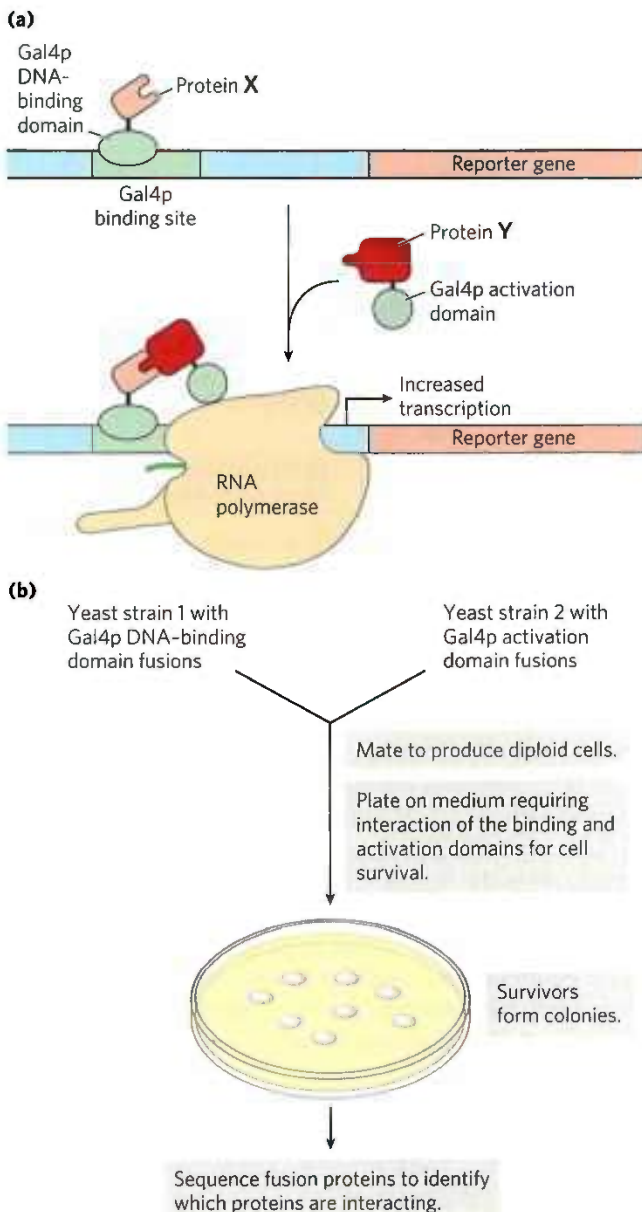
Yeast Two-Hybrid Analysis A sophisticated genetic approach to defining protein-protein interactions is based on the properties of the Gal4 protein (Gal4p; see Fig. 28-32), which activates the transcription of *GAL* genes (encoding

the enzymes of galactose metabolism) in yeast. Gal4p has two domains: one that binds a specific DNA sequence, and another that activates RNA polymerase to synthesize mRNA from an adjacent gene. The two domains of Gal4p are stable when separated, but activation of RNA polymerase requires interaction with the activation domain, which in turn requires positioning by the DNA-binding domain. Hence, the domains must be brought together to function correctly.

In **yeast two-hybrid analysis**, the protein-coding regions of the genes to be analyzed are fused to the yeast gene for either the DNA-binding domain or the activation domain of Gal4p, and the resulting genes express a series of fusion proteins (Fig. 9-20). If a protein fused to the DNA-binding domain interacts with a protein fused to the activation domain, transcription is activated. The reporter gene transcribed by this activation is generally one that yields a protein required for growth or an enzyme that catalyzes a reaction with a colored product. Thus, when grown

on the proper medium, cells that contain a pair of interacting proteins are easily distinguished from those that do not.

A library can be set up with a particular yeast strain in which each cell in the library has a gene fused to the Gal4p DNA-binding domain gene, and many such genes are represented in the library. In a second yeast strain, a gene of interest is fused to the gene for the Gal4p activation domain. The yeast strains are mated, and individual diploid cells are grown into colonies. The only cells that grow on the selective medium, or that produce the appropriate color, are those in which the gene of interest is binding to a partner, allowing transcription of the reporter gene. This allows large-scale screening for cellular proteins that interact with the target protein. The interacting protein that is fused to the Gal4p DNA-binding domain present in a particular selected colony can be quickly identified by DNA sequencing of the fusion protein's gene. Some false positive results occur, due to the formation of multiprotein complexes.



The Effect of Deleting or Altering a Protein Can Suggest Its Function

P2 P3 One of the most informative paths to understanding the function of a gene is to change (mutate) the gene or delete it. An investigator can then examine how the genomic alteration affects cell growth or function. The methods available to modify genomes grow more sophisticated every year. The most common approach is to cut the gene of interest at a site that is functionally critical, generating a double-strand break. In eukaryotes, such breaks are most commonly repaired by cellular systems that promote nonhomologous end joining (NHEJ), a process described in Chapter 25. NHEJ seals the double-strand break, but the process is imprecise. Nucleotides are often deleted or added during the repair, inactivating the gene. In bacteria, introduced double-strand breaks are usually repaired more accurately, by homologous recombination systems (Chapter 25), but inactivating mutations can appear. Many traditional approaches to targeting a gene in this way were supplanted by the advent of CRISPR/Cas systems in 2011.

CRISPR/Cas Systems “CRISPR” stands for clustered, regularly interspaced short palindromic repeats; as the name suggests, these consist of a series of regularly spaced short repeats in the bacterial genome. A Cas (CRISPR-associated) protein is a nuclease. The CRISPR sequences and Cas protein are components of a kind of immune system that evolved to allow bacteria to survive infection

FIGURE 9-20 Yeast two-hybrid analysis. (a) The goal is to bring together the DNA-binding domain and the activation domain of the yeast Gal4 protein (Gal4p) through the interaction of two proteins, X and Y, to which one or other of the domains is fused. This interaction is accompanied by the expression of a reporter gene. (b) The two gene fusions are created in separate yeast strains, which are then mated. The mated mixture is plated on a medium on which the yeast cannot survive unless the reporter gene is expressed. Thus, all surviving colonies have interacting fusion proteins. Sequencing of the fusion proteins in the survivors reveals which proteins are interacting.

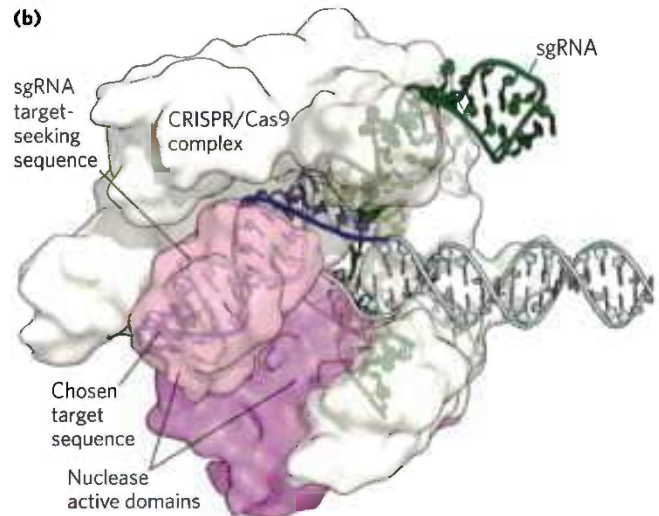
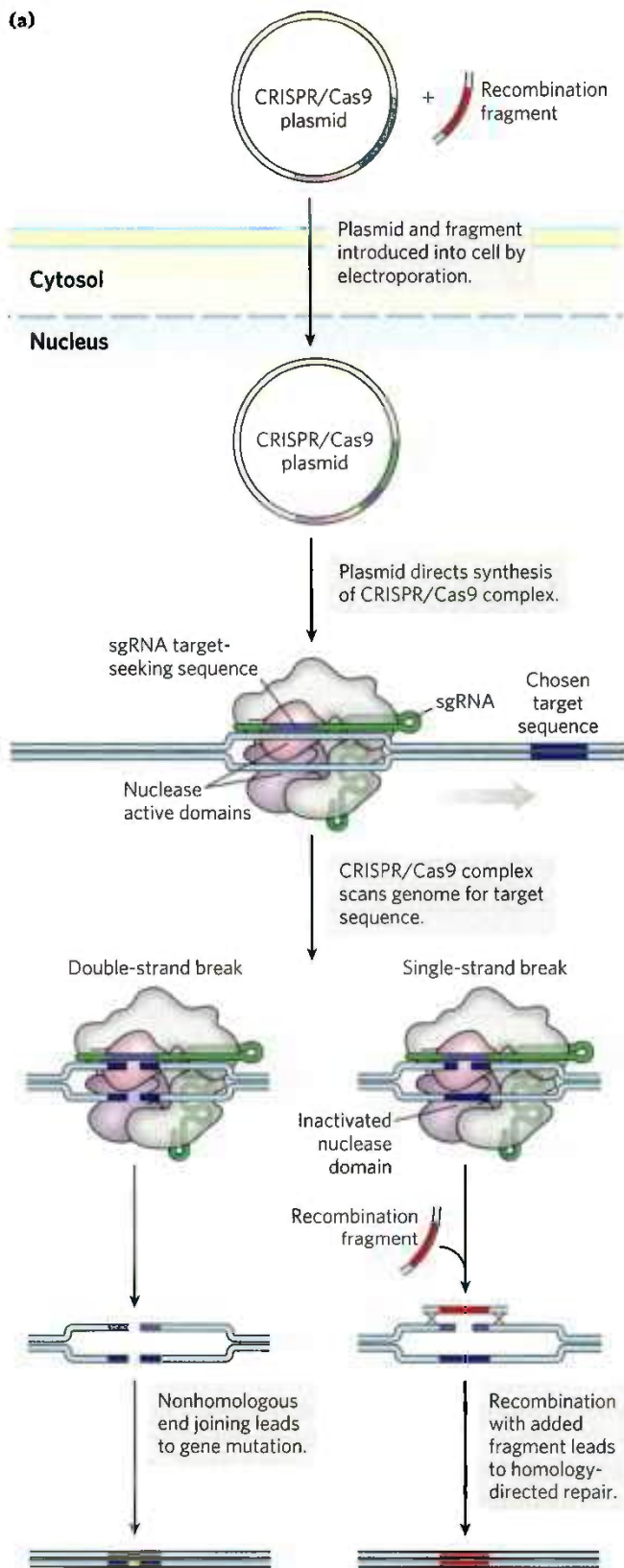


FIGURE 9-21 The CRISPR/Cas9 system for genomic engineering.

(a) The genes encoding the Cas9 protein and sgRNA are introduced into a cell in which a targeted genomic change is planned. The sgRNA has a region complementary to the chosen genomic target sequence (purple); this region can be engineered to include any desired sequence. A complex consisting of the CRISPR sgRNA and the Cas9 protein forms within the cell and binds to the chosen target site in the DNA. The structure of the bound complex is shown in (b). In the pathway shown on the left in (a), two nuclease active sites in the Cas9 protein separately cleave each DNA strand in the target, producing a double-strand break. The double-strand break is usually repaired by nonhomologous end joining, which generally deletes or alters the nucleotides at the site where joining occurs. Alternatively, as shown in the pathway on the right, if one nuclease site is inactivated, Cas9 nuclease activity creates a single-strand break in the target sequence. In the presence of a recombination donor DNA fragment, identical to the target sequence but incorporating the desired sequence change (fragment shown in red), homologous DNA recombination will sometimes change the sequence at the site of the break to match that of the donor DNA. [Data from PDB ID 4UN3, C. Anders et al., *Nature* 513:569, 2014.]

corresponding CRISPR/Cas system, the CRISPR sequence and Cas protein act together to destroy the viral DNA. First, the CRISPR sequences are transcribed to RNA, and individual viral spacer sequences are cleaved to form products called **guide RNAs (gRNAs)**, which include some adjacent repeat RNA. A gRNA forms a complex with one or more Cas proteins and, in some cases, with another RNA called a **trans-activating CRISPR RNA**, or **tracrRNA**. The resulting complex binds specifically to the invading bacteriophage DNA, cleaving and destroying it through the nuclease activities associated with the Cas proteins.

The current technology was made possible by discovery of a relatively simple CRISPR/Cas system in *Streptococcus pyogenes*. This system requires only a single Cas protein, Cas9, to cleave DNA. Work in many laboratories, particularly those of Jennifer Doudna and Emmanuelle Charpentier, has produced a streamlined CRISPR/Cas9 system composed of just one protein (Cas9) and one associated RNA, consisting of gRNA and tracrRNA fused into a **single guide RNA (sgRNA)**. **P2 P3** The power of the system is embedded in this sgRNA, in which the guide sequence can be altered to specifically and efficiently target almost any genomic sequence (**Fig. 9-21**). Cas9 has two separate nuclease

by bacteriophages. CRISPR sequences are embedded in the bacterial genome, surrounding sequences derived from phage pathogens that previously infected the bacterium without killing it. The viral sequences are, in effect, spacer sequences separating the CRISPR sequences. When the same bacteriophage again attacks a bacterium that has the

domains: one domain cleaves the DNA strand paired with the sgRNA, and the other cleaves the opposite DNA strand. Inactivating one domain creates an enzyme that cleaves just one strand, forming a single-strand break, or nick. The sgRNA is needed both to pair with the target sequence in the DNA and to activate the nuclease domains for cleavage.

Plasmids expressing the required protein and RNA components of CRISPR/Cas9 can be introduced into microbial cells by electroporation (p. 306). For mammalian cells, the genes encoding the CRISPR/Cas9 components can be incorporated into engineered viruses that subsequently deliver them to the cell nuclei. For many organisms, the targeted gene is inactivated in a high percentage of the treated cells. If a genomic change (mutation) rather than a simple gene inactivation is required, it can be introduced by recombination when a DNA fragment encompassing the cleavage site and including the desired change enters the cell with the CRISPR/Cas9 plasmids. This recombination is often inefficient, but success can be improved somewhat by introducing a nick rather than a double-strand break at the target site (Fig. 9-21).

CRISPR/Cas9 can be combined with other approaches to extract additional information. For example, a particular gene can be inactivated with CRISPR/Cas9. Then, the effect of that gene inactivation on the transcription of other genes can be probed with RNA-Seq at the level of tissues, cell populations, or single cells.

New applications for CRISPR/Cas9 are being developed rapidly, both for basic research and for medicine. Genetic screens based on CRISPR are described in the next section. CRISPR is being used to enhance food production, provide new approaches to combat bacterial infections, and eliminate nonnative pest species that can harbor diseases (**Box 9-1**). New CRISPR-based treatments for genetic diseases are being cautiously advanced to clinical trials for vision loss due to inherited retinal dystrophies, Duchenne muscular dystrophy, β -thalassaemia, and many other conditions. Uncertainties remain, particularly the potential for occasional cleavage at unintended chromosomal sites (off-target cleavage). The impact of CRISPR/Cas9 will continue to grow as problems are overcome, current applications mature, and new applications are imagined and created.

Many Proteins Are Still Undiscovered

For most biological processes, ranging from intermediary metabolism to neurological function to DNA metabolism, the list of known participating enzymes and proteins is far from complete. **P2** Genetic screening for new gene functions has been underway for many decades. The goal is to efficiently interrogate large numbers of genes, sometimes the entire genome, for genes that affect a particular

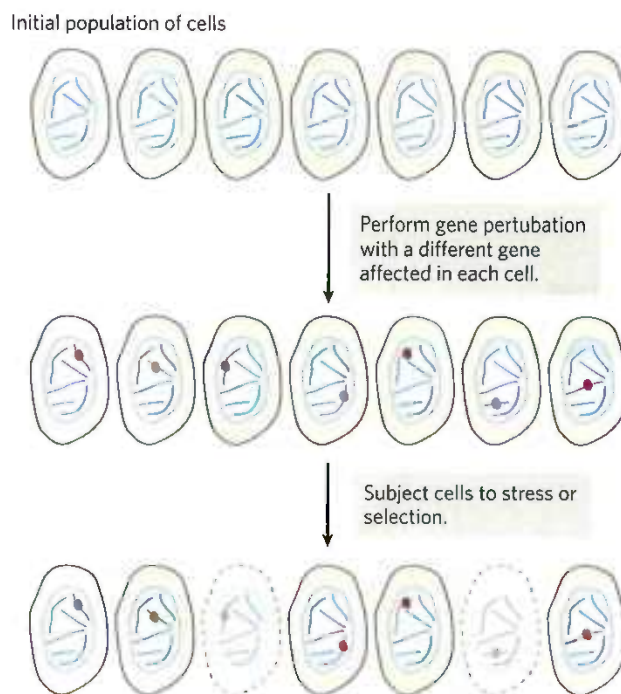


FIGURE 9-22 High-throughput genetic screening. A gene perturbation, either inactivation or activation, is introduced to a population of cells such that only one gene in each cell is affected. However, all or most genes are affected in one cell or another within the population. The population is then subjected to a selection that requires a response from some cellular genes. Cells that lack the required genes or that have those genes activated will drop out or be enriched within the population, respectively. In the example shown, two cells drop out.

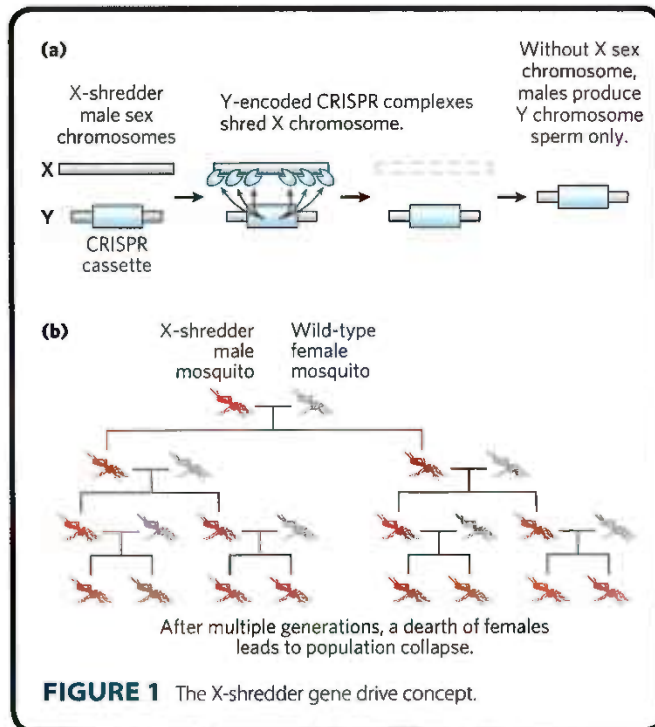
cellular reaction or process. A gene perturbation—a treatment that inactivates a gene or activates its expression—is introduced under conditions in which just one gene is affected in each cell, but most or all of the genes are affected in one or more cells within the population (**Fig. 9-22**). The population is then subjected to a stress or selection. Cells in which a gene required to respond to the selection is altered may drop out of the population or be enriched in the population, depending upon the goals and design of the screen.

CRISPR-based technologies increasingly play a central role in large-scale screening protocols (**Fig. 9-23**). Libraries of sgRNAs have been generated to target virtually all genes in a mammalian genome, or specialized subsets of them. The targeting sequence in each sgRNA is 20 bp long. In addition to targeting a particular gene, each targeting sequence acts as a kind of unique bar code identifier that is readily recognized by computer programs after sequencing. The sgRNAs are packaged in a DNA cassette set up to also express the Cas9 protein or a Cas9 variant. The cassettes are incorporated into carefully engineered lentiviral vectors derived from HIV (with genes required for HIV multiplication eliminated). The viral vectors deliver the cassette to the nucleus as a single-stranded RNA, convert it to double-stranded

Getting Rid of Pests with Gene Drives

Invasive introduced plant and animal species can wreak havoc on any natural environment, and can also spread human disease. Mosquitoes that harbor Zika virus and other diseases in many parts of the world, introduced rats on almost every continent, cane toads and rabbits in Australia, the kudzu vine in the southern United States—represent just a few examples of invasive species that cause human misery and annual financial damage totaling billions of dollars. Traditional methods of control such as poisoning or trapping are often unsuccessful and can have detrimental effects on native species that become unintended targets.

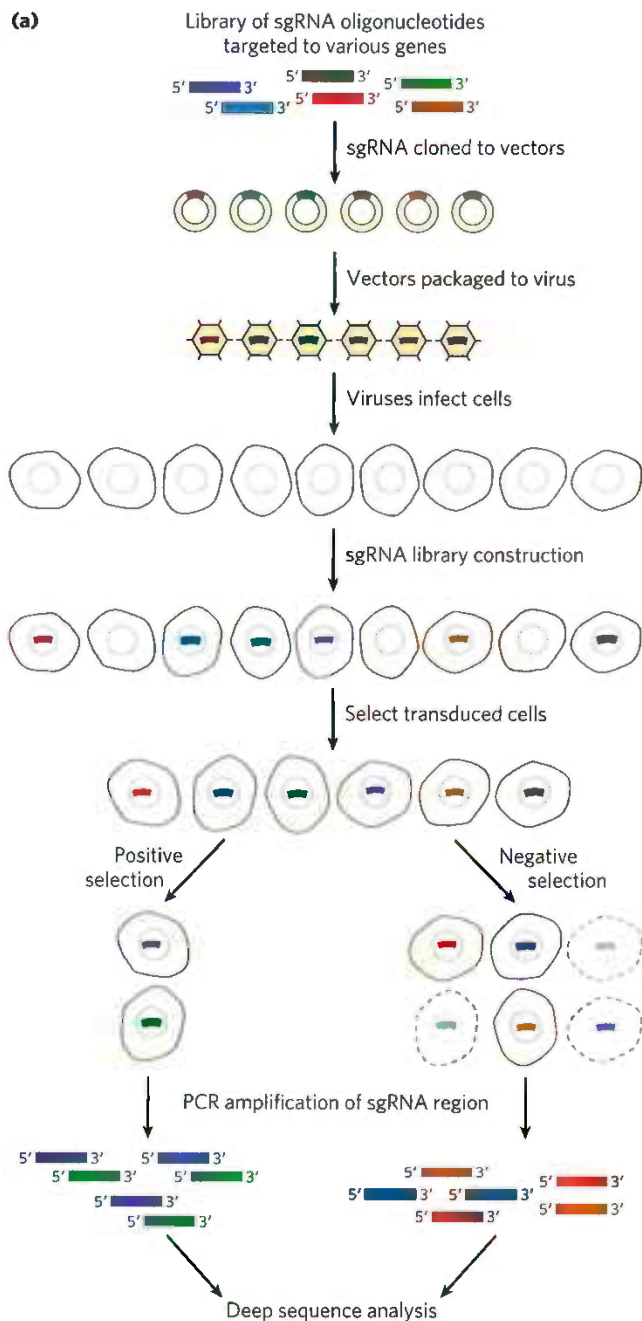
The discovery of selfish DNA elements such as homing endonucleases and transposons that can spread through a population gave rise to the concept of **gene drives** as a new approach to the control of invasive species. The most recent and promising iteration of this idea involves synthetic gene drives based on CRISPR/Cas9. The overall idea is to set up a system that skews the male to female ratio in a target species far away from the favored 1:1, resulting in population collapse. A strategy called X shredder, already proven in the laboratory with mosquitoes, is highlighted in Figure 1. A cassette that includes genes expressing Cas9, as well as several sgRNAs targeted to different unique sites on the X chromosome, is inserted into an intergenic region on the Y chromosome. The cassette is engineered into males, and is controlled by a gene regulatory system that is expressed only during spermatogenesis. Thus, during spermatogenesis, the cassette is expressed so that the X chromosome is cleaved at multiple locations, basically destroying it. This ensures that the only viable sperm have Y chromosomes. All the offspring of any cross with a female are all males, and all of them possess Y chromosomes containing the X-shredder cassette. When those male offspring mate with other females later, the same result ensues. As these males mate and spread the cassette through the population, a dearth of females occurs and the population collapses. In principle, this same strategy could be applied to rats, cane toads, and many other invasive species.



To date, gene drives have been restricted to the laboratory. The potential for a gene drive escaping to species that are not intended targets is not yet clear. Resistance in the target species could occur by mutating the sgRNA target sites, although the use of multiple sites makes this less likely. The gene drive approach is a good illustration of the power and potential of CRISPR. However, once males with a gene drive are released, it would be essentially impossible to call a halt to the effects. Nature has a way of imposing consequences, both unintended and unexpected. The potential positive effects on health and agriculture continue to drive research to improve the technology and address potential problems.

DNA with the viral-encoded reverse transcriptase, and integrate the DNA into a chromosome. The CRISPR/Cas9 components are expressed to perturb the target gene specified by the particular sgRNA delivered to that cell. The effect produced depends upon the Cas9 variant used. The unmodified Cas9 nuclease will create a double-strand break that inactivates the gene. A modified Cas9 that lacks the nuclease activity will simply bind to its target and block transcription. Cas9 fused to a protein transcription inhibitor or activator may more effectively block or activate transcription, respectively (Fig. 9-23b).

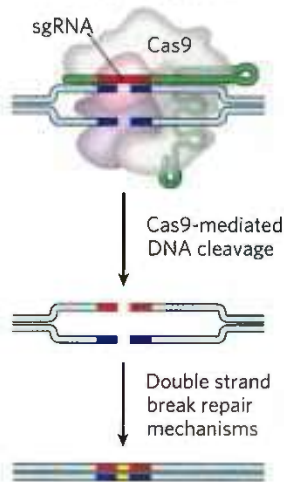
Whatever strategy is used, a different gene is affected in each cell. Once the population has been treated with a stress or selection, cells in which genes required to survive the treatment are inactivated or activated by the CRISPR/Cas9 variant will die or thrive. The decreased or increased presence of the relevant bar code sequences can be detected by deep DNA sequencing, using a universal priming sequence incorporated into the cassette near the sgRNA sequence. The strategies outlined here only hint at the variety of protocols in use, limited only by the imagination of the investigators.



SUMMARY 9.2 Exploring Protein Function on the Scale of Cells or Whole Organisms

- Proteins can be studied at the level of phenotypic, cellular, or molecular function.
- Comparative genomics can elucidate protein function by identifying structural motifs within the encoded protein and comparing gene sequences from different organisms.
- A determination of when and where a protein appears in a cell can offer functional clues. RNA-Seq provides information on what genes are being expressed in a cell. Mass spectrometry can define cellular proteomes. By fusing a gene of interest with genes that encode green fluorescent protein or epitope tags, researchers can

(b) Gene knockout through frameshift, insertion of stop codon



(c) Transcriptional repression

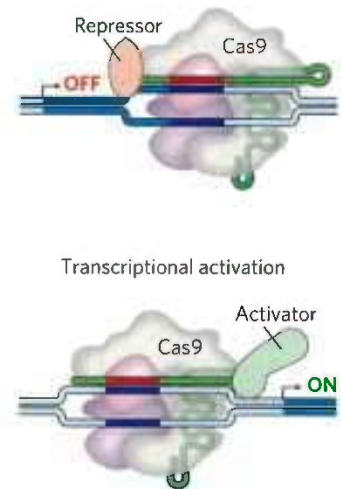


FIGURE 9-23 Use of CRISPR/Cas9 in high-throughput screening.

CRISPR/Cas9 provides the gene perturbation in many screening protocols. (a) In a typical screen, a library of sgRNAs is constructed so as to target all known genes in a genome of interest. These are cloned into viral vectors. The vectors infect cells at a multiplicity of infection (MOI) small enough so that most cells will gain only one vector. The vector RNA is converted to DNA and integrated into the genome. When expressed, it will affect one target gene, with most genes affected in one or more cells in the population. After selection, some cells drop out or are enriched in the population, depending on the nature of the screen. (b) Several variations of Cas9 are shown to illustrate a few of the ways genes can be affected. Unaltered Cas9 will cleave the DNA at the target site. (c) If engineered to lack nuclease activity, and fused with a gene repressor or activator, the modified Cas9 will bind to the target site and either decrease or increase gene transcription, respectively.

visualize the cellular location of the gene product, either directly or by immunofluorescence.

- The interactions of a protein with other proteins or RNA can be investigated with epitope tags and immunoprecipitation or affinity chromatography. Yeast two-hybrid analysis probes molecular interactions in vivo.
- The cellular effects of inactivating a gene can be conveniently explored using the CRISPR/Cas9 programmable nuclease. CRISPR/Cas9 can also be used to alter gene sequences in a targeted manner.
- Screens for new genes increasingly employ variants of the CRISPR/Cas9 system.

9.3 Genomics and the Human Story

Since the report of the first complete human genomes in 2001, human genome sequencing has become routine. The genomes of tens of thousands of other species have now been sequenced and made publicly available, providing a look at genomic complexity throughout the three domains of living organisms: Bacteria, Archaea, and Eukarya. Whereas many early sequencing projects focused on species commonly used in research laboratories, the projects now include species of practical, medical, agricultural, and evolutionary interest. Genomes from

every known bacterial family have been sequenced. Completed eukaryotic genome sequences number in the tens of thousands. Genomes of extinct species such as *Homo neanderthalensis* and of humans who died in past millennia have also been sequenced. Personal genomes are playing an ever-increasing role in medicine.

Each genome sequence becomes an international resource for researchers. Collectively, the sequences provide a source for broad comparisons that help pinpoint both variable and highly conserved gene segments, and they allow the identification of genes that are unique to a species or group of species. Efforts to map genes, identify new proteins and disease-related genes, elucidate genetic patterns of medical interest, and trace our evolutionary history are among the many initiatives under way.

The Human Genome Contains Many Types of Sequences

The rapidly growing genome databases have the potential not only to fuel advances in all realms of biochemistry but also to change the way we think about ourselves. What does our own genome, and comparisons with those of other organisms, tell us?

In some ways, we are not as complicated as we once imagined. Humans have only about 20,000 protein-coding genes—less than twice the number in a fruit fly (13,600 genes), not many more than in a nematode worm (19,700 genes), and fewer than in a rice plant (38,000 genes).

In other ways, we are more complex than we previously realized. Many, if not most, eukaryotic genes contain one or more segments of DNA that do not code for the amino acid sequence of a polypeptide product. These nontranslated segments interrupt the otherwise colinear relationship between the gene's nucleotide sequence and the amino acid sequence of the encoded polypeptide. Such nontranslated DNA segments are called **introns**, and the coding segments are called **exons** (Fig. 9-24). Few bacterial genes contain introns. The introns are spliced from a precursor RNA transcript to generate a transcript that can be translated contiguously into a protein product (see Chapter 26). An exon often (but not

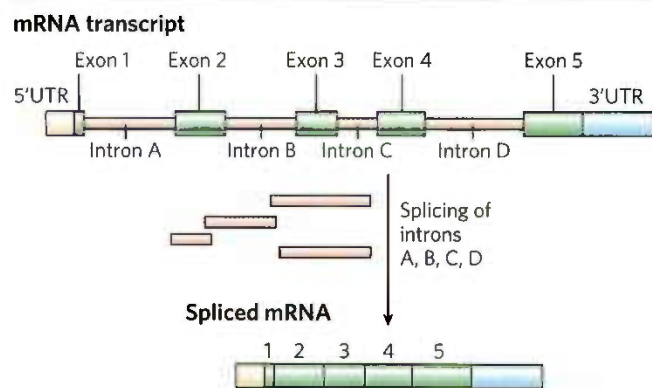
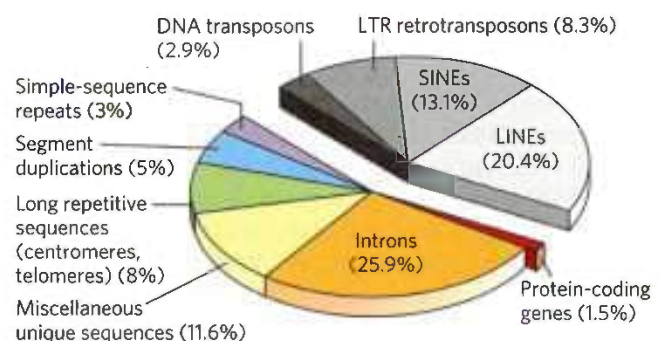


FIGURE 9-24 Introns and exons. This gene transcript contains five exons and four introns, along with 5' and 3' untranslated regions (5'UTR and 3'UTR). Splicing removes the introns to create an mRNA product for translation into protein.

always) encodes a single domain of a larger, multidomain protein. Humans share many protein domain types with plants, worms, and flies, but the domains in the human genome are mixed and matched in more complex ways, increasing the variety of proteins found in our proteome. Alternative modes of gene expression and RNA splicing permit alternative combinations of exons, leading to the production of more than one protein from a single gene. Alternative splicing (Chapter 26) is far more common in humans and other vertebrates than in worms or bacteria, allowing greater complexity in the number and kinds of proteins generated.

In mammals and some other eukaryotes, the typical gene has a much higher proportion of intron DNA than exon DNA; in most cases, the function of introns is not clear. Less than 1.5% of human DNA is “protein-coding” or exon DNA, carrying information for protein products (Fig. 9-25a). However, when introns are included in the accounting, as much as 30% of the human genome consists of genes that encode proteins. Several efforts are

(a) Human genome: DNA sequence types



(b) Human genome: Protein-coding genes

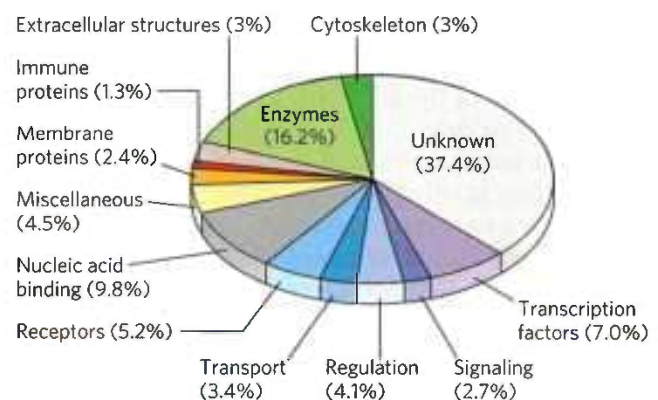


FIGURE 9-25 A snapshot of the human genome. (a) This pie chart shows the proportions of various types of sequences in our genome. The classes of transposons that represent nearly half of the total genomic DNA are indicated in shades of gray. LTR retrotransposons are retrotransposons with long terminal repeats (see Fig. 26-33). Long interspersed nuclear elements (LINEs) and short interspersed nuclear elements (SINEs) are special classes of particularly common DNA transposons. (b) The approximately 20,000 protein-coding genes in the human genome can be classified by the type of protein encoded. [Information from (a) T. R. Gregory, *Nature Rev. Genet.* 6:699, 2005; (b) www.pantherdb.org.]

under way to categorize protein-coding genes by type of function (Fig. 9-25b).

The relative paucity of protein-coding genes in the human genome leaves a lot of DNA unaccounted for. Much of the DNA that does not encode proteins (exons or introns) is in the form of repeated sequences of several kinds. Perhaps most surprising is that about half the human genome is made up of moderately repeated sequences that are derived from **transposons**, segments of DNA, ranging from a few hundred to several thousand base pairs long, that can move from one location to another in the genome. Originally discovered in corn by Barbara McClintock, who called them transposable elements, transposons are a kind of molecular parasite. They make their home in the genomes of essentially every organism. Many transposons contain genes encoding the proteins that catalyze the transposition process itself, as described in more detail in Chapters 25 and 26. There are several classes of transposons in the human genome. Many are strictly DNA segments, which have slowly increased in number over the millennia as a result of replication events coupled to the transposition process. Some, called retrotransposons, are closely related to retroviruses, transposing from one genomic location to another through RNA intermediates that are reconverted to DNA by reverse transcription. Some transposons in the human genome are active elements, moving at a low frequency, but most are inactive, evolutionary relics altered by mutations. Transposon movement can lead to the redistribution of other genomic sequences, and this has played a major role in human evolution.

Once the protein-coding genes (including exons and introns) and transposons are accounted for, perhaps 25% of the total DNA remains. As a follow-up to the Human Genome Project, the ENCODE initiative was launched by the National Human Genome Research Institute in 2003 to identify functional elements in the human genome.

P1 The work of the worldwide consortium of research groups engaged in the ENCODE initiative has revealed that the vast majority (>80%, including protein-coding genes, most transposons, and more) of the DNA in the human genome is either transcribed into RNA in at least one type of cell or tissue or is involved in some functional aspect of chromatin structure. Much of the noncoding (nontranscribed) DNA in the remaining 20% contains regulatory elements that affect the expression of the 20,000 protein-coding genes and the many additional genes encoding functional RNAs. Many mutations (SNPs; described below) associated with human genetic diseases lie in this noncoding DNA, probably affecting regulation of one or more genes. As described in Chapters 26 and 27, new classes of functional RNAs are being discovered at a rapid pace. **P1** Many of these functional RNAs, now being identified by a variety of screening methods, are produced by RNA-coding genes whose existence was previously unsuspected.

About 3% or so of the human genome consists of highly repetitive sequences referred to as **simple-sequence repeats (SSRs)**. Generally less than 10 bp long, an SSR is sometimes repeated millions of times per cell, distributed in short segments of tandem repeats. The most prominent examples of SSR DNA are found in centromeres and telomeres (see Chapter 24). Human telomeres, for example, consist of up to 2,000 contiguous repeats of the sequence GGTTAG. Additional, shorter repeats of simple sequences also occur throughout the genome. These isolated segments of repeated sequences, often containing up to a few dozen tandem repeats of a simple sequence, are called **short tandem repeats (STRs)**. Such sequences are the targets of the technologies used in forensic DNA analysis (see Box 8-1).

What does all this information tell us about the similarities and differences among individual humans? Within the human population there are millions of single-base variations, called **single nucleotide polymorphisms**, or **SNPs** (pronounced “snips”). Each person differs from the next by, on average, 1 in every 1,000 bp. Many of these variations are in the form of SNPs, but the human population also has a wide range of larger deletions, insertions, and small rearrangements. From these often subtle genetic differences comes the human variety we are all aware of—such as differences in hair color, stature, foot size, eyesight, allergies to medication, and (to some unknown degree) behavior.

The process of genetic recombination during meiosis tends to mix and match these small genetic variations so that different combinations of genes are inherited (see Chapter 25). However, groups of SNPs and other genetic differences that are close together on a chromosome are rarely affected by recombination and are usually inherited together; such a grouping of multiple SNPs is known as a **haplotype**. Haplotypes provide convenient markers for certain human populations and for individuals within populations.

Defining a haplotype requires several steps. First, positions that contain SNPs in the human population are identified in genomic DNA samples from multiple individuals (Fig. 9-26a). Each SNP in a prospective haplotype may be separated from the next SNP by several thousand base pairs and still be regarded as “nearby” in the context of chromosomes that extend for millions of base pairs. Second, a set of SNPs typically inherited together is chosen as a defined haplotype (Fig. 9-26b); each haplotype consists of the particular bases found at the various SNP positions within the defined set. Finally, tag SNPs—a subset of SNPs that define an entire haplotype—are chosen to uniquely identify each haplotype (Fig. 9-26c). By sequencing just these tag positions in genomic samples from human populations, researchers can quickly identify which of the haplotypes are present in each individual. Especially stable haplotypes exist in the mitochondrial genome (which does not undergo meiotic recombination) and on the Y chromosome (only 3%

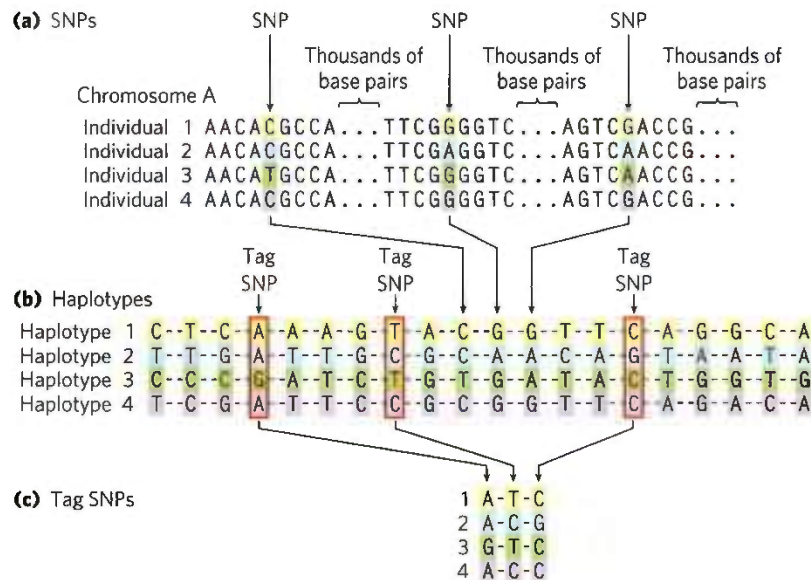


FIGURE 9-26 Haplotype identification. (a) The positions of SNPs in the human genome can be identified in genomic samples. The SNPs can be in any part of the genome, whether or not it is part of a known gene. (b) Groups of SNPs are compiled into a haplotype. The SNPs vary in the overall human population, as in the four fictitious individuals shown here, but the SNPs chosen to define a haplotype are often the same in most individuals of a particular population. (c) A few SNPs are chosen as haplotype-defining (tag

SNPs, outlined in red), and these are used to simplify the process of identifying an individual's haplotype (by sequencing 3 instead of 20 loci). (c) For example, if the positions shown here were sequenced, an A-T-C haplotype might be characteristic of a population native to one location in northern Europe, whereas G-T-C might be the prevailing sequence in a population in Asia. Multiple haplotypes of this kind are used to trace prehistoric human migrations. [Information from International HapMap Consortium, *Nature* 426:789, 2003, Fig. 1.]

of which is homologous to the X chromosome and thus subject to recombination). As we will see, haplotypes can be used as markers to trace human migrations.

Genome Sequencing Informs Us about Our Humanity

The human genome is very closely related to other mammalian genomes over large segments of every chromosome. However, for a genome measured in billions of base pairs, differences of just a few percent can add up to millions of genetic distinctions. Searching among these, and making use of comparative genomics techniques, researchers can begin to explore the molecular basis of definably human characteristics.

The genome sequences of our closest biological relatives, the chimpanzee (*Pan troglodytes*) and bonobo (*Pan paniscus*), offer some important clues, and we can use them to illustrate the comparative process. Human and chimpanzee shared a common ancestor about 7 million years ago. Genomic differences between the species, including SNPs and larger genomic rearrangements such as inversions, deletions, and fusions, can be used to construct a phylogenetic tree (Fig. 9-27a). Over the course of evolution, segments of chromosomes may become inverted as a result of a segmental duplication, transposition of one copy to another arm of the same chromosome, and recombination between them (Fig. 9-27b); such inversions have occurred in the human lineage on chromosomes 1, 12, 15, 16, and 18. Two chromosomes found in other primate lineages have been fused to form human

chromosome 2 (Fig. 9-27c). The human lineage thus has 23 chromosome pairs rather than the 24 pairs typical of simians. Once this fusion appeared in the line leading to humans, it would have represented a major barrier to interbreeding with other primates that lacked it.

If we look only at base-pair changes, the published human and chimpanzee genomes differ by only 1.23% (compared with the 0.1% variance from one human to another). Some variations are at positions where there is a known polymorphism in either the human population or the chimpanzee population, and these are unlikely to reflect a species-defining evolutionary change. When we ignore these positions, the differences amount to about 1.06%, or about 1 in 100 bp. This small fraction translates into more than 30 million base-pair differences, some of which affect protein function and gene regulation. Humans are approximately as closely related to bonobos as to chimpanzees.

The genomic rearrangements that help distinguish chimpanzee and human include 5 million short insertions or deletions involving a few base pairs each, as well as a substantial number of larger insertions, deletions, inversions, and duplications that can involve many thousands of base pairs. When transposon insertions—a major source of genomic variation—are added to the list, the differences between the human and chimpanzee genomes increase. The chimpanzee genome has two classes of retrotransposons that are not present in the human genome (see Chapter 26). Other types of rearrangements, especially segmental duplications, are also common in primate

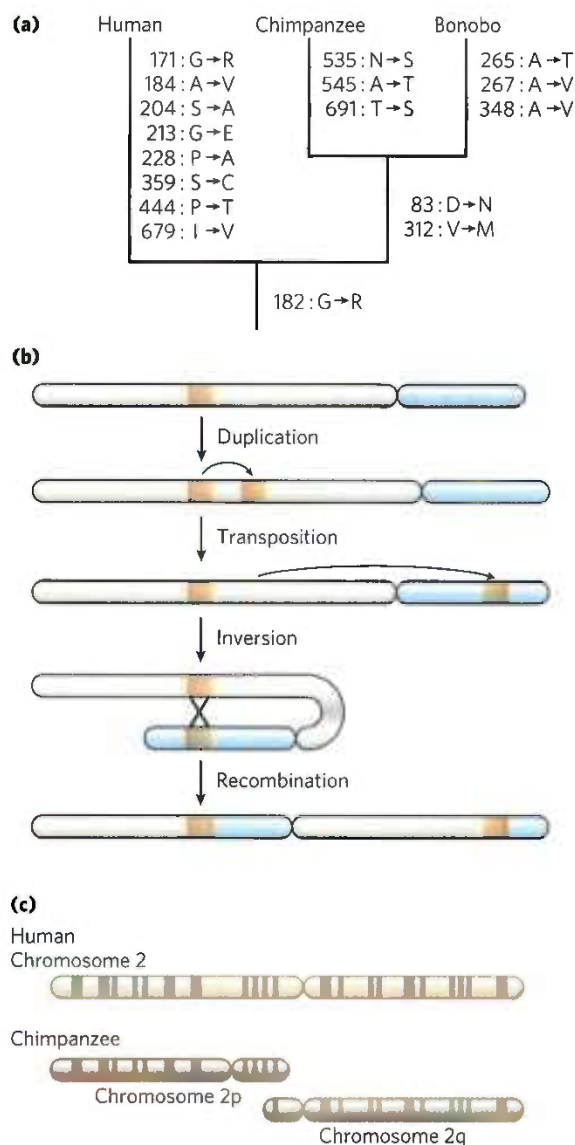


FIGURE 9-27 Genomic alterations in the human lineage.

(a) This evolutionary tree is for the progesterone receptor, which helps regulate many events in reproduction. The gene encoding this protein has undergone more evolutionary alterations than most. Amino acid changes associated uniquely with human, chimpanzee, and bonobo are listed beside each branch (with the residue number). (b) One of the multistep processes that can lead to the inversion of a chromosome segment. A gene or a chromosome segment is duplicated, then moved to another chromosomal location by transposition. Recombination of the two segments may result in inversion of the DNA between them. (c) The genes on chimpanzee chromosomes 2p and 2q are homologous to those on human chromosome 2, implying that two chromosomes fused at some point in the line leading to humans. Homologous regions can be visualized as bands created in metaphase by certain dyes, as shown here. [(a) Information from C. Chen, *Mol. Phylogenet. Evol.* 47:637, 2008.]

lineages. Duplications of chromosomal segments can lead to changes in the expression of genes contained in these segments. There are about 90 million bp of such differences between human and chimpanzee, representing another 3% of these genomes. Each species has segments of DNA, constituting 40 million to 45 million bp, that are

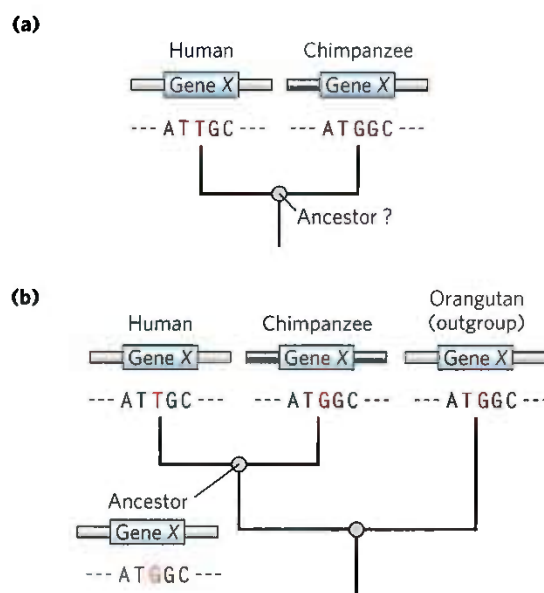


FIGURE 9-28 Determination of sequence alterations unique

to one ancestral line. (a) Sequences from the same hypothetical gene in human and chimpanzee are compared. The sequence of this gene in the two species' last common ancestor is unknown. (b) The orangutan genome is used as an outgroup. Because the sequence of the orangutan gene is identical to that of the chimpanzee gene, the mutation causing the difference between human and chimpanzee almost certainly occurred in the line leading to modern humans, and the common ancestor of human and chimpanzee (and orangutan) had the variant now found in chimpanzees.

entirely unique to that particular genome, with larger chromosomal insertions, duplications, and other rearrangements affecting more base pairs than do single-nucleotide changes. Thus, in all, chimpanzee and human differ over about 4% of their genomes.

Sorting out which genomic distinctions are relevant to features that are uniquely human is a daunting task. If one assumes a similar rate of evolution in the chimpanzee and human lines after they diverged from their common ancestor, half the changes represent chimpanzee lineage changes and half represent human lineage changes. By comparing both genome sequences with those of more distantly related species referred to as **outgroups**, we can determine which variant was present in the common ancestor. Consider a locus, X, where there is a difference between the human and chimpanzee genomes (**Fig. 9-28a**). The lineage of the orangutan, an outgroup, diverged from that of chimpanzee and human prior to the common ancestor of chimpanzee and human (**Fig. 9-28b**). If the sequence at locus X is identical in orangutan and chimpanzee, this sequence was probably present in the chimpanzee and human ancestor, and the sequence seen in humans is specific to the human lineage. Sequences that are identical in human and orangutan can be eliminated as candidates for human-specific genomic features. The importance of comparisons with closely related

outgroups has given rise to new efforts to sequence the genomes of orangutan, macaque, and many other primate species. Comparison of the human and bonobo genomes is refining the analysis of genes and alleles of special significance to humans.

The search for the genetic underpinnings of special human characteristics, such as our enhanced brain function, can benefit from two complementary approaches. The first approach searches for genomic regions where extreme changes have occurred, such as genes that have been duplicated many times or large genomic segments not present in other primates. The second approach looks at genes known to be involved in relevant human disease conditions. For brain function, for example, one would examine genes that, when mutated, contribute to cognitive or mental disorders.

Notably, analyses of the human lineage have not detected an increased rate of genetic change in protein-coding genes involved in brain development or size. In primates, most genes that function uniquely in the brain are even more highly conserved than genes functioning in other tissues, perhaps due to some special constraints related to brain biochemistry. However, there are some differences in gene expression patterns between humans and other primates that may affect brain function. For example, the gene encoding the enzyme glutamate dehydrogenase, which plays an important role in neurotransmitter synthesis, has been subjected to gene duplication events, so that there are now multiple copies of it. Genomic regions related to gene regulation have disproportionately high numbers of changes in genes involved in neural development and nutrition. Our brains have become larger as a result, and additional functional effects may eventually be defined. A variety of RNA-coding genes, some with expression concentrated in the brain, also show evidence of accelerated evolution (Fig. 9-29). Many of

these are probably involved in regulating the expression of other genes. As we continue to discover many new classes of RNA (see Chapter 26), we are likely to radically change our perspective on how evolution alters the workings of living systems.

Genome Comparisons Help Locate Genes Involved in Disease



One of the motivations for the Human Genome Project was its potential for accelerating the discovery of genes underlying genetic diseases. That promise has been fulfilled: **P1** more than 6,000 human mutation phenotypes, mostly associated with genetic diseases, have been mapped to particular genes or groups of genes.

For the last two decades, the main approach to gene mapping has been **linkage analysis**, yet another approach derived from evolutionary biology. In brief, the gene involved in a disease condition is mapped relative to well-characterized genetic polymorphisms that occur throughout the human genome. We can illustrate this approach by describing the search for one gene involved in early-onset Alzheimer disease.

About 10% of all cases of Alzheimer disease in the United States result from an inherited predisposition. Several different genes have been discovered that, when mutated, can lead to early onset of the disease. One such gene, *PS1*, encodes the protein presenilin-1, and its discovery made heavy use of linkage analysis. The search begins with large families having multiple individuals affected by a particular disease—in this case, Alzheimer disease. Two of the many family pedigrees used to search for this gene in the early 1990s are shown in Figure 9-30a. In studies of this type, DNA samples are collected from both affected and unaffected family members. Researchers first localize the region associated

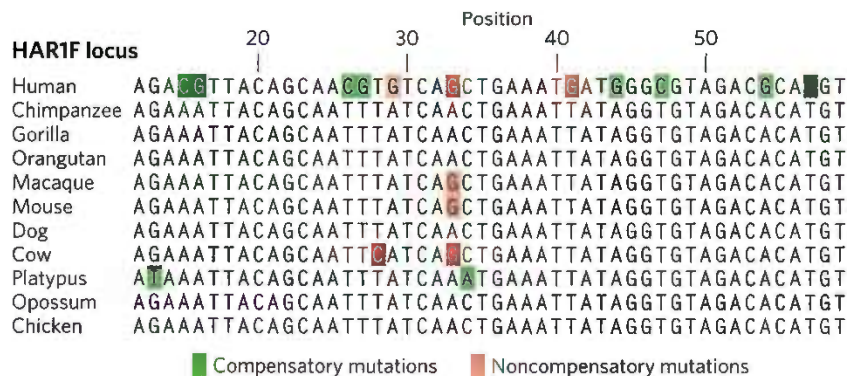


FIGURE 9-29 Accelerated evolution in some human genes. The HAR1F locus specifies a noncoding RNA that is highly conserved in vertebrates. The human HAR1F gene has an unusual number of substitutions (highlighted by color shading), providing evidence of accelerated

evolution. HAR1F RNA functions in the brain during neurodevelopment. Compensatory substitutions are those that retain complementarity where strand segments are paired. [Information from T. Marques-Bonet, *Annu. Rev. Genomics Hum. Genet.* 10:355, 2009.]

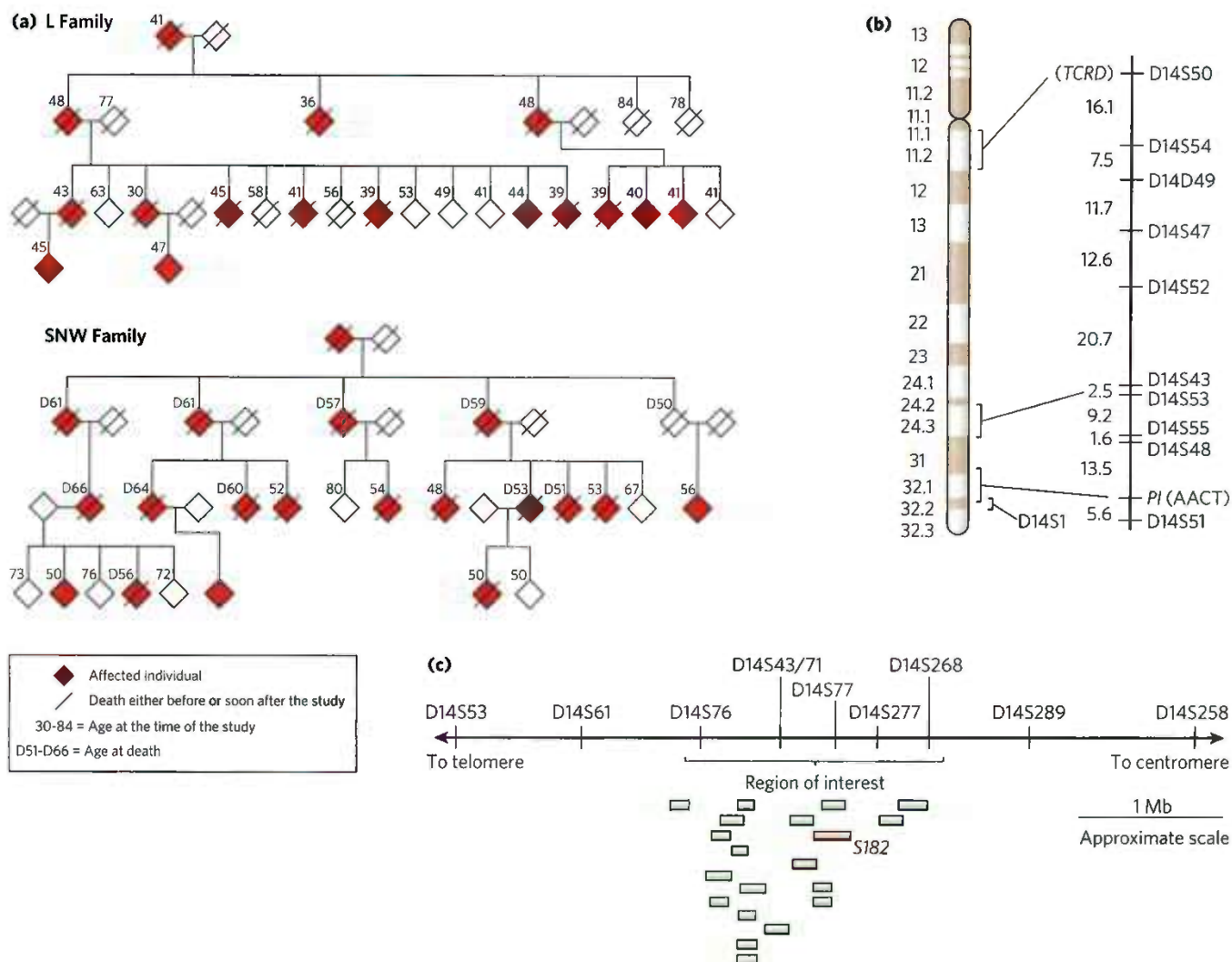


FIGURE 9-30 Linkage analysis in the discovery of disease genes.

(a) These pedigrees for two families affected by early-onset Alzheimer disease are based on the data available at the time of the study. To protect family privacy, gender is not indicated. (b) Chromosome 14, with bands created by certain dyes. Chromosome marker positions are shown at the right, with the genetic distance between them in centimorgans, a genetic distance measurement that reflects the frequency of recombination between markers. *TCRD*

(T-cell receptor delta) and *PI* (AACT (α 1-antichymotrypsin)), two genes with alterations in the human population, were used along with SNPs as markers in chromosome mapping. (c) By comparing DNA from affected and unaffected family members, researchers eventually defined a region of interest near marker D14S43 that contains 19 expressed genes. The gene labeled *S182* (red) encodes presenilin-1. (1 Mb = 10^6 base pairs.) [Information from (a, b) G. D. Schellenberg et al., *Science* 258:668, 1992; (c) R. Sherrington et al., *Nature* 375:754, 1995.]

with a disease to a specific chromosome by comparing the genotypes of individuals with and without the disease, focusing especially on close family members. The specific points of comparison are sets of well-characterized SNP loci mapped to each chromosome, as identified by the Human Genome Project. By identifying the SNPs that are most often inherited with the disease-causing gene, investigators can gradually localize the responsible gene to a single chromosome. In the case of the *PSI* gene, co-inheritance was strongest with markers on chromosome 14 (Fig. 9-30b).

Chromosomes are very large DNA molecules, and localizing the gene to one chromosome is only a small part of the battle. Localizing the gene to one chromo-

some (in this case, chromosome 14) is only the beginning of narrowing the search for a gene. Chromosomes are very large molecules; each one houses thousands of SNPs and other changes. Simply sequencing the entire chromosome would be unlikely to reveal the SNP or other change associated with the disease. Instead, investigators rely on statistical methods that correlate the inheritance of additional, more closely spaced polymorphisms with the occurrence of the disease, focusing on a denser panel of polymorphisms known to occur on the chromosome of interest. The more closely a marker is located to a disease gene, the more likely it is to be inherited along with that gene. This process can pinpoint a region of the chromosome that contains the

gene. However, the region may still encompass many genes. In our example, linkage analysis indicated that the disease-causing gene, *PS1*, was somewhere near the SNP locus D14S43 (Fig. 9-30c).

The final steps in identifying the gene use the human genome databases. The local region containing the gene is examined, and the genes within it are identified. DNA from many individuals, some who have the disease and some who do not, is sequenced over this region. As the DNA in this region is sequenced from increasing numbers of individuals, gene variants that are consistently present in individuals with the disease and absent in unaffected individuals can be identified. An understanding of the function of the genes in the target region can aid the search, because particular metabolic pathways may be more likely than others to produce the disease state. In 1995, the chromosome 14 gene associated with Alzheimer disease was identified as *S182*. The product of this gene was given the name presenilin-1, and the gene was subsequently renamed *PS1*.

Many human genetic diseases are caused by mutations in a single gene or in sequences involved in regulation of that gene. Several different mutations in a particular gene, all leading to the same or related genetic condition, may be present in the human population. For example, there are several variants of *PS1*, all giving rise to a much-increased risk of early-onset Alzheimer disease. Another, more extreme example is the several genes encoding different hemoglobins: more than 1,000 known mutational variants are present in the human population. Some of these variants are innocuous; some cause diseases ranging from sickle cell disease to thalassaemias. The inheritance of particular mutant genes may be concentrated in families or in isolated populations.

More complex are cases in which a disease condition is caused by mutations in two different genes (neither of which, alone, causes the disease), or in which a particular condition is enhanced by an otherwise innocuous mutation in another gene. Identifying the genes and mutations responsible for these digenic diseases is exceedingly difficult, and sometimes such diseases can be documented only within small, isolated, and highly inbred populations.

Genome databases provide alternative paths to the identification of disease genes. In many cases, we already have biochemical information about the disease. In the case of early-onset Alzheimer disease, an accumulation of the amyloid- β protein in limbic and association cortices of the brain is at least partly responsible for the symptoms. Defects in presenilin-1 (and in a related protein, presenilin-2, encoded by a gene on chromosome 1) lead to the elevated cortical levels of amyloid- β protein. Focused databases are being developed that catalog such functional information on the

protein products of genes and on protein-interaction networks and SNP locations, along with other data. The result is a streamlined path to the identification of candidate genes for a particular disease. If a researcher knows a little about the kinds of enzymes or other proteins likely to contribute to disease symptoms, these databases can quickly generate a list of genes known to encode proteins with relevant functions, a list of additional uncharacterized genes with orthologous or paralogous relationships to these genes, a list of proteins known to interact with the target proteins or orthologs in other organisms, and a map of gene positions. Often, with the aid of data from some selected family pedigrees, a short list of potentially relevant genes can be rapidly determined.

These approaches are not limited to human diseases. The same methods can be used to identify the genes involved in diseases—or genes that produce desirable characteristics—in other animals and in plants. Of course, they can also be used to track down genes involved in any observable trait that a researcher might be interested in. ■

Genome Sequences Inform Us about Our Past and Provide Opportunities for the Future

Anatomically modern humans arose in Africa between 250,000 and 350,000 years ago. About 100,000 to 120,000 years ago, humans in Africa looked out across the Red Sea to Asia. Perhaps encouraged by some innovation in small boat construction, or driven by conflict or famine, or simply curious, they crossed the water barrier. That initial colonization began a journey that did not stop until humans reached Tierra del Fuego (at the southern tip of South America), many thousands of years later. As *Homo sapiens* populations moved into more northern parts of Europe and Asia about 45,000 years ago, established populations from previous hominid expansions into Eurasia, including *Homo neanderthalensis* and a group now called the Denisovans, were displaced. The Neanderthals and Denisovans disappeared, just as other hominid lines had disappeared before them.

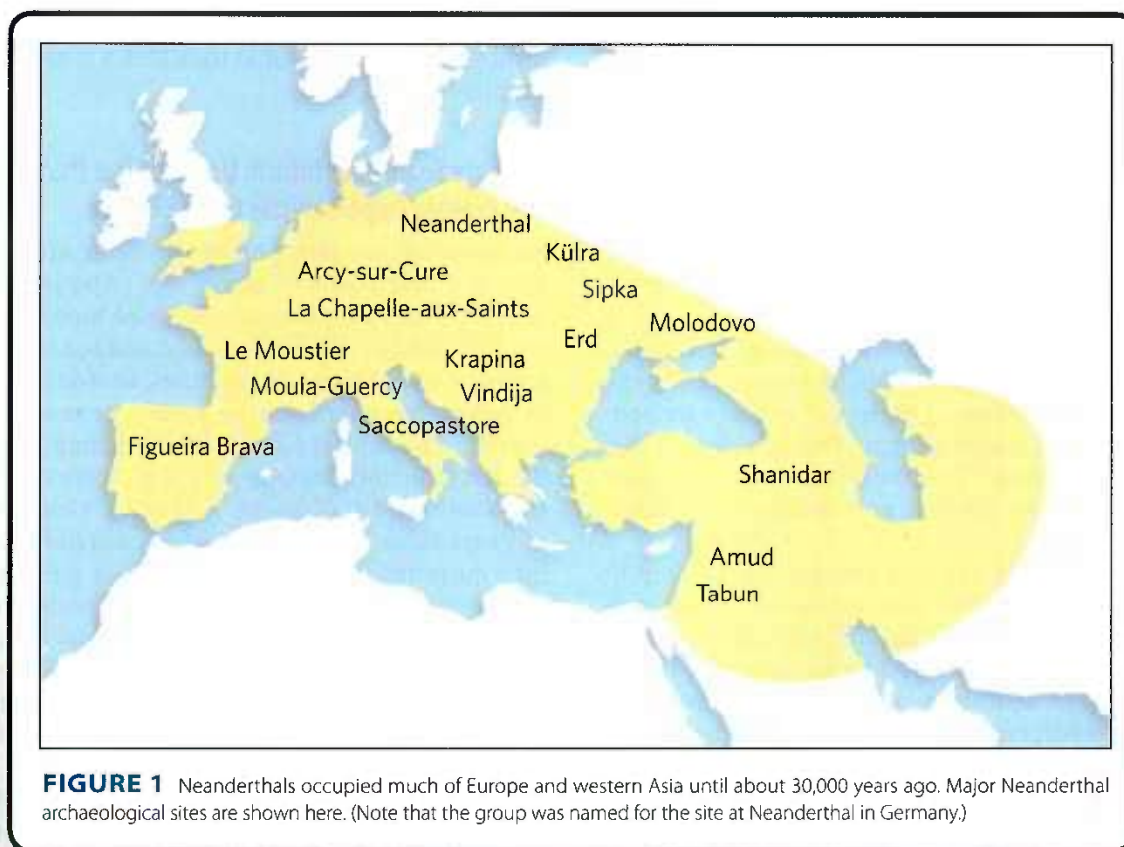
The story of how modern humans first appeared in Africa a few hundred thousand years ago, and their migrations as they eventually radiated out of Africa, is written in our DNA. Genomic sequences from multiple species have brought both primate and hominid evolution into sharper focus. Using haplotypes present in extant human populations, we can trace the migrations of our intrepid ancestors across the planet (Fig. 9-31a). The Neanderthals were not simply displaced. Some mingling occurred (Fig. 9-31b). Using sensitive PCR-based methods, we now possess multiple complete sequences of the Neanderthal genome (Box 9-2). We know that up to 5% of the genome of most

Getting to Know Humanity's Next of Kin

Modern humans and Neanderthals coexisted in Europe and Asia as recently as 30,000 years ago. The human and Neanderthal ancestral populations diverged about 370,000 years ago, somewhat before the appearance of anatomically modern humans. Neanderthals used tools, lived in small groups, and buried their dead. Of the known hominid relatives of modern humans, Neanderthals are the closest. For hundreds of millennia, they inhabited large parts of Europe and western Asia (Fig. 1). If the chimpanzee genome can tell us something about what it is to be human, the Neanderthal genome can tell us more.

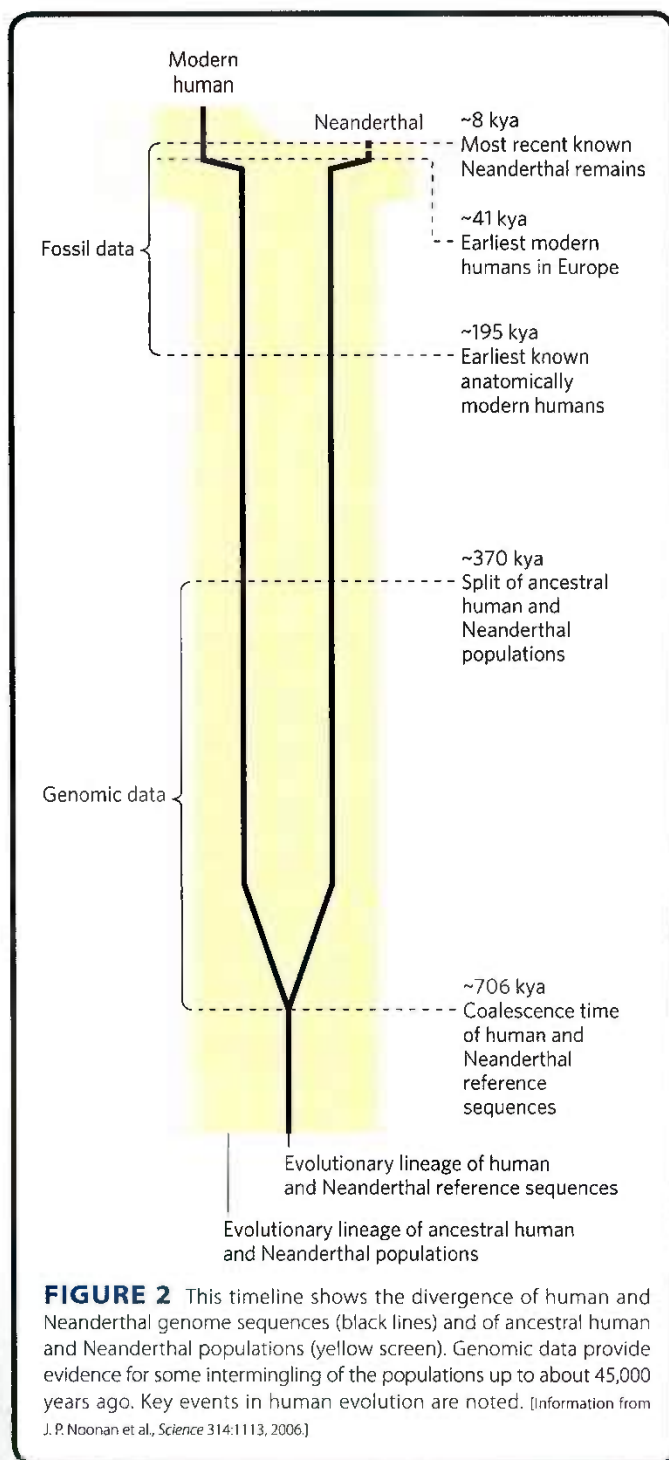
Buried in the bones and other remains taken from burial sites are fragments of Neanderthal genomic DNA. Technologies developed for use in forensic science (see Box 8-1) and studies of ancient DNA have been combined in the Neanderthal genome project.

This endeavor is unlike the genome projects aimed at extant species. The Neanderthal DNA is present in small amounts, and it is contaminated with DNA from other animals and bacteria. How does one get at it, and how can one be certain that the sequences really came from Neanderthals? The answers have been revealed by



non-African humans is derived from Neanderthals. Anatomically modern human remains up to 45,000 years old have been sequenced, beginning an effort to pinpoint the period of interbreeding. Human populations native to Melanesia and Australia acquired up to 6% of their genomic DNA from the Denisovans. Neanderthal DNA

gave humans a more complex immune system, making us more resistant to infection but also a little more susceptible to autoimmune diseases. The story of our past is gradually taking shape as more genomes, of humans alive today and those who lived in past millennia, are being assembled.



innovative applications of biotechnology. In essence, the small quantities of DNA fragments found in Neanderthal bone or other remains are cloned into a library, and the cloned DNA segments are sequenced at random, contaminants and all. The sequencing results are compared with the existing human genome and chimpanzee genome databases. Segments derived from Neanderthal DNA have sequences closely related to human DNA and chimpanzee DNA and thus are readily distinguished by computerized analysis from segments derived from bacteria or insects. Once these segments are sequenced, they can be used as probes to identify sequence fragments in ancient samples that overlap with these known fragments. The potential problem of contamination with the closely related modern human DNA can be controlled for by examining mitochondrial DNA. Human populations have readily identifiable haplotypes (distinctive sets of genomic differences; see Fig. 9-26) in their mitochondrial DNA, and analysis of Neanderthal samples has shown that Neanderthals' mitochondrial DNA has its own distinct haplotypes. The presence in the Neanderthal samples of some base-pair differences that are found in the chimpanzee database but not in the human database is more evidence that nonhuman hominid sequences are being found.

Multiple high-quality Neanderthal genomic sequences have been completed. The data provide evidence that modern humans and the Neanderthals who were the source of this DNA shared a common ancestor about 700,000 years ago (Fig. 2). Analysis of mitochondrial DNA suggests that the two groups continued on the same track, with some gene flow between them, for about 300,000 more years. The lines split with the appearance of anatomically modern humans, although evidence now exists for some intermingling of the lines somewhat later as humans spread through Eurasia.

Expanded libraries of Neanderthal DNA from different sets of remains are revealing Neanderthal genetic diversity, and may eventually tell us about Neanderthal migrations, providing a fascinating look at our hominid past.

The medical promise of personal genomic sequences grows as sequencing costs continue to decline and more genes underlying inherited diseases are defined.

P3 Knowledge of genomic sequences also provides the prospect of altering them. It is now commonplace to engineer the DNA sequences of organisms ranging from bacteria

and yeast to plants and mammals, for research and commercial purposes. Efforts to cure inherited human diseases by human gene therapy have not yet lived up to their potential, but technologies for gene delivery are constantly being improved. Few scientific disciplines will affect the future of our species more than modern genomics.

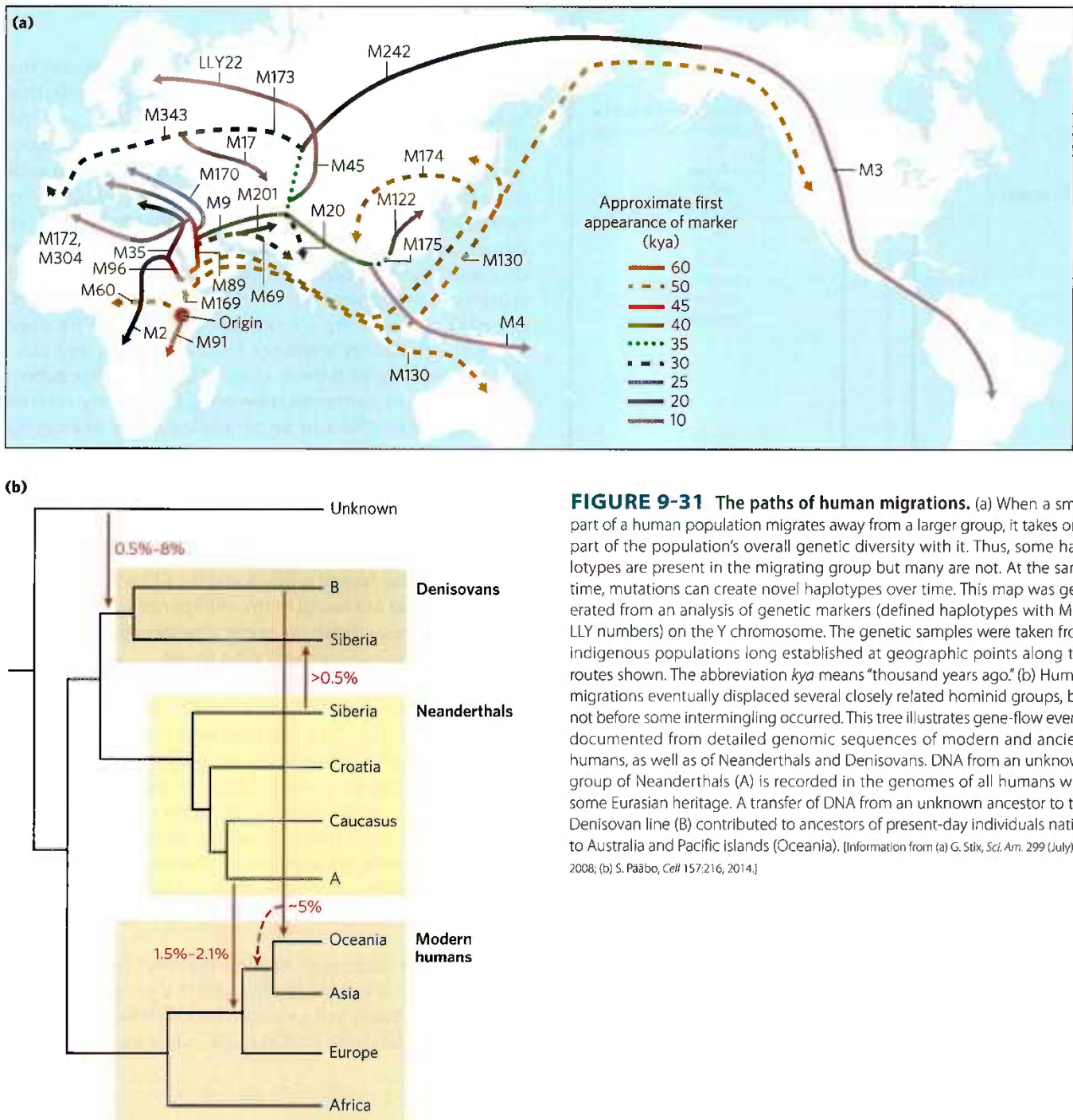


FIGURE 9-31 The paths of human migrations. (a) When a small part of a human population migrates away from a larger group, it takes only part of the population's overall genetic diversity with it. Thus, some haplotypes are present in the migrating group but many are not. At the same time, mutations can create novel haplotypes over time. This map was generated from an analysis of genetic markers (defined haplotypes with M or LLY numbers) on the Y chromosome. The genetic samples were taken from indigenous populations long established at geographic points along the routes shown. The abbreviation *kya* means "thousand years ago." (b) Human migrations eventually displaced several closely related hominid groups, but not before some intermingling occurred. This tree illustrates gene-flow events documented from detailed genomic sequences of modern and ancient humans, as well as of Neanderthals and Denisovans. DNA from an unknown group of Neanderthals (A) is recorded in the genomes of all humans with some Eurasian heritage. A transfer of DNA from an unknown ancestor to the Denisovan line (B) contributed to ancestors of present-day individuals native to Australia and Pacific Islands (Oceania). [Information from (a) G. Stix, *Sci. Am.* 299 (July):56, 2008; (b) S. Pääbo, *Cell* 157:216, 2014.]

SUMMARY 9.3 Genomics and the Human Story

■ About 30% of the DNA in the human genome is in the exons and introns of genes that encode proteins. Nearly half of the DNA is derived from parasitic transposons. Much of the rest encodes RNAs of many types. Simple-sequence repeats make up the centromere and telomeres.

■ The gene alterations that define humanity can be discerned in part through comparative genomics, using other primates.

■ Comparative genomics is also used to locate the gene alterations that define inherited diseases.

■ Human genomics can be used to study the evolution and migration of our human ancestors over many millennia.

KEY TERMS

Terms in bold are defined in the glossary.

genome 301	DNA library 315
genomics 301	complementary DNA (cDNA) 316
systems biology 301	cDNA library 316
cloning 302	transcriptome 317
vector 302	proteome 317
recombinant DNA 302	transcriptomics 317
recombinant DNA technology 302	proteomics 317
genetic engineering 302	comparative genomics 317
restriction endonucleases 302	genome annotation 317
DNA ligases 303	orthologs 317
restriction-modification system 303	paralogs 317
multiple cloning site (MCS) 304	synteny 318
plasmid 305	RNA-Seq 318
bacterial artificial chromosome (BAC) 307	single cell RNA-Seq (scRNA-Seq) 318
yeast artificial chromosome (YAC) 307	green fluorescent protein (GFP) 319
expression vector 309	epitope tag 320
baculovirus 311	yeast two-hybrid analysis 322
bacmid 311	CRISPR/Cas 322
site-directed mutagenesis 312	guide RNA (gRNA) 323
fusion protein 313	trans-activating CRISPR RNA (tracrRNA) 323
tag 313	single guide RNA (sgRNA) 323
reverse transcriptase PCR (RT-PCR) 314	single nucleotide polymorphism (SNP) 328
quantitative PCR (qPCR) 314	haplotype 328

PROBLEMS

1. Engineering Cloned DNA When joining two or more DNA fragments, a researcher can adjust the sequence at the junction in a variety of subtle ways, as seen in these exercises.

(a) Write the sequence of each end of a linear DNA fragment produced by an EcoRI restriction digest (include those sequences remaining from the EcoRI recognition sequence).

(b) Write the sequence resulting from the reaction of this end sequence with DNA polymerase I and the four deoxynucleoside triphosphates (see Fig. 8-34).

(c) Write the sequence produced at the junction that arises if two ends with the structure derived in (b) are ligated (see Fig. 25-15).

(d) Write the sequence produced if the structure derived in (a) is treated with a nuclease that degrades only single-stranded DNA.

(e) Write the sequence of the junction produced if an end with structure (b) is ligated to an end with structure (d).

(f) Write the sequence of the end of a linear DNA fragment that was produced by a PvuII restriction digest (include those sequences remaining from the PvuII recognition sequence).

(g) Write the sequence of the junction produced if an end with structure (b) is ligated to an end with structure (f).

(h) Suppose you can synthesize a short duplex DNA fragment with any sequence you desire. With this synthetic fragment and the procedures described in (a) through (g), design a protocol that would remove an EcoRI restriction site from a DNA molecule and incorporate a new BamHI restriction site at approximately the same location. (See Fig. 9-2.)

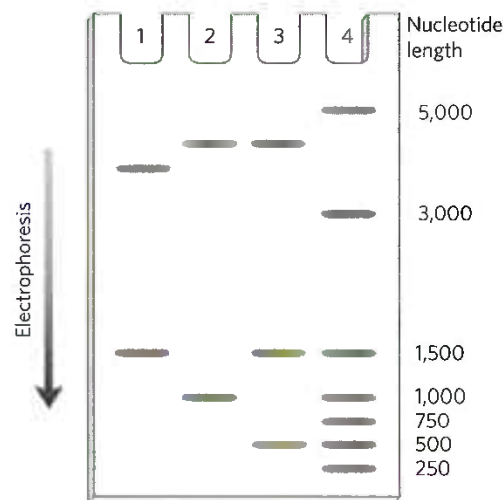
(i) Design four different short synthetic double-stranded DNA fragments that would permit ligation of structure (a) with a DNA fragment produced by a PstI restriction digest. In one of these fragments, design the sequence so that the final junction contains the recognition sequences for both EcoRI and PstI. In the second and third fragments, design the sequence so that the junction contains only the EcoRI and only the PstI recognition sequence, respectively. Design the sequence of the fourth fragment so that neither the EcoRI nor the PstI sequence appears in the junction.

2. Selecting for Recombinant Plasmids When cloning a foreign DNA fragment into a plasmid, it is often useful to insert the fragment at a site that interrupts a selectable marker (such as the tetracycline-resistance gene of pBR322). The loss of function of the interrupted gene can be used to identify clones containing recombinant plasmids with foreign DNA. With a yeast artificial chromosome (YAC) vector, a researcher can distinguish vectors that incorporate large foreign DNA fragments from those that do not, without interrupting gene function. How are these recombinant vectors identified?

3. DNA Cloning The restriction endonuclease PstI cleaves the plasmid cloning vector pBR322 (see Fig. 9-3). A researcher ligates an isolated DNA fragment from a eukaryotic genome (also produced by PstI cleavage) to the prepared vector. She then uses the mixture of ligated DNAs to transform bacteria and selects plasmid-containing bacteria by growth in the presence of tetracycline.

(a) In addition to the desired recombinant plasmid, what other types of plasmids might be found among the transformed bacteria that are tetracycline-resistant? How can the types be distinguished?

(b) The cloned DNA fragment is 1,000 bp long and has an EcoRI site 250 bp from one end. The researcher cleaves three different recombinant plasmids with EcoRI and analyzes them by gel electrophoresis, with the results shown in the image. What does each pattern say about the cloned DNA? Note that in pBR322, the PstI and EcoRI restriction sites are about 750 bp apart. The entire plasmid with no cloned insert is 4,361 bp. Size markers in lane 4 have the number of nucleotides noted.



visualization, and it binds to the first antibody. In principle, one could simply label the first antibody and skip one step. Why use two successive antibodies?

10. Yeast Two-Hybrid Analysis You are a researcher who has just discovered a new protein in a fungus. Design a yeast two-hybrid experiment to identify the other proteins in the fungal cell with which your protein interacts, and explain how this could help you determine the function of your protein.

11. RNA-Seq RNA-Seq is a next-generation sequencing method used to quantitatively profile the cellular transcriptome. Researchers use RNA-Seq to compare the expression of genes under different environmental conditions or between different types of cells. There are three general steps in an RNA-Seq workflow:

1. Generate a cDNA library from cellular RNA.
2. Add oligonucleotide adapters to the fragments of the cDNA library.
3. Use next-generation sequencing to identify transcriptionally active genes from the cDNA library.

What is the role of the enzyme reverse transcriptase in an RNA-Seq workflow?

12. Cellular RNA Suppose that an investigative team conducted an RNA-Seq experiment on mouse liver cells. The team found many sequences that contained no open reading frames (Chapter 27)—long stretches of consecutive triplet codons that could be translated into a protein and therefore suggest the presence of a gene. Suggest a reason for this observed lack of ORFs.

13. Use of Outgroups in Comparative Genomics A hypothetical protein found in human, orangutan, and chimpanzee has the following sequences (red indicates amino acid residue M differences; dashes indicate a deletion, meaning the residues are missing in that sequence):

Human:	ATSAAGYDEWEGGKVLHL – – KLQNRGALL ELDIGAV
Orangutan:	ATSAAGWDEWEGGKVLHLDGKLQNRGALL ELDIGAV
Chimpanzee:	ATSAAGWDEWEGGKILHLDGKLQNRGALL ELDIGAV

What is the most likely sequence of the protein present in the last common ancestor of human and chimpanzee?

14. Human Migrations I Native American populations in North America and South America have mitochondrial DNA haplotypes that can be traced to populations in northeast Asia. The Aleut and Eskimo populations in the far northern parts of North America possess a subset of the same haplotypes that link other Native Americans to Asia, and the Aleut and Eskimo populations also have several additional haplotypes that can be traced to Asian origins but are not found in native populations in other parts of the Americas. Provide a possible explanation.

15. Human Migrations II DNA (haplotypes) originating from the Denisovans can be found in the genomes of Indigenous Australians and Melanesian Islanders. However, the same DNA markers are not found in the genomes of people native to Africa. Explain.

16. Finding Disease Genes You are a gene hunter, trying to find the genetic basis for a rare inherited disease. Examination

of six pedigrees of families affected by the disease provides inconsistent results. For two of the families, the disease is co-inherited with markers on chromosome 7. For the other four families, the disease is co-inherited with markers on chromosome 12. Explain how this difference might have arisen.

17. RT-PCR Primer Design Investigators can use sequences of transcribed mRNA as a PCR template to produce a corresponding DNA sequence. Reverse transcriptase, an enzyme that works like DNA polymerase, amplifies the mRNA template as DNA in the first PCR cycle. After making the DNA strands from the RNA template, the investigator can carry out the remaining cycles with DNA polymerase, using standard PCR protocols. She can then compare the detected amplified sequences to the genome to analyze transcriptional activity. Thus, reverse transcriptase PCR (RT-PCR) is a powerful experimental technique used to detect RNA from living cells, which transcribe their DNA into RNA, as opposed to dead tissues, which do not. Consider the mRNA transcript shown.

5'–AUAUCGCUCCACGUAACUGAAAGAAAAGUGUGGAGCUA
GCAGUCGAGA–3'

Which DNA oligonucleotide pair could serve as a suitable primer in an RT-PCR amplification of this transcript? The oligonucleotides are written in the 5' to 3' direction.

- (a) Primer 1: GGAGACCTTGACT; Primer 2: AGTCAAGGTCTCC
- (b) Primer 1: GACTGCTAGCTCC; Primer 2: GTTACGTGGAGCG
- (c) Primer 1: GCCGCGCGCGCGC; Primer 2: CCCC GCCGCGCGC
- (d) Primer 1: CACGATTCAACGTG; Primer 2: TTCGCATTG CCGAA

DATA ANALYSIS PROBLEM

18. HincII: The First Restriction Endonuclease Discovery of the first restriction endonuclease to be of practical use was reported in two papers published in 1970. In the first paper, Smith and Wilcox described the isolation of an enzyme that cleaved double-stranded DNA. They initially demonstrated the enzyme's nuclease activity by measuring the decrease in viscosity of DNA samples treated with the enzyme.

(a) Why does treatment with a nuclease decrease the viscosity of a solution of DNA?

The authors determined whether the enzyme was an endonuclease or an exonuclease by treating ^{32}P -labeled DNA with the enzyme, then adding trichloroacetic acid (TCA). Under the conditions used in their experiment, single nucleotides would be TCA-soluble and oligonucleotides would precipitate.

(b) No TCA-soluble ^{32}P -labeled material formed upon treatment of the ^{32}P -labeled DNA with the nuclease. Based on this finding, is the enzyme an endonuclease or is it an exonuclease? Explain your reasoning.

When a polynucleotide is cleaved, the phosphate usually is not removed but remains attached to the 5' or 3' end of the resulting DNA fragment. Smith and Wilcox determined the location of the phosphate on the fragment formed by the nuclease in three steps:

1. Treat unlabeled DNA with the nuclease.
2. Treat a sample (A) of the product with γ - ^{32}P -labeled ATP and polynucleotide kinase (which can attach the γ -phosphate of ATP to a 5' OH but not to a 5' phosphate or to a 3' OH or 3' phosphate). Measure the amount of ^{32}P incorporated into the DNA.

3. Treat another sample (B) of the product of step 1 with alkaline phosphatase (which removes phosphate groups from free 5' and 3' ends), followed by polynucleotide kinase and γ - ^{32}P -labeled ATP. Measure the amount of ^{32}P incorporated into the DNA.

(c) Smith and Wilcox found that sample A had 136 counts/min of ^{32}P ; sample B had 3,740 counts/min. Did the nuclease cleavage leave the phosphate on the 5' end or the 3' end of the DNA fragments? Explain your reasoning.

(d) Treatment of bacteriophage T7 DNA with the nuclease gave approximately 40 specific fragments of various lengths. How is this result consistent with the enzyme's recognizing a specific sequence in the DNA as opposed to making random double-strand breaks?

At this point, there were two possibilities for the site-specific cleavage: the cleavage occurred either (1) at the site of recognition or (2) near the site of recognition but not within the sequence recognized. To address this issue, Kelly and Smith determined the sequence of the 5' ends of the DNA fragments generated by the nuclease, in five steps:

1. Treat phage T7 DNA with the enzyme.
2. Treat the resulting fragments with alkaline phosphatase to remove the 5' phosphates.
3. Treat the dephosphorylated fragments with polynucleotide kinase and γ - ^{32}P -labeled ATP to label the 5' ends.

4. Treat the labeled molecules with DNases to break them into a mixture of mono-, di-, and trinucleotides.
5. Determine the sequence of the labeled mono-, di-, and trinucleotides by comparing them with oligonucleotides of known sequence on thin-layer chromatography.

The labeled products were identified as follows: mononucleotides—A and G; dinucleotides—(5')ApA(3') and (5')GpA(3'); trinucleotides—(5')ApApC(3') and (5')GpApC(3').

(e) Which model of cleavage is consistent with these results? Explain your reasoning.

Kelly and Smith went on to determine the sequence of the 3' ends of the fragments. They found a mixture of (5')TpC(3') and (5')TpT(3'). They did not determine the sequence of any trinucleotides at the 3' end.

(f) Based on these data, what is the recognition sequence for the nuclease, and where in the sequence is the DNA backbone cleaved? Use Table 9-2 as a model for your answer.

References

- Kelly, T. J., and H. O. Smith. 1970.** A restriction enzyme from *Haemophilus influenzae*: II. Base sequence of the recognition site. *J. Mol. Biol.* 51:393–409.
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LIPIDS

10.1 Storage Lipids 341

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10.4 Working with Lipids 361

Biological lipids are a chemically diverse group of compounds, the common and defining feature of which is their insolubility in water. The biological functions of the lipids are as diverse as their chemistry. Fats and oils are the principal stored forms of energy in many organisms. Phospholipids and sterols are major structural elements of biological membranes. Other lipids, although present in relatively small quantities, play crucial roles as enzyme cofactors, electron carriers, light-absorbing pigments, hydrophobic anchors for proteins, “chaperones” to help membrane proteins fold, emulsifying agents in the digestive tract, hormones, and intracellular messengers. This chapter introduces representative lipids of each type, organized according to their functional roles, with emphasis on their chemical structure and physical properties. Although we follow a functional organization for our discussion, the thousands of different lipids can also be organized into eight general categories of chemical structure (listed in Table 10-2, at the end of this chapter). We discuss the energy-yielding oxidation of lipids in Chapter 17 and their synthesis in Chapter 21. Here, we will focus on four principles of cellular lipid function:

P1 Fatty acids are water-insoluble hydrocarbons used for cellular energy storage. Fatty acids are highly reduced and thus provide a rich source of stored chemical energy for cells. Storage of hydrophobic fats as triacylglycerols is also highly efficient because water is not needed to hydrate the stored fats.

P2 Membrane lipids are composed of hydrophobic tails attached to polar head groups. Cellular membranes are composed of a variety of lipids, including glycerophospholipids

and sterols. These lipids are used for structuring membranes as well as for displaying molecules on the membrane surfaces for signaling and molecular recognition.

P3 Lipids have uses in the cell beyond energy storage and construction of membranes.

Many lipids are present in the cell at smaller amounts than those making up membranes or being stored as fat. These lipids can function as cellular messengers, hormones, electron carriers, or pigments.

P4 The chemical properties of lipids are related to their structure and composition.

As in studies of other biomolecules, methods such as enzymatic, chromatographic, and mass spectrometry can all be used to identify lipids and determine their atomic structure.

10.1 Storage Lipids

P1 The fats and oils almost universally used as stored forms of energy in living organisms are derivatives of **fatty acids**. The fatty acids are hydrocarbon derivatives, at about the same low oxidation state (that is, as highly reduced) as the hydrocarbons in fossil fuels. The cellular oxidation of fatty acids (to CO_2 and H_2O), like the controlled, rapid burning of fossil fuels in internal combustion engines, is highly exergonic.

We introduce here the structures and nomenclature of the fatty acids most commonly found in living organisms. We describe two types of fatty acid-containing compounds, triacylglycerols and waxes, to illustrate the diversity of structures and physical properties in this family of compounds.

Fatty Acids Are Hydrocarbon Derivatives

Fatty acids are carboxylic acids with hydrocarbon chains ranging from 4 to 36 carbons long (C_4 to C_{36}). In some fatty acids, this chain is unbranched and fully saturated (contains no double bonds); in others, the chain contains one or more double bonds (**Table 10-1**). A few contain three-carbon rings, hydroxyl groups, or methyl-group branches.

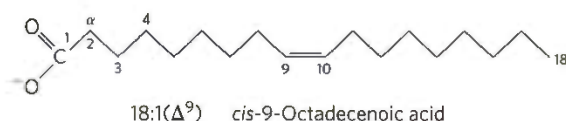
TABLE 10-1 Some Naturally Occurring Fatty Acids: Structure, Properties, and Nomenclature

Carbon skeleton	Structure ^a	Systematic name ^b	Common name (derivation)	Melting point (°C)	Solubility at 30 °C (mg/g solvent)	
					Water	Benzene
12:0	$\text{CH}_3(\text{CH}_2)_{10}\text{COOH}$	<i>n</i> -Dodecanoic acid	Lauric acid (Latin <i>laurus</i> , "laurel plant")	44.2	0.063	2,600
14:0	$\text{CH}_3(\text{CH}_2)_{12}\text{COOH}$	<i>n</i> -Tetradecanoic acid	Myristic acid (Latin <i>Myristica</i> , nutmeg genus)	53.9	0.024	874
16:0	$\text{CH}_3(\text{CH}_2)_{14}\text{COOH}$	<i>n</i> -Hexadecanoic acid	Palmitic acid (Latin <i>palma</i> "palm tree")	63.1	0.0083	348
18:0	$\text{CH}_3(\text{CH}_2)_{16}\text{COOH}$	<i>n</i> -Octadecanoic acid	Stearic acid (Greek <i>stear</i> , "hard fat")	69.6	0.0034	124
20:0	$\text{CH}_3(\text{CH}_2)_{18}\text{COOH}$	<i>n</i> -Eicosanoic acid	Arachidic acid (Latin <i>Arachis</i> , legume genus)	76.5		
24:0	$\text{CH}_3(\text{CH}_2)_{22}\text{COOH}$	<i>n</i> -Tetracosanoic acid	Lignoceric acid (Latin <i>lignum</i> , "wood" + <i>cera</i> , "wax")	86.0		
16:1(Δ^9)	$\text{CH}_3(\text{CH}_2)_5\text{CH}=\text{CH}(\text{CH}_2)_7\text{COOH}$	<i>cis</i> -9-Hexadecenoic acid	Palmitoleic acid	1 to -0.5		
18:1(Δ^9)	$\text{CH}_3(\text{CH}_2)_7\text{CH}=\text{CH}(\text{CH}_2)_7\text{COOH}$	<i>cis</i> -9-Octadecenoic acid	Oleic acid (Latin <i>oleum</i> , "oil")	13.4		
18:2($\Delta^{9,12}$)	$\text{CH}_3(\text{CH}_2)_4\text{CH}=\text{CHCH}_2\text{CH}=\text{CH}(\text{CH}_2)_7\text{COOH}$	<i>cis</i> -, <i>cis</i> -9,12-Octadecadienoic acid	Linoleic acid (Greek <i>linon</i> , "flax")	1-5		
18:3($\Delta^{9,12,15}$)	$\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_2\text{CH}=\text{CHCH}_2\text{CH}=\text{CH}(\text{CH}_2)_7\text{COOH}$	<i>cis</i> -, <i>cis</i> -, <i>cis</i> -9,12,15-Octadecatrienoic acid	α -Linolenic acid	-11		
20:4($\Delta^{5,8,11,14}$)	$\text{CH}_3(\text{CH}_2)_4\text{CH}=\text{CHCH}_2\text{CH}=\text{CHCH}_2\text{CH}=\text{CHCH}_2\text{CH}=\text{CH}(\text{CH}_2)_3\text{COOH}$	<i>cis</i> -, <i>cis</i> -, <i>cis</i> -, <i>cis</i> -5,8,11,14-Icosatetraenoic acid	Arachidonic acid	-49.5		

^aAll acids are shown in their nonionized form. At pH 7, all free fatty acids have an ionized carboxylate. Note that numbering of carbon atoms begins at the carboxyl carbon.

^bThe prefix *n*- indicates the "normal" unbranched structure. For instance, "dodecanoic" simply indicates 12 carbon atoms, which could be arranged in a variety of branched forms; "*n*-dodecanoic" specifies the linear unbranched form. For unsaturated fatty acids, the configuration of each double bond is indicated; in biological fatty acids the configuration is almost always *cis*.

KEY CONVENTION A simplified nomenclature for unbranched fatty acids specifies the chain length and number of double bonds, separated by a colon. For example, the 16-carbon saturated palmitic acid is abbreviated 16:0, and the 18-carbon oleic (octadecenoic) acid, with one double bond (shown below), is 18:1. Each line segment of the zigzag in the structure represents a single bond between adjacent carbons. The carboxyl carbon is assigned the number 1 (C-1), and the α carbon next to it is C-2. The positions of any double bonds, designated Δ (delta), are specified relative to C-1 by a superscript number indicating the lower-numbered carbon in the double bond. By this convention, oleic acid, with a double bond between C-9 and C-10, is designated 18:1(Δ^9); a 20-carbon fatty acid with one double bond between C-9 and C-10 and another between C-12 and C-13 is designated 20:2($\Delta^{9,12}$).

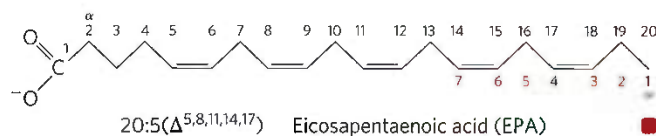


The most commonly occurring fatty acids have even numbers of carbon atoms in an unbranched chain

of 12 to 24 carbons (Table 10-1). As we shall see in Chapter 21, the even number of carbons results from the mode of synthesis of these compounds, which involves successive condensations of two-carbon acetyl units.

There is also a common pattern in the location of double bonds; in most monounsaturated fatty acids the double bond is between C-9 and C-10 (Δ^9), and the other double bonds of polyunsaturated fatty acids are generally Δ^{12} and Δ^{15} . (Arachidonic acid is an exception to this generalization; see Table 10-1.) The double bonds of polyunsaturated fatty acids are almost never conjugated (alternating single and double bonds, as in $-\text{CH}=\text{CH}-\text{CH}=\text{CH}-$), but are separated by a methylene group: $-\text{CH}=\text{CH}-\text{CH}_2-\text{CH}=\text{CH}-$. In nearly all naturally occurring unsaturated fatty acids, the double bonds are in the *cis* configuration. Notable exceptions include the *trans* fatty acids produced by fermentation in the rumen of dairy animals. Humans can ingest these *trans* fats from dairy products and meat.

KEY CONVENTION The family of **polyunsaturated fatty acids (PUFAs)** with a double bond between the third carbon and the fourth carbon from the methyl end of the chain are of special importance in human nutrition. Because the physiological role of PUFAs is related more to the position of the first double bond near the *methyl* end of the chain than to that near the carboxyl end, an alternative nomenclature is sometimes used for these fatty acids. The carbon of the methyl group—that is, the carbon most distant from the carboxyl group—is called the ω (omega; the last letter in the Greek alphabet) carbon and is given the number 1 (C-1); the carboxyl carbon in this convention has the highest number. The positions of the double bonds are indicated relative to the ω carbon. In this convention, PUFAs with a double bond between C-3 and C-4 are called **omega-3 (ω -3) fatty acids**, and those with a double bond between C-6 and C-7 are **omega-6 (ω -6) fatty acids**. Shown below is eicosapentaenoic acid, which can be designated as 20:5($\Delta^{5,8,11,14,17}$) by the standard nomenclature but is also referred to as an omega-3 fatty acid, emphasizing the biologically important double bond in the omega-3 position.



Humans require the omega-3 PUFA α -linolenic acid (ALA; 18:3($\Delta^{9,12,15}$), in the standard convention), but do not have the enzymatic capacity to synthesize it and must therefore obtain ALA from the diet. Humans also use ALA to synthesize two other omega-3 PUFAs important in cellular function: eicosapentaenoic acid (EPA; 20:5($\Delta^{5,8,11,14,17}$), shown in the Key Convention above) and docosahexaenoic acid (DHA; 22:6($\Delta^{4,7,10,13,16,19}$)). An imbalance of omega-6 and omega-3 PUFAs in the diet is associated with an increased risk of cardiovascular disease. The optimal dietary ratio of omega-6 to omega-3 PUFAs is between 1:1 and 4:1, but the ratio in the diets of most North Americans is closer to 10:1 to 30:1. The “Mediterranean diet,” which has been associated with lowered cardiovascular risk, is richer in omega-3 PUFAs, obtained in leafy vegetables (salads) and fish oils. Fish oils especially rich in EPA and DHA are a common dietary supplement even though their exact role in preventing cardiovascular disease is controversial. ■

P1 The physical properties of the fatty acids, and of compounds that contain them, are largely determined by the length and degree of unsaturation of the hydrocarbon chain. The nonpolar hydrocarbon chain accounts for the poor solubility of fatty acids in water. Lauric acid (12:0, M_r 200), for example, has a solubility in water of 0.063 mg/g—much less than that of glucose (M_r 180), which is 1,100 mg/g. The longer the fatty acyl chain and the fewer the double bonds, the lower the solubility in water. The carboxylic acid group is polar (and ionized at neutral pH) and accounts for the slight solubility of short-chain fatty acids in water.

Melting points are also strongly influenced by the length and degree of unsaturation of the hydrocarbon chain. At room temperature (25 °C), the saturated fatty acids from 12:0 to 24:0 have a waxy consistency, whereas unsaturated fatty acids of these lengths are oily liquids. This difference in melting points is due to different degrees of packing of the fatty acid molecules (Fig. 10-1). In the fully saturated compounds, free rotation around each carbon–carbon bond gives the hydrocarbon chain great flexibility; the most stable conformation is the fully extended form, in which the steric hindrance of neighboring atoms is minimized. These molecules can pack together tightly in nearly crystalline arrays, with atoms along their lengths in van der Waals contact with the atoms of neighboring molecules. In unsaturated fatty acids, a *cis* double bond forces a kink in the hydrocarbon chain. Fatty acids with one or several such kinks cannot pack together as tightly as fully saturated fatty acids, and their interactions with each other are therefore weaker. Because less thermal energy is needed to disorder these poorly ordered arrays of unsaturated fatty acids, they have markedly lower melting points than saturated fatty acids of the same chain length (Table 10-1).

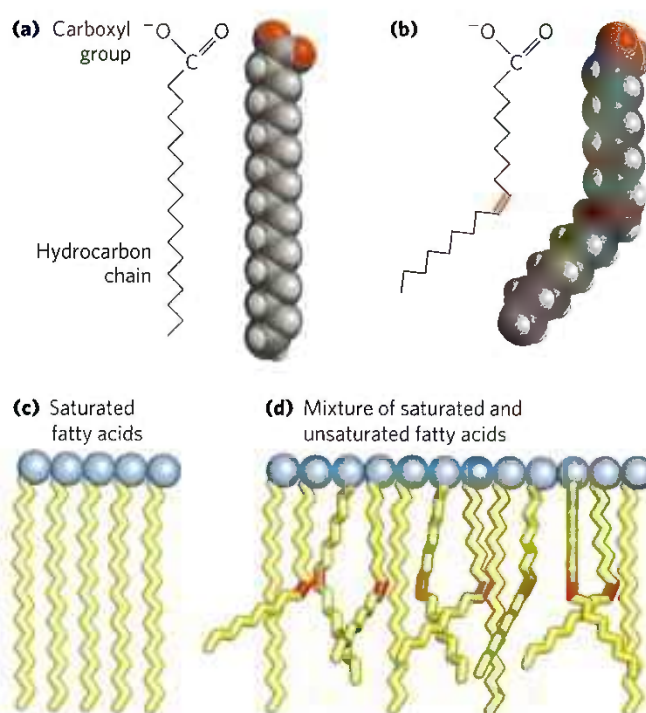


FIGURE 10-1 The packing of fatty acids into stable aggregates.

The extent of packing depends on the degree of saturation. (a) Two representations of the fully saturated acid stearic acid, 18:0 (stearate at pH 7), in its usual extended conformation. (b) The *cis* double bond (red) in oleic acid, 18:1(Δ^9) (oleate), restricts rotation and introduces a rigid bend in the hydrocarbon tail. All other bonds in the chain are free to rotate. (c) Fully saturated fatty acids in the extended form pack into nearly crystalline arrays, stabilized by extensive hydrophobic interaction. (d) The presence of one or more fatty acids with *cis* double bonds (red) interferes with this tight packing and results in less stable aggregates.

In vertebrates, free fatty acids (unesterified fatty acids, with a free carboxylate group) can circulate in the blood bound noncovalently to carrier proteins such as serum albumin. However, fatty acids are present in blood plasma mostly as carboxylic acid derivatives such as esters or amides. Lacking the charged carboxylate group, these fatty acid derivatives are generally even less soluble in water than are the free fatty acids and are transported through the blood primarily as lipoprotein particles (see Chapter 21).

Triacylglycerols Are Fatty Acid Esters of Glycerol

The simplest lipids constructed from fatty acids are the **triacylglycerols**, also referred to as triglycerides, fats, or neutral fats. Triacylglycerols are composed of three fatty acids, each in ester linkage with a single glycerol (**Fig. 10-2**). Those containing the same kind of fatty acid in all three positions are called simple triacylglycerols and are named after the fatty acid they contain. Simple triacylglycerols of 16:0, 18:0, and 18:1, for example, are tripalmitin, tristearin, and triolein, respectively. Most naturally occurring triacylglycerols are mixed; they contain two or three different fatty acids. To name these

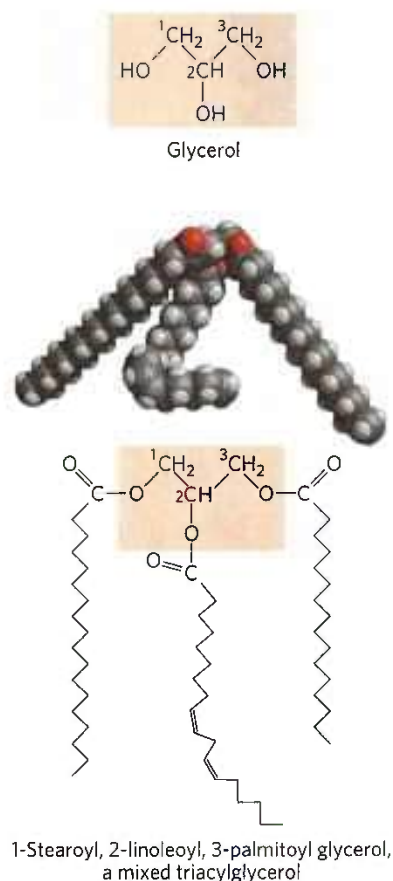


FIGURE 10-2 Glycerol and a triacylglycerol. The mixed triacylglycerol shown here has three different fatty acids attached to the glycerol backbone. When glycerol has different fatty acids at C-1 and C-3, C-2 is a chiral center.

compounds unambiguously, the name and position of each fatty acid must be specified.

Because the polar hydroxyls of glycerol and the polar carboxylates of the fatty acids are bound in ester linkages, triacylglycerols are nonpolar, hydrophobic molecules, essentially insoluble in water. Lipids have lower specific gravities than water, which explains why mixtures of oil and water (oil-and-vinegar salad dressing, for example) have two phases: oil, with the lower specific gravity, floats on the aqueous phase.

Triacylglycerols Provide Stored Energy and Insulation

In most eukaryotic cells, triacylglycerols form a separate phase of microscopic, oily droplets in the aqueous cytosol, serving as depots of metabolic fuel. In vertebrates, specialized cells called adipocytes, or fat cells, store large amounts of triacylglycerols as fat droplets that nearly fill the cell (**Fig. 10-3a**). Triacylglycerols are also stored as oils in the seeds of many types of plants, providing energy and biosynthetic precursors during seed germination

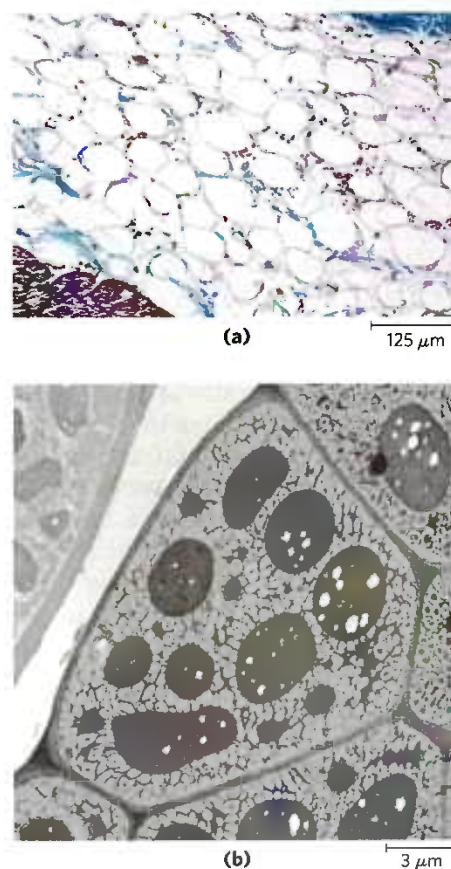


FIGURE 10-3 Fat stores in cells. (a) Cross section of human white adipose tissue. Each cell contains a fat droplet (white) so large that it squeezes the nucleus (stained red) against the plasma membrane. (b) Cross section of a cotyledon cell from a seed of the plant *Arabidopsis*. The large dark structures are protein bodies, which are surrounded by stored oils in the light-colored oil bodies. [(a) Biophoto Associates/Science Source. (b) Courtesy Howard Goodman, Department of Genetics, Harvard Medical School.]

(Fig. 10-3b). Adipocytes and germinating seeds contain **lipases**, enzymes that catalyze the hydrolysis of stored triacylglycerols, releasing fatty acids for export to sites where they are required as fuel.

P1 There are two significant advantages to using triacylglycerols as stored fuels, rather than polysaccharides such as glycogen and starch. First, the carbon atoms of fatty acids are more reduced than those of sugars, so oxidation of triacylglycerols yields more than twice as much energy, gram for gram, as the oxidation of carbohydrates. Second, because triacylglycerols are hydrophobic and therefore unhydrated, the organism that carries stored fuel in the form of fat does not have to carry the extra weight of water of hydration that is associated with stored polysaccharides (2 g per gram of polysaccharide). Humans have fat tissue (composed primarily of adipocytes) under the skin, in the abdominal cavity, and in the mammary glands. Moderately obese people with 15 to 20 kg of triacylglycerols deposited in their adipocytes could meet their energy needs for months by drawing on their fat stores. In contrast, the human body can store less than a day's energy supply in the form of glycogen. Carbohydrates such as glucose do offer certain advantages as quick sources of metabolic energy; one of those advantages is their ready solubility in water.

In some animals, triacylglycerols stored under the skin serve not only as energy stores but also as insulation against low temperatures. Seals, walruses, penguins, and other warm-blooded polar animals are amply padded with triacylglycerols. In hibernating animals (bears, for example), the huge fat reserves accumulated before hibernation serve the dual purposes of insulation and energy storage (see Box 17-1).

Partial Hydrogenation of Cooking Oils Improves Their Stability but Creates Fatty Acids with Harmful Health Effects

Most natural fats, such as those in vegetable oils, dairy products, and animal fat, are complex mixtures of simple and mixed triacylglycerols. These contain a variety of fatty acids differing in chain length and degree of saturation (Fig. 10-4). Vegetable oils such as corn (maize) oil and olive oil are composed largely of triacylglycerols with unsaturated fatty acids and thus are liquids at room temperature. Triacylglycerols containing only saturated fatty acids, such as tristearin, the major component of beef fat, are white, greasy solids at room temperature.

When lipid-rich foods are exposed too long to the oxygen in air, they may spoil and become rancid. The unpleasant taste and smell associated with rancidity result from the oxidative cleavage of double bonds in unsaturated fatty acids, which produces aldehydes and carboxylic acids of shorter chain length and therefore higher volatility; these compounds pass readily through

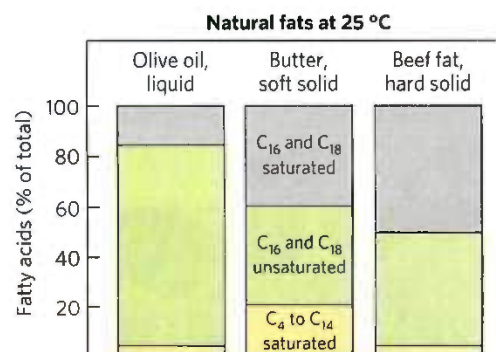
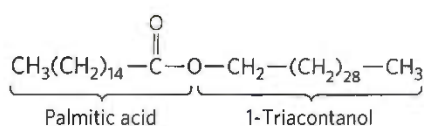


FIGURE 10-4 Fatty acid composition of three food fats. Olive oil, butter, and beef fat consist of mixtures of triacylglycerols differing in their fatty acid composition. The melting points of these fats — and hence their physical state at room temperature (25 °C) — are a direct function of their fatty acid composition. Olive oil has a high proportion of long-chain (C₁₆ and C₁₈) unsaturated fatty acids, which accounts for its liquid state at 25 °C. The higher proportion of long-chain (C₁₆ and C₁₈) saturated fatty acids in butter increases its melting point, so butter is a soft solid at room temperature. Beef fat, with an even higher proportion of long-chain saturated fatty acids, is a hard solid.

the air to your nose. Throughout the twentieth century, partial hydrogenation of commercial vegetable oils was used to improve shelf life and to increase the stability of cooking oils at the high temperatures used in deep-frying. This process converts many of the *cis* double bonds in the fatty acids to single bonds and increases the melting temperature of the oils so that they are more nearly solid at room temperature (margarine is produced from vegetable oil in this way). Partial hydrogenation, however, has another, undesirable, effect: some *cis* double bonds are converted to *trans* double bonds. There is now strong evidence that dietary intake of *trans* fatty acids (often referred to simply as “*trans* fats”) leads to a higher incidence of cardiovascular disease, and that avoiding these fats in the diet substantially reduces the risk of coronary heart disease. Dietary *trans* fatty acids raise the level of triacylglycerols and of LDL (“bad”) cholesterol in the blood, and lower the level of HDL (“good”) cholesterol, and these changes alone are enough to increase the risk of coronary heart disease. But *trans* fatty acids may have further adverse effects. They seem, for example, to increase the body's inflammatory response, which is another risk factor for heart disease. (See Chapter 21 for descriptions of LDL and HDL — low-density and high-density lipoprotein — cholesterol and their health effects.) Regulatory agencies around the world now limit or ban the use of *trans* fatty acids in prepared and packaged foods. ■

Waxes Serve as Energy Stores and Water Repellents

Biological waxes are esters of long-chain (C₁₄ to C₃₆) saturated and unsaturated fatty acids with long-chain (C₁₆ to C₃₀) alcohols (Fig. 10-5). Their melting points (60 to



(a)



(b)

FIGURE 10-5 Biological wax. (a) Triacontanoylpalmitate, the major component of beeswax, is an ester of palmitic acid with the alcohol triacontanol. (b) The beeswax of a honeycomb is firm at 25 °C and completely impervious to water. The term “wax” originates in the Old English *weax*, meaning “the material of the honeycomb.” [(b) Irochka_T/Getty Images]

100 °C) are generally higher than those of triacylglycerols. In plankton, the free-floating microorganisms at the bottom of the food chain for marine animals, waxes are the chief storage form of metabolic fuel.

Waxes also serve a diversity of other functions related to their water-repellent properties and their firm consistency. Certain skin glands of vertebrates secrete waxes to protect hair and skin and keep it pliable, lubricated, and waterproof. Birds, particularly waterfowl, secrete waxes from preen glands to keep their feathers water-repellent. The shiny leaves of holly, rhododendrons, poison ivy, and many tropical plants are coated with a thick layer of waxes, which prevents excessive evaporation of water and protects against parasites.

Biological waxes find a variety of applications in the pharmaceutical, cosmetic, and other industries. Lanolin (from lamb's wool), beeswax (Fig. 10-5), carnauba wax (from a Brazilian palm tree), and wax extracted from the seeds of the jojoba bush are widely used in the manufacture of lotions, ointments, and polishes.

SUMMARY 10.1 Storage Lipids

■ Lipids are water-insoluble cellular components of diverse structure that can be extracted from tissues by nonpolar solvents. Almost all fatty acids, the hydrocarbon components of many lipids, have an even number of

carbon atoms (usually 12 to 24); they are either saturated or unsaturated, with double bonds almost always in the *cis* configuration.

■ Triacylglycerols contain three fatty acid molecules esterified to the three hydroxyl groups of glycerol. Simple triacylglycerols contain only one type of fatty acid; mixed triacylglycerols contain two or three types.

■ Triacylglycerols are present in many foods and are primarily used for storing energy and providing insulation in animals. Lipases release fatty acids from storage so they can be used as fuel.

■ Since many natural fats can easily turn rancid, partial hydrogenation has been used to extend their shelf life. This process can produce *trans* fatty acids, which increase risk for coronary heart disease. As a result, their use in prepared and processed foods has become highly regulated.

■ Waxes are esters of long-chain fatty acids and long-chain alcohols with diverse uses in biology and industry.

10.2 Structural Lipids in Membranes

The central architectural feature of biological membranes is a double layer of lipids, which acts as a barrier to the passage of polar molecules and ions. **P2** Membrane lipids are amphipathic: one end of the molecule is hydrophobic, the other hydrophilic. The association of their hydrophobic regions with each other and their hydrophilic interactions with water direct their packing into sheets called membrane bilayers. In this section, we describe four general types of membrane lipids: **phospholipids**, which have hydrophobic regions composed of two fatty acids joined to glycerol or sphingosine; **glycolipids**, which contain a simple sugar or a complex oligosaccharide at the polar ends; **archaeal tetraether lipids**, which have two very long alkyl chains ether-linked to glycerol at both ends; and **sterols**, compounds characterized by a rigid system of four fused hydrocarbon rings (Fig. 10-6). Within these groups of membrane lipids, enormous diversity results from various combinations of fatty acid “tails” and polar “heads.” The arrangement of these lipids in membranes, and their structural and functional roles therein, are considered in the next chapter.

Glycerophospholipids Are Derivatives of Phosphatidic Acid

P2 **Glycerophospholipids**, also called phosphoglycerides, are membrane lipids in which two fatty acids are attached in ester linkage to the first and second carbons of glycerol, and a highly polar or charged group is attached through a phosphodiester linkage to the third carbon. Glycerol is prochiral; it has no asymmetric carbons, but attachment of phosphate at one

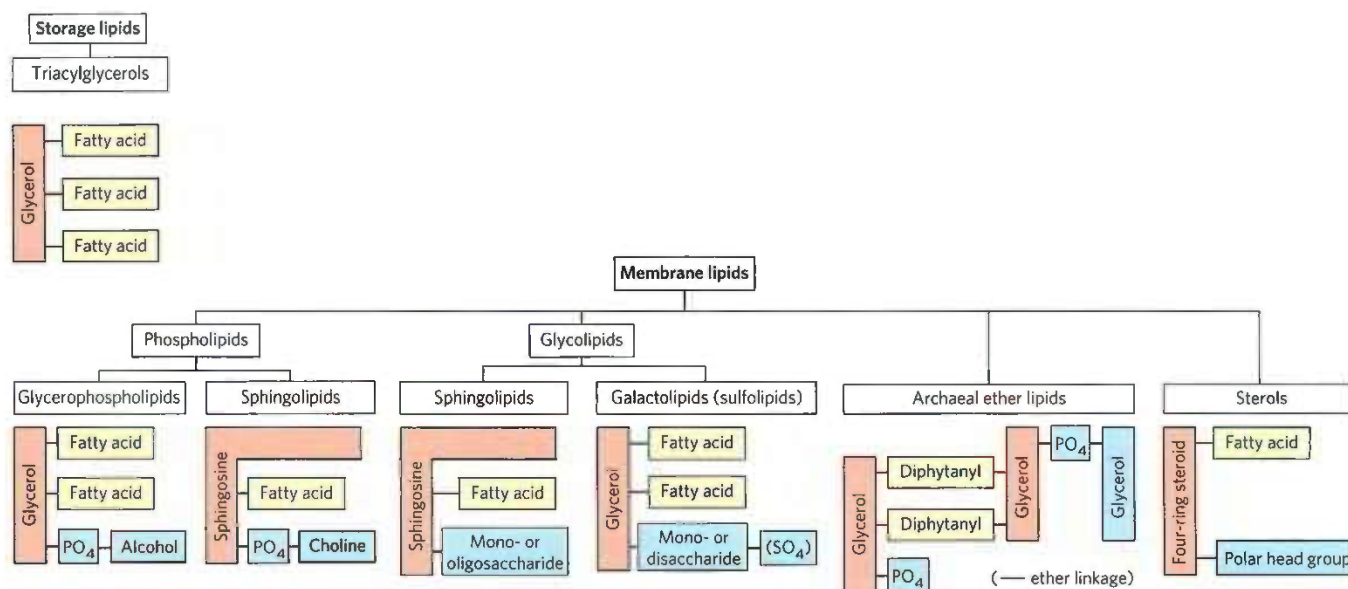


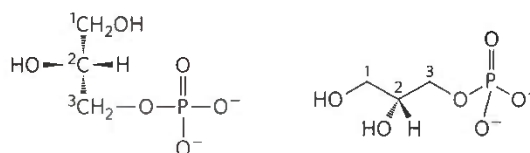
FIGURE 10-6 Some common types of storage and membrane lipids. The triacylglycerol, phospholipid, glycolipid, and archaeal ether lipid types shown here have either glycerol or sphingosine as the backbone (light red screen), to which are attached one or more long-chain alkyl groups (yellow) and a polar head group (blue). Sterols have a core composed of four fused carbocyclic rings called the steroid nucleus. In triacylglycerols, glycerophospholipids, galactolipids, and sulfolipids, the alkyl groups are fatty acids in ester linkage. Sphingolipids contain a single fatty acid, in amide linkage to

the sphingosine backbone. The membrane lipids of archaea are variable; that shown here has two very long, branched alkyl chains, each end in ether linkage with a glycerol moiety. Sterols have many different types of alkyl chains that can be attached to the steroid D ring at C-17. In phospholipids, the polar head group is joined through a phosphodiester, whereas glycolipids have a direct glycosidic linkage between the head-group sugar and the backbone glycerol. Many sterols contain a polar head group such as a carbonyl or hydroxyl group attached at C-3 of the steroid A ring.

end converts it into a chiral compound, which can be correctly named either L-glycerol 3-phosphate, D-glycerol 1-phosphate, or *sn*-glycerol 3-phosphate (Fig. 10-7). Glycerophospholipids are named as derivatives of the parent compound, phosphatidic acid (Fig. 10-8), according to the polar alcohol in the head group. Phosphatidylcholine and phosphatidylethanolamine have choline and ethanolamine as their polar head groups, for example. Cardiolipin is a two-tailed glycerophospholipid in which two phosphatidic acid moieties share the same glycerol as their head group (Fig. 10-8). Cardiolipin is found in most bacterial membranes; in eukaryotic cells, cardiolipin is located almost exclusively in the inner mitochondrial membrane (where it is synthesized), a location consistent with the endosymbiosis hypothesis for the origin of these organelles (see Fig. 1-37).

In all glycerophospholipids, the head group is joined to glycerol through a phosphodiester bond, in which the phosphate group bears a negative charge at neutral pH. The polar alcohol may be negatively charged (as in phosphatidylinositol 4,5-bisphosphate), neutral (phosphatidylserine), or positively charged (phosphatidylcholine, phosphatidylethanolamine). As we shall see in Chapter 11, these charges contribute greatly to the surface properties of membranes.

The fatty acids in glycerophospholipids can be any of a wide variety, so a given phospholipid (phosphatidylcholine, for example) may consist of several molecular



L-Glycerol 3-phosphate (*sn*-glycerol 3-phosphate)

FIGURE 10-7 L-Glycerol 3-phosphate, the backbone of phospholipids. Glycerol itself is not chiral, as it has a plane of symmetry through C-2. However, glycerol is prochiral—it can be converted to a chiral compound by adding a substituent such as phosphate to either of the $-\text{CH}_2\text{OH}$ groups. One unambiguous nomenclature for glycerol phosphate is the D, L system (described on p. 73), in which the isomers are named according to their stereochemical relationships to glyceraldehyde isomers. By this system, the stereoisomer of glycerol phosphate found in most lipids is correctly named either L-glycerol 3-phosphate or D-glycerol 1-phosphate. Another way to specify stereoisomers is the *sn* (stereospecific numbering) system, in which C-1 is, by definition, the group of the prochiral compound that occupies the pro-S position. The common form of glycerol phosphate in phospholipids is, by this system, *sn*-glycerol 3-phosphate (in which C-2 has the R configuration). In archaea, the glycerol in lipids has the other configuration; it is D-glycerol 3-phosphate.

species, each with its unique complement of fatty acids. The distribution of molecular species is specific to the organism, to the particular tissue within the organism, and to the particular glycerophospholipids in the same cell or tissue. In general, glycerophospholipids contain a C_{16} or C_{18} saturated fatty acid at C-1 and a C_{18} or C_{20} unsaturated fatty acid at C-2. With few exceptions, the

Saturated fatty acid (e.g., palmitic acid) Unsaturated fatty acid (e.g., linoleic acid) Fatty acids Glycerol Head-group substituent			
Name of glycerophospholipid	Head group	Formula of —X	Net charge (at pH 7)
Phosphatidic acid	—	—H	—2*
Phosphatidylethanolamine	Ethanolamine		0
Phosphatidylcholine	Choline		0
Phosphatidylserine	Serine		—1
Phosphatidylglycerol	Glycerol		—1
Phosphatidylinositol 4,5-bisphosphate	<i>myo</i> -Inositol 4,5-bisphosphate		—4*
Cardiolipin	Phosphatidyl-glycerol		—2

FIGURE 10-8 Glycerophospholipids. The common glycerophospholipids are diacylglycerols linked to head-group alcohols through a phosphodiester bond. Phosphatidic acid, a phosphomonoester, is the parent compound. Each derivative is named for the head-group alcohol, with the

prefix "phosphatidyl-." In cardiolipin, two phosphatidic acids share a single glycerol (R^1 and R^2 are fatty acyl groups), resulting in a symmetric molecule. *Note that phosphate esters each have a charge of about -1.5 ; one of their —OH groups is only partially ionized at pH 7.

biological significance of the variation in fatty acids and head groups is not yet understood.

Some Glycerophospholipids Have Ether-Linked Fatty Acids

Some animal tissues and some unicellular organisms are rich in **ether lipids**, in which one of the two acyl chains is attached to glycerol in ether, rather than ester, linkage. The ether-linked chain may be saturated, as in the alkyl

ether lipids, or it may contain a double bond between C-1 and C-2, as in **plasmalogens** (Fig. 10-9). Vertebrate heart tissue is uniquely enriched in ether lipids; about half of the heart phospholipids are plasmalogens. The membranes of halophilic bacteria, ciliated protists, and certain invertebrates also contain high proportions of ether lipids. The functional significance of ether lipids in these membranes is unknown; perhaps their resistance to the phospholipases that cleave ester-linked fatty acids from membrane lipids is important in some roles.

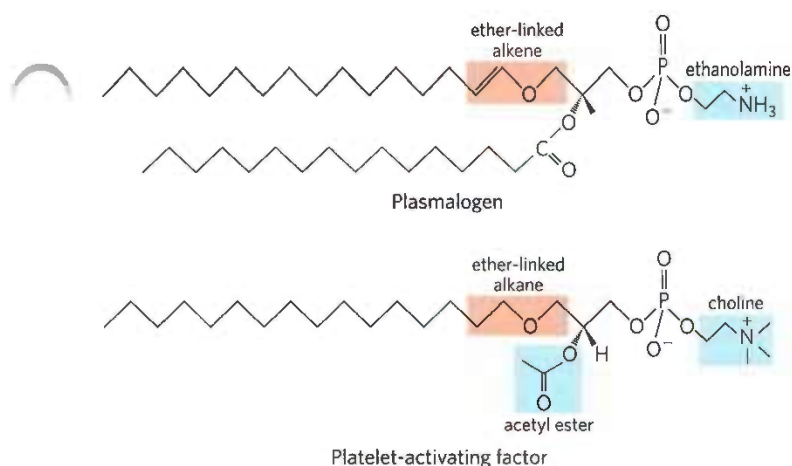


FIGURE 10-9 Ether lipids. Plasmalogens have an ether-linked alkenyl chain where most glycerophospholipids have an ester-linked fatty acid (compare to Fig. 10-8). Platelet-activating factor has a long ether-linked alkyl chain at C-1 of glycerol, but C-2 is ester-linked to acetic acid, which makes the compound much more water-soluble than most glycerophospholipids and plasmalogens. The head-group alcohol is ethanolamine in plasmalogens and choline in platelet-activating factor.



At least one ether lipid, **platelet-activating factor**, is a potent molecular signal. It is released from leukocytes called basophils and stimulates platelet aggregation and the release of serotonin (a vasoconstrictor) from platelets. It also exerts a variety of effects on liver, smooth muscle, heart, uterine, and lung tissues and plays an important role in inflammation and the allergic response. ■

Galactolipids of Plants and Ether-Linked Lipids of Archaea Are Environmental Adaptations

The second group of membrane lipids in Figure 10-6, the **glycolipids**, includes those that predominate in plant cells: the **galactolipids**, in which one or two galactose residues are connected by a glycosidic linkage to C-3 of a 1,2-diacylglycerol (Fig. 10-10). Galactolipids are localized in the thylakoid membranes (internal membranes) of chloroplasts; they make up 70% to 80% of the total membrane lipids of a vascular plant, and are therefore probably the most abundant membrane lipids in the biosphere.

Phosphate is often the limiting plant nutrient in soil, and perhaps the evolutionary pressure to conserve phosphate for more critical roles favored plants that made phosphate-free lipids. Plant membranes also contain sulfolipids, in which a sulfonated glucose residue is joined to a diacylglycerol in glycosidic linkage. The sulfonate group bears a negative charge like that of the phosphate group in phospholipids.

Some archaea that live in ecological niches with extreme conditions—for example, high temperatures (boiling water), low pH, or high ionic strength—have membrane lipids containing long-chain (32 carbons) branched hydrocarbons linked at each end to glycerol (Fig. 10-6). These linkages are through ether bonds, which are much more stable to hydrolysis at low pH and high temperature than are the ester bonds found in the lipids of bacteria and eukaryotes. In their fully extended form, these archaeal lipids are twice the length of phospholipids and sphingolipids, and can span the full width of the plasma membrane.

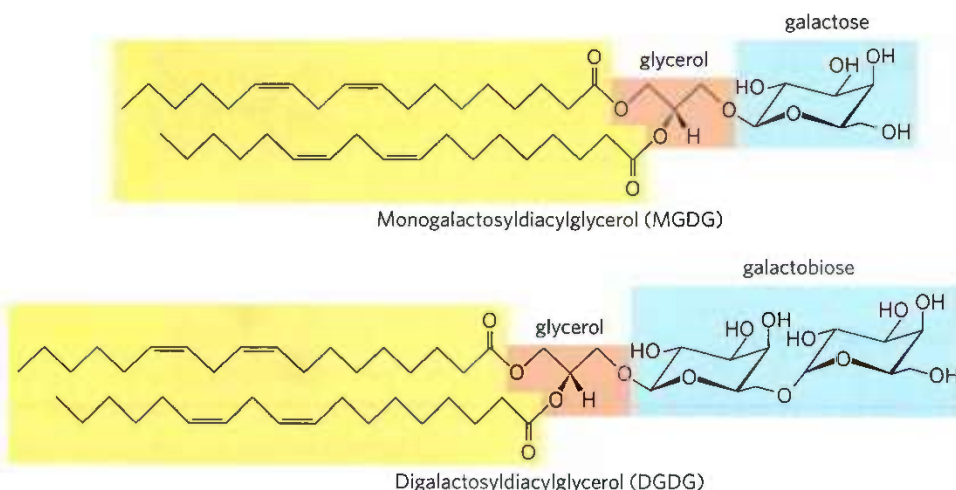


FIGURE 10-10 Two galactolipids of chloroplast thylakoid membranes. In monogalactosyldiacylglycerols (MGDGs) and digalactosyldiacylglycerols (DGDGs), both acyl groups are polyunsaturated and the head groups are uncharged.

Sphingolipids Are Derivatives of Sphingosine

P2 Sphingolipids, a large class of membrane phospholipids and glycolipids, have a polar head group and two non-polar tails, but unlike glycerophospholipids and galactolipids they contain no glycerol (Fig. 10-6). Sphingolipids are composed of one molecule of the long-chain amino alcohol sphingosine (also called 4-sphingenine) or one of its derivatives, one molecule of a long-chain fatty acid, and a polar head group that is joined by a glycosidic linkage in some cases and a phosphodiester in others (Fig. 10-11).

Carbons C-1, C-2, and C-3 of the sphingosine molecule are structurally analogous to the three carbons of glycerol in glycerophospholipids. When a fatty acid is attached in amide linkage to the —NH_2 on C-2, the resulting compound is a **ceramide**, which is structurally similar to a diacylglycerol. Ceramides are the structural parents of all sphingolipids.

There are three subclasses of sphingolipids, all derivatives of ceramide but differing in their head

groups: sphingomyelins, neutral (uncharged) glycolipids, and gangliosides. **Sphingomyelins** contain phosphocholine or phosphoethanolamine as their polar head group and are therefore classified along with glycerophospholipids as phospholipids (Fig. 10-6). Indeed, sphingomyelins resemble phosphatidylcholines in their general properties and three-dimensional structure, and in having no net charge on their head groups (Fig. 10-12). Sphingomyelins are present in the plasma membranes of animal cells and are especially prominent in myelin, a membranous sheath that surrounds and insulates the axons of some neurons—thus the name “sphingomyelins.”

Glycosphingolipids, which occur largely in the outer face of plasma membranes, have head groups with one or more sugars connected directly to the —OH at C-1 of the ceramide moiety; they do not contain phosphate. **Cerebrosides** have a single sugar linked to ceramide; those with galactose are characteristically found in the plasma membranes of cells in neural tissue, and those

Sphingosine		
Name of sphingolipid	Head group	Formula of —X
Ceramide	—	—H
Sphingomyelin	Phosphocholine	$\begin{array}{c} \text{O} \\ \parallel \\ \text{—P—O—CH}_2\text{—CH}_2\text{—N}^+(\text{CH}_3)_3 \\ \\ \text{O}^- \end{array}$
Neutral glycolipids Glucosylceramide	Glucose	
Lactosylceramide (a globoside)	Di-, tri-, or tetrasaccharide	
Ganglioside GM2	Complex oligosaccharide	

FIGURE 10-11 Sphingolipids. The first three carbons at the polar end of sphingosine are analogous to the three carbons of glycerol in glycerophospholipids. The amino group at C-2 bears a fatty acid in amide linkage. The fatty acid is usually saturated or monounsaturated, with 16, 18, 22,

or 24 carbon atoms. Ceramide is the parent compound for this group. Other sphingolipids differ in the polar head group attached at C-1. Gangliosides have very complex oligosaccharide head groups. Standard symbols for sugars are used in this figure, as shown in Table 7-1.

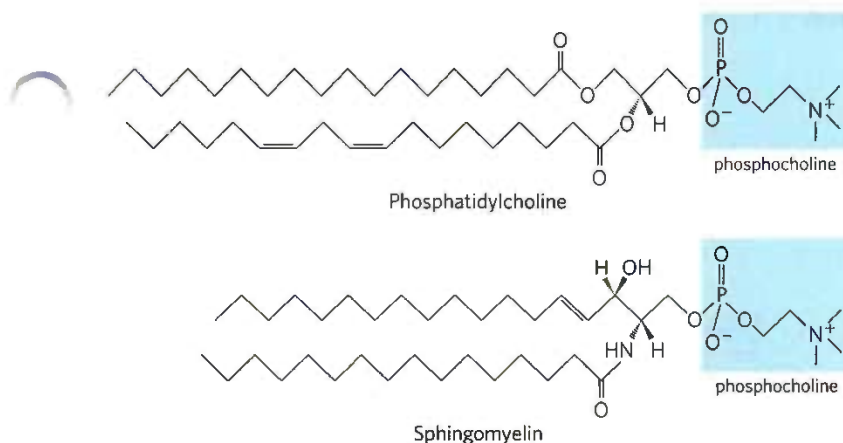
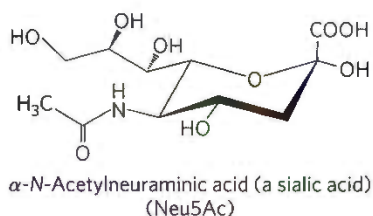


FIGURE 10-12 The similar molecular structures of two classes of membrane lipid. Phosphatidylcholine (a glycerophospholipid) and sphingomyelin (a sphingolipid) have similar dimensions and physical properties, but presumably play different roles in membranes.

with glucose are found in the plasma membranes of cells in nonneural tissues. **Globosides** are glycosphingolipids with two or more sugars, usually D-glucose, D-galactose, or N-acetyl-D-galactosamine. Cerebrosides and globosides are sometimes called **neutral glycolipids**, as they have no charge at pH 7.

Gangliosides, the most complex sphingolipids, have oligosaccharides as their polar head groups and one or more residues of N-acetylneuraminic acid (Neu5Ac), a sialic acid (often simply called “sialic acid”), at the termini. Deprotonated sialic acid gives gangliosides the negative charge at pH 7 that distinguishes them from globosides. Gangliosides with one sialic acid residue are in the GM (*M* for mono-) series, those with two sialic acid residues are in the GD (*D* for di-) series, and so on (GT, three sialic acid residues; GQ, four).



Sphingolipids at Cell Surfaces Are Sites of Biological Recognition

When sphingolipids were discovered more than a century ago by the physician-chemist Johann Thudichum, their biological role seemed as enigmatic as the Sphinx, for which he therefore named them. In humans, at least 60 different sphingolipids have been identified in cellular membranes. **P2** Many of these are especially prominent in the plasma membranes of neurons, and some are clearly recognition sites on the cell surface, but a specific function for only a few sphingolipids has been discovered thus far. The carbohydrate moieties of certain sphingolipids define the human blood groups and therefore determine the type of blood that individuals can safely receive in blood transfusions (Fig. 10-13).

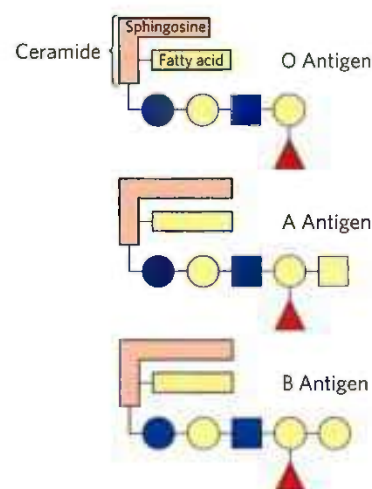


FIGURE 10-13 Glycosphingolipids as determinants of blood groups. The human blood groups (O, A, B) are determined in part by the oligosaccharide head groups of these glycosphingolipids. The same three oligosaccharides are also found attached to certain blood proteins of individuals of blood types O, A, and B, respectively. Standard symbols for sugars are used here (see Table 7-1).



Gangliosides are concentrated on the outer face of cell surface plasma membranes where they present points of recognition for extracellular molecules or the surfaces of neighboring cells. The kinds and amounts of gangliosides in the plasma membrane change dramatically during embryonic development. Tumor formation induces the synthesis of a new complement of gangliosides, and very low concentrations of a specific ganglioside have been found to induce differentiation of cultured neuronal tumor cells. Guillain-Barré syndrome is a serious autoimmune disorder in which the body makes antibodies against its own gangliosides, including those in neurons. The resulting inflammation damages the peripheral nervous system, leading to temporary (or sometimes permanent) paralysis. In cholera, cholera toxin produced by the intestinal bacterium *Vibrio cholerae* enters sensitive cells after attaching to specific gangliosides on the intestinal epithelial cell surface. Investigation of the biological roles of diverse gangliosides remains fertile ground for future research. ■

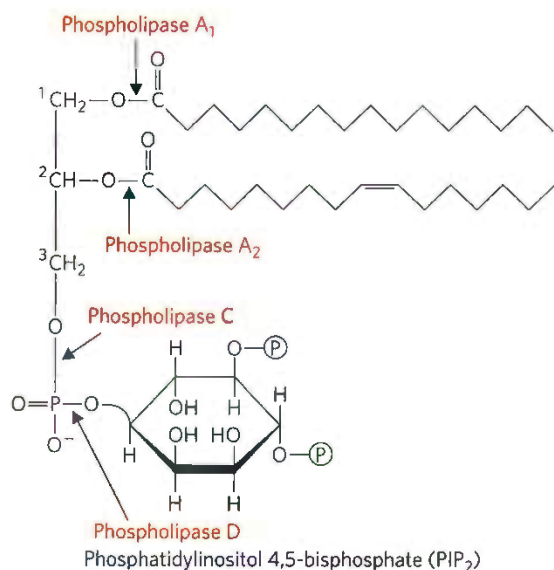


FIGURE 10-14 The specificities of phospholipases. Phospholipases A₁ and A₂ hydrolyze the ester bonds of intact glycerophospholipids at C-1 and C-2 of glycerol, respectively. When one of the fatty acids has been removed by a type A phospholipase, the second fatty acid is removed by a lysophospholipase (not shown). Phospholipases C and D each split one of the phosphodiester bonds in the head group. Some phospholipases act on only one type of glycerophospholipid, such as phosphatidylinositol 4,5-bisphosphate (PIP₂, shown here) or phosphatidylcholine; others are less specific.

Phospholipids and Sphingolipids Are Degraded in Lysosomes

Most cells continually degrade and replace their membrane lipids. For each hydrolyzable bond in a glycerophospholipid, there is a specific hydrolytic enzyme in the lysosome (Fig. 10-14). Phospholipases of the A type remove one of the two fatty acids, producing a lysophospholipid. (These esterases do not attack the ether link of plasmalogens.) Lysophospholipases remove the remaining fatty acid.

Gangliosides are degraded by a set of lysosomal enzymes that catalyze the stepwise removal of sugar units, finally yielding a ceramide. A genetic defect in any of these hydrolytic enzymes leads to the accumulation of gangliosides in the cell, with severe medical consequences (Box 10-1).

Sterols Have Four Fused Carbon Rings

P2 Sterols are structural lipids present in the membranes of most eukaryotic cells. The characteristic structure of this group of membrane lipids is the steroid nucleus, consisting of four fused rings, three with six carbons and one with five (Fig. 10-15). The steroid nucleus is almost planar and is relatively rigid; the fused rings do not allow rotation about C—C bonds. **Cholesterol**, the major sterol in animal tissues, is amphipathic, with a polar head group (the hydroxyl group at C-3) and a nonpolar hydrocarbon body (the steroid nucleus and the hydrocarbon side chain at C-17) about as long as a 16-carbon fatty acid in its extended form. Similar sterols are found

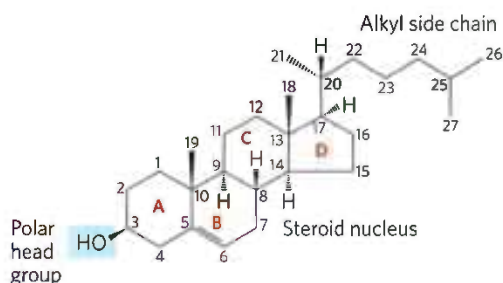
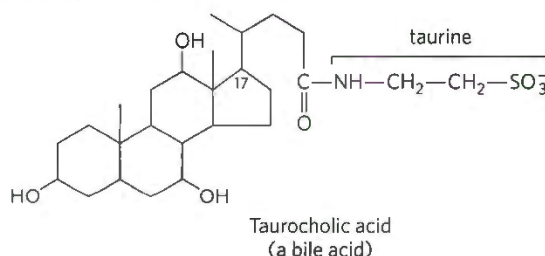


FIGURE 10-15 Cholesterol. In this chemical structure of cholesterol, the rings are labeled A through D to simplify reference to derivatives of the steroid nucleus. The C-3 hydroxyl group (shaded blue) is the polar head group. For storage and transport of the sterol, this hydroxyl group condenses with a fatty acid to form a sterol ester.

in other eukaryotes: stigmasterol in plants and ergosterol in fungi, for example. Bacteria cannot synthesize sterols; a few bacterial species, however, can incorporate exogenous sterols into their membranes. The sterols of all eukaryotes are synthesized from simple five-carbon isoprene subunits, as are the fat-soluble vitamins, quinones, and dolichols described in Section 10.3.

P3 In addition to their roles as membrane constituents, the sterols serve as precursors for a variety of products with specific biological activities. Steroid hormones, for example, are potent biological signals that regulate gene expression. **Bile acids** are polar derivatives of cholesterol that act as detergents in the intestine, emulsifying dietary fats to make them more readily accessible to digestive lipases.



We return to cholesterol and other sterols in later chapters, to consider the structural role of cholesterol in biological membranes (Chapter 11), signaling by steroid hormones (Chapter 12), and the remarkable biosynthetic pathway to cholesterol and transport of cholesterol by lipoprotein carriers (Chapter 21).

SUMMARY 10.2 Structural Lipids in Membranes

■ Lipids with polar heads and nonpolar tails are major components of membranes. The most abundant are the glycerophospholipids, which contain fatty acids esterified to two of the hydroxyl groups of glycerol. The third hydroxyl of glycerol is esterified with the polar head group via a phosphodiester bond. Common glycerophospholipids such as phosphatidylethanolamine and phosphatidylcholine differ in the structure of their head group.



Abnormal Accumulations of Membrane Lipids: Some Inherited Human Diseases

The polar lipids of membranes undergo constant metabolic turnover, the rate of their synthesis normally counterbalanced by the rate of breakdown. The breakdown of lipids is promoted by hydrolytic enzymes in lysosomes, with each enzyme capable of hydrolyzing a specific bond. When sphingolipid degradation is impaired by a defect in one of these enzymes (Fig. 1), partial breakdown products accumulate in the tissues, causing serious disease. More than 50 distinct lysosomal storage diseases have been discovered, each the result of a single mutation in one of the genes for a lysosomal protein.

For example, Niemann-Pick disease is caused by a rare genetic defect in the enzyme sphingomyelinase, the enzyme that cleaves phosphocholine from sphingomyelin. Sphingomyelin accumulates in the brain, spleen, and liver. The disease

becomes evident in infants and causes intellectual disability and early death. More common is Tay-Sachs disease, in which ganglioside GM2 accumulates in the brain and spleen (Fig. 2) owing to lack of the enzyme hexosaminidase A. The symptoms of Tay-Sachs disease are progressive developmental delay and disability, paralysis, blindness, and death by the age of 3 or 4 years.

Genetic counseling can predict and avert many inheritable diseases. Tests on prospective parents can detect abnormal enzymes; then DNA testing can determine the exact nature of the defect and the risk it poses for offspring. Once a pregnancy occurs, fetal cells obtained by sampling a part of the placenta (chorionic villus sampling) or the fluid surrounding the fetus (amniocentesis) can be tested in the same way.

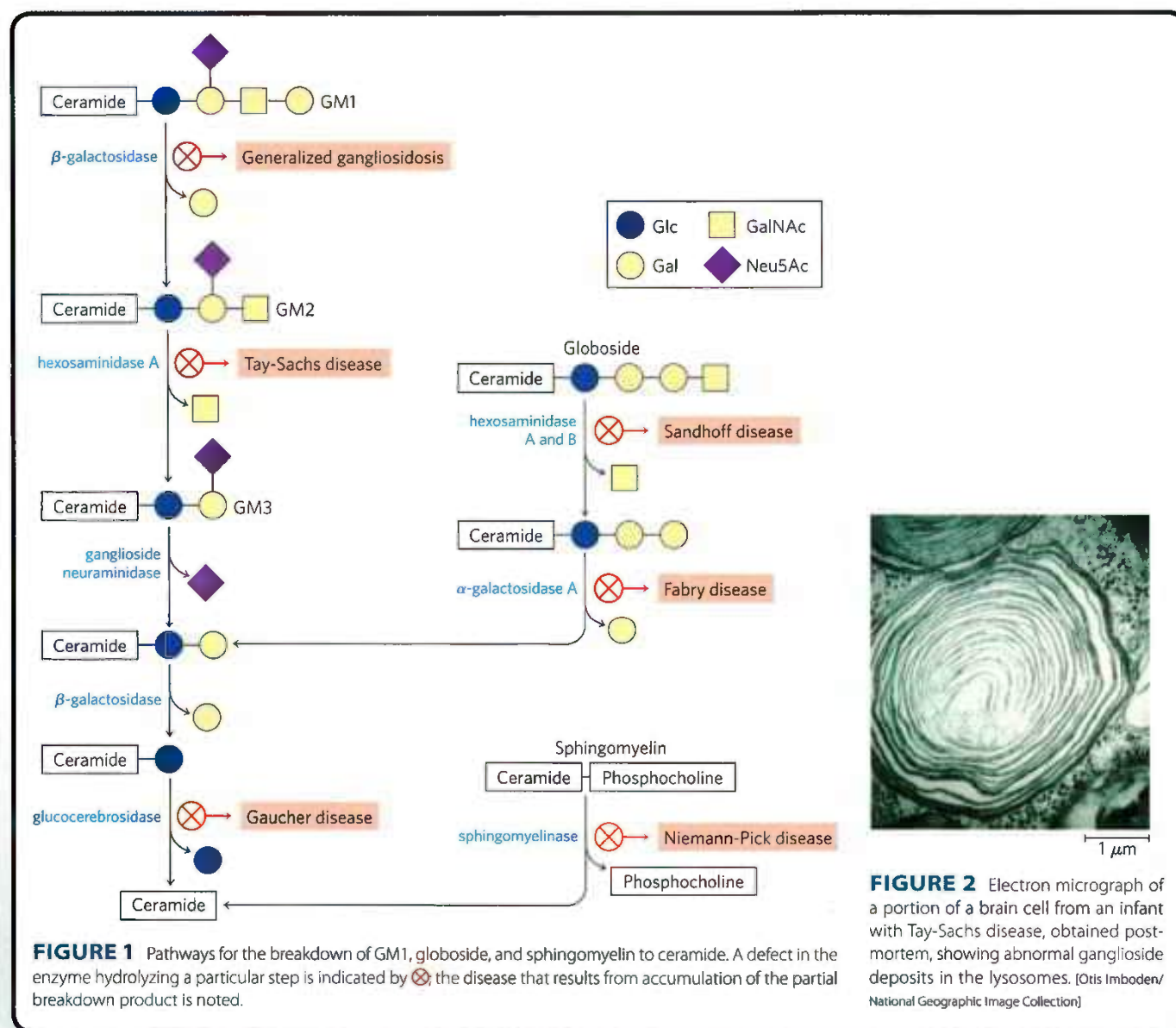


FIGURE 1 Pathways for the breakdown of GM1, globoside, and sphingomyelin to ceramide. A defect in the enzyme hydrolyzing a particular step is indicated by $\otimes$; the disease that results from accumulation of the partial breakdown product is noted.

FIGURE 2 Electron micrograph of a portion of a brain cell from an infant with Tay-Sachs disease, obtained post-mortem, showing abnormal ganglioside deposits in the lysosomes. [Otis Imboden/National Geographic Image Collection]

- Heart tissue plasmalogens and platelet-activating factor are examples of glycerophospholipids containing ether-linked acyl chains.
- Chloroplast membranes are rich in galactolipids, composed of a diacylglycerol with one or two linked galactose residues, and sulfolipids, diacylglycerols with a linked sulfonated sugar residue. Some archaea have unique membrane lipids, with long-chain alkyl groups ether-linked to glycerol at each end. These lipids are stable under the harsh conditions in which these archaea live.
- The sphingolipids contain sphingosine instead of glycerol. The three subclasses of sphingolipids are all ceramide derivatives: sphingomyelins, neutral glycolipids, and gangliosides. Sphingomyelins have choline head groups; the other classes have sugar components.
- Sphingolipids are abundant in the plasma membranes of neurons, and they define human blood groups.
- Phospholipases degrade glycerophospholipids and catalyze hydrolysis at specific positions within the structure.
- Sterols have four fused rings and a hydroxyl group. Cholesterol, the major sterol in animals, is both a structural component of membranes and precursor to a wide variety of steroids.

10.3 Lipids as Signals, Cofactors, and Pigments

The two functional classes of lipids considered thus far (storage lipids and structural lipids) are major cellular components; membrane lipids make up 5% to 10% of the dry mass of most cells, and storage lipids make up more than 80% of the mass of an adipocyte. With some important exceptions, these lipids play a *passive* role in the cell; lipid fuels are stored until oxidized by enzymes, and membrane lipids form impermeable barriers around cells and cellular compartments. **P3** Another group of lipids, present in much smaller amounts, includes those with *active* roles in metabolic traffic as metabolites and messengers. Some serve as potent signals—as hormones, carried in the blood from one tissue to another, or as intracellular messengers generated in response to an extracellular signal (hormone or growth factor). Others function as enzyme cofactors in electron-transfer reactions in chloroplasts and mitochondria, or in the transfer of sugar moieties in a variety of glycosylation reactions. A third group consists of lipids with a system of conjugated double bonds: pigment molecules that absorb visible light. Some of these act as light-capturing pigments in vision and photosynthesis; others produce natural colorations, such as the orange of pumpkins and carrots and the yellow of canary feathers. Finally, a very large group of volatile lipids produced in plants consists of signaling molecules that pass through the air, allowing plants to communicate with each other and to invite

animal friends and deter foes. In this section, we describe a few representatives of these biologically active lipids. In later chapters, we consider their synthesis and biological roles in more detail.

Phosphatidylinositols and Sphingosine Derivatives Act as Intracellular Signals

P3 Phosphatidylinositol (PI) and its phosphorylated derivatives (Fig. 10-16) act at several levels to regulate cell structure and metabolism. Phosphatidylinositol 4,5-bisphosphate (PIP₂; Fig. 10-14) in the cytoplasmic (inner) face of plasma membranes serves as a reservoir of messenger molecules that are released inside the cell in response to extracellular signals interacting with specific surface receptors. Extracellular signals such as the hormone vasopressin activate a specific phospholipase C in the membrane, which hydrolyzes PIP₂ to release two products that act as intracellular messengers: inositol 1,4,5-trisphosphate (IP₃), which is water soluble, and diacylglycerol, which remains associated with the plasma membrane. IP₃ triggers release of Ca²⁺ from the endoplasmic reticulum, and the combination of diacylglycerol and elevated cytosolic Ca²⁺ activates the enzyme protein kinase C. By phosphorylating specific proteins, this enzyme brings about the cell's response to the extracellular signal. This signaling mechanism is described more fully in Chapter 12 (see Fig. 12-15).

P3 Inositol phospholipids also serve as points of nucleation for supramolecular complexes involved in signaling or in exocytosis. Certain signaling proteins bind specifically to phosphatidylinositol 3,4,5-trisphosphate

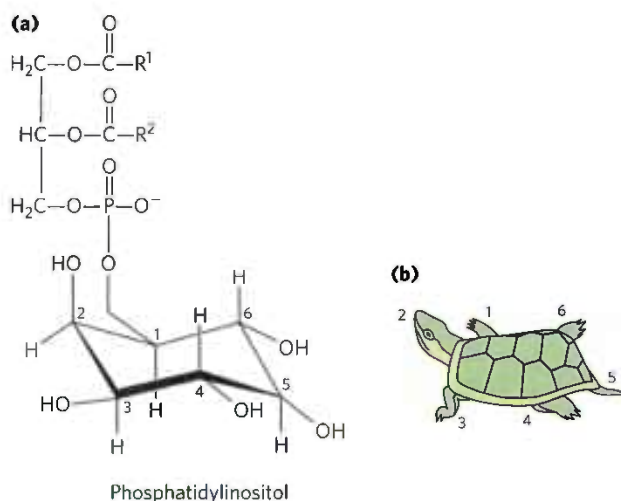


FIGURE 10-16 Phosphatidylinositol and its derivatives. (a) In phosphatidylinositol, the glycerol phospholipid is attached at C-1 of inositol. Phosphorylation of the remaining inositol hydroxyl groups forms signaling molecules such as phosphatidylinositol 4,5-bisphosphate (PIP₂; see Fig. 10-14), inositol 1,4,5-bisphosphate (IP₃), and phosphatidylinositol 3,4,5-trisphosphate (PIP₃). (b) A useful trick for remembering the inositol carbon-numbering scheme is to compare the chair conformation of inositol to a turtle. Begin numbering with C-1 as the right front flipper and move counterclockwise around the structure, with C-2 the head and C-5 the tail.

(PIP₃) in the plasma membrane, initiating the formation of multienzyme complexes at the membrane's cytosolic surface. Thus, formation of PIP₃ in response to extracellular signals brings the proteins together in signaling complexes at the surface of the plasma membrane (see Figs. 12-11 and 12-23).

Membrane sphingolipids also can serve as sources of intracellular messengers. Both ceramide and sphingomyelin (Fig. 10-11) are potent regulators of protein kinases, and ceramide or its derivatives are involved in the regulation of cell division, differentiation, migration, and programmed cell death (also called apoptosis; see Chapter 12).

Eicosanoids Carry Messages to Nearby Cells

P3 Eicosanoids are paracrine hormones, substances that act only on cells near the point of hormone synthesis instead of being transported in the blood to act on cells in other tissues or organs. These fatty acid derivatives have a variety of dramatic effects on vertebrate tissues. They are involved in reproductive function; in the inflammation, fever, and pain associated with injury or disease; in the formation of blood clots and the regulation of blood pressure; in gastric acid secretion; and in various other processes important in human health or disease.

Eicosanoids are derived from arachidonate (arachidonic acid; 20:4($\Delta^{5,8,11,14}$)) and eicosapentaenoic acid (EPA; 20:5($\Delta^{5,8,11,14,17}$)), from which they take their general name (Greek *eikosi*, “twenty”). There are four major classes of eicosanoids: prostaglandins, thromboxanes, leukotrienes, and lipoxins (Fig. 10-17). Eicosanoid names include letter designations for the functional groups on the ring and numbers indicating the number of double bonds in the hydrocarbon chain.

Prostaglandins (PG) contain a five-carbon ring, and their name derives from the prostate gland from which they were first isolated. PGE₂ and other series 2 prostaglandins are synthesized from arachidonate; series 3 prostaglandins are derived from EPA (see Fig. 21-12). Prostaglandins have an array of functions. Some stimulate contraction of the smooth muscle of the uterus during menstruation and labor. Others affect blood flow to specific organs, the wake-sleep cycle, and the responsiveness of certain tissues to hormones such as epinephrine and glucagon. Prostaglandins in a third group elevate body temperature (producing fever) and cause inflammation and pain.

The **thromboxanes (TX)** have a six-membered ring containing an ether. They are produced by platelets (also called thrombocytes) and act in the formation of blood clots and reduction of blood flow to the site of a clot. Nonsteroidal anti-inflammatory drugs (NSAIDs)—aspirin, ibuprofen, and meclofenamate, for example—inhibit the enzyme cyclooxygenase or COX (also called prostaglandin H₂ synthase), which catalyzes an early step in the pathway from arachidonate to series 2 prostaglandins and thromboxanes (Fig. 10-17) and from EPA to series 3 prostaglandins and thromboxanes (see Fig. 21-12).

Leukotrienes (LT), first found in leukocytes, contain three conjugated double bonds. They are powerful biological signals. For example, leukotriene D₄, derived from leukotriene A₄, induces contraction of the smooth muscle lining the airways to the lung. Overproduction of leukotrienes causes asthmatic attacks, and leukotriene synthesis is one target of antiasthmatic drugs such as prednisone. The strong contraction of the smooth muscle of the lungs that occurs during anaphylactic shock is part of the potentially fatal allergic reaction in individuals hypersensitive to bee stings, penicillin, or other agents.

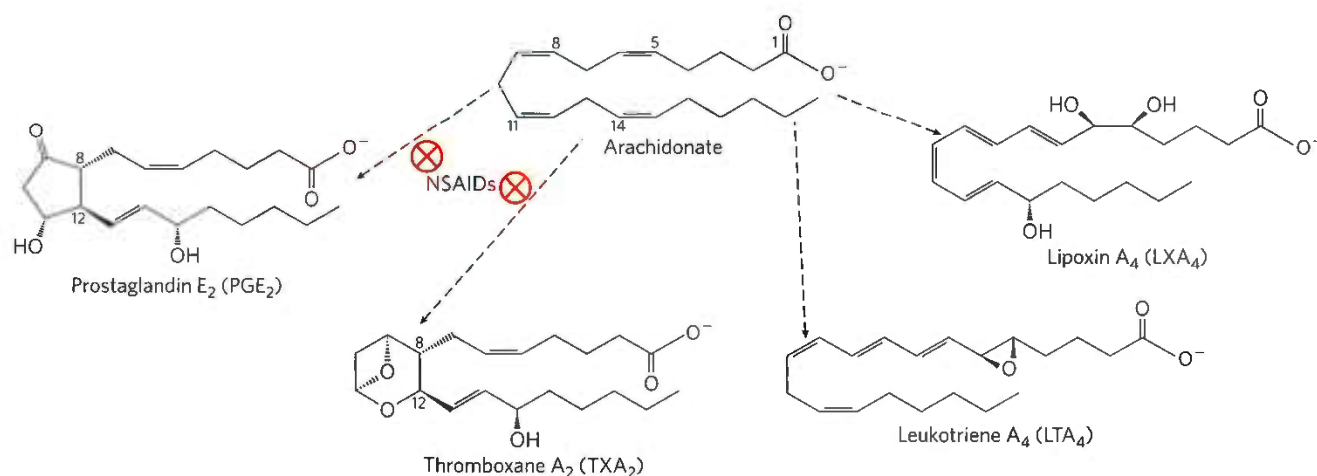


FIGURE 10-17 Arachidonic acid and some eicosanoid derivatives. Arachidonic acid (arachidonate at pH 7) is the precursor of eicosanoids, including the prostaglandins, thromboxanes, leukotrienes, and lipoxins. In prostaglandin E₂, C-8 and C-12 of arachidonate are joined to form the characteristic five-membered ring. In thromboxane A₂, the C-8 and C-12 are joined and an oxygen atom is added to form the six-membered ring.

Nonsteroidal anti-inflammatory drugs (NSAIDs) such as aspirin and ibuprofen block the formation of prostaglandins and thromboxanes from arachidonate by inhibiting the enzyme cyclooxygenase (prostaglandin H₂ synthase). Leukotriene A₄ has a series of three conjugated double bonds, and no cyclic moiety. Lipoxins are also noncyclic derivatives of arachidonate, with several hydroxyl groups.

Lipoxins (LX), like leukotrienes, are linear eicosanoids. Their distinguishing feature is the presence of several hydroxyl groups along the chain (Fig. 10-17). These compounds are potent anti-inflammatory agents. Because their synthesis is stimulated by low doses (81 mg) of aspirin taken daily, this low dose is commonly prescribed for individuals with cardiovascular disease. ■

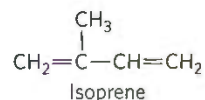
Steroid Hormones Carry Messages between Tissues

 Steroids are oxidized derivatives of sterols; they have the sterol nucleus but lack the alkyl chain attached to ring D of cholesterol, and they are more polar than cholesterol. **P3** Steroid hormones move through the bloodstream (on protein carriers) from their site of production to target tissues, where they enter cells, bind to highly specific receptor proteins in the nucleus, and trigger changes in gene expression and thus metabolism. Because hormones have very high affinity for their receptors, very low concentrations of hormones (nanomolar or less) are sufficient to produce responses in target tissues. The major groups of steroid hormones are the male and female sex hormones and the hormones produced by the adrenal cortex, cortisol and aldosterone (Fig. 10-18). Prednisone is a steroid drug with strong anti-inflammatory activity, mediated in part by the inhibition of arachidonate release by phospholipase A₂ and consequent inhibition of the synthesis of prostaglandins, thromboxanes, leukotrienes, and lipoxins. Prednisone and similar drugs have a variety of medical applications, including the treatment of asthma and rheumatoid arthritis. ■

Vascular plants contain the steroidlike brassinolide (Fig. 10-18), a potent growth regulator that increases the rate of stem elongation and affects the orientation of cellulose microfibrils in the cell wall during growth.

Vascular Plants Produce Thousands of Volatile Signals

P3 Plants produce thousands of different lipophilic compounds, volatile substances that are used to attract pollinators, repel herbivores, attract organisms that defend the plant against herbivores, and communicate with other plants. Jasmonate, for example, derived from the fatty acid 18:3($\Delta^{9,12,15}$) in membrane lipids, triggers the plant's defenses in response to insect-inflicted damage. The methyl ester of jasmonate gives the characteristic fragrance of jasmine oil, which is widely used in the perfume industry. Many plant volatiles, including geraniol (the characteristic scent of geraniums), β -pinene (pine trees), limonene (limes), and menthol, are derived from fatty acids or from compounds made by the condensation of five-carbon isoprene units.



Vitamins A and D Are Hormone Precursors

 During early decades of the twentieth century, a major focus of research in physiological chemistry was the identification of **vitamins**, compounds that are essential to the health of humans and other vertebrates

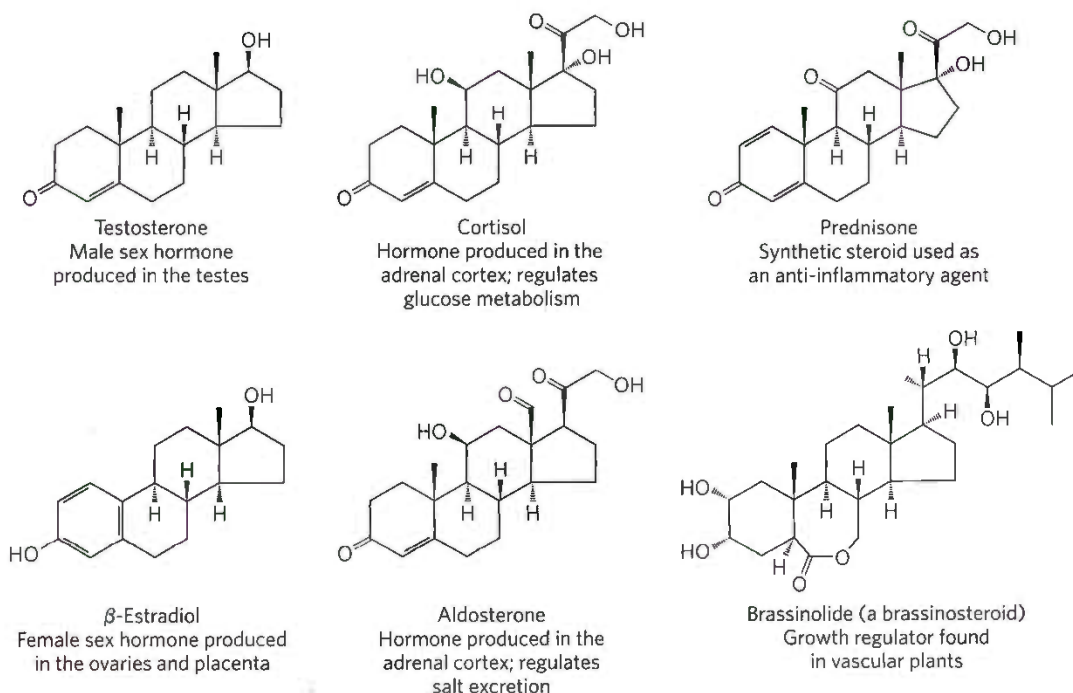


FIGURE 10-18 Steroids derived from cholesterol.

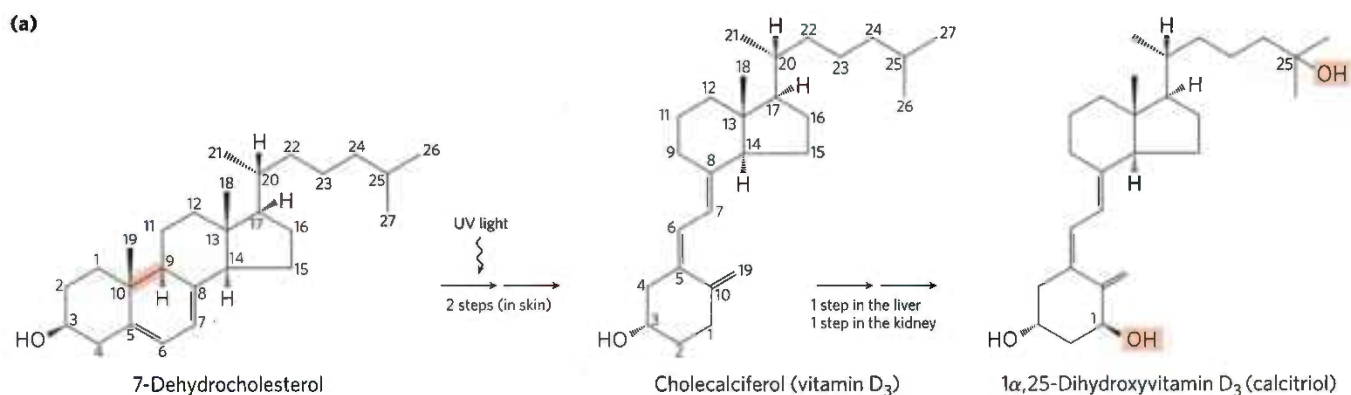


FIGURE 10-19 Vitamin D₃ production and metabolism.

(a) Cholecalciferol (vitamin D₃) is produced in the skin by UV irradiation of 7-dehydrocholesterol, which breaks the bond shaded light red. In the liver, a hydroxyl group is added at C-25; in the kidney, a second hydroxylation at C-1 produces the active hormone, 1 α ,25-dihydroxyvitamin D₃. This hormone regulates the metabolism of Ca²⁺ in kidney, intestine, and bone. (b) Dietary vitamin D prevents rickets, a disease once common in cold climates where heavy clothing blocks the UV component of sunlight necessary for the production of vitamin D₃ in skin. Rickets results in weak or soft bones in children; it can often be identified by bowed legs and other bone deformities. [Biophoto Associates/Science Source]



but cannot be synthesized by these animals and must therefore be obtained in the diet. Early nutritional studies identified two general classes of such compounds: those soluble in nonpolar organic solvents (fat-soluble vitamins) and those that could be extracted from foods with aqueous solvents (water-soluble vitamins). **P3** Eventually, the fat-soluble group was resolved into the four vitamin groups A, D, E, and K, all of which are isoprenoid compounds synthesized by the condensation of multiple isoprene units. Two of these (D and A) serve as hormone precursors.

Vitamin D₃, also called **cholecalciferol**, is normally formed in the skin from 7-dehydrocholesterol in a photochemical reaction driven by the UV component of sunlight (Fig. 10-19a). Vitamin D₃ is not itself biologically active, but it is converted by enzymes in the liver and kidney to 1 α ,25-dihydroxyvitamin D₃ (calcitriol), a hormone that regulates calcium uptake in the intestine and calcium levels in kidney and bone. Deficiency of vitamin D leads to defective bone formation and the disease rickets (Fig. 10-19b), for which administration of vitamin D produces a dramatic cure. Vitamin D₂ (ergocalciferol) is a commercial product formed by UV irradiation of the ergosterol of yeast. Vitamin D₂ is structurally similar to D₃, with slight modification to the side chain attached to the sterol D ring. Both have the same biological effects, and D₂ is commonly added to milk and butter as a dietary supplement. The product of vitamin D metabolism, 1 α ,25-dihydroxyvitamin D₃, regulates gene expression by interacting with specific nuclear receptor proteins. We discuss such regulation of gene expression in more detail in Chapter 28.

Vitamin A₁ (all-*trans*-retinol) and its oxidized metabolites retinoic acid and retinal act in the processes of development, cell growth and differentiation, and vision (Fig. 10-20). Vitamin A₁ or β -carotene in the diet can be converted enzymatically to all-*trans*-retinoic acid, a retinoid hormone that acts through a family of nuclear receptor proteins (RAR, RXR, PPAR) to regulate gene expression central to embryonic development, stem cell differentiation, and cell proliferation. All-*trans*-retinoic acid is used to treat certain types of leukemia, and it is the active ingredient in the drug tretinoin (Retin-A), used to treat severe acne and wrinkled skin. In the vertebrate eye, retinal bound to the protein opsin forms the photoreceptor pigment rhodopsin. The photochemical conversion of 11-*cis*-retinal to all-*trans*-retinal is the fundamental event in vision (see Fig. 12-19).

Unlike most vitamins, vitamin A can be stored for some time in the body (primarily as its ester with palmitic acid, in the liver). Vitamin A was first isolated from fish liver oils; eggs, whole milk, and butter are also good dietary sources. Another source is β -carotene (Fig. 10-20), the pigment that gives carrots, sweet potatoes, and other yellow vegetables their characteristic color. Carotene is one of a very large number (>700) of **carotenoids**, natural products with a characteristic extensive system of conjugated double bonds, which makes possible their strong absorption of visible light (450–470 nm).

Vitamin A deficiency in a pregnant woman can lead to congenital malformations and growth retardation in the infant. In adults, vitamin A is also essential to vision, immunity, and reproduction. Deficiency of vitamin A leads to a variety of symptoms, including dryness of the

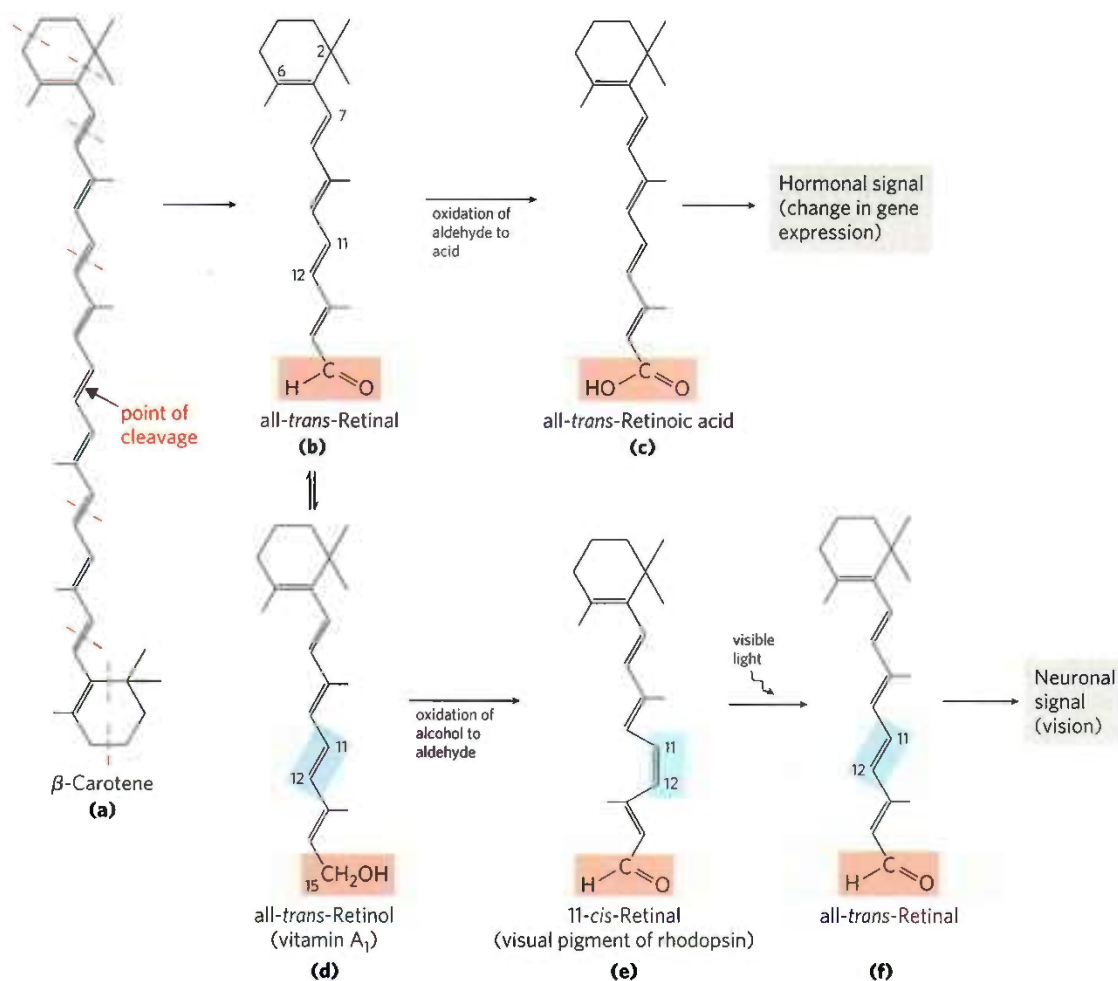


FIGURE 10-20 Dietary β -carotene and vitamin A₁ as precursors of the retinoids. (a) β -Carotene is shown with its isoprene structural units set off by dashed red lines. Symmetric cleavage of β -carotene yields two molecules of all-trans-retinal (b), which can be either further oxidized to all-trans-retinoic acid, a retinoid hormone (c), or reduced to all-trans-retinol, vitamin A₁ (d). In the visual pathway, all-trans-retinol from this reaction, or obtained directly through the diet, can be converted to the aldehyde 11-cis-retinal

(e). This product combines with the protein opsin to form rhodopsin (not shown), a visual pigment widespread in nature. In the dark, the retinal of rhodopsin is in the 11-cis form. When a rhodopsin molecule is excited by visible light, the 11-cis-retinal undergoes a series of photochemical reactions that convert it to all-trans-retinal (f), forcing a change in the shape of the entire rhodopsin molecule. This transformation in the rod cell of the vertebrate retina sends an electrical signal to the brain that is the basis of visual transduction.



FIGURE 10-21 Carotene-enriched rice. Worldwide, more than 250 million children and pregnant women suffer from vitamin A deficiency, which causes at least 250,000 cases of irreversible blindness each year in children. Half of these children end up dying within a year of losing their sight. This deficiency is particularly prevalent where rice is a staple food. An international humanitarian effort — the Golden Rice Project — has made great strides in addressing this health crisis. Wild-type rice grains (left) do not produce β -carotene, the metabolic precursor of vitamin A. Rice plants have been genetically engineered to produce β -carotene in the grain, which takes on the yellow color of the carotene (right). A diet supplemented with Golden Rice provides enough β -carotene to prevent vitamin A deficiency and its tragic health consequences. [© Golden Rice Humanitarian Board (www.goldenrice.org)]

skin, eyes, and mucous membranes, and night blindness, an early symptom commonly used in diagnosing vitamin A deficiency. In the developing world, vitamin A deficiency causes an estimated 250,000 or more cases of blindness or death in children each year. One effective strategy for providing vitamin A is the metabolic engi-

neering of rice strains to overproduce β -carotene. Rice has all the enzymatic machinery to produce β -carotene in its leaves, but these enzymes are less active in the grain. Introduction of two genes into the rice has resulted in "golden rice" with grains enriched in β -carotene (Fig. 10-21). ■

Vitamins E and K and the Lipid Quinones Are Oxidation-Reduction Cofactors

 **Vitamin E** is the collective name for a group of closely related lipids called **tocopherols**, all of which contain a substituted aromatic ring and a long isoprenoid side chain (**Fig. 10-22a**). Because they are hydrophobic, tocopherols associate with cell membranes, lipid deposits, and lipoproteins in the blood. Tocopherols are biological antioxidants. The aromatic ring reacts with and destroys the most reactive forms of oxygen radicals and other free radicals, protecting unsaturated fatty acids from oxidation and preventing oxidative damage to membrane lipids, which can cause cell fragility. Tocopherols are found in eggs and vegetable oils and are especially abundant in wheat germ. Laboratory animals fed

diets depleted of vitamin E develop scaly skin, muscular weakness and wasting, and sterility. Vitamin E deficiency in humans is very rare; the principal symptom is fragile erythrocytes.

The aromatic ring of **vitamin K** (Fig. 10-22b) undergoes a cycle of oxidation and reduction during the formation of active prothrombin, a blood plasma protein essential in blood clotting. Prothrombin is a proteolytic enzyme that splits peptide bonds in the blood protein fibrinogen to convert it to fibrin, the insoluble fibrous protein that holds blood clots together (see Fig. 6-43). Vitamin K deficiency, which slows blood clotting, is extremely uncommon in humans, aside from a small percentage of infants who suffer from hemorrhagic disease of the newborn, a potentially fatal disorder. In the United

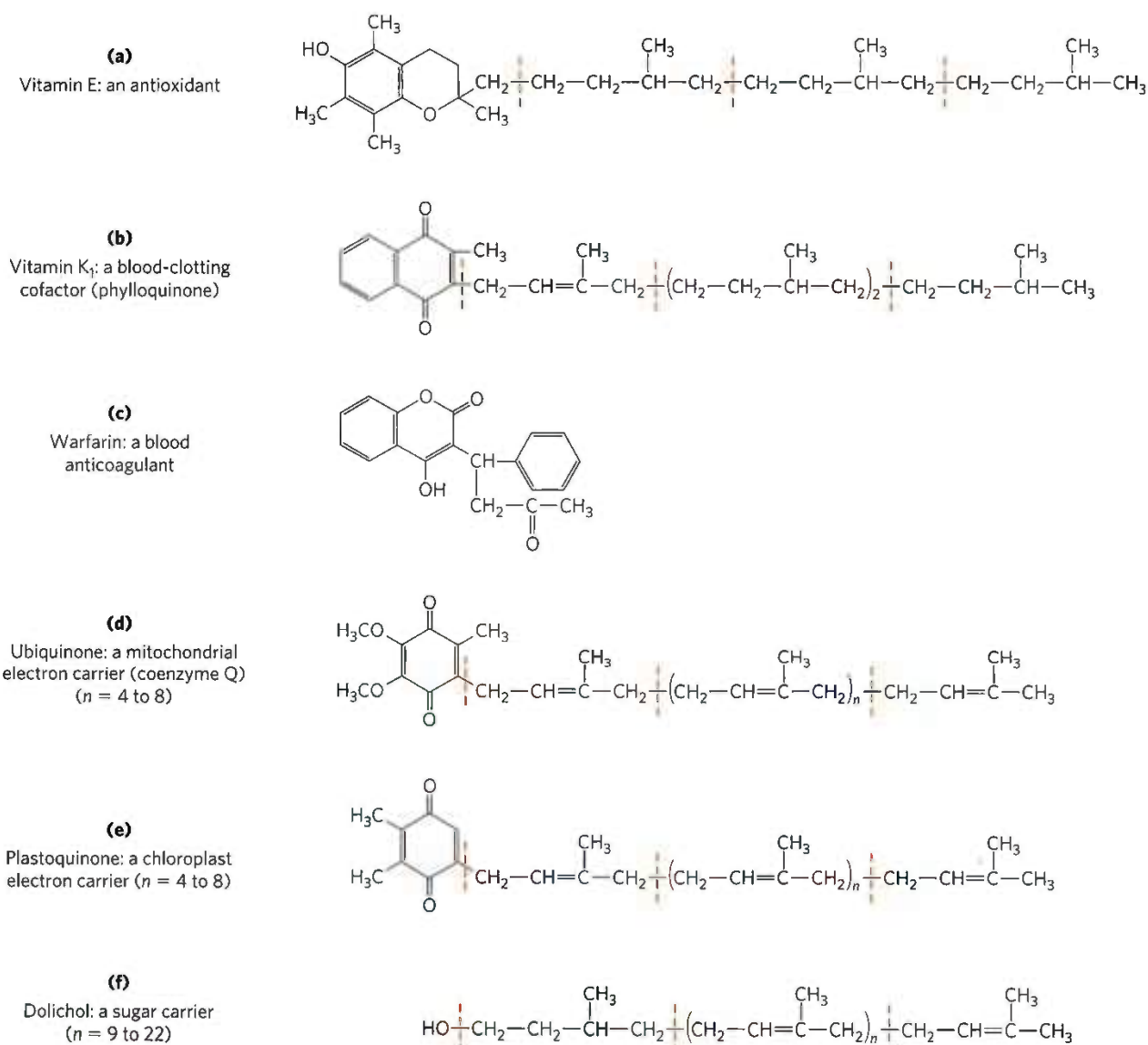


FIGURE 10-22 Some other biologically active isoprenoid compounds or derivatives. Units derived from isoprene are set off by dashed red lines. In most mammalian tissues, ubiquinone (also called coenzyme Q)

has 10 isoprene units. Dolichols of animals have 17 to 21 isoprene units (85 to 105 carbon atoms), bacterial dolichols have 11, and dolichols of plants and fungi have 14 to 24.

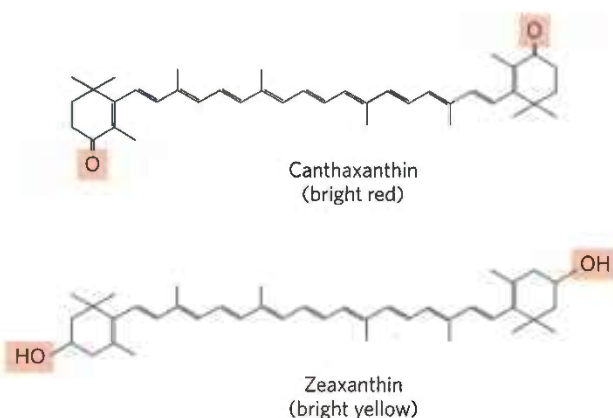


FIGURE 10-23 Lipids as pigments in plants and bird feathers.

Compounds with long conjugated systems absorb light in the visible region of the spectrum. Subtle differences in the chemistry of these compounds produce pigments of strikingly different colors. Birds acquire the pigments



that color their feathers red or yellow by eating plant materials that contain carotenoid pigments, such as canthaxanthin and zeaxanthin. The differences in pigmentation between male and female birds are the result of differences in intestinal uptake and processing of carotenoids. [Gaertner/Alamy Stock Photo]

States, newborns are routinely given a 1 mg injection of vitamin K. Vitamin K₁ (phylloquinone) is found in green plant leaves; a related form, vitamin K₂ (menaquinone), is formed by bacteria living in the vertebrate intestine.

Warfarin (Fig. 10-22c) is a synthetic compound that inhibits the formation of active prothrombin. It is particularly poisonous to rats, causing death by internal bleeding. Ironically, this potent rodenticide is also an invaluable anticoagulant drug for treating humans at risk for excessive blood clotting, such as surgical patients and those with coronary thrombosis. ■

Ubiquinone (also called coenzyme Q) and plastoquinone (Fig. 10-22d, e) are isoprenoids that function as lipophilic electron carriers in the oxidation-reduction reactions that drive ATP synthesis in mitochondria and chloroplasts, respectively. Both ubiquinone and plastoquinone can accept either one or two electrons and either one or two protons (see Fig. 19-3).

Dolichols Activate Sugar Precursors for Biosynthesis

During assembly of the complex carbohydrates of bacterial cell walls, and during the addition of polysaccharide units to certain proteins (glycoproteins) and lipids (glycolipids) in eukaryotes, the sugar units to be added are chemically activated by attachment to isoprenoid alcohols called **dolichols** (Fig. 10-22f). These compounds have strong hydrophobic interactions with membrane lipids, anchoring the attached sugars to the membrane, where they participate in sugar-transfer reactions.

Many Natural Pigments Are Lipidic Conjugated Dienes

Conjugated dienes have carbon chains with alternating single and double bonds. Because this structural arrangement allows the delocalization of electrons, the compounds can be excited by low-energy electromagnetic radiation (visible light), giving them colors visible

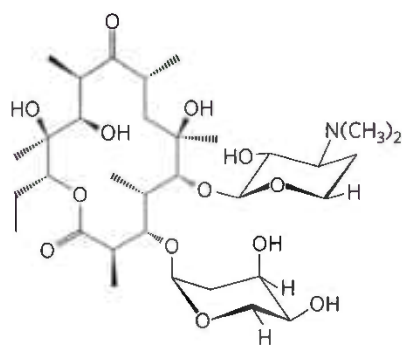
to humans and other animals. Carotene (Fig. 10-20) is yellow-orange; similar compounds give bird feathers their striking reds, oranges, and yellows (Fig. 10-23). Like sterols, steroids, dolichols, vitamins A, E, D, and K, ubiquinone, and plastoquinone, these pigments are synthesized from five-carbon isoprene derivatives; the biosynthetic pathway is described in detail in Chapter 21.

Polyketides Are Natural Products with Potent Biological Activities

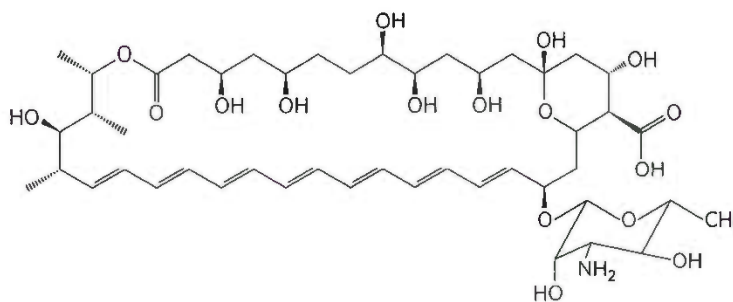
Polyketides are a diverse group of lipids with biosynthetic pathways (Claisen condensations) similar to those for fatty acids. They are **secondary metabolites**, compounds that are not central to an organism's metabolism but serve some subsidiary function that gives the organism an advantage in some ecological niche. Many polyketides find use in medicine as antibiotics (erythromycin), antifungals (amphotericin B), or inhibitors of cholesterol synthesis (lovastatin) (Fig. 10-24). ■

SUMMARY 10.3 Lipids as Signals, Cofactors, and Pigments

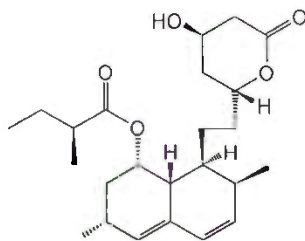
- Some types of lipids, although present in relatively small quantities, play critical roles as cofactors or signals.
- Membrane sphingolipids and derivatives of phosphatidylinositol such as PIP₂ and PIP₃ are used as signaling molecules or for nucleating formation of multiprotein complexes.
- Prostaglandins, thromboxanes, leukotrienes, and lipoxins, all of which are eicosanoids derived from arachidonate, are extremely potent hormones involved in reproduction, inflammation, regulating blood pressure, and other bodily processes. NSAIDs inhibit formation of some prostaglandins and thromboxanes.
- Steroid hormones, such as the sex hormones, are derived from sterols. They serve as powerful biological



Erythromycin (antibiotic)



Amphotericin B (antifungal)



Lovastatin (statin)

FIGURE 10-24 Three polyketide natural products used in human medicine.

signals and move through the bloodstream to alter gene expression in target cells.

- Plants produce volatile lipids to attract or repel other organisms and for communication. Many of these lipids are used as fragrances in perfumes.

- Vitamins D, A, E, and K are fat-soluble compounds made up of isoprene units. All play essential roles in the metabolism or physiology of animals. Vitamin D is precursor to a hormone that regulates calcium metabolism. Vitamin A furnishes the visual pigment of the vertebrate eye and is a regulator of gene expression during epithelial cell growth.

- Vitamins E and K and quinones can be oxidized or reduced by cells. Vitamin E functions in the protection of membrane lipids from oxidative damage, and vitamin K is essential for blood clotting. Quinones are essential for carrying electrons during the reactions that drive ATP synthesis in mitochondria and chloroplasts.

- Dolichols activate and anchor sugars to cellular membranes; the sugar groups are then used in the synthesis of complex carbohydrates, glycolipids, and glycoproteins.

- Lipidic conjugated dienes serve as pigments in flowers and fruits and give bird feathers their striking colors.

- Polyketides are natural products widely used in medicine.

10.4 Working with Lipids

Because lipids are insoluble in water, their extraction and subsequent fractionation require the use of organic solvents and some techniques not commonly used in the purification of water-soluble molecules such as proteins and carbohydrates. **P4** In general, complex mixtures of lipids are separated by differences in polarity or solubility

in nonpolar solvents. Lipids that contain ester- or amide-linked fatty acids can be hydrolyzed by treatment with acid or alkali or with specific hydrolytic enzymes (phospholipases, glycosidases) to yield their components for analysis. Some methods commonly used in lipid analysis are shown in **Figure 10-25** and discussed below.

Lipid Extraction Requires Organic Solvents

Neutral lipids (triacylglycerols, waxes, pigments, and so forth) are readily extracted from tissues with ethyl ether, chloroform, or benzene, solvents that do not permit lipid clustering driven by the hydrophobic effect. Membrane lipids are more effectively extracted by more polar organic solvents, such as ethanol or methanol, which reduce the hydrophobic interactions among lipid molecules while also weakening the hydrogen bonds and electrostatic interactions that bind membrane lipids to membrane proteins. A commonly used extractant is a mixture of chloroform, methanol, and water, initially in volume proportions (1:2:0.8) that are miscible, producing a single phase. After tissue is homogenized in this solvent to extract all lipids, more water is added to the resulting extract, and the mixture separates into two phases: methanol/water (top phase) and chloroform (bottom phase). The lipids remain in the chloroform layer, and the more polar molecules such as proteins and sugars partition into the methanol/water layer (Fig. 10-25a).

Adsorption Chromatography Separates Lipids of Different Polarity

P4 Complex mixtures of tissue lipids can be fractionated by chromatographic procedures based on the different polarities of each class of lipid (Fig. 10-25b).

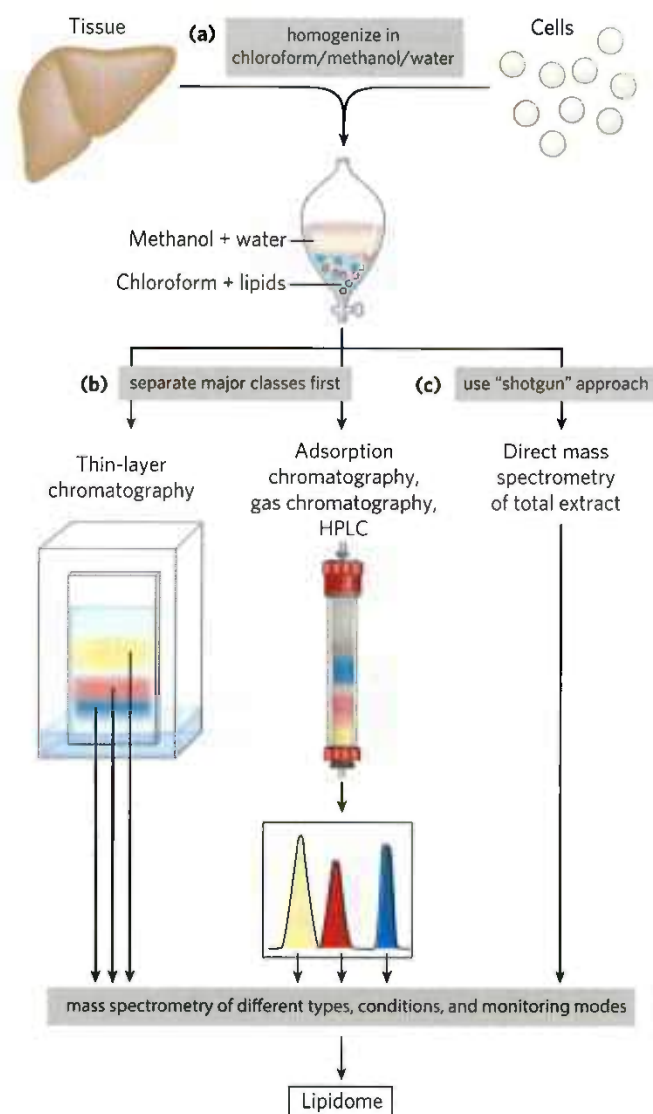


FIGURE 10-25 Common procedures in the extraction, separation, and identification of cellular lipids. (a) Tissue is homogenized in a chloroform/methanol/water mixture, which upon addition of water and removal of unextractable sediment by centrifugation yields two phases. (b) Major classes of extracted lipids in the chloroform phase may first be separated by thin-layer chromatography (TLC), in which lipids are carried up a silica gel-coated plate by a rising solvent front, with less-polar lipids traveling farther than more-polar or charged lipids, or by adsorption chromatography on a column of silica gel, through which solvents of increasing polarity are passed. For example, column chromatography with appropriate solvents can be used to separate closely related lipid species such as phosphatidylserine, phosphatidylglycerol, and phosphatidylinositol. Once separated, each lipid's complement of fatty acids can be determined by mass spectrometry. (c) Alternatively, in the "shotgun" approach, an unfractionated extract of lipids can be directly subjected to high-resolution mass spectrometry of different types and under different conditions to determine the total composition of all the lipids — that is, the lipidome.

In adsorption chromatography, an insoluble, polar material such as silica gel (a form of silicic acid, $\text{Si}(\text{OH})_4$) is packed into a glass column, and the lipid mixture (in chloroform solution) is applied to the top of the column. (In high-performance liquid chromatography, or HPLC, the column

is of smaller diameter and solvents are forced through the column under high pressure.) The polar lipids bind tightly to the polar silicic acid, but the neutral lipids pass directly through the column and emerge in the first chloroform wash. The polar lipids are then eluted, in order of increasing polarity, by washing the column with solvents of progressively higher polarity. Uncharged but polar lipids (cerebrosides, for example) are eluted with acetone, and very polar or charged lipids (such as glycerophospholipids) are eluted with methanol.

Thin-layer chromatography (TLC) on silicic acid employs the same principle (Fig. 10-25b). A thin layer of silica gel is spread onto a glass plate, to which it adheres. A small sample of lipids dissolved in chloroform is applied near one edge of the plate, which is dipped in a shallow container of an organic solvent or solvent mixture; the entire setup is enclosed in a chamber saturated with the solvent vapor. As the solvent rises on the plate by capillary action, it carries lipids with it. The less polar lipids move farthest, as they have less tendency to bind to the silicic acid. The separated lipids can be detected by spraying the plate with a dye (rhodamine) that fluoresces when associated with lipids, or by exposing the plate to iodine fumes. Iodine reacts reversibly with the double bonds in fatty acids, such that lipids containing unsaturated fatty acids develop a yellow or brown color. Several other spray reagents are also useful in detecting specific lipids. For subsequent analysis, regions containing separated lipids can be scraped from the plate and the lipids recovered by extraction with an organic solvent.

Gas Chromatography Resolves Mixtures of Volatile Lipid Derivatives

P4 Gas chromatography (GC) separates volatile components of a mixture according to their relative tendencies to dissolve in the inert material packed in the chromatography column or to volatilize and move through the column, carried by a current of an inert gas such as helium. Some lipids are naturally volatile, but most must first be derivatized to increase their volatility (that is, lower their boiling point). For an analysis of the fatty acids in a sample of phospholipids, the lipids are first transesterified: heated in a methanol/HCl or methanol/NaOH mixture to convert fatty acids esterified to glycerol into their methyl esters. These fatty acyl methyl esters are then loaded onto the GC column, and the column is heated to volatilize the compounds. Those fatty acyl esters that are most soluble in the column material will partition into (dissolve in) that material; the less-soluble lipids are carried by the stream of inert gas and emerge first from the column. The order of elution depends on the nature of the solid adsorbent in the column and on the boiling point of the components of the lipid mixture. With these techniques, mixtures of fatty acids of various chain lengths and various degrees of unsaturation can be completely resolved.

Specific Hydrolysis Aids in Determination of Lipid Structure

P4 Certain classes of lipids are susceptible to degradation under specific conditions. For example, all ester-linked fatty acids in triacylglycerols, phospholipids, and sterol esters are released by mild acid or alkaline treatment, and somewhat harsher hydrolysis conditions release amide-bound fatty acids from sphingolipids. Enzymes that specifically hydrolyze certain lipids are also useful in the determination of lipid structure. Phospholipases A, C, and D (Fig. 10-14) each split particular bonds in phospholipids and yield products with characteristic solubilities and chromatographic behaviors. Phospholipase C, for example, releases a water-soluble phosphoryl alcohol (such as phosphocholine from phosphatidylcholine) and a chloroform-soluble diacylglycerol, each of which can be characterized separately to determine the structure of the intact phospholipid. The combination of specific hydrolysis with characterization of the products by TLC, GC, or HPLC often allows determination of a lipid structure.

Mass Spectrometry Reveals Complete Lipid Structure

P4 To establish unambiguously the length of a hydrocarbon chain or the position of double bonds, mass spectrometric analysis of lipids or their volatile derivatives is

invaluable. The chemical properties of similar lipids (for example, two fatty acids of similar length unsaturated at different positions, or two isoprenoids with different numbers of isoprene units) are very much alike, and their order of elution from the various chromatographic procedures often does not distinguish between them. When the eluate from a chromatography column is sampled by mass spectrometry, however, the components of a lipid mixture can be simultaneously separated and identified by their unique pattern of fragmentation (Fig. 10-26). With the increased resolution of mass spectrometry, it is possible to identify individual lipids in very complex mixtures without first fractionating the lipids in a crude extract. This “shotgun” method (Fig. 10-25c) avoids losses during the preliminary separation of lipid subclasses, and it is faster.

Lipidomics Seeks to Catalog All Lipids and Their Functions

As lipid biochemists have become aware of the thousands of different naturally occurring lipids, they have created a database analogous to the Protein Data Bank. The LIPID MAPS Lipidomics Gateway (www.lipidmaps.org) has its own classification system that places each lipid species in one of eight chemical categories, each designated by two letters (Table 10-2). Within each category, finer distinctions are indicated by numbered classes and subclasses.

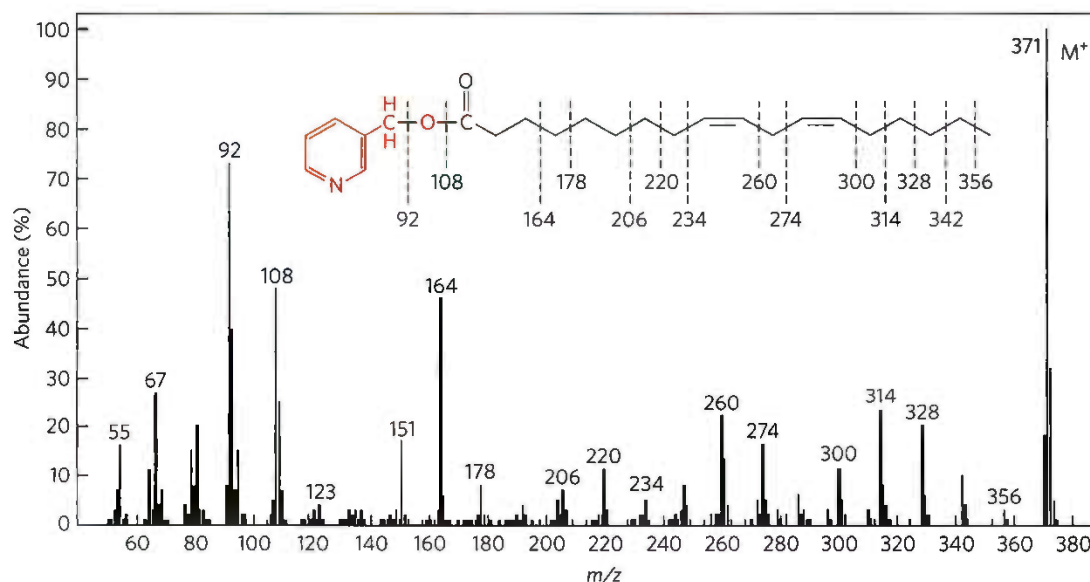


FIGURE 10-26 Determination of fatty acid structure by mass spectrometry. The fatty acid is first converted to a derivative that minimizes migration of the double bonds when the molecule is fragmented by electron bombardment. The derivative shown here is a picolinyl ester of linoleic acid — 18:2($\Delta^{9,12}$) (M_r 371) — in which the alcohol is picolinol (red). When bombarded with a stream of electrons, this molecule is volatilized and converted to a parent ion (M^+ ; M_r 371), in which the N atom bears the positive charge, and a series of smaller fragments produced by breakage of C—C bonds in the fatty acid. The mass spectrometer separates these charged fragments according to their mass/charge ratio (m/z).

The prominent ions at m/z = 92, 108, 151, and 164 contain the pyridine ring of the picolinol and various fragments of the carboxyl

group, showing that the compound is indeed a picolinyl ester. The molecular ion, M^+ (m/z = 371), confirms the presence of a C_{18} fatty acid with two double bonds. The uniform series of ions 14 atomic mass units (u) apart represents loss of each successive methyl and methylene group from the methyl end of the acyl chain (beginning at C-18; the right end of the molecule as shown here), until the ion at m/z = 300 is reached. This is followed by a gap of 26 u for the carbons of the terminal double bond, at m/z = 274; a further gap of 14 u for the C-11 methylene group, at m/z = 260; and so forth. By this means, the entire structure is determined, although these data alone do not reveal the configuration (cis or trans) of the double bonds. [Information from W. W. Christie, *Lipid Technol.* 8:64, 1996.]

TABLE 10-2 Eight Major Categories of Biological Lipids

Category	Category code	Examples
Fatty acids	FA	Oleate, stearoyl-CoA, palmitoylcarnitine
Glycerolipids	GL	Di- and triacylglycerols
Glycerophospholipids	GP	Phosphatidylcholine, phosphatidylserine, phosphatidylethanolamine
Sphingolipids	SP	Sphingomyelin, ganglioside GM2
Sterol lipids	ST	Cholesterol, progesterone, bile acids
Prenol lipids	PR	Farnesol, geraniol, retinol, ubiquinone
Saccharolipids	SL	Lipopolysaccharide
Polyketides	PK	Tetracycline, erythromycin, aflatoxin B ₁

For example, all glycerophosphocholines are GP01. The subgroup of glycerophosphocholines with two fatty acids in ester linkage is designated GP0101; the subgroup with one fatty acid ether-linked at position 1 and one ester-linked at position 2 is GP0102. The specific fatty acids are designated by numbers that give every lipid its own unique identifier, so that each individual lipid, including lipid types not yet discovered, can be unambiguously described in terms of a 12-character identifier, the LM_ID. One factor used in this classification system is the nature of the biosynthetic precursor. For example, prenyl lipids (such as dolichols and vitamins E and K) are formed from isoprenyl precursors.

The eight chemical categories in Table 10-2 do not coincide perfectly with the less formal categorization according to biological function that we have used in this chapter. For example, the structural lipids of membranes include both glycerophospholipids and sphingolipids, which are separate categories in Table 10-2. Each method of classification has its advantages.

The application of mass spectrometric techniques with high throughput and high resolution can provide quantitative catalogs of all the lipids present in a specific cell type under particular conditions—the **lipidome**—and of the ways in which the lipidome changes with differentiation, disease such as cancer, or drug treatment. An animal cell contains more than a thousand different lipid species, each presumably having a specific function. These functions are known for a growing number of lipids, but the still largely unexplored lipidome offers a rich source of new problems for the next generation of biochemists and cell biologists to solve.

SUMMARY 10.4 Working with Lipids

- In the determination of lipid composition, lipids can first be extracted from tissues with organic solvents.
- Lipids in mixtures can be separated on the basis of their polarity and interactions with polar materials such as silica, using adsorption chromatography methods such as HPLC or TLC.
- GC volatilizes lipids so that they can be carried by a stream of inert gas and resolved based on their ability to partition into a soluble column material.

■ Phospholipases specific for one of the bonds in a phospholipid can be used to generate simpler compounds for subsequent analysis.

■ High-resolution mass spectrometry allows the analysis of crude mixtures of lipids without prefractionation—the “shotgun” approach.

■ Lipidomics combines powerful analytical techniques to determine the full complement of lipids in a cell or a tissue (the lipidome) and to assemble annotated databases that allow comparisons between lipids of different cell types and under different conditions.

KEY TERMS

Terms in bold are defined in the glossary.

fatty acid 341	sterol 352
polyunsaturated fatty acid (PUFA) 343	cholesterol 352
omega-3 (ω -3) fatty acids 343	bile acids 352
triacylglycerol 344	eicosanoid 355
lipases 345	prostaglandin (PG) 355
phospholipid 346	thromboxane (TX) 355
sterols 346	leukotriene (LT) 355
glycerophospholipid 346	lipoxin (LX) 356
ether lipid 348	vitamin 356
plasmalogen 348	vitamin D ₃ 357
glycolipid 349	cholecalciferol 357
galactolipid 349	vitamin A ₁ (all- <i>trans</i> -retinol) 357
sphingolipid 350	carotenoids 357
ceramide 350	vitamin E 359
sphingomyelin 350	tocopherol 359
glycosphingolipid 350	vitamin K 359
cerebroside 350	dolichol 360
globoside 351	polyketide 360
ganglioside 351	lipidome 364

PROBLEMS

- Operational Definition of Lipids** How is the definition of “lipid” different from the types of definitions used for other biomolecules such as amino acids, nucleic acids, and proteins?
- Structure of an Omega-3 Fatty Acid** The omega-3 fatty acid docosahexaenoic acid (DHA, 22:6($\Delta^{4,7,10,13,16,19}$)) is the